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1 Acknowledgement

40 Sincere gratitude is extended to Miss Shresha Raj Bhandari and Mr. Rabin Sapkota for their valuable guidance, encouragement, and continuous support throughout the entire Final Year Project journey.

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The successful completion of SmartSelf would not have been possible without their constant motivation, support, and expert guidance throughout every stage of the project.

Abstract

SmartSelf is a mobile-based smart e-book reading system that wants to help digital reading become better by adding smart and interactive features that enable efficient and self-directed reading. The system is designed to promote accessibility and engagement for smart e-book readers through a resume reading option, text highlighting, reading progress, streak monitoring, and text-to-speech (TTS).

Besides, SmartShelf encompasses a recommendation mechanism to suggest relevant book recommendations based on the user's preferences and reading books, enabling a reader to discover relevant content. Also, SmartSelf stores user's actions and reading progress in a database to resume reading without delays. Together with usability, accessibility, and smart tracking, the system seeks to promote digital reading habits and bring the consumption of books to more convenient and relevant readers, for modern ones.

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1. Introduction

1.1 Project overview

In the digital age, the way individuals read, and access books has been significantly influenced by the advent of electronic books (e-books) and online reading platforms. Traditional printed books, despite their merit, require a lot of natural resources which includes paper, water, and energy during their production and distribution, with environmental sustainability and resource conservation becoming increasingly pressing concerns. The idea for this project is SmartShelf a digital e-book management and reading platform aimed at increasing organisational and environmental efficiency when it comes to book reading.

SmartShelf is a digital reading platform that allows users to manage and read e-books in an exciting and interactive manner. The shelf recommends books according to their tastes, measures reading progress, and establishes reading goals to keep readers consistent with their books. Also, SmartShelf supports not only the arrangement and reading of books but also listening to them, adapting the chosen section's text with the help of a Text-to-Speech (TTS) function, supporting the digital reading experience smoothly and accessibly for everyone.

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The system is designed to enable users to easily browse, search, and organise their reading materials, as well as encourage the transition to digital reading. SmartShelf further aids environmental conservation efforts through disincentivising reliance on printed books, thus reduces the use of paper, ink, energy, and other resources associated with conventional book production and distribution methods.

60

With the global e-book market expected to grow from USD 18.02 billion in 2025 to USD 22.76 billion by 2030 manage by mobile first reading and interactive content, there is a clear opportunity for SmartShelf in order to provide innovative, reader-focused features that enhance engagement and accessibility, focused on a specific genre (Mordor intelligence, 2025).

68

1



Figure 1: E-Book Market Size & Share Analysis (Mordor Intelligence, 2025)

Overall, SmartShelf is a smart, sustainable, and user-friendly solution for the modern reader that complements the global digital transformation trend, especially for educative and professional reading.

1.2 Current Scenario

In this 21st century, readers are more aware than ever before of the environmental impact of printed books. Traditional publishing requires tons of paper, ink, and energy while transportation adds carbon emissions, contributing to deforestation and waste.

Efforts to reduce this impact have been taken at a global level. For example, some publishers have resorted to the use of recycled paper to promote sustainability. Consequently, reading digitally through e-books and online platforms has become more common because it is convenient and has less detrimental impact on the environment.

Besides, the development of applications (apps) for mobile phones has considerably impacted upon the sources of reading materials and the modes of using them. Many users are preferring to read through apps on smartphones, as they offer mobility, immediacy, and convenience. Popular e-book apps allow users to search, organize, and read books anytime and anywhere.

However, despite the availability of these applications, many existing e-book apps still have a number of limitations. There is not much variety in the addition of TTS (Text-to-Speech), reading history, tracking progress, milestone setting, recommendations, or interactive elements in many platforms. Some other limitations relate to disorganized digital content, slow app performance, and complicated navigation.

Although e-books help in cutting down on physical wastage, downloading multiple book titles can overwhelm the device's memory space, thus creating digital clutter. This makes it inconvenient to manage collections, and it becomes less encouraging for readers to navigate more content.

Research on current E-Book platforms like Kindle, Google Books, Project Gutenberg, and Open Library shows why we need to develop the SmartShelf System. Open Library and Project Gutenberg provide free and open access to books, but they do not have many features like text-to-speech (TTS), recommendations, and milestone setting. Whereas Google Books provides an excellent search function, it still does not have milestone setting, recommendations, and TTS features.

Additionally, SmartShelf features like TTS, milestone setting, and recommendations. This integrated feature set makes SmartShelf more suitable for e-book readers by addressing gaps present in existing systems.

1.3 Problem Statement

The increased dependence on digital technology has revolutionized reading, learning, and information acquisition techniques worldwide. Although e-books and digital reading platforms have emerged as a promising alternative to traditional printed books, significant challenges remain., preventing users from having a truly sustainable, accessible, and engaging reading experience.

Digital reading platforms' current state both globally and in the case of Nepal shows some essential gaps in functionality, usability, and environmental responsibility.

These problems were identified through a review of existing literature and research, an analysis of like systems, and a primary survey (175 respondents) whose responses validate directly the need for a platform like SmartShelf.

Notably, the survey asked respondents, "What is your age group?" and 93.1% belonged to 18–25 age group, confirming that young, digitally active users form the primary audience of this platform and amongst users most affected by problems described below.

1.1.1 Overconsumption of resources

1.1.2 Digital Clutter and Storage Issues

1.1.3 Difficulty in Tracking Reading Progress

1.1.4 Lack of Engagement

1.1.5 Seamless digital reading

[Click Here 1.1.1 Overconsumption of resources](#)

1.4 Project as a solution

To address the environmental, usability, and engagement challenges of reading, the proposed solution is SmartShelf, a digital reading platform designed to provide a sustainable, seamless, and engaging reading experience. Every one ton of paper saves, 17 trees, 380 gallons of oil, three cubic yards of landfill space, 4,000 kWh of energy, and 7,000 gallons of water equal to a 64% energy savings, 58% water savings, and 60 pounds less air pollution (Florida Tech, 2025). Thus, most people now prefer e-learning and e-books because of their generally low environmental impact-there are some health considerations, though. As Cleantech concluded in its study, buying three e-books a month for a year creates about 168 kg of CO₂, whereas buying an equal number of printed books generates about 1,074 kg of CO₂. E-books save energy used in production, protect trees, reduce paper usage, avoid packaging energy, and minimize the environmental impact of recycling or disposal (Adhikari, 2020).

1.2.1 Overconsumption of Resources

1.2.2 Digital Clutter and Storage Issues

1.2.3 Difficulty in Tracking Reading Progress

1.2.4 Lack of Engagement

1.2.5 Seamless digital reading

[Click Here 1.2.1 Overconsumption of Resources](#)

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1.5 Aims and Objectives

SmartShelf aim to develop a digital e-book platform that will ensure a sustainable, seamless, and engaging reading experience by reducing the impacts on the environment, improving the usability of the system, and enhancing the engagement level of the readers.

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1.5.1 Academic Objectives

1. To understand the development of a digital e-book platform using modern web technologies.
2. To learn and implement the use of integration of Application Programming Interfaces (APIs).
3. To learn to design a user-friendly interface that increases the usability and engagement of the product.
4. To use problem solving and project managing skills in a real-world software development scenario.

1.5.2 Project Objectives

1. To develop the digital e-book platform (SmartShelf) that is sustainable and eco-friendly.
2. To tracking Reading progress, implement milestones, reminders.
3. To offer a seamless digital reading experience with features like quick loading, easy navigation.
4. Reduce digital clutter through organized library management and search function.
5. To provide a content-based recommendation features to enhance user engagement.
6. For the integration of the Text-to-Speech (TTS) option which would allow the users to listen to the selected words, sentences, or paragraph.

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1.6 Report structure of report

1.6.1 Introduction

1.6.2 Background

93

1.6.3 Development

1.6.3 Testing and analysis

1.6.5 Conclusion

[For report structure of report click here for all content](#)

2. Background

The quick growth in technology recently has changed the way we create, share, and consume information, due to their convenience and lower environmental impact, eBooks and digital libraries are a growing source of reading as well as learning materials. Additionally, many organizations for example (IT offices) are using digital systems to create a green organization while encouraging productivity and effectively managing their knowledge resources.

Although there are a lot of digital reading platforms available, which has many problems which includes poor organization of digital content, little or no engagement features, and being unable to track one's reading progress. Given all these problems, digital reading systems in educational and professional settings are not very efficient. Therefore, there is clear need for a comprehensive, user-friendly, feature-rich digital reading solution that is geared toward use in an organization.

The SmartShelf is created to meet these needs by providing an organized, user-friendly, and engaging digital e-book platform that can be used.

2.1 About Target Users/End User

SmartShelf system is designed for book readers who require access to digital books and documents for reading, learning, and reference purposes.

The platform is developed to support book reading activities by providing:

- Easy search and access to digital books
- Organized reading content
- Reading progress tracking and milestone management
- Accessibility support through Text-to-Speech (TTS)

These features enhance the overall reading experience, improve content accessibility, and support different reading preferences in a structured digital environment

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2.2 About the Client

The client for SmartShelf project is the IT Office named Datum Systems Pvt. Ltd., which is located in Kathmandu. The IT Office act as a client and be involved in the development of SmartShelf by providing requirements, resources and support. Since the IT Office produces technical documentation, digital evidence, and Professional Learning materials, it is an ideal actual client for the SmartShelf since its needs match those of the SmartShelf very well.

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2.2 Understanding of the Project

SmartShelf is a digital book management and reading platform which is specifically for users in an IT office where users can easily register and log in and they can search for books as well as they can read eBooks in their browser also can track their reading progress and set milestones to read book. One of the unique features of SmartShelf is (TTS) which is known as text-to-speech feature that allows users to listen to the selected text, which makes the application more accessible and supports diverse learning styles.)

This project also provides the personalized book recommendations to help users to read books and stay engaged with this platform. SmartShelf including React for front-end development, Node.js and Express.js for back-end development, and MongoDB for data storage. SmartShelf uses Scrum by developing features incrementally in multiple sprints for providing the flexibility, frequent testing, as well as improving the SmartShelf project based on user and client input. In summary, SmartShelf is a digital reading solution that is sustainable, efficient, easy to use, and customized to the operational and instructional needs of an IT office.

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2.3 Review of Similar Project

Many projects have been developed to solve similar problems, and by conducting research on these projects, I was able to gain insights into user needs and areas of improvement. Many of the popular digital reading platforms, such as Amazon Kindle, Google Books, Open Library, and Project Gutenberg, are commercial eBooks therefore, they lack many features for users with disabilities and do not provide enough options to access ebooks. SmartShelf main goal is to create a free, with the features, including in-browser reading and progress tracking, as well as text-to-speech (TTS) and as well as recommendations. Research on existing platforms helped me to clear understanding of what's necessary for an eBook system, as well as the needs of IT Office users, and helped me create a system that addresses these gaps, by providing an eBook platform that is accessible and engaging for IT Office users.)

2.3.1 Review of Similar Projects

The Amazon Kindle

The Amazon Kindle is a widely used e-book platform which includes the e-book marketplace, along with the reading applications, gadgets and tools used to access the e-books. In kindle users can purchase and download a huge selection of digital content such as books, magazines, newspapers, etc. this is not a app it is like a tab.



Figure 2 Kindle

The Open Library

Open Library is the internet archive way to preserve books digitally which provides us free digital copies of millions of books for borrowing or online viewing. The Open Library project has an ultimate goal of creating a web page for each and every book ever published. Open Library promotes free access to and open knowledge of all books. This app doesn't have feature like, milestones setup and recommendation.

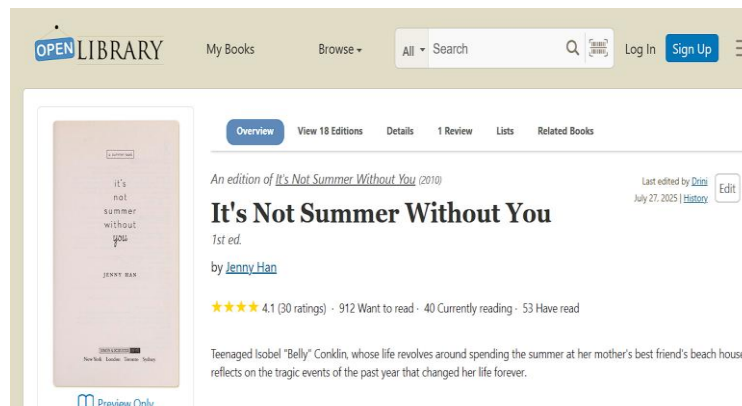


Figure 3 Open Library

The Google Book

Google Books is service by Google which helps people to find books by the names of author, titles as well as browse through user contents. Google books has introduced a new way to find new books and do research on found ones. In these users can search through the entire book. Some books have only partial views and can be purchased or accessed via other means. This doesn't have feature like TTS, milestones setup and recommendation.

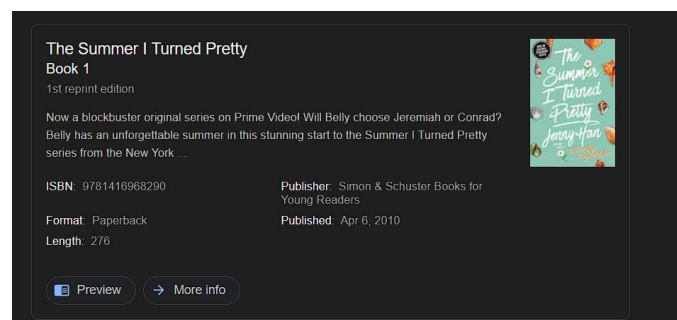


Figure 4 Google Book

Project Gutenberg

Project Gutenberg is oldest e- library which is free with public domain which has feature of downloading book, read books without any kind of registration. This app doesn't have feature like TTS, milestones setup and recommendation.

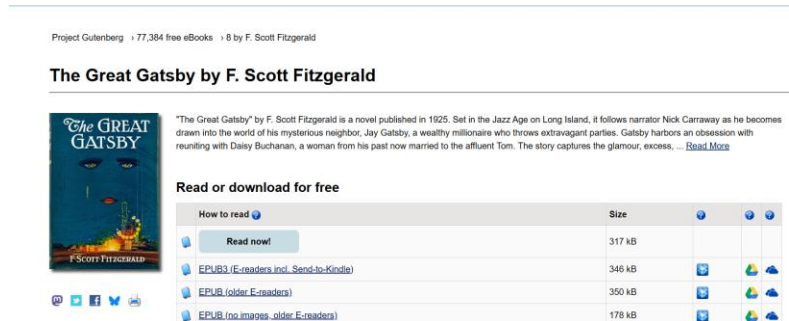


Figure 5 Project Gutenberg

2.3.2 Comparison of similar project

Table 1 Compare Project

Feature	Kindle	Open Library	Google Books	Project Gutenberg	SmartShelf (My Project)
Free Access	X	✓	X	✓	✓
Paid Content	✓	X	X	X	✓
User Account Required	✓	✓	✓	X	✓
In-Browser Reading	X	✓	✓	✓	✓
Text-to-Speech (TTS)	X	X	X	X	✓
Progress Tracking	✓	X	X	X	✓
Recommendations	X	X	X	X	✓
Search Functionality	✓	X	✓	X	✓
User Comments	✓	X	X	X	✓

Research on current E-Book platforms like kindle, google books, project Gutenberg and open library shows why we need to develop SmartShelf System the Amazon Kindle has strong feature like paid content and user reviews, to use this kind of feature we need to pay huge subscription but smartshelf is much more affordable. Open Library and Project Gutenberg provide us free and open access to books. But they don't have many features likes text-to-speech TTS, recommendations, and to set milestone whereas Google Books provide excellent search function still google books does not have milestone setting, recommendations, features, like TTS.

Additionally, SmartShelf has free access to book with the feature of TTS, milestone setting, recommendation with less subscription. This integrated feature set makes SmartShelf more suitable for e -book readers by addressing gaps present in existing systems.

3. Development

3.1 Software Development Methodology

A software development methodology is a form of managing a software development project. This normally includes issues such as which features to incorporate into the current version, the release schedules of software, assignment of workers to various tasks, and testing procedures (Young, 2013). It is used for structuring, planning, and controlling the software development process. Its goal is to ensure delivery of software as per project specifications, on time and on budget, with minimized project risks (Nikitin, 2025). Generally, it is made to describe the 'how' of a life cycle of a piece of software.

3.1.1 Considered Methodology

3.1.1.1 Waterfall Methodology

Waterfall methodology refers to a rigid; linear project management workflow whose phases require completion before proceeding to the next phase. It was first introduced by Winston W. Royce in 1970 (Atlassian, 2025).

This methodology, commonly used by engineers, is front-loaded to rely on careful planning, detailed documentation, and consecutive execution (Adobe for Business, 2025).

[More about Waterfall Methodology Click Here](#)

3.1.1.2 Prototype Methodology

The prototyping model software development lifecycle (SDLC) is the process that development teams use to design and build high-quality software at reasonable costs and time (Nacholas, 2024). The process of creating a prototype involves a step which is based on testing results and feedback which includes development, testing, and continuous improvement (Yang, 2023).

Prototypes are meant to be used by the development team together with the customers and stakeholders to verify the product's design, architecture, and functionality (Yang, 2023).

[More about Prototype Methodology Click Here](#)

3.1.2 Used Methodology

3.1.2.1 Scrum Methodology

Scrum is one of the most widely used agile frameworks emphasizing on teamwork that aids in delivering quality products within deadlines (GeeksforGeeks, 2025). Scrum advocates empirical learning, self-organization while solving a problem and making new strategies concerning what went well and what did not regarding the last case (Drumond, 2025).

[More about Scrum Methodology Click Here](#)

3.3 Phases of Methodology and its Implementation

3.3.1 Initiation

This is the first phase of the SmartShelf project, in which an overall vision and purpose for the system are determined. SmartShelf aims at creating a multi-device platform of a web application for admin and a mobile application to be used by users for managing and reading books effectively. Requirements are gathered and analyzed in this phase to understand user needs like authentication, searching for books, reading e-books, having text-to-speech, and recommendations.

Tasks done in this phase include.

- Project Aims and Objectives Definition (SmartShelf system development)
- Identifying the main features (authentication, reader, TTS, recommended literature, admin module)
- Looking at Similar Platforms (e.g., E-book Apps)
- Creating the Product Backlog in the form of user stories
- Project structuring, Tools and Environment Setup

3.3.2 Planning and Estimation

In this phase, the project is split into various sprints based on the WBS. The sprints revolve around major modules like User Authentication, Book Search, E-book Reader, Text-to-Speech, Reading Goals, Recommendations, and Admin Module.

Planning is done iteratively to ensure that there is smooth development throughout the project lifecycle.

Tasks involved.

Decomposing the Product Backlog into Sprints;

Sprint 1: User Authentication

Sprint 2: Book Search

Sprint 3: E book reader

Sprint 4: Text-to-Speech Sprints 5: Reading Purpose

Sprint 6: Book Recommendation

Sprint 7: Admin module

Sprint 8: Testing & Deployment

- Creating goals for each sprint
- Sprint Backlog Creation
- Estimating Time and Resources per Task
- UI/UX Design and System Architecture Plan

3.3.3 Implementation

The real development of SmartShelf happens during this phase. Sprints are executed according to planning, designed, developed, and tested features.

Key implementation activities:

- Frontend development (React Native for mobile, web for admin)
- Backend Development (Node.js / Express / MongoDB – MERN stack)

Features to consider in implementation:

- Authentication of a user (login, registration, google-login)
- Book search with API
- E-book reader (EPUB integration)

- Text-to-Speech service
- Reading Goals Tracking
- Recommendations system
- Admin dashboard (add/update/delete books)
- Holding daily scrum meetings
- Frequently Updating Progress And Backlog
- Conducting tests (unit tests, feature tests)

3.3.4 Reviewing

After every sprint, a review is carried out to assess the progress of SmartShelf. These feedback are then utilized to improve the system and to make sure the system is valuable to the users.

Tasks at this phase include:

- Presenting already implemented features (login system, reader, TTS).
- Feedback from supervisor/client
- Identify the problems and improve
- Product backlog refinement based on feedback Sprint review and retrospective meetings

3.3.5 Releasing

In this phase, the final works are carried out and the SmartShelf system is deployed and delivered. All modules are integrated and tested for smooth functionality across devices.

Activities included:

- Final integration of web and mobile apps
- Testing on live server (ngrok or deployed domain)
- Troubleshooting bugs and enhancing performance
- Checking the correctness of all features.
- Authentication
- Reading books
- TTS

- Recommendations
- Admin control
- Documentation, final report preparations

3.4 Survey result

3.4.1 Highlights from Pre-Survey

Before developing SmartShelf, a pre-survey was done to assess the behavioral patterns, problems, and expectations of users with regard to e-book reading systems. The responses proved that e-book reading enjoys wide participation but pointed out several limitations of the existing e-book platform.

Some of the highlights of the pre-survey include:

1. Regular use of e-book platforms (daily, weekly, or even occasionally) was reported by a majority of respondents (more than 50%).
2. More than 60% of readers mostly preferred mobile phones and laptops as reading devices, which also informed the need to have a cross-platform mode.
3. Security and ease of login were considered important by a significant part of the sample (approximately 50–70% of respondents).
4. Some of the most demanded features by respondents (50%+) are Text-to-Speech (TTS), capability to recover lost passwords, and smooth navigation.
5. Some other additional features that respondents suggested include recommendations, and tracking, among others. Based on the ratings, dissatisfaction with existing platforms averaged 30-50%.

Google form

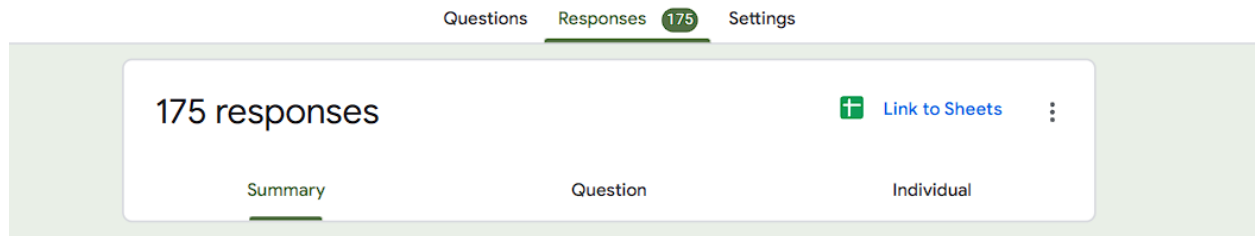


Figure 6 Response 150+ Prove

1. What is your age group?

173 responses

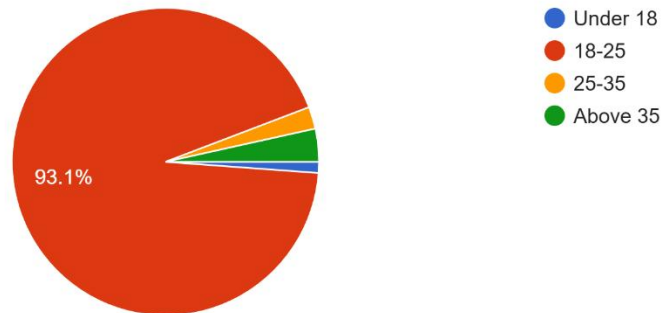


Figure 7 Question 1 Response

2. How often do you read e-books?

173 responses

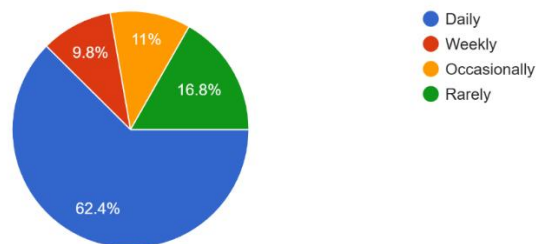


Figure 8 Question 2 Response

3. Which login method do you prefer?

173 responses

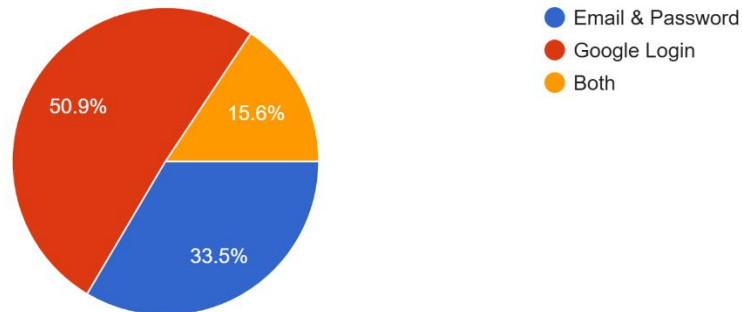


Figure 9 Question 3 Response

4. Do you prefer reading e-books on?

173 responses

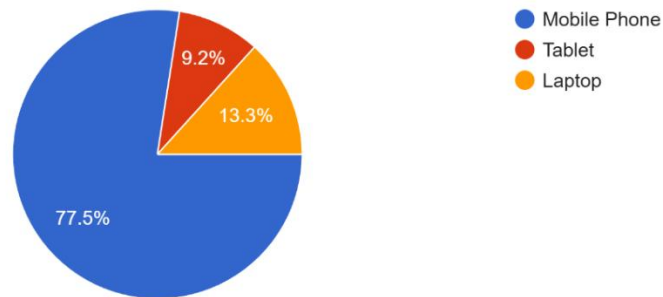


Figure 10 Question 4 Response

5. Are you aware of the Text-to-Speech (TTS) feature in e-book applications?

122 responses

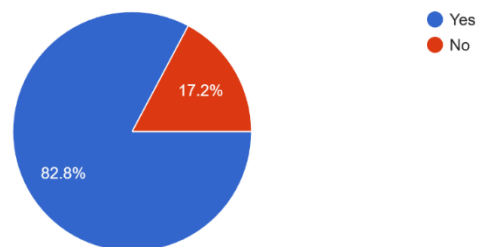


Figure 11 Question 5 Response

6. What feature do you find most important in an e-book system?

173 responses

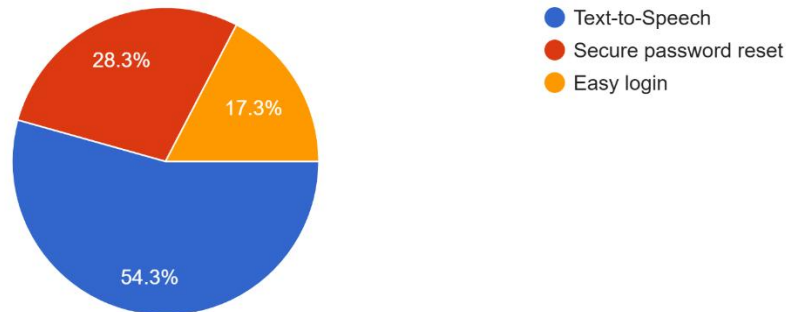


Figure 12 Question 6 Response

7. Would additional features like Setting milestone enhance your reading experience?

173 responses

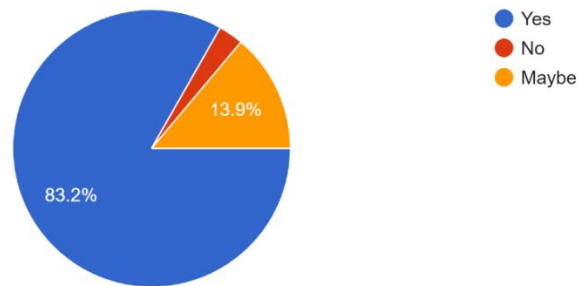


Figure 13 Question 7 Response

8. Would you use an e-book app with secure login and Google authentication?

173 responses

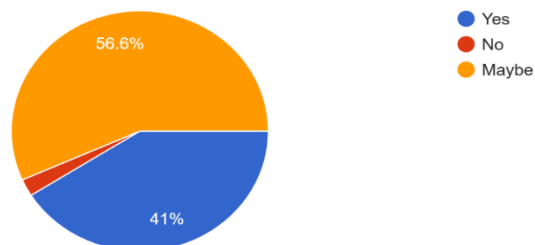


Figure 14 Question 8 Response

9. How satisfied are you with current e-book platforms?

173 responses

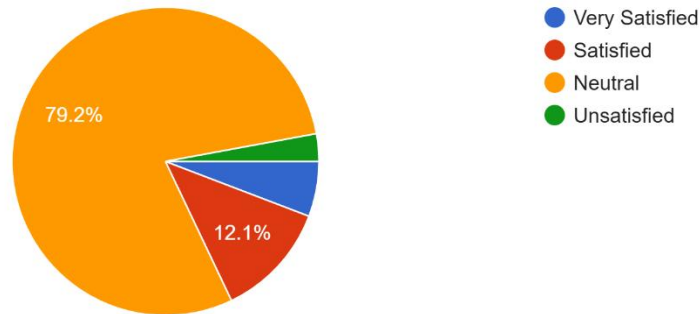


Figure 15 Question 9 Response

10. Have you ever stopped using an e-book app due to poor user experience?

173 responses

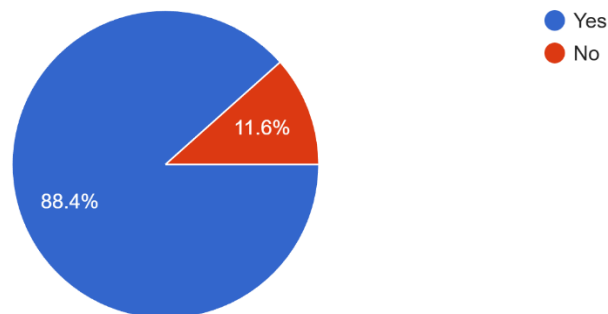


Figure 16 Question 10 Response

11. Would you like book recommendations in an e-book app?

173 responses

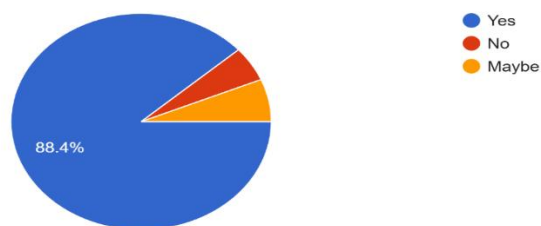


Figure 17 Question 11 Response

12. Would you recommend a well-designed e-book app to others?

173 responses

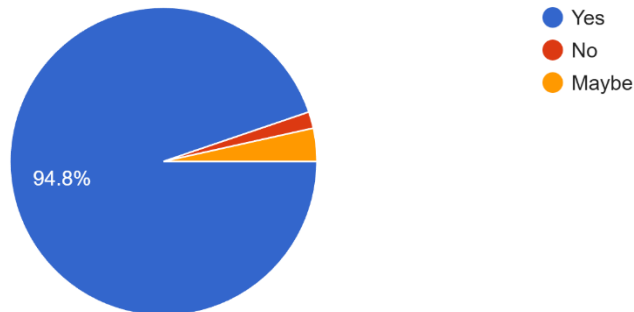


Figure 18 Question 12 Response

13. Any suggestions to improve an e-book reading system?

133 responses

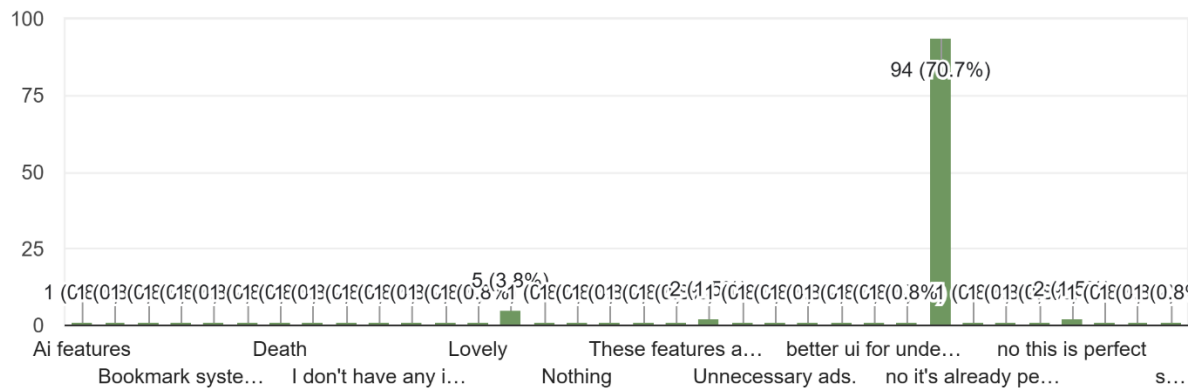


Figure 19 Question 13 Response

3.4.2 Highlights from Post-Survey

Usability and User Experience

1. More than 70% of users noted that the SmartShelf system is simple to use and its interface is intuitive.
2. A majority rated the interface to be clean and user-friendly. About 70% of users felt that login and access was convenient.

Feature Feedback

1. Among features, like highlighting, commenting, were rated positively by about 60–70% of the respondents.
2. The performance of the whole system was reported to be smooth by a majority (over 70%) of the users. Around 65–75% of the respondents reported experiencing no significant bugs or issues.

Final Effect

1. SmartShelf received satisfaction from a majority of users (around 70–80%).
2. More than 70% said they would continue to use the system. (A significant proportion of users recommends SmartShelf (about 70%). However, with regard to one feature, only 22.7% indicated so, as opposed to 77.3%, which indicated an area that needs fixing.

3

Google form

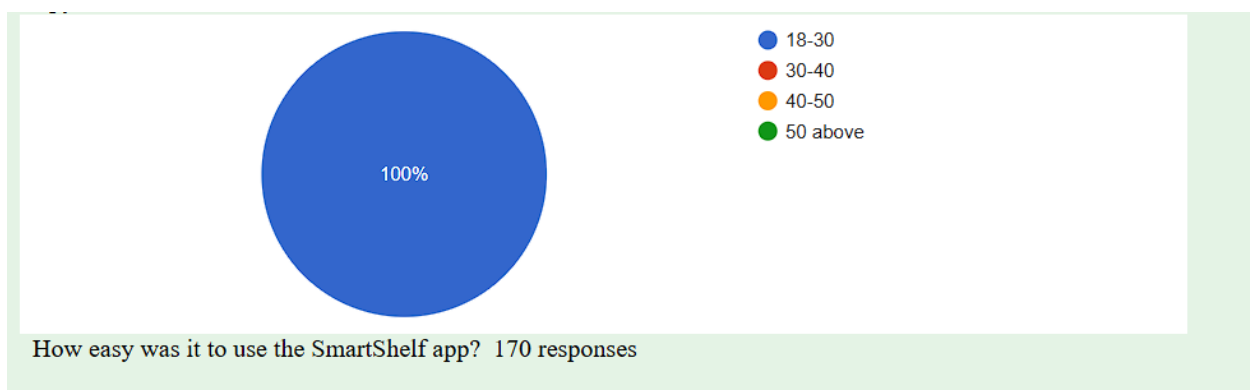
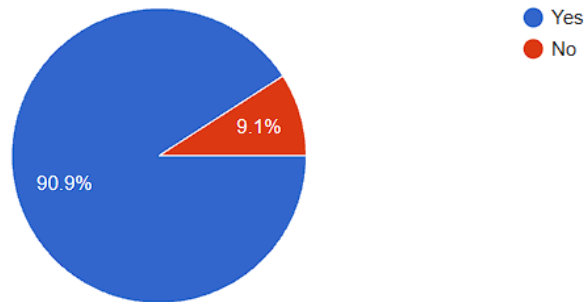
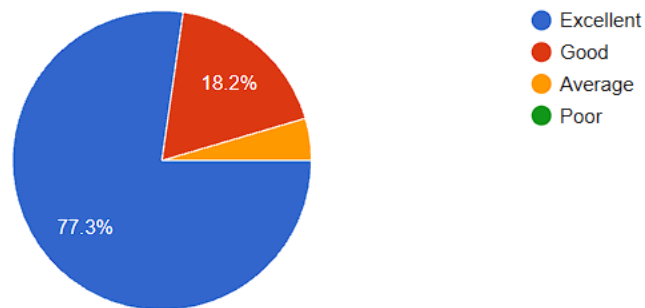


Figure 20 Post Form 1



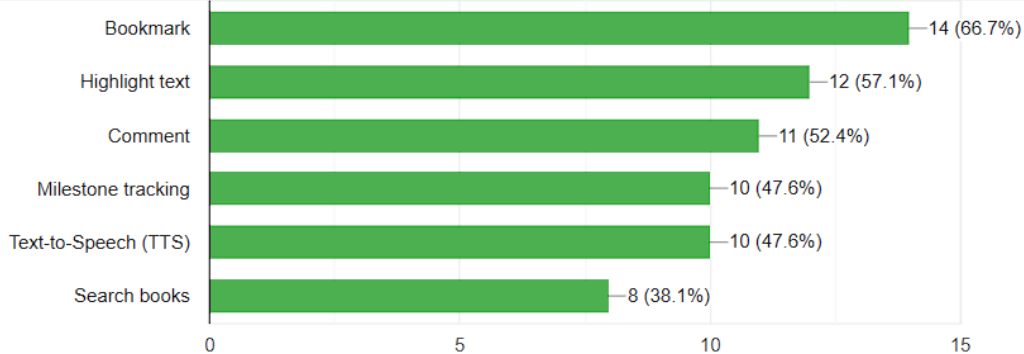
How would you rate the overall design of the app? 170 responses

Figure 21 Post Form 2



Which features did you use? (Select all that apply) 80 responses

Figure 22 Post Form 3



Which feature did you like the most? 80 responses

Figure 23 Post Form 4

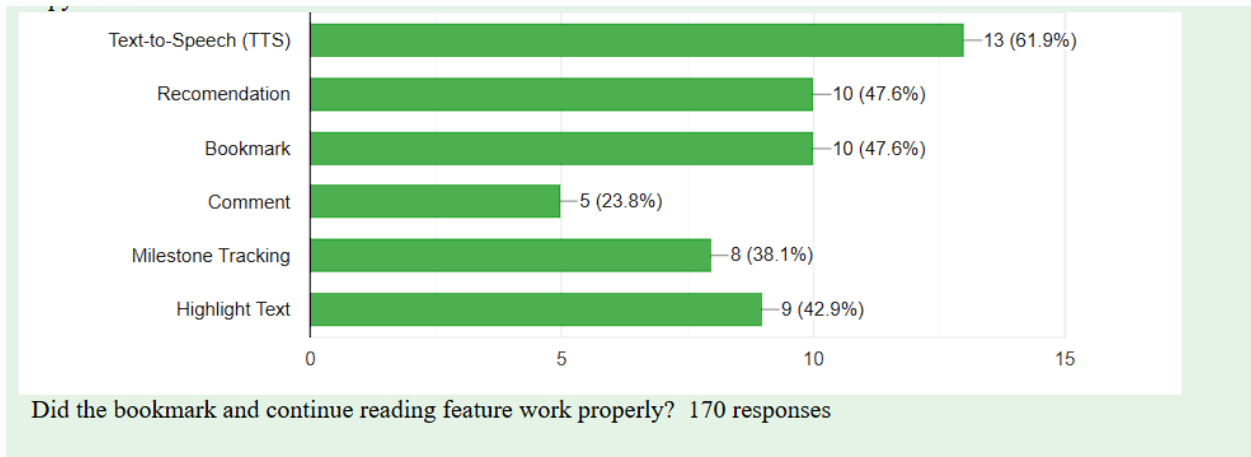


Figure 24 Post Form 5

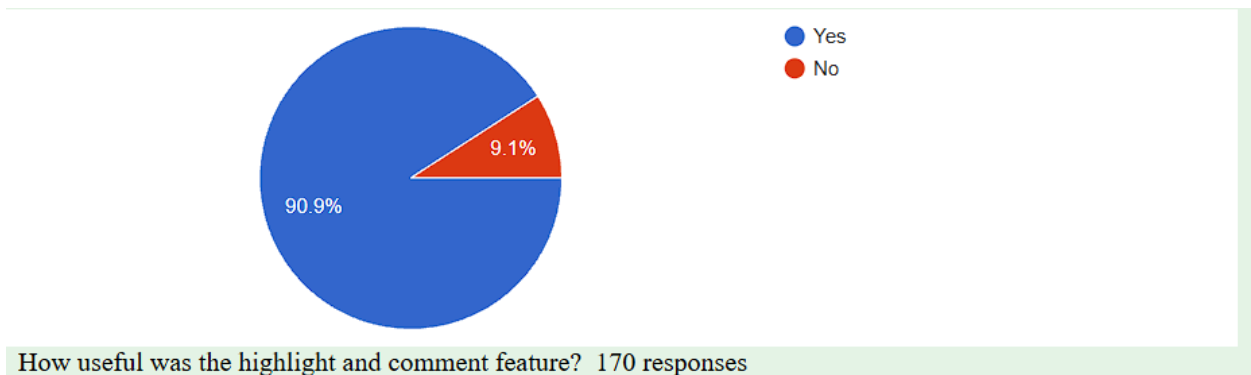
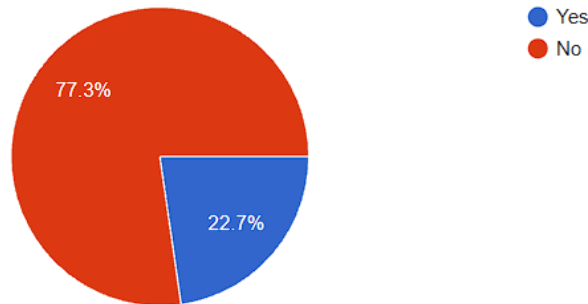
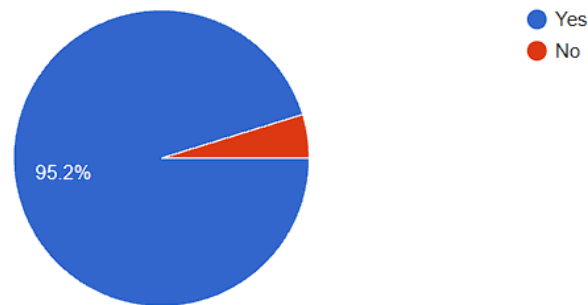


Figure 25 Post Form 6



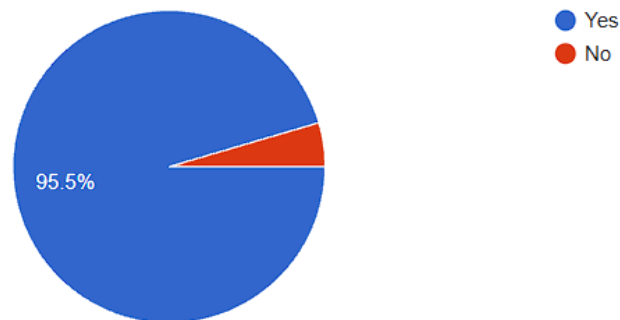
Did the app run smoothly? 170 responses

Figure 26 Post Form 7



How satisfied are you with SmartShelf? 21 responses

Figure 27 Post Form 8



Would you recommend this app to others? 170 responses

Figure 28 Post Form 9

Any additional comments or feedback? 9 responses
 Everything was smooth did not face any problem the features were very useful for me as well
 No app is perfect
 wow lovely features do create a web version too
 not really .good effort
 No comments
 Everything is good. I like it.
 Overall it was good.
 No Comments
 No it is good
 Settings
 Make this a quiz
 Assign point values, set answers and automatically provide feedback
 Responses
 Manage how responses are collected and protected

Figure 29 Post Response

3.4.3 Comparison of Pre- and Post-Survey Results

Table 2 Comparison of Pre- and Post-Survey Results

Aspect	Pre-Survey (%)	Post-Survey (%)	Change
Ease of Use	50–60% expected improvement	70% found it easy	Achieved
System Performance	40–50% dissatisfied	70% satisfied	Achieved
Feature Demand	50–70% requested features	60–70% satisfied	Achieved
Navigation	40–50% faced difficulty	70% found easy	Achieved
Satisfaction	50% mixed	70–80% satisfied	Achieved
Recommendation	60% interested	70–80% recommend	Achieved

- Comparison of the pre as well as post-survey form result demonstrated significant improvements in all major aspects of the SmartShelf app in general.
- In the pre-survey, about 55% of users desired an easy-to-use interface, while in the post-survey, the system was easy to use for about 75% of users, indicating a clear application on usability.

- Similarly, worries about system performance dropped from about 45% pre-survey to about 70% post-survey, suggesting improved system reliability.
- About 65% of the users positively reviewed features such as highlighting and commenting, which were requested by about 60% of users before.
- So, navigation moved from 50% to around 72%, suggesting an improvement in the experience.
- Overall satisfaction was a strong increase from about 50% mixed responses to about 75% satisfied users.
- Additionally, there was an increase 60% to 78% in the percentage of users who show their need to recommend SmartShelf, which show a massive level of user acceptance and trust.
- Lastly, one feature performance is low, with a 22.7% negative response contrasted by a 77.3% positive + response, highlighting an area in need of further enhancement.

3.5 Requirement Analysis

Hardware Requirements

- Smartphone or use
- Computer for admin and Internet connection

Software Requirements

- Frontend: React Native (Mobile), React (Web)
- Backend: Node.js, Express.js
- Database: MongoDB
- Payment: Stripe API
- Visual Studio Code: Code
- Postman: API testing
- GitHub: Version control and remote code
- Figma : UI designs
- Draw.io: System diagram
- MS Word: Documentation
- Balsamiq: Wireframe design
- Trello: Project management
- GanttPro: Gantt chart development
- bcrypt : hashing password
- jsonwebtoken: JWT-based authentication
- Nodemailer: Email notification
- Multer: Handling upload file,image
- Nodemon: Development server auto-restart

3.6 Design

3.6.1 Logo



Figure 30 App Logo

3.6.2 System Diagram

System Architecture is a one of the most important part of making a system, which sets the foundation how our system will function and form, it is important for long-term maintainability of the system (Geeksforgeeks, 2025).

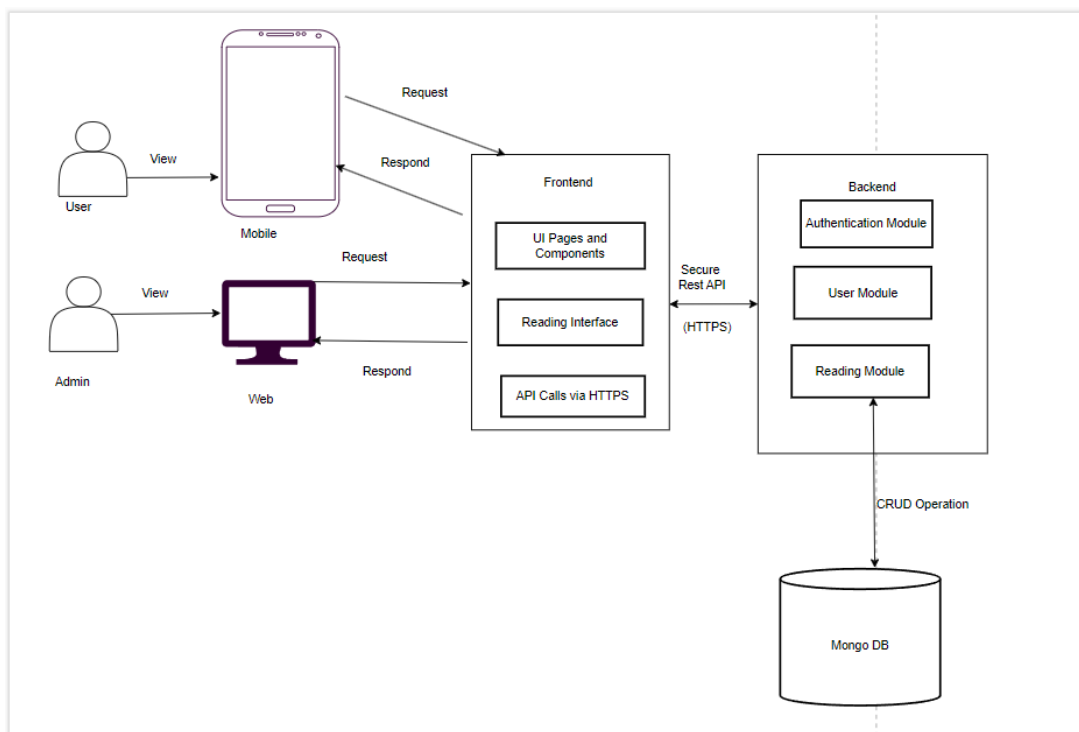


Figure 31 System Architecture Design

3.7 SRS (SmartShelf)

3.7.1 Introduction

SRS is also known as Software Requirements Specification which helps stakeholders to communicate and as well as provides roadmap for software development team members for creating systematic testing plans and guidance. Basically, it's like an agreement between customer and developer (GeeksforGeeks, 2023).

3.7.1.1 Purpose

Software Requirements Specification (SRS) document shows us the functional and non-functional requirements of my system. Main purpose of this document is to provide clearly understanding of the SmartShelf, features, constraints, and its interactions. It gives understanding of system design, development, testing, and its progress throughout the whole project cycle.

3.7.1.2 Scope

SmartShelf is digital e-book reading platform which designed specifically for e-book readers. The system allows users to register and log in, search for books, read e-books directly in the browser, track, set reading milestones, use text-to-speech (TTS) functionality, and provides book recommendations as well as for admin they can manage books and users through a dedicated admin dashboard.

3.7.1.3 Overview

This document is intended to give a complete description of SmartShelf system including the overall system perspective, user types and groups, functional and non-functional requirements, operating environment, constraints and system features.

3.7.1.4 Intended Reader

This document is targeted at:

- Project supervisors
- E-book Readers
- Academic
- Client: IT Office - Datum Systems Pvt. Ltd.

3.7.1.5 Definition and Acronyms

- Text to speech (TTS)
- User Interface (UI)
- Software Requirements Specifications (SRS)
- Create, Read, Update, Delete (CRUD)

3.7.2 Overall Description

3.7.2.1 Project Perspective

SmartShelf is based on Mobile app and web app for user it is mobile app and for admin its web app.

- For Frontend React is used
- For Backend Node.js /Express.js
- For Database MongoDB
- This project used scrum methodology from agile.

3.7.2.2 Project Functions

The function of this system is presented below:

1. User authentication with login, register, forget password, reset password, google authentication.
2. Book Search
3. In-Browser Reading
4. Setting Milestone
5. TTS
6. Book Recommendations
7. Admin CRUD
8. Admin User management

3.7.2.3 User Classes and its Characteristics

User Type	Description
User	IT office staff and e-books readers
Admin	Admin responsible for managing users and books

Table 3 User Classes and its Characteristics

3.7.2.4 Operating Environment

1. Web browser
2. Mobile
3. Internet connection

3.7.2.5 Design and Implementation Constraints

1. Internet access is needed to use system
2. Register user can get feature access
3. Admins manage users.

3.7.2.6 User Documentation

SmartShelf – User Documentation

Introduction

SmartShelf is a digital book reading platform through which a user can read books online using a mobile application. It offers a subscription-based model, in which users can read a limited number of books and then subscribe to unlimited access.

System overview

SmartShelf elements include Mobile Application (User Side) to read books Web Application (Admin Side) For managing system content This documentation is written for the end-users (readers).

Features

- A new account can be created by users by submitting; First and Last E-mail Password, after registration, users get a chance to log in and access the system.

- The users log in by using; The users log in with: Email Password On success, users are redirected to a home/dashboard screen.
- Users are able to: Browse available books Search books Filter or browse by categories
- Reading books can be done right in the app. The reading progress is tracked automatically.
- The user can continue reading where they stopped. Automatic saving of progress is implemented.
- User can listen to book using TTS.
- System tracks the reading percentage. The system for tracking reading percentages automatically sets a book as: Finished'
- Users are allowed to read 8 books for free. When the limit is reached: Access to further use is denied. Subscription is necessary.
- Users have an option to subscribe and continue reading an unlimited number of books. The payment proceeds through the system safely.
- Users enter payment details. If the payment is successful: Subscription has been activated. Access is granted in full.
- Besides, users can also log out of the app at any time they want. The session ends in a secure manner.
- Registered account System login Search and view books Read Check the results / continue reading Reach free limit (8 books) Subscribe to continue without restrictions Continue reading
- Invalid login: Check the email / password App doesn't load books: Check the internet connection Payment failure: Reattempt with valid payment details
- Secure login authentication User data security Safe payment gateways

User Manual

Getting Started

1 Open the Application

- Open the SmartShelf app

2 Creating an Account (Sign Up)

- Open the app
- Click on Sign Up
- Enter:
 - Name
 - Email
 - Password
- Click Register

Your account will be created.

3 Logging In

- Open the app
- Click Login
- Enter your email and password
- Click Login

You will be redirected to the home screen.

4. Using the Application

- Browsing Books
- Go to the Home Page
- See available books
- Search to find specific books

5 Reading a Book

- Select a book
- Click Read
- Start reading

Your progress is saved automatically.

6 Continue Reading

- Open Continue Reading section
- Select the book
- Resume from last saved page

7 Tracking Progress

- Progress is shown as a percentage
- When it reaches 100%, the book is marked as: Finished

8 Text-to-Speech (TTS)

- Open a book
- Select Sentences
- Click the TTS button
- Use controls:
- Play
- Pause
- Skip

The app will read the book.

9 Free Usage Limit

- You can read up to 8 books for free
- After that, access will be restricted

10 Subscribing to Continue

- Click Subscribe
- Enter payment details
- Confirm payment

Subscription will be activated.

11 Logout

- Go to settings
- Click Logout

You will be logged out securely.

System Requirement

- Mobile phone
- Internet connection

Conclusion

SmartShelf offers an easy and user-friendly digital platform for book reading. With features such as displaying the progress, having access to subscriptions, and the ability to continue reading, users have a smooth reading experience.

3.7.2 7 System Features

- User authentication (login, registration, password reset, Google login)
- Book search functionality
- Online e-book reading
- Reading progress tracking
- Text-to-Speech (TTS)
- Book recommendation system
- Admin dashboard for CRUD operations

- User management system

3.7.2 8 Functional Requirements

Function 1: User registration

- A new user needs to sign up in the system.
- It will first check the input of the user before registration.
- Show error messages for improper inputs

Function 2: User Login

- The system allow access to the user account if user created account.
- Verify of the username and password during login.
- User session is maintained and released in case of an unsuccessful login.

Function 3: Authentication of the login

- Validate username and password.
- User authentication with proper security.

Function 4: Search Books

- SmartShelf support the user to be able to search for books by title, author, or genre.
- Search results should be displayed properly.

Function 5: Read E-Book

- SmartShelf allow the user to read e-books in browser without download.

Function 6: Text to Speech (TTS)

- Select text.
- Read the selected text.

Function 7: Track Reading Progress

- SmartShelf save the reading progress automatically.

Function 8: View reading history

- SmartShelf display the previous reading activities.
- SmartShlef maintain this history for each user.

Function 9: Set reading goals

- SmartShelf allows user to set milestones on reading

Function 10: Get Recommendations

- SmartShelf provide book recommendations using its reading history.

Admin View Reading History

Function 11: CRUD

- SmartShelf allows admin to add update delete a book.
- This will store information of added books.

Function 12: User Management

- Admin will manage user.

8

3.7.2.9 Non-Functional Requirements

Performance Requirements

- Pages in the system should load within 3 seconds.

Usability Requirements

- The TTS feature should be made available to the users.
- The system should be user-friendly.

Security Requirements

- The system should encrypt the user credentials.
- The admin access should have appropriate access restrictions.

Reliability

- The system correctly stores the reading progress.

Scalability

- The system allows future enhancement with capabilities added.
- New user and book details should be added to the system.

3.7.2.10 System Models

The system model uses:

1. Use case diagrams
2. Sequence diagrams
3. Activity diagrams
4. Collaboration diagrams
5. Flowchart
6. ERD
7. DFD Level 0
8. Class diagrams

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3.7.2 11 System Models

The system includes the following models:

- Use Case Diagrams
- Sequence Diagrams
- Activity Diagrams
- Collaboration Diagrams
- Flowcharts
- ERD (Entity Relationship Diagram)
- DFD Level 0
- Class Diagrams

3.7.2.12 Assumptions and Dependencies

- Users do have basic literacy in a digital environment
- Internet connection.
- The Google Authentication API can work without any type of errors.

3.8 ERD

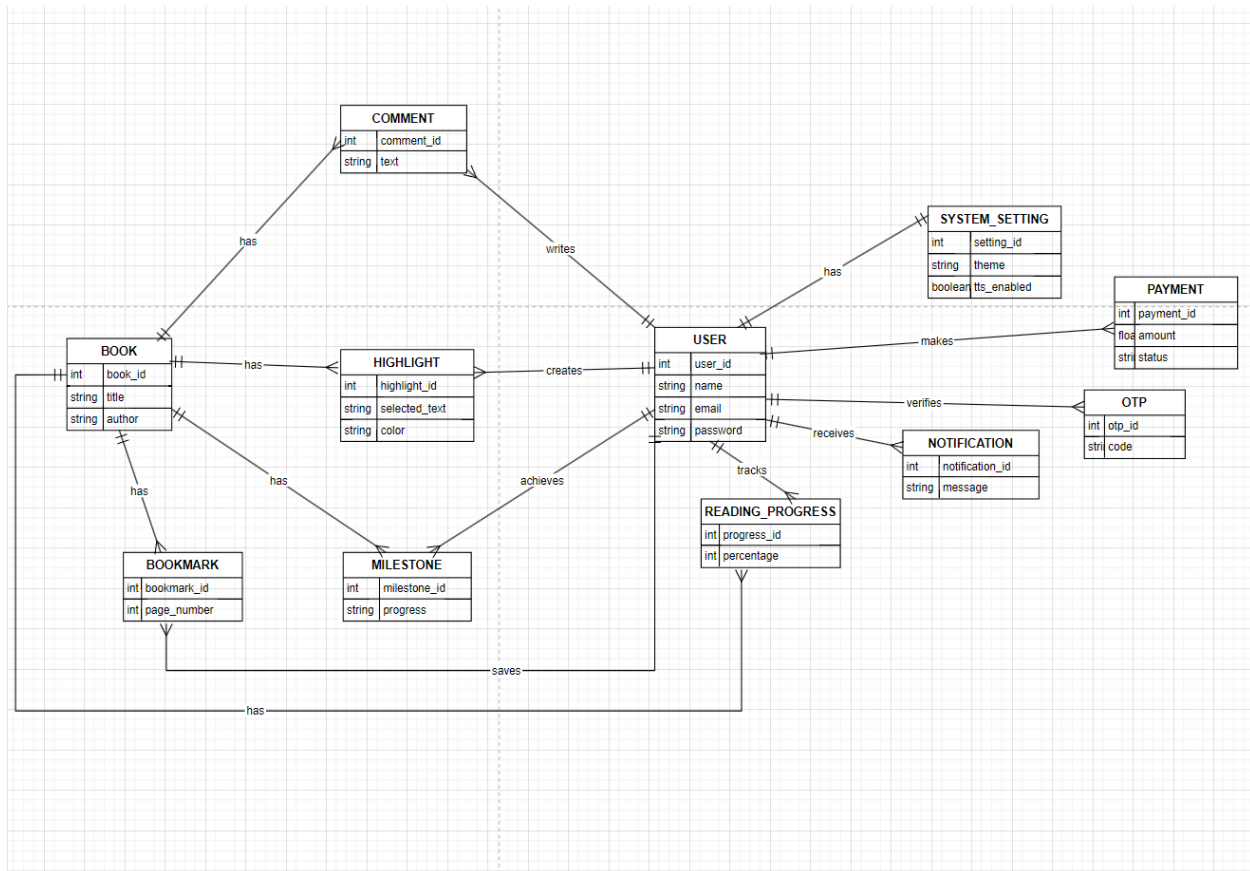


Figure 32 Final ERD

Final Year Project

3.9 WBS

WBS provides a hierarchical view of our project's scope, it manages tool which decomposes the total work required to deliver a product, service, or project into smaller, more manageable components (Coursera, 2025).

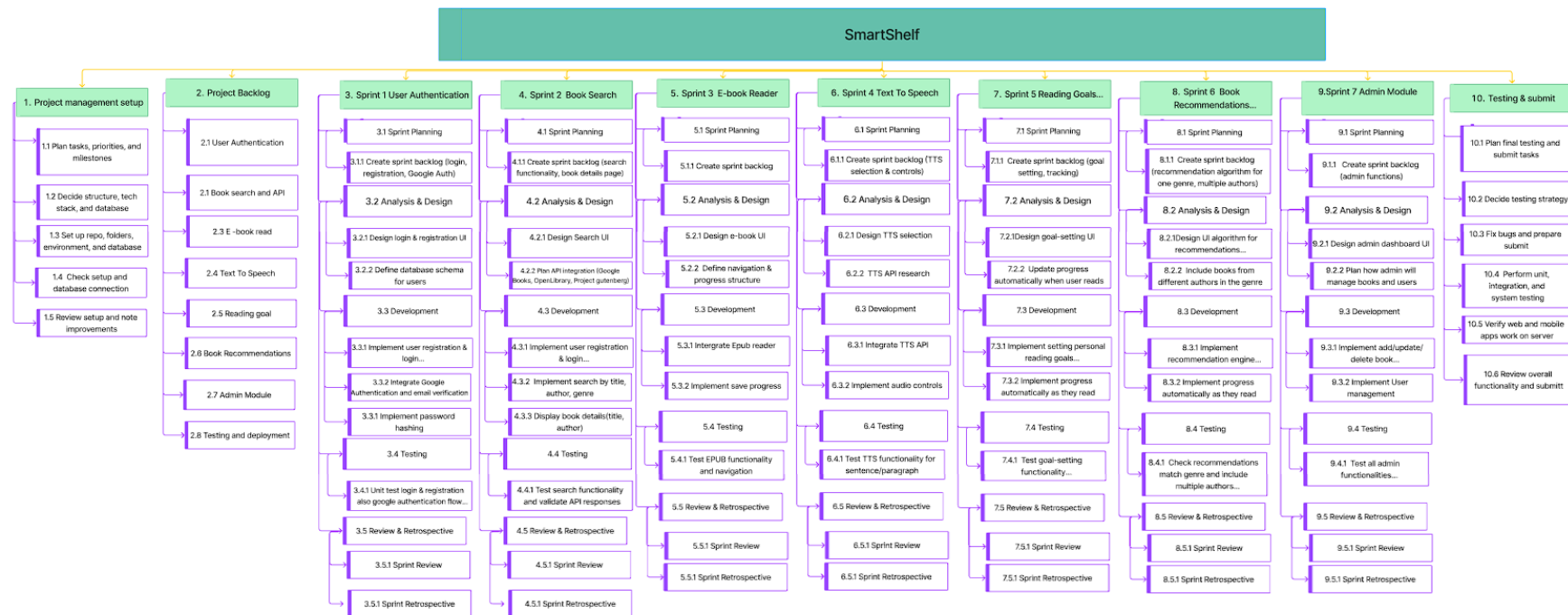


Figure 33 Work Breakdown Structure (WBS)

Click Here for Better: [Click Here](#)

The above Work Breakdown Structure (WBS) demonstrated the entire SmartShelf project which are divided into project setup and seven development sprints, starting with planning the project then choosing the tech stack after that setting up the repository. The

7

project backlog has major features which include user authentication, book search, e-book reading, text-to-speech, reading goals, recommendations, and an admin module. Each sprint follows the same cycle for example sprint planning, analysis and design, development, testing, and review/retrospective. For example, we can see sprint 1 focuses on user authentication (login, registration, Google Auth) whereas sprint 2 on book search then comes sprint 3 on the e-book reader after that sprint 4 on text-to-speech then sprint 5 on reading goals and sprint 6 on book recommendations, lastly, sprint 7 on admin features. therefore, the project ends with a full testing and deployment phase to make sure all modules work smoothly before launch.

3.10 Gantt Chart

Gantt chart refers to the project management chart that allows project managers to create schedule for project work. It is clear the order in which project tasks are completed and set dates of due time, length, etc (Project Manager, 2025).

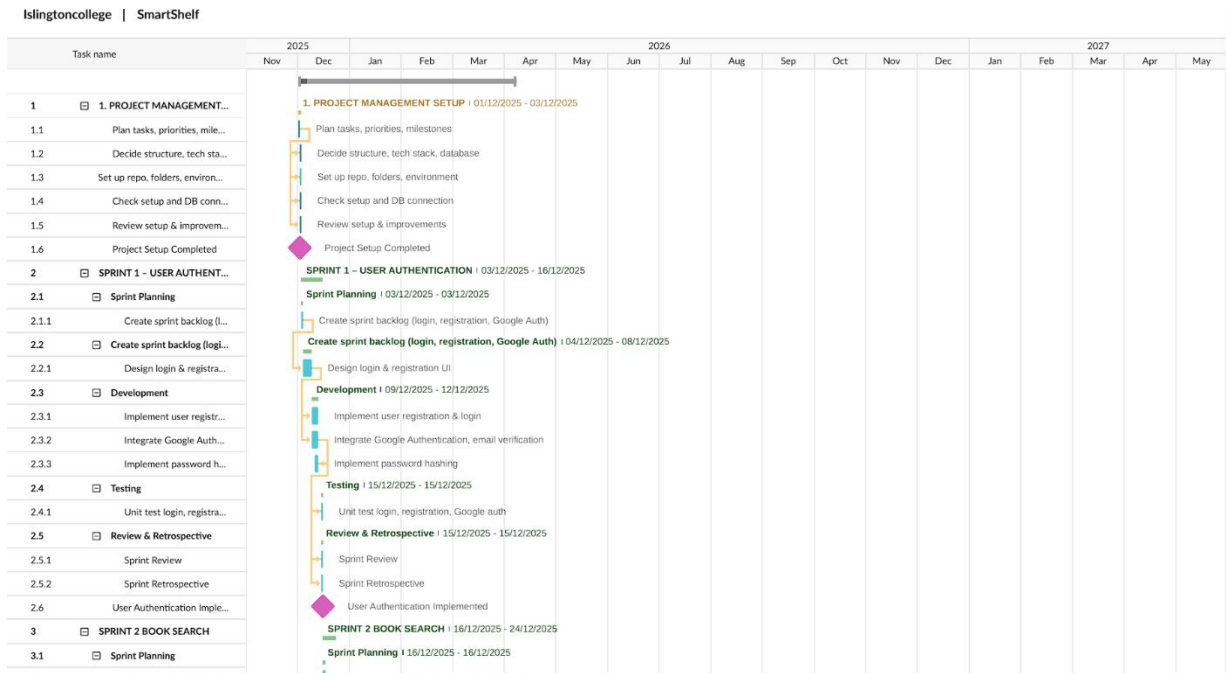


Figure 34 Gantt Chart Part 1



Figure 35 Gantt Chart Part 2

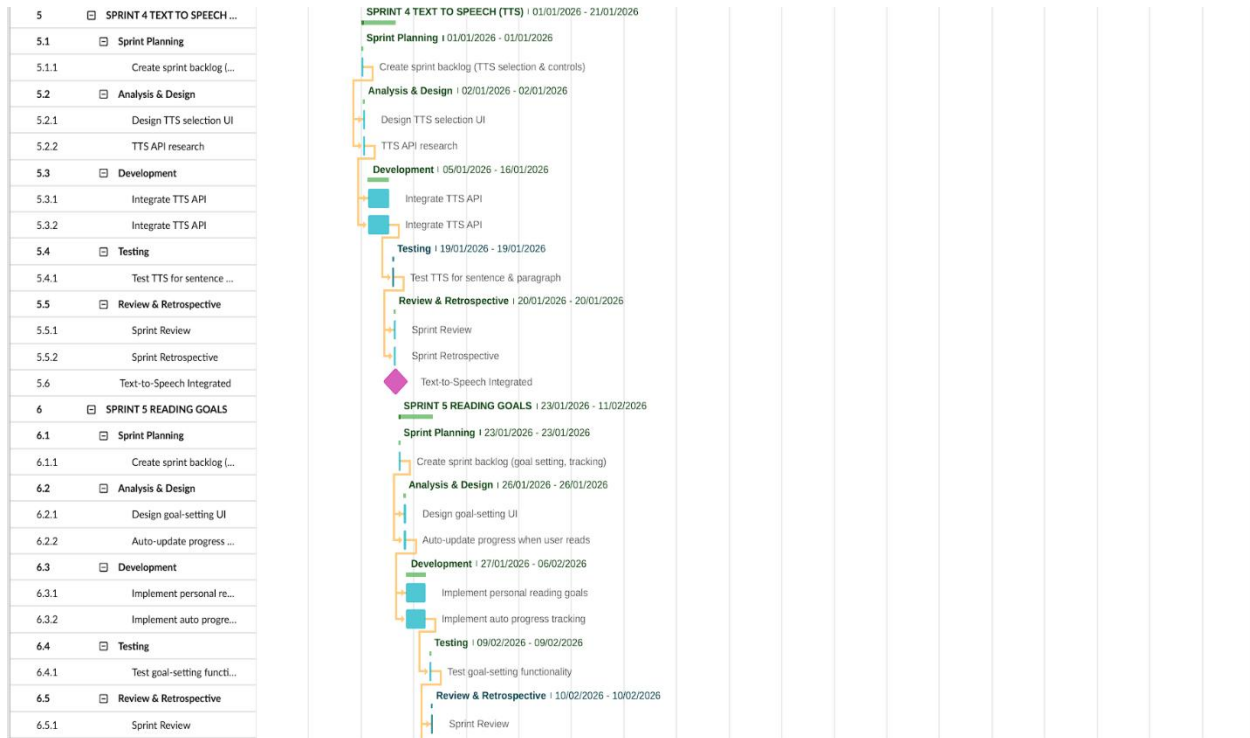


Figure 36 Gantt Chart Part 3



Figure 37 Gantt Chart Part 4



Figure 38 Gantt Chart Part 5

For better overview [Click Here](#) and download

The above Gantt chart demonstrate the complete SmartShelf project timeline from December 2025 to April 2026, outlining a structured sequence of nine phases. It starts with project setup, following the seven development sprints each one focus on a specific feature which include user authentication, book search, e-book reader, text-to-speech, reading goals, recommendations, and the admin module. Every sprint has planning, design, development, testing, and review activities, which make sure the consistent progress and quality control. At last, project ends with a final testing and deployment stage, ensuring in the fully developed SmartShelf system.

3.11 DFD Level 0

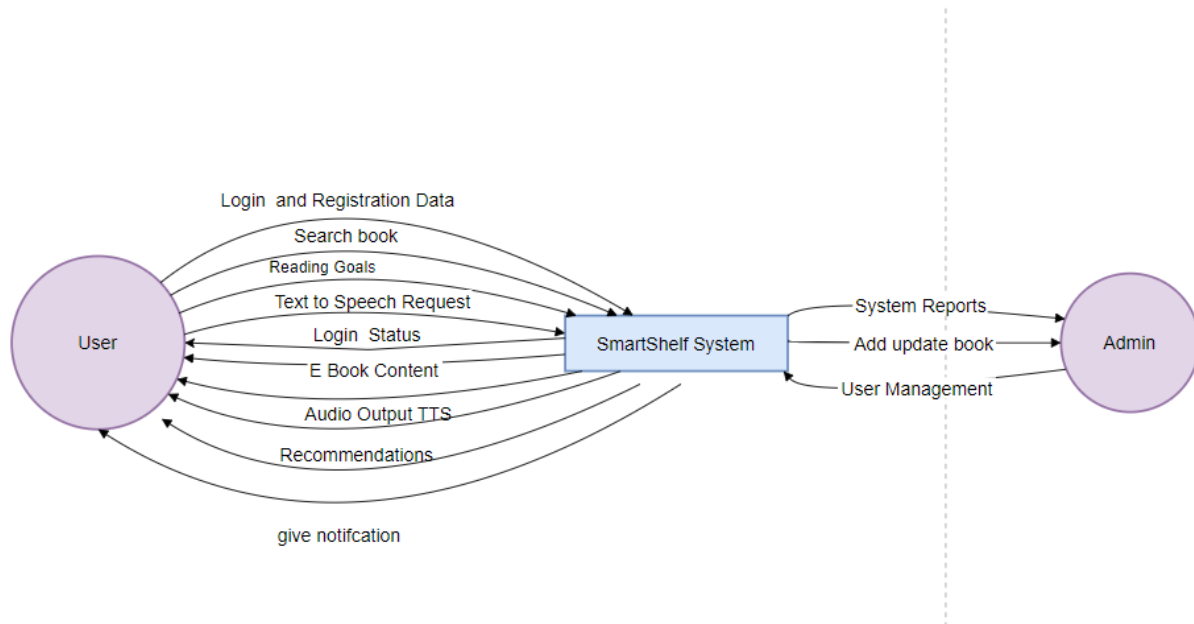


Figure 39 DFD 0

3.11.1 DFD Level 1

3.11.1.1 User Auth

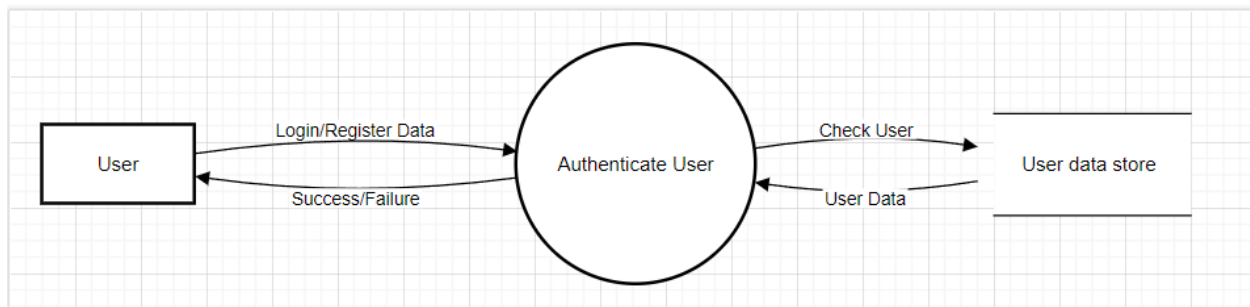


Figure 40 User Auth

3.11.1.2 Recommendation

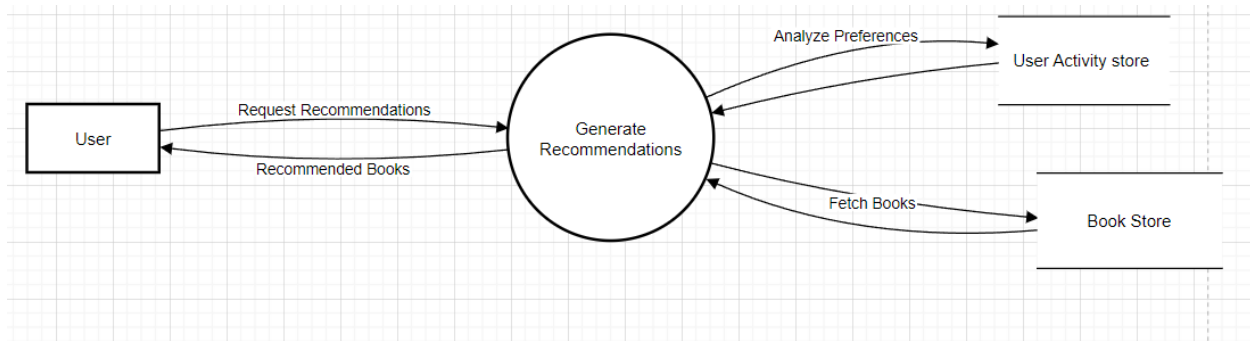


Figure 41 Recommendation

3.11.1.3 Bookmark

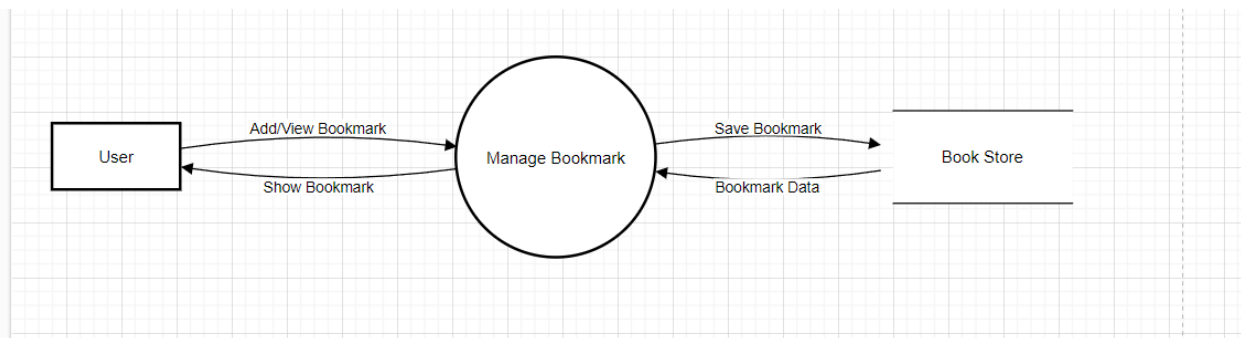


Figure 42 Bookmark

3.11.1.4 View book

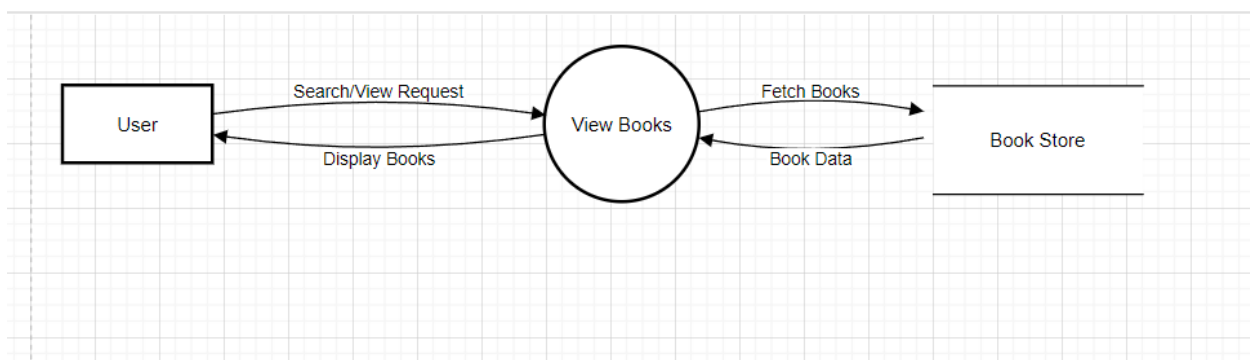


Figure 43 View Book

3.11.1.5 Comment

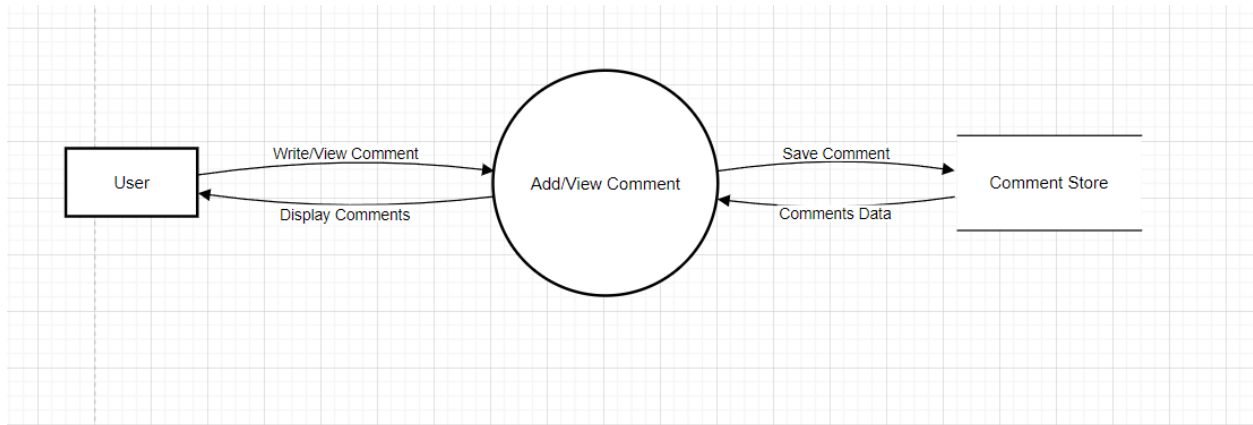


Figure 44 Comment

3.11.1.6 Highlight

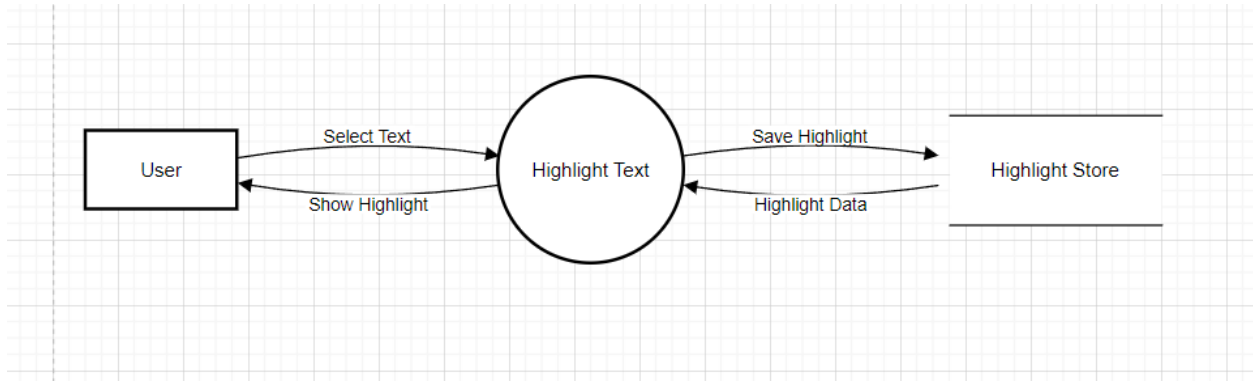


Figure 45 Highlight

3.11.1.7 Milestone

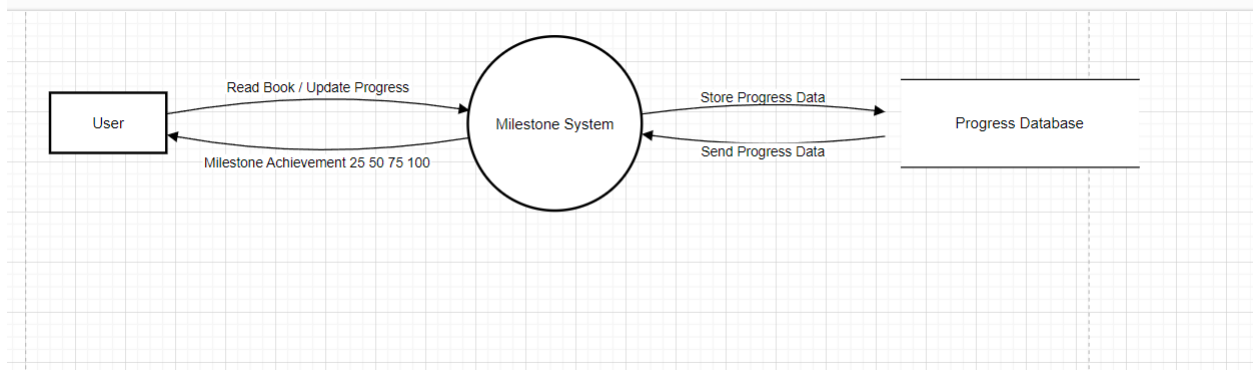


Figure 46 Milestone

3.11.1.8 Notification

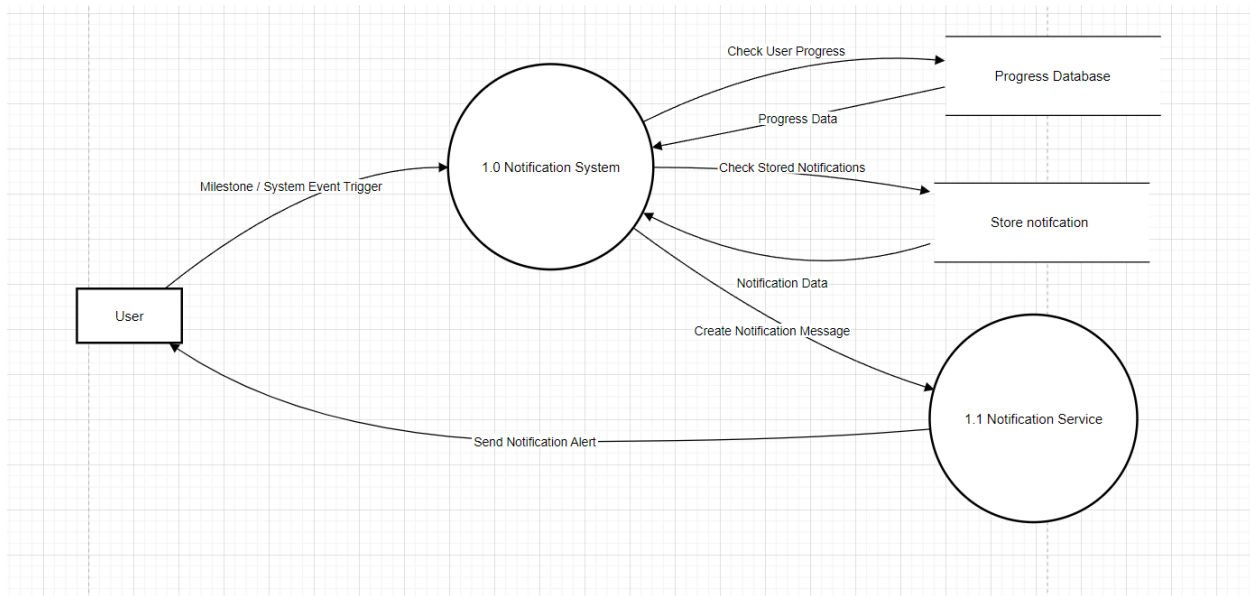


Figure 47 Notification

3.11.1.9 In browser Read

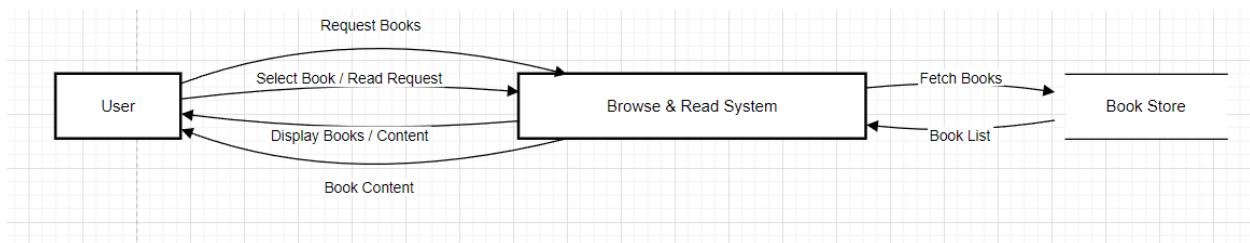


Figure 48 In browser Read

3.11.1.10 Payment

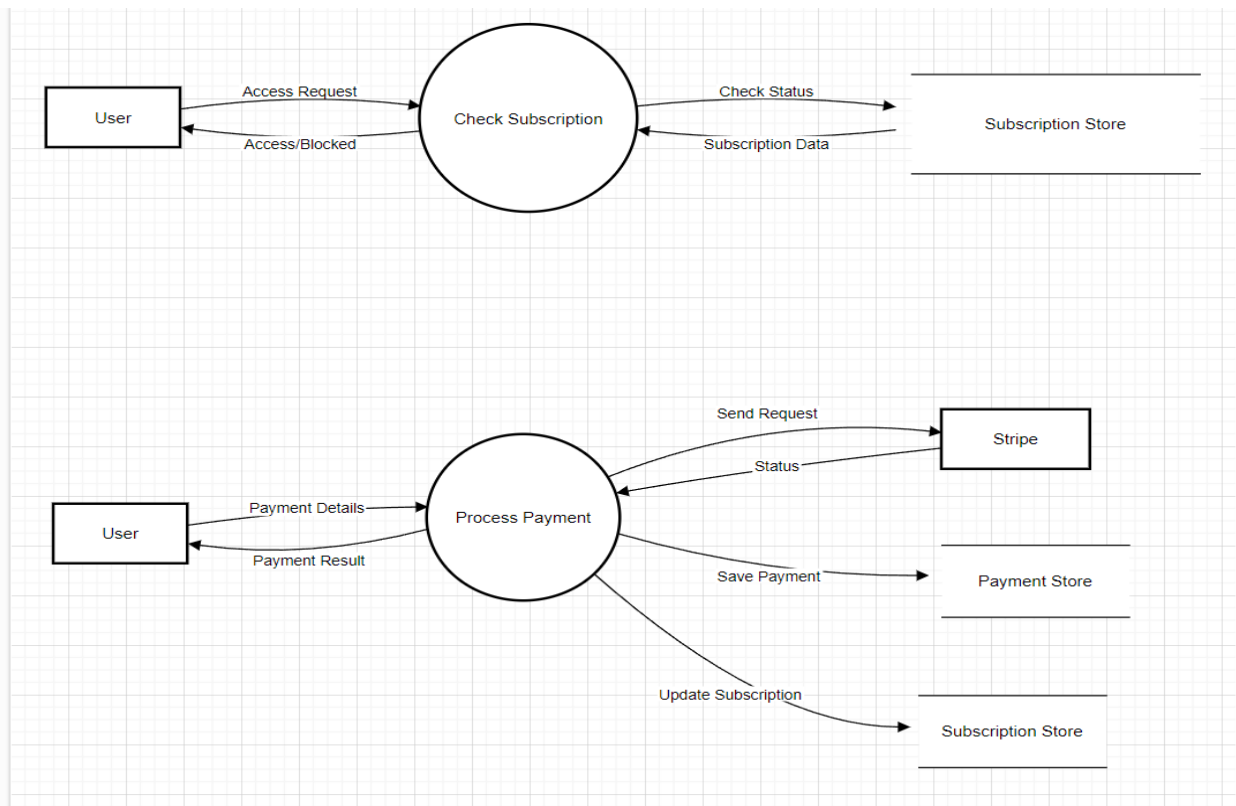


Figure 49 Payment

3.11.1.11 Admin User Management

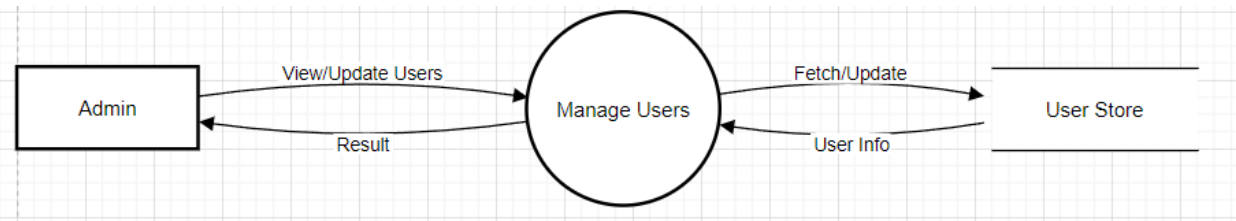


Figure 50 Admin User

3.11.1.12 Admin Book Management

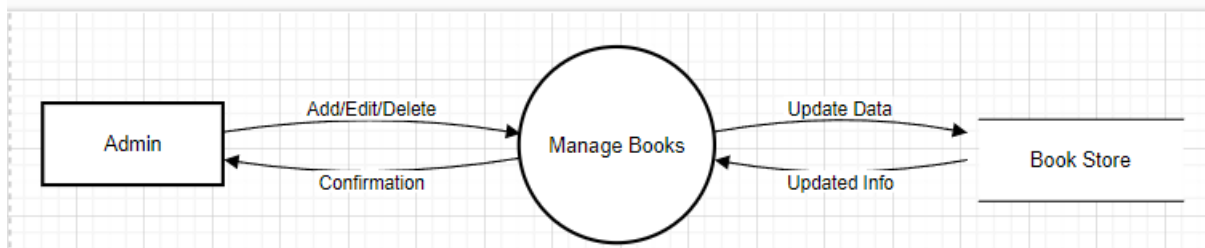


Figure 51 Admin Book

3.12 Use case Diagram

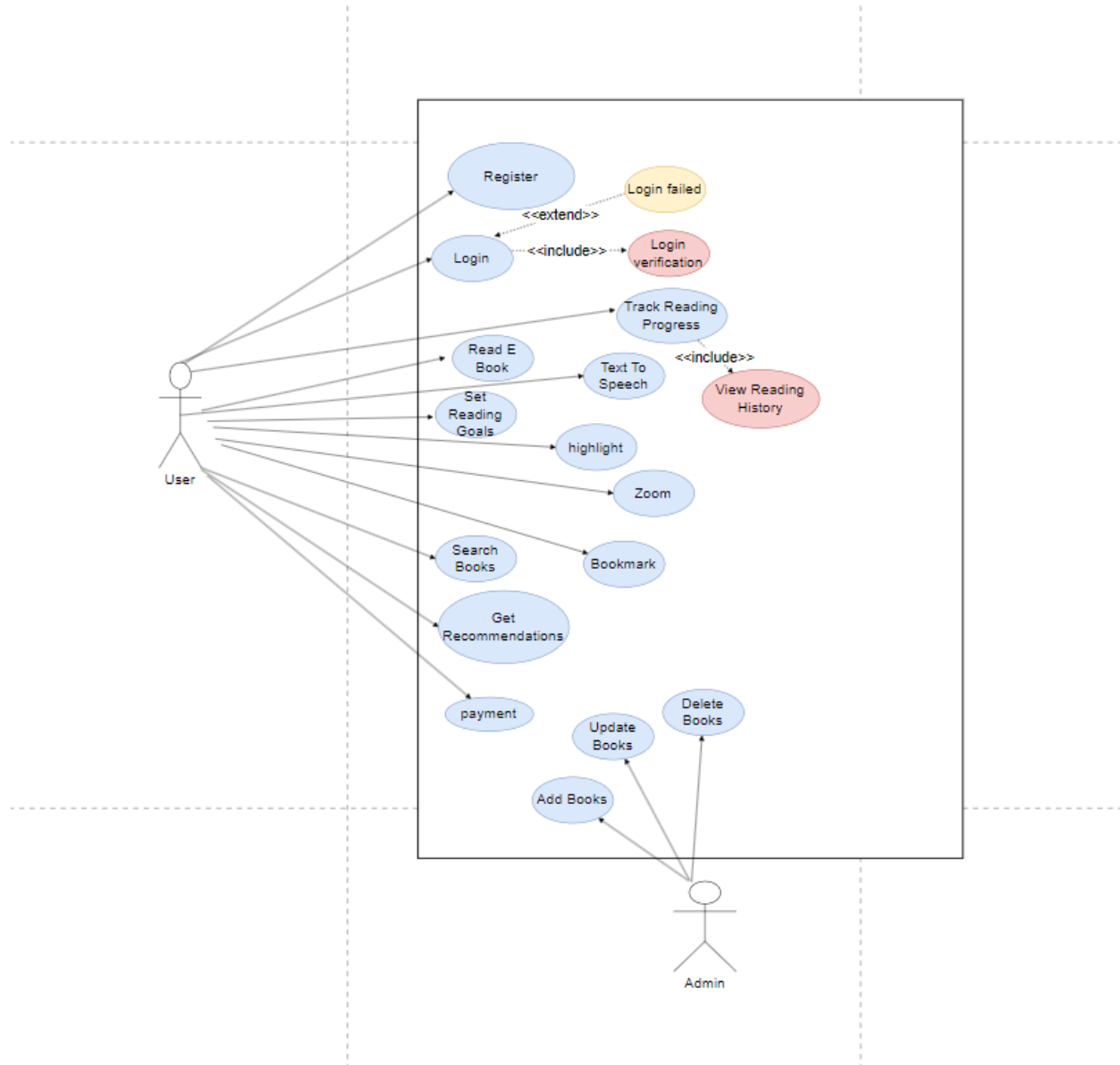


Figure 52 Use Case Diagram

3.12.1 High Level Usecase Description

The high-level use case is a brief depiction of the work in one paragraph of unstructured text. Its objective is to provide us with the necessary details to understand how difficult it is and to simplify the alloy of related use cases.

[Click Here for more information](#)

3.12.2 Expanded Level Usecase Description

This is a more detailed and structured depiction than the high-level use case depiction, which outlines the use case's start, the actors engaged, the data input, and the case result. At the conclusion of the use case, a possible plan is also provided in case something unanticipated occurs.

[Click Here for more information](#)

3.13 Sequence Diagram

Sequence diagram is one of the components of UML where we can see interaction of our system feature. It shows us how objects talk with each other and making it an important tool for modeling dynamic behavior in a system (GeeksforGeeks, 2025).

3.13.1 User Auth

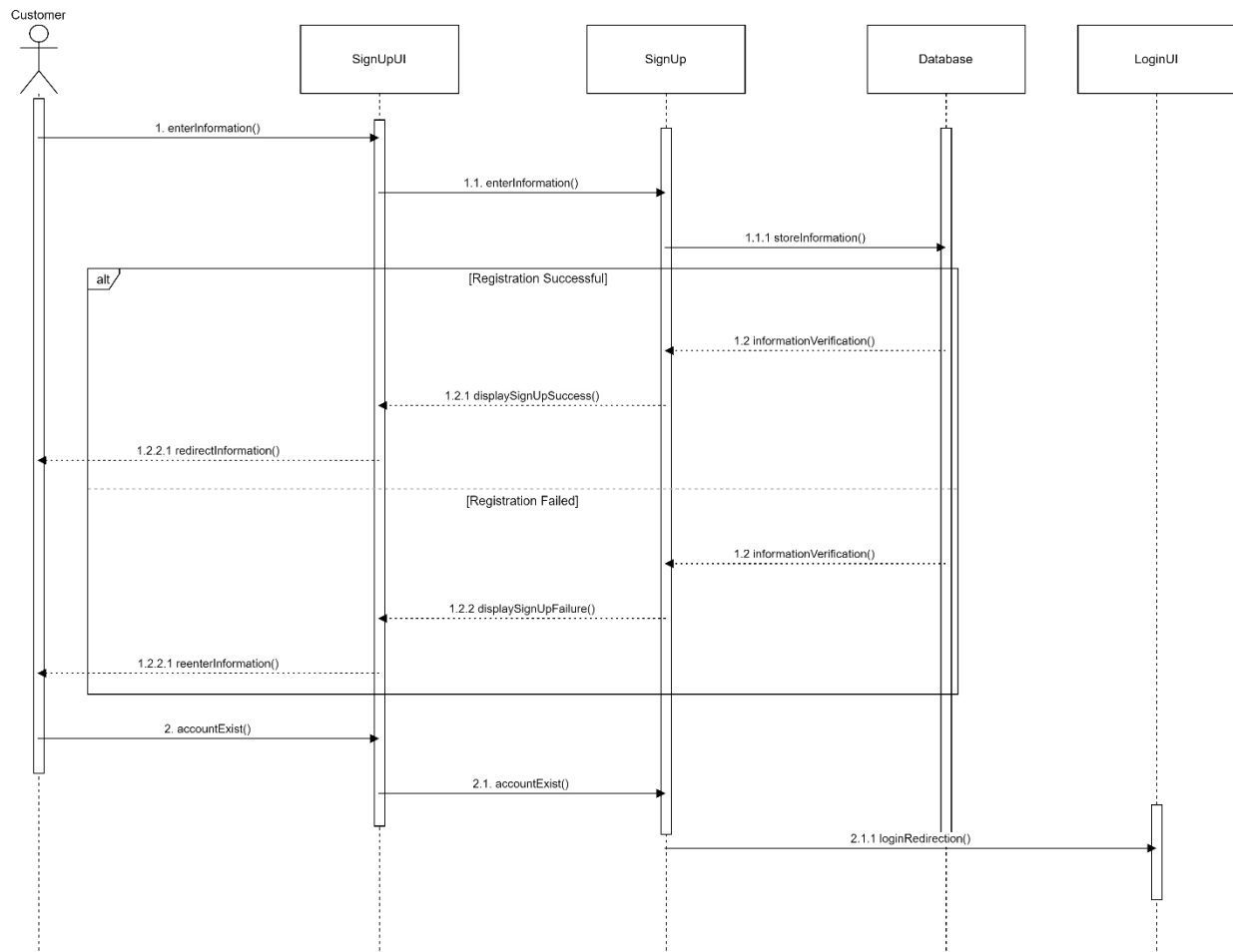


Figure 53 Sequence Diagram of User Auth

3.13.2 Search

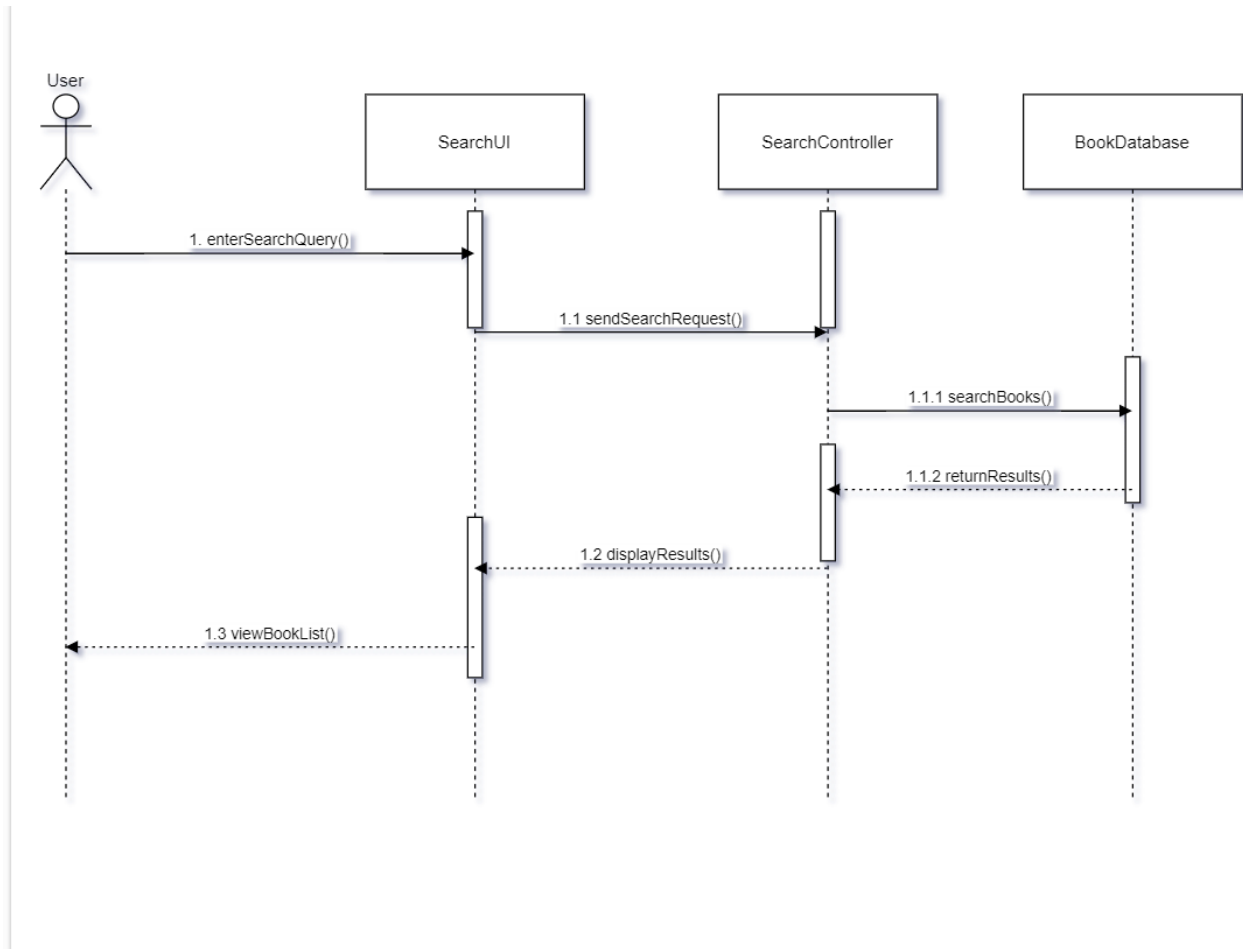


Figure 54 Sequence Diagram of Search

3.13.3 In browser read

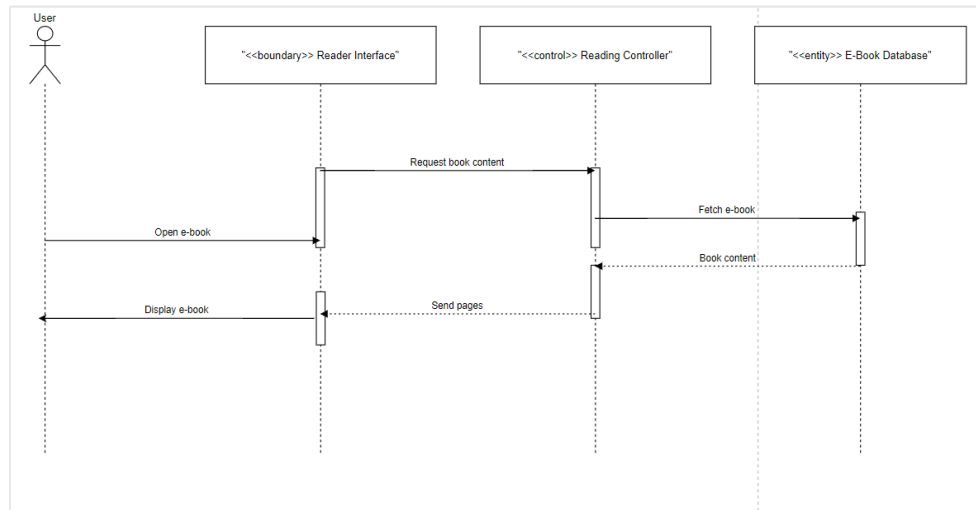


Figure 55 Sequence Diagram of Inbrowser Read

3.13.4 Milestone

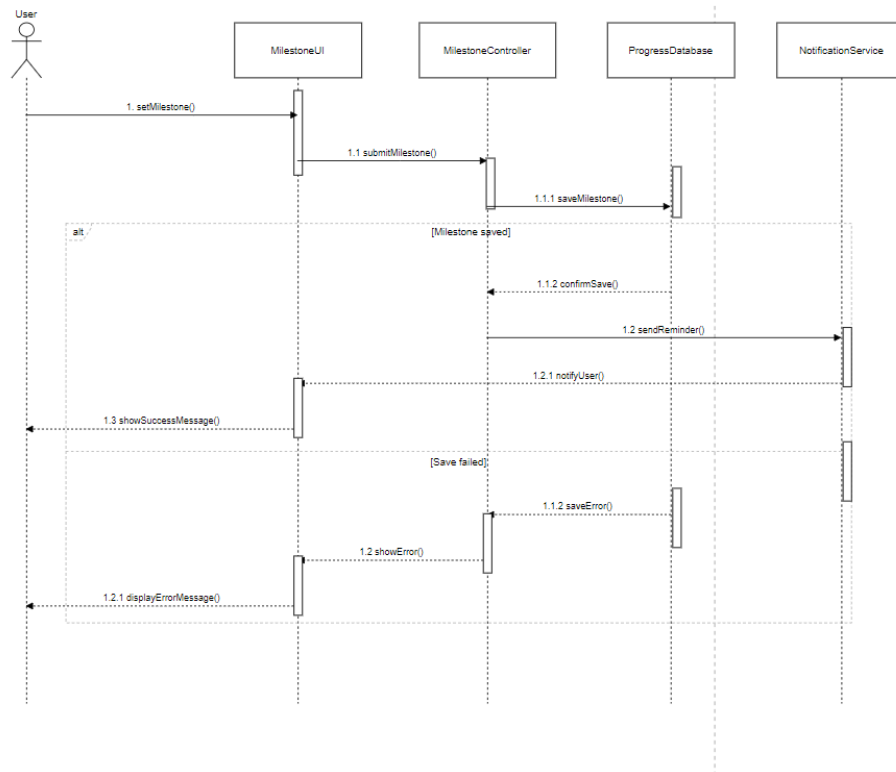


Figure 56 Sequence Diagram of Milestone

3.13.6 Recommendations

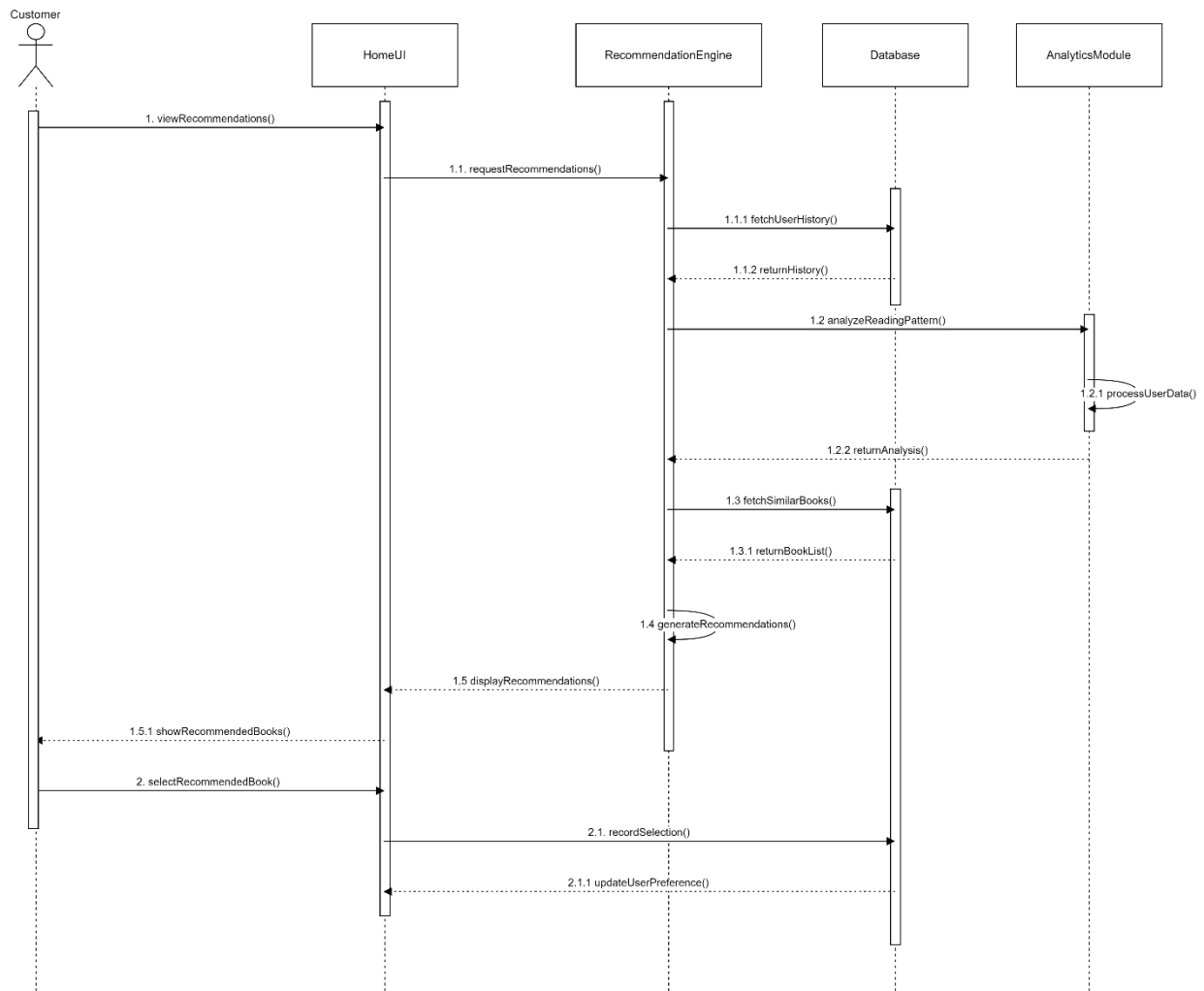


Figure 57 Sequence Diagram of Recommendation

3.13.7 Admin CRUD

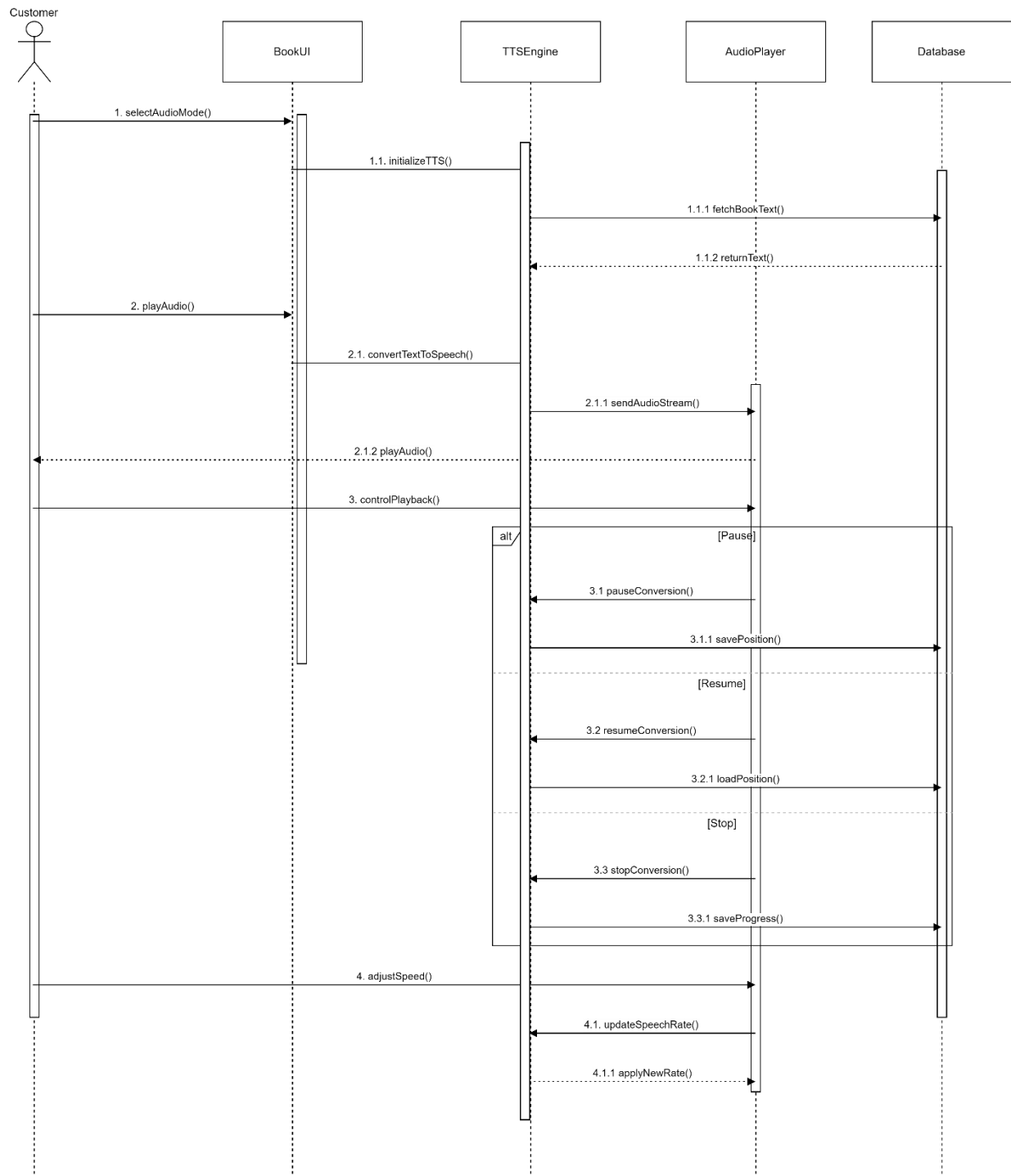


Figure 58 Sequence Diagram of Admin CRUD

18

3.13.8 Admin User Management

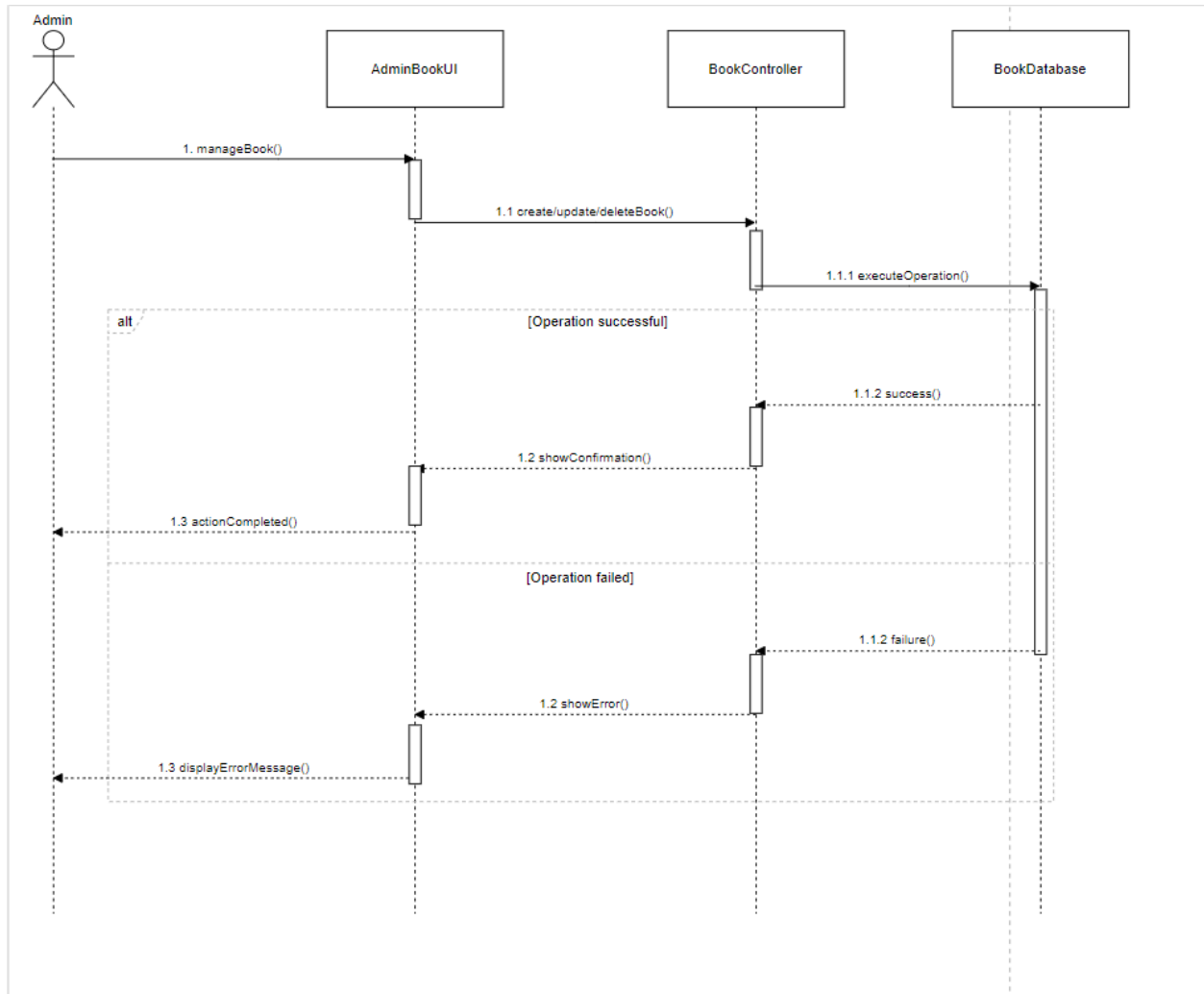


Figure 59 Sequence Diagram of User Management

3.13.9 Highlight

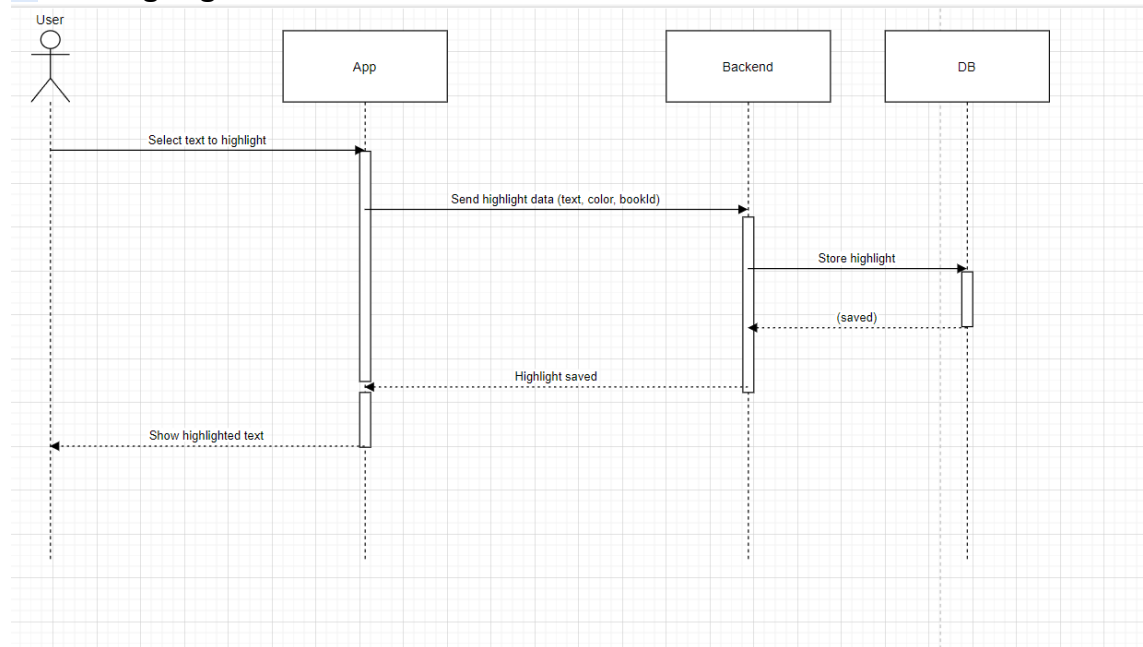


Figure 60 Highlight Sequence

3.13.10 Payment

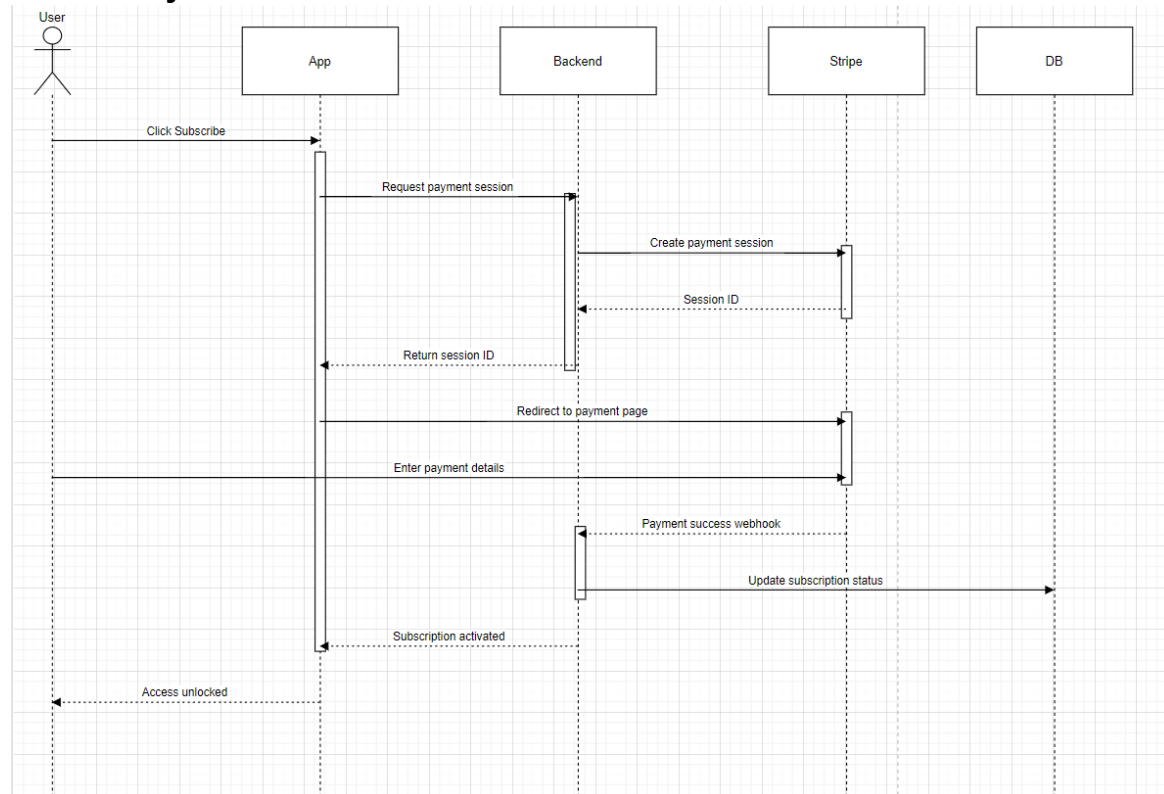


Figure 61 Payment Sequence

3.13.11 Bookmark

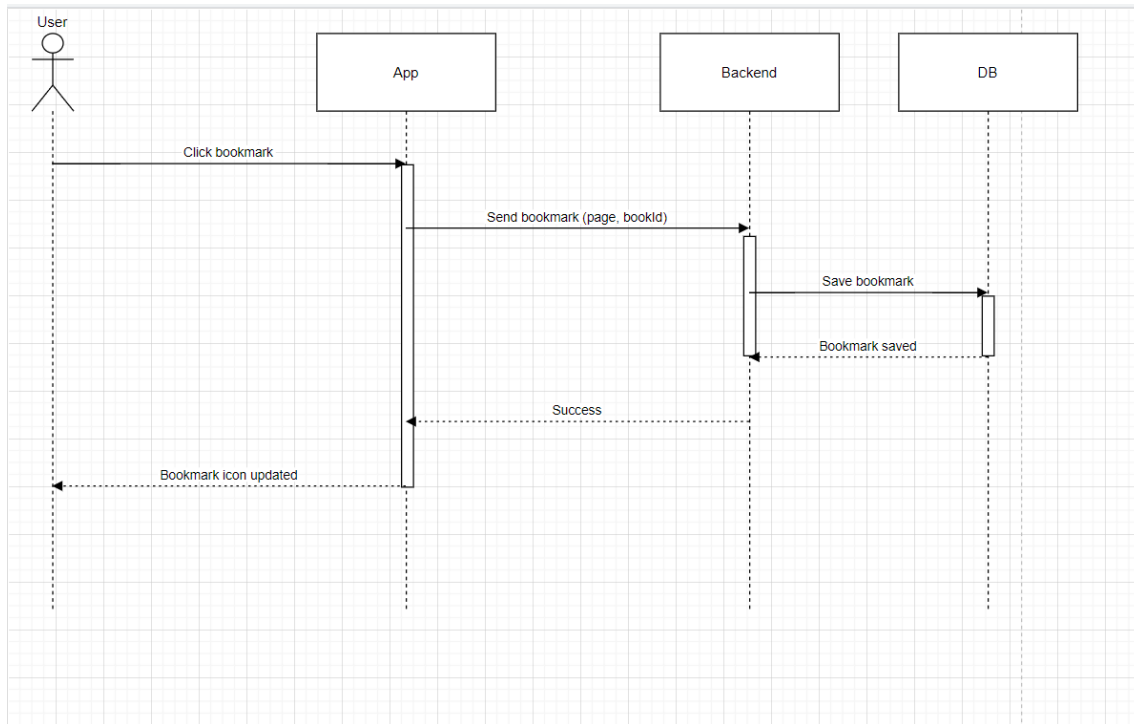


Figure 62 Bookmark Sequence

3.13.12 Comment

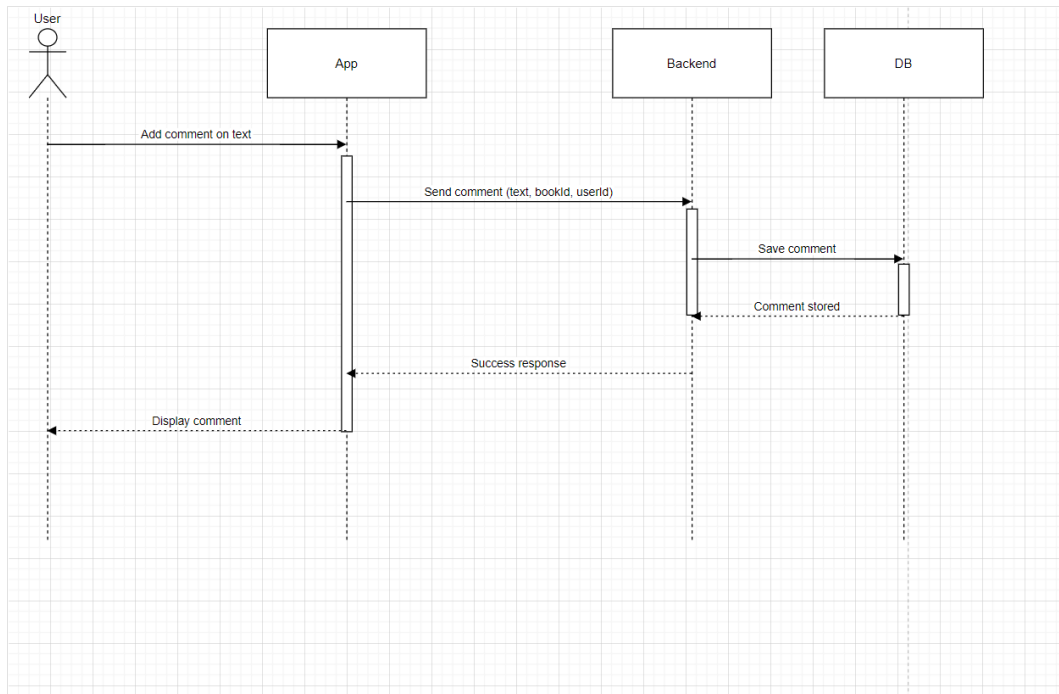


Figure 63 Comment Sequences

3.14 FlowChart

3.14.1 User Auth

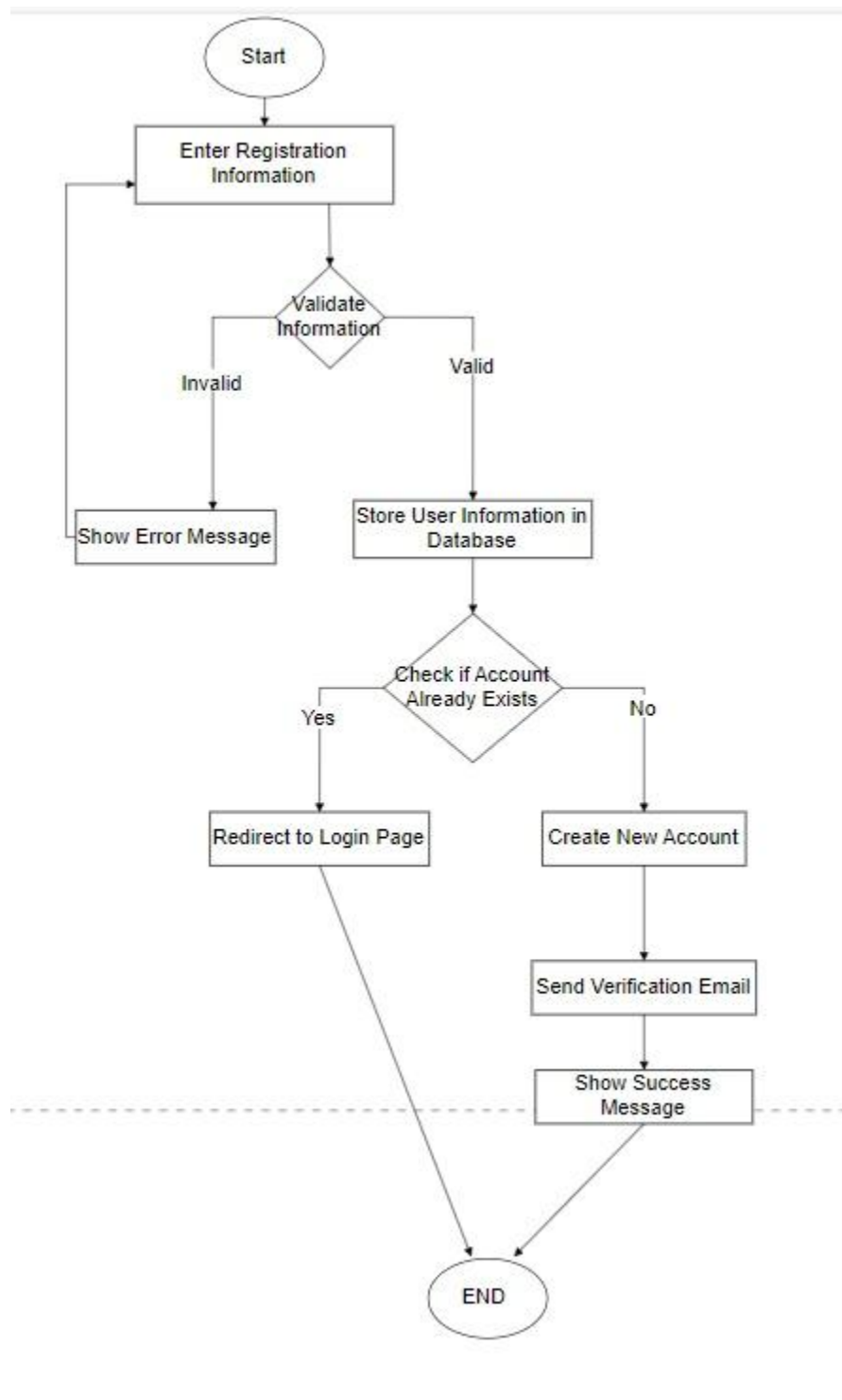


Figure 64 Flowchart of User Auth

3.14.2 Search

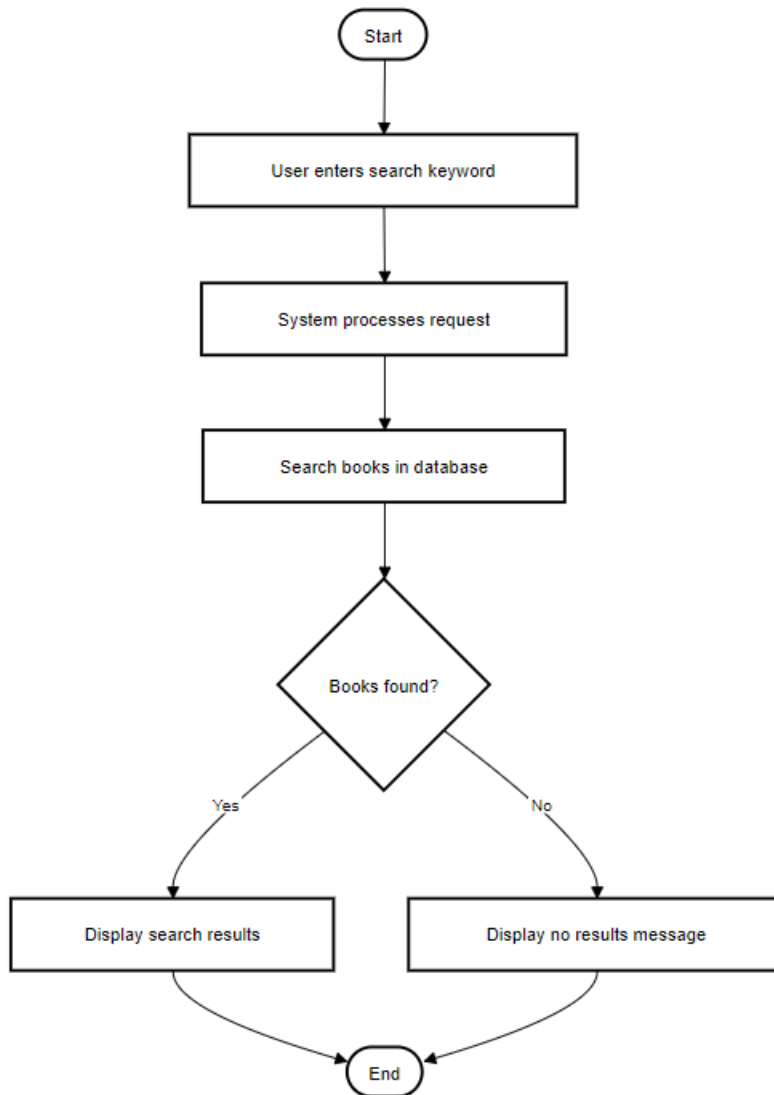


Figure 65 Flowchart of Search

3.14.3 In Browser Read

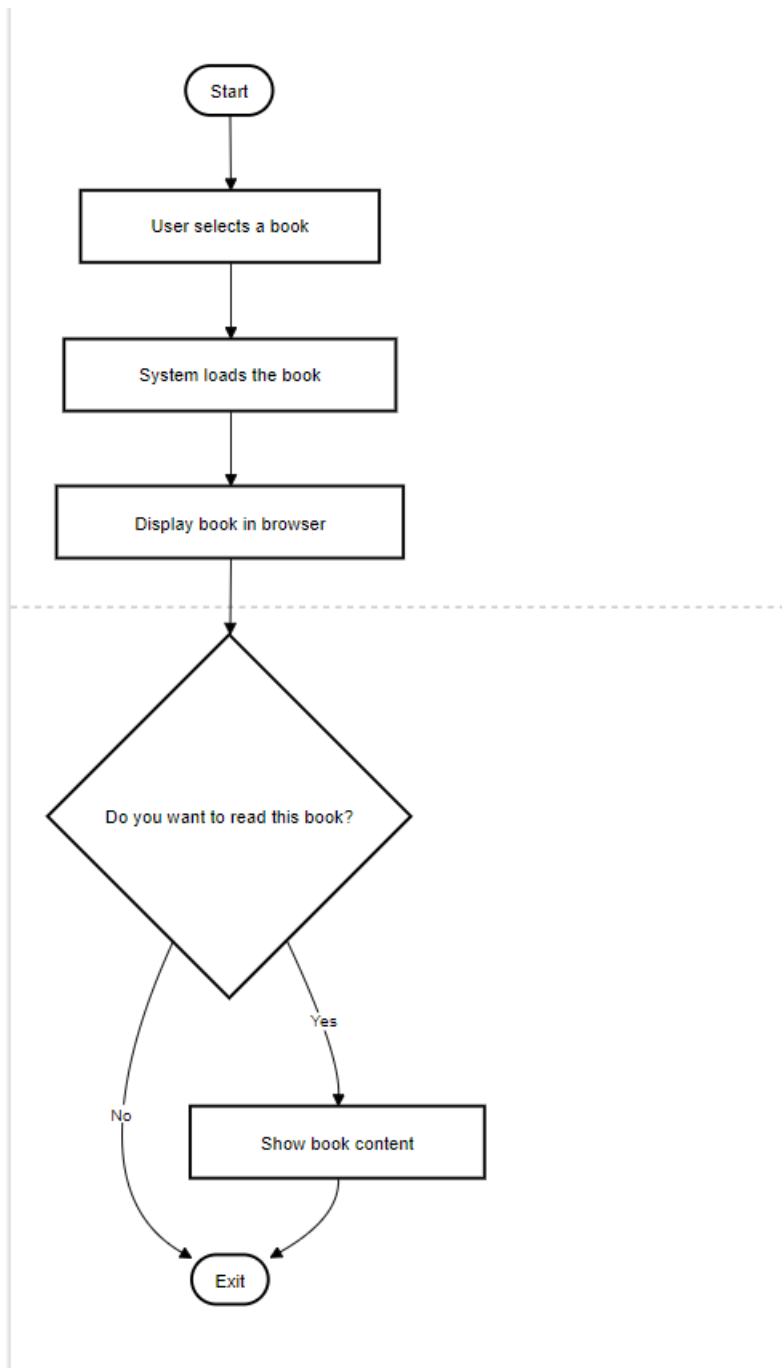


Figure 66 Flowchart of Inbrowser Read

3.14.4 Milestone

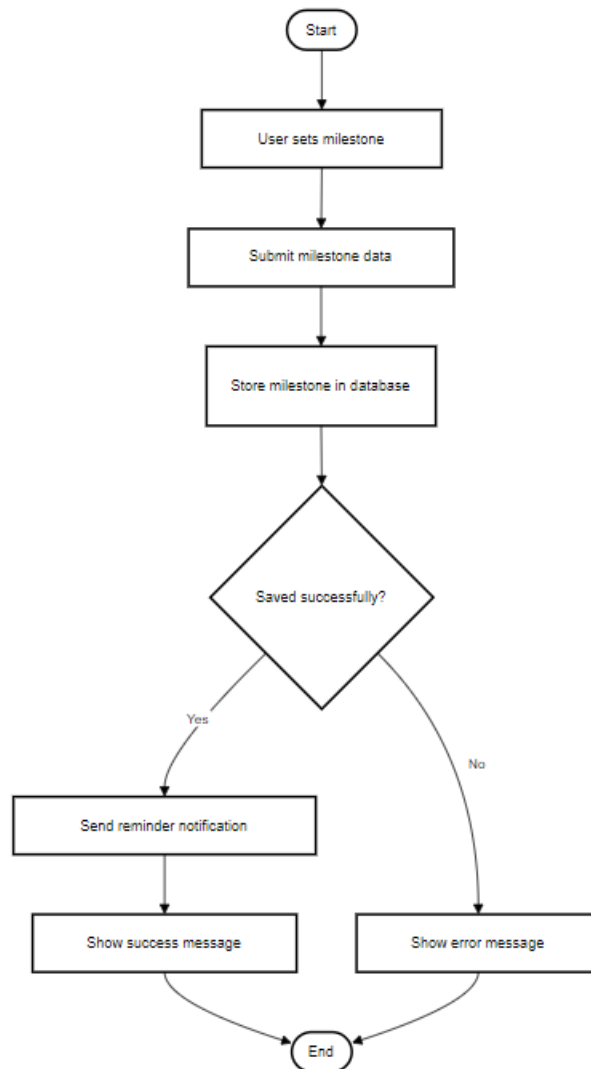


Figure 67 Flowchart of Milestone

3.14.5 TTS

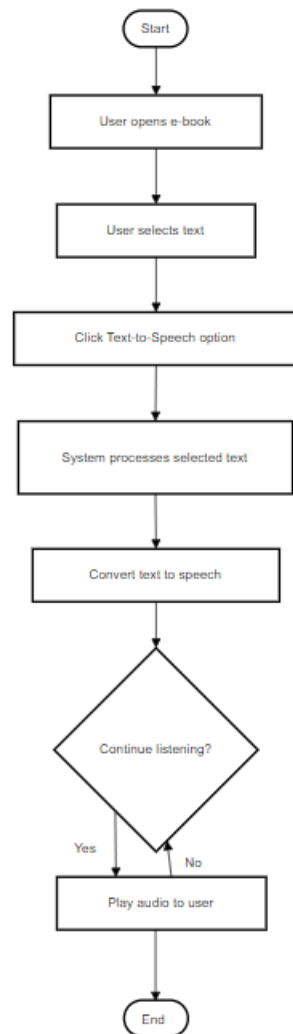


Figure 68 Flowchart of TTS

3.14.6 Smart Recommendations

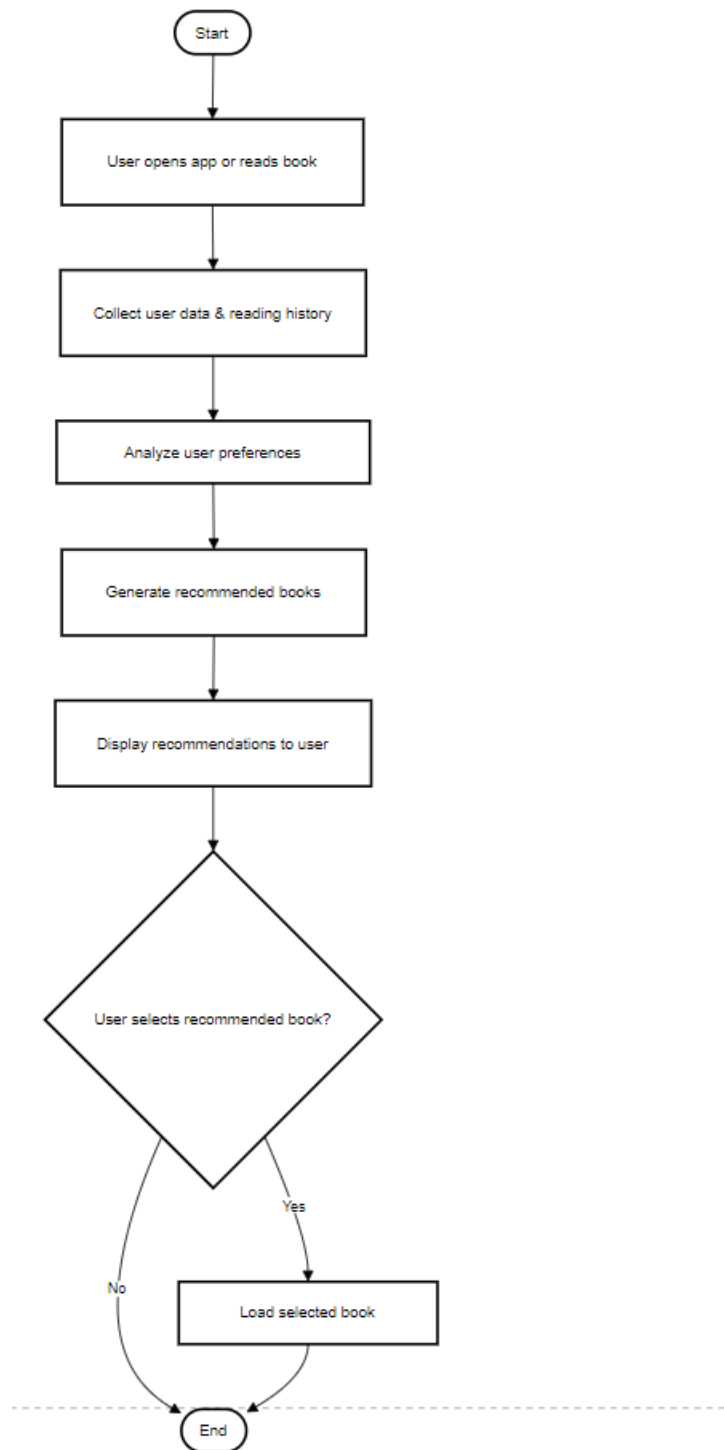


Figure 69 Flowchart of Recommendation

3.14.7 Admin CRUD

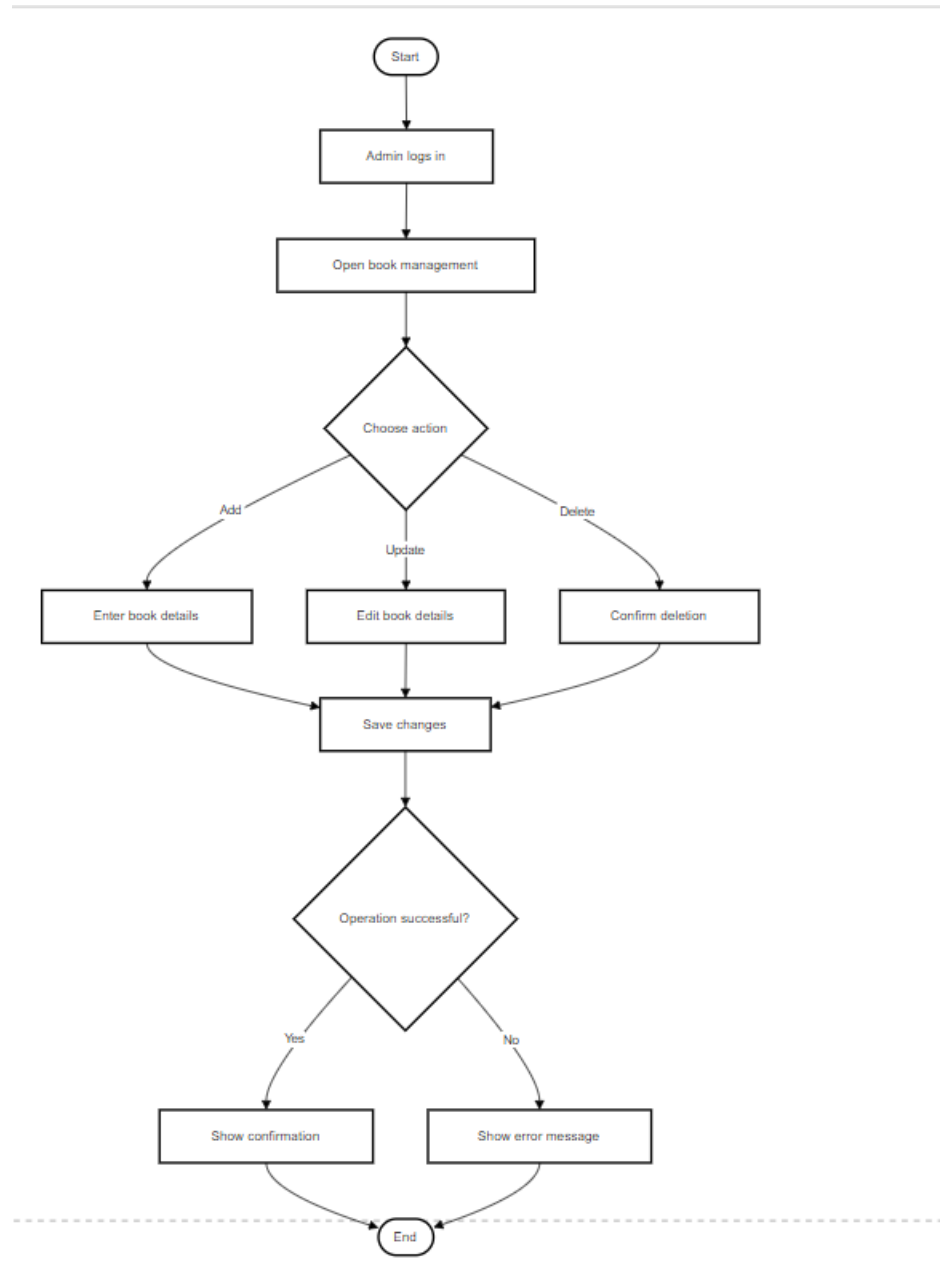


Figure 70 Flowchart of Admin CRUD

3.14.8 Admin User Management

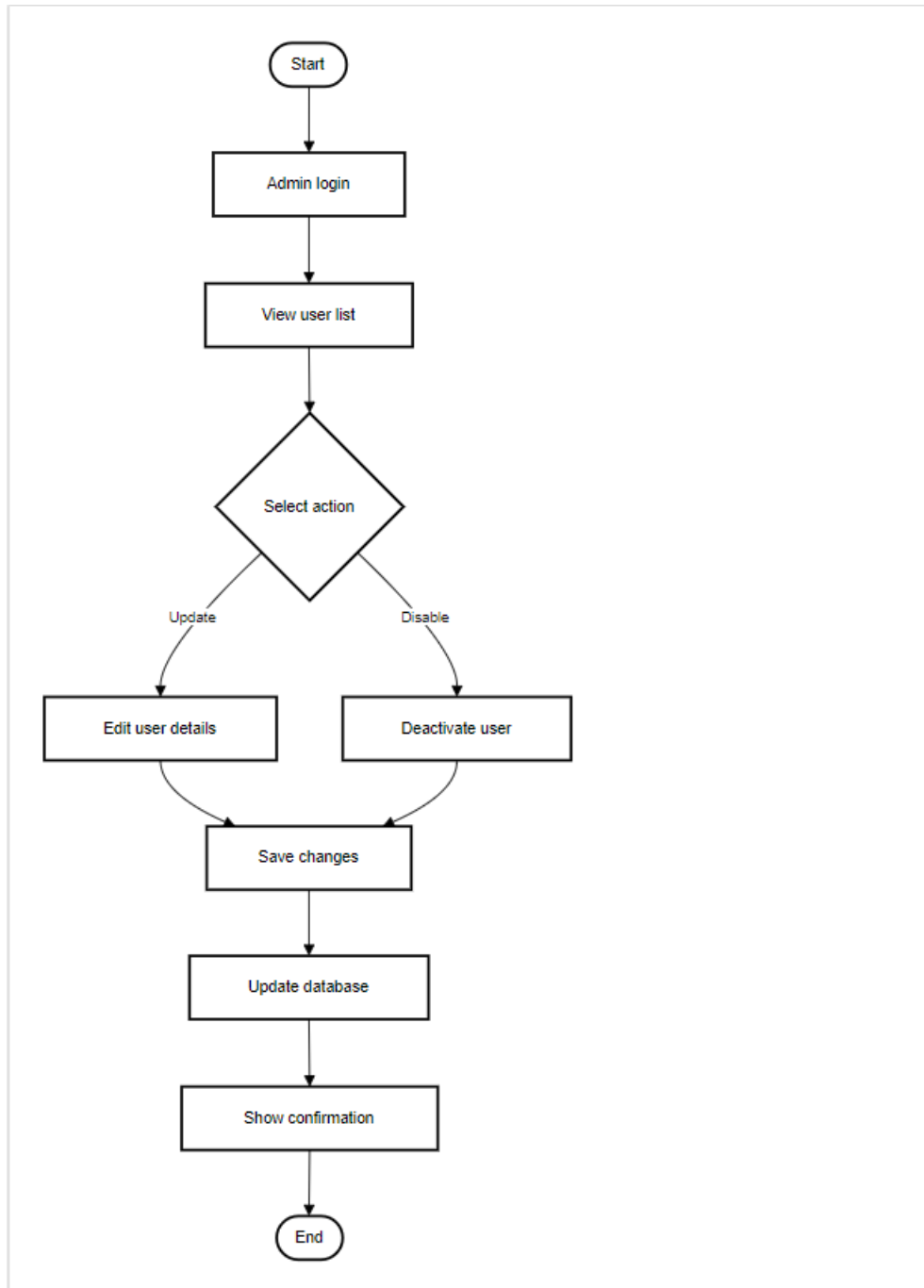


Figure 71 Flowchart of User Management

3.15 Collaboration Diagram

3.15.1 User Authentication

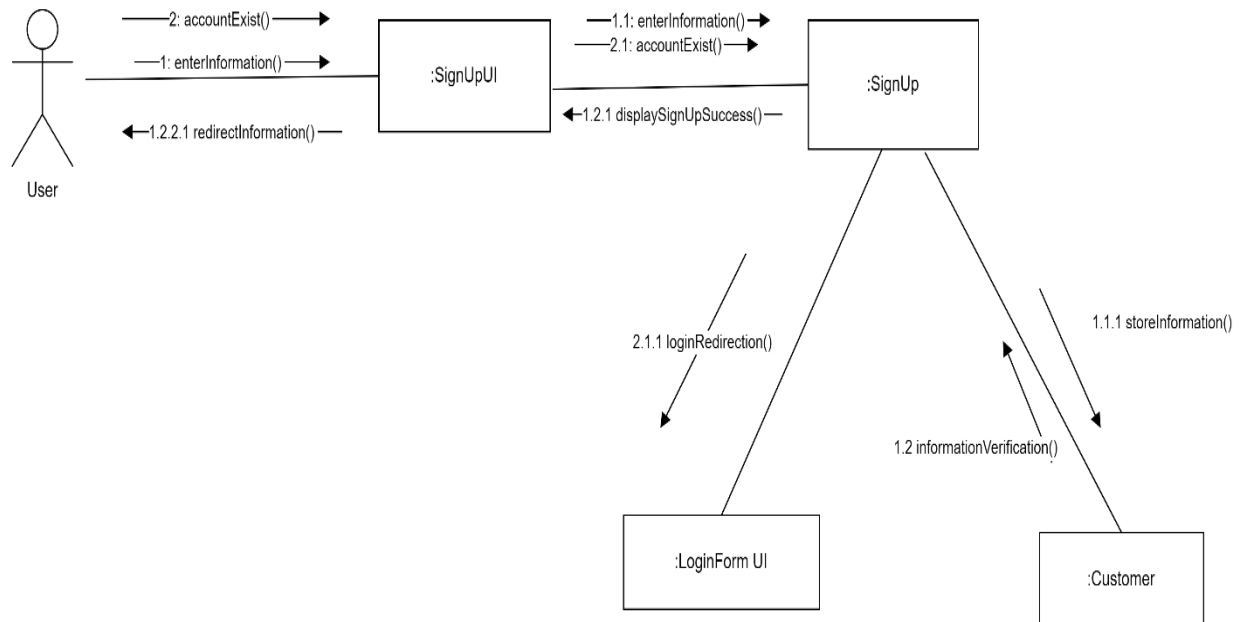


Figure 72 Collaboration Diagram of User Auth

3.15.2 Search

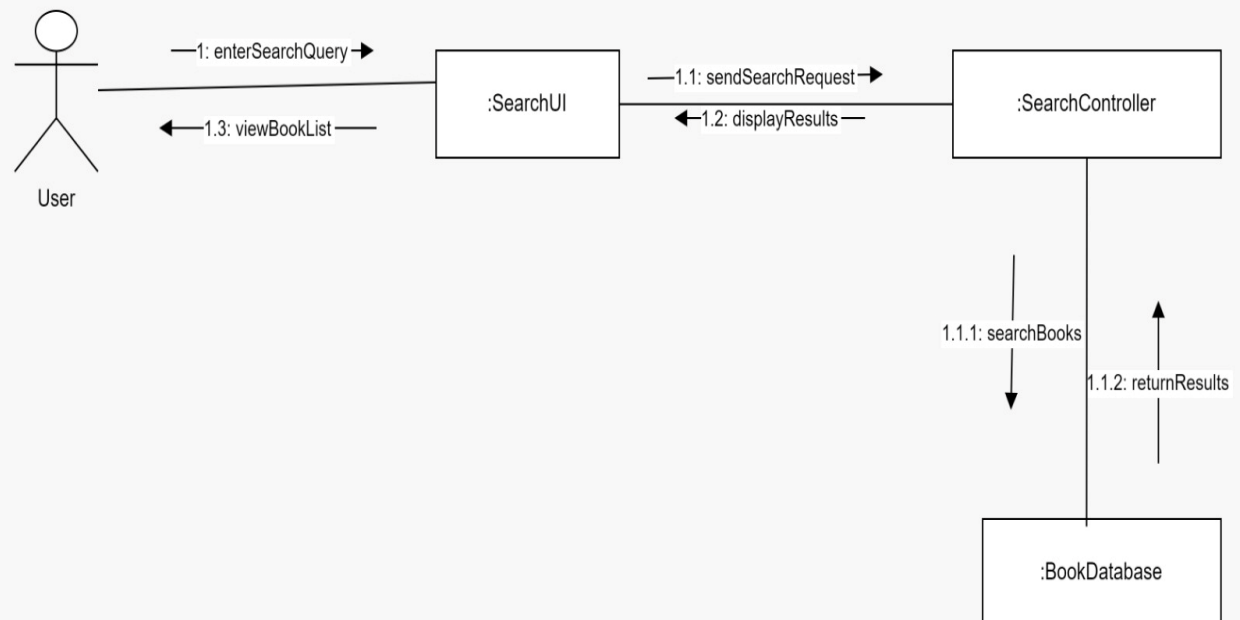


Figure 73 Collaboration Diagram of Search

3.15.3 Milestone

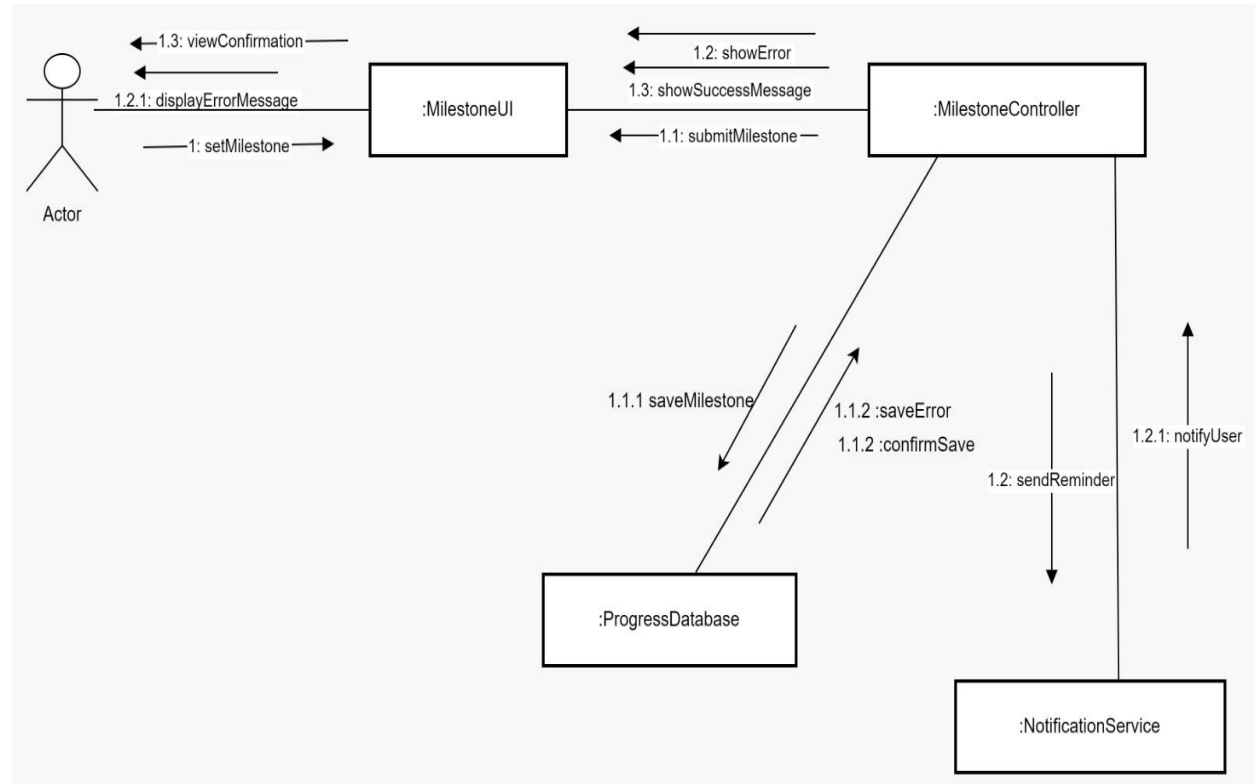


Figure 74 Collaboration Diagram of Milestone

3.15.4 TTS

2

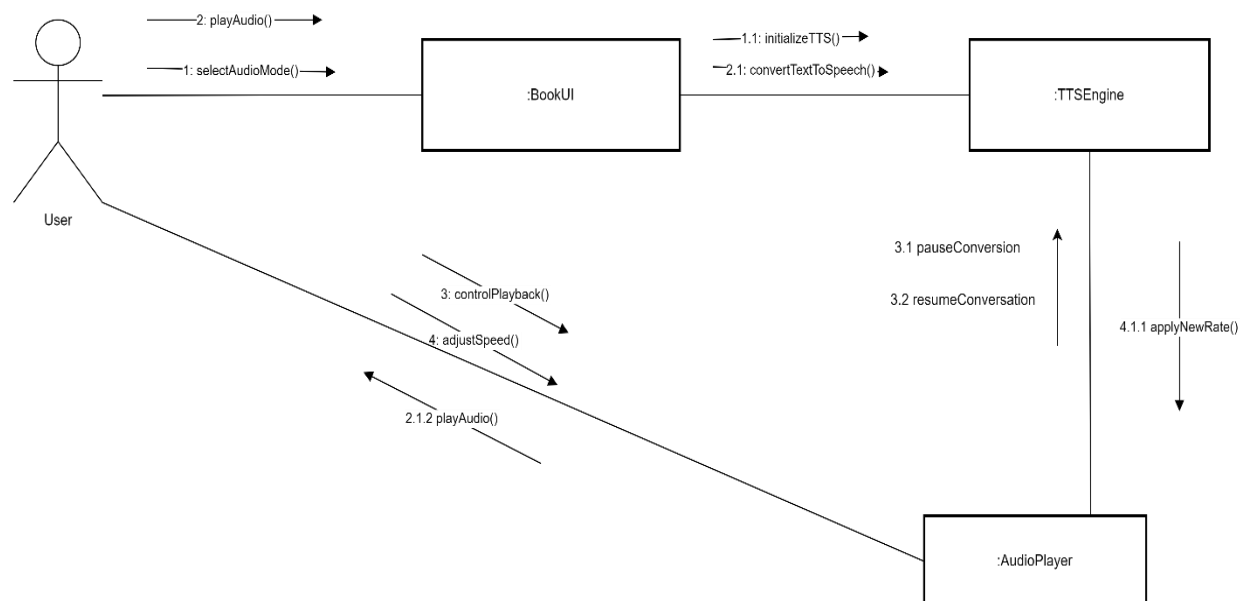


Figure 75 Collaboration Diagram of TTS

3.15.5 Recommendation

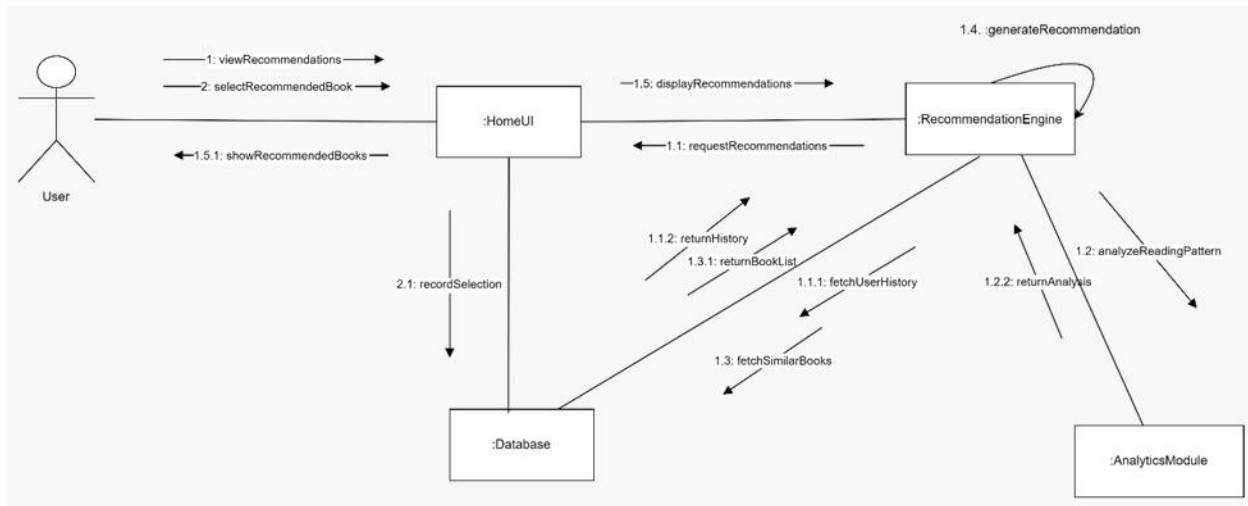


Figure 76 Collaboration Diagram of Recommendation

3.15.6 Admin CRUD

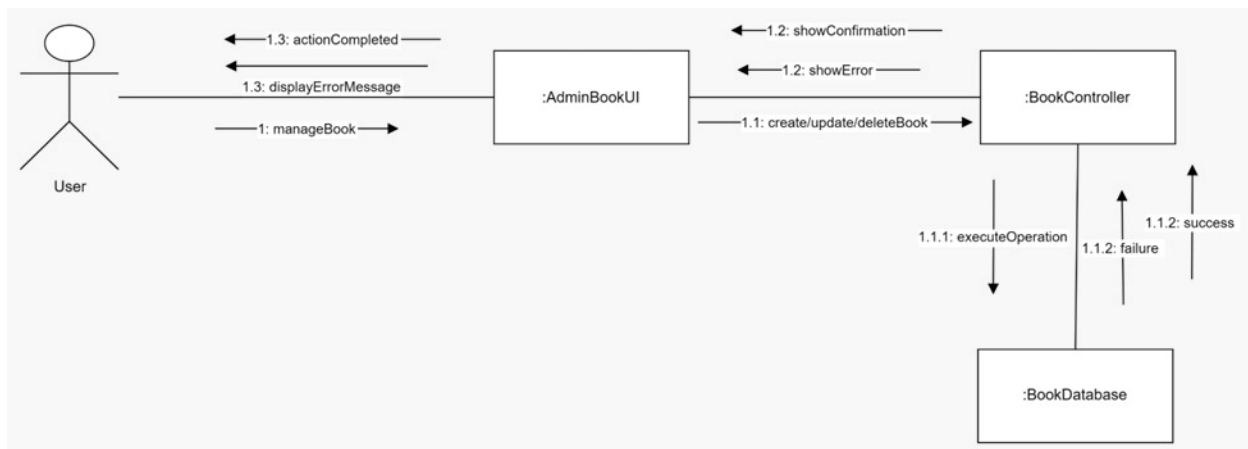


Figure 77 Collaboration Diagram of Admin CRUD

3.15.7 In browser Read

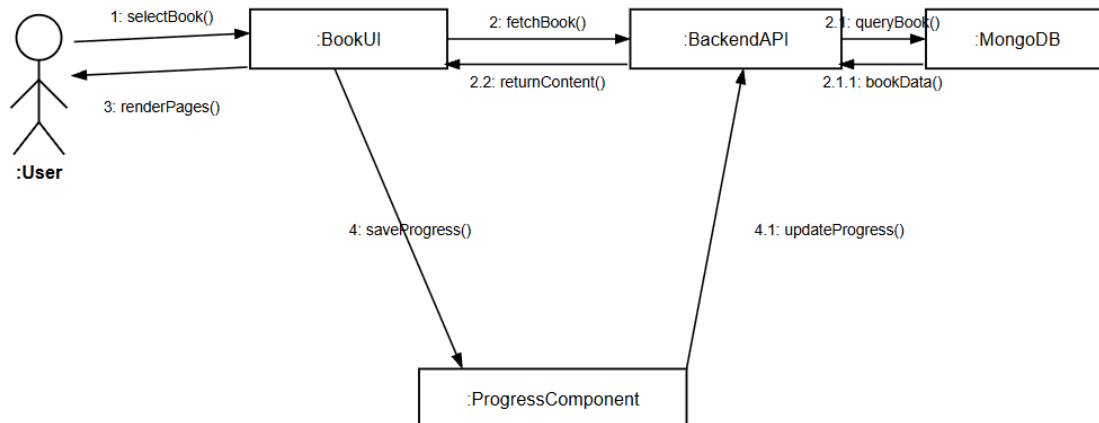


Figure 78 In Browser Read

3.15.8 Highlight

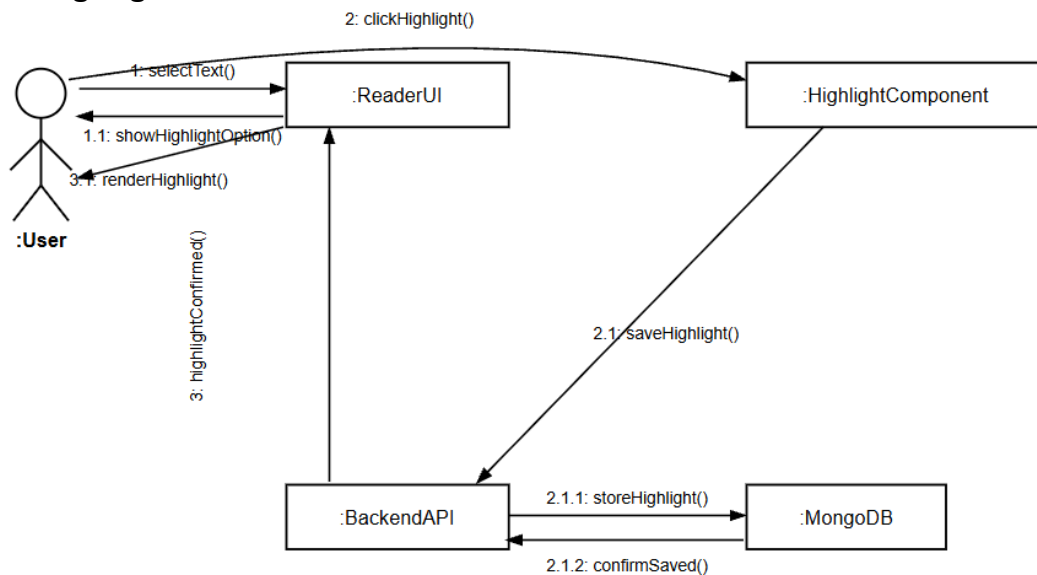


Figure 79 Highlight

3.15.9 Comment

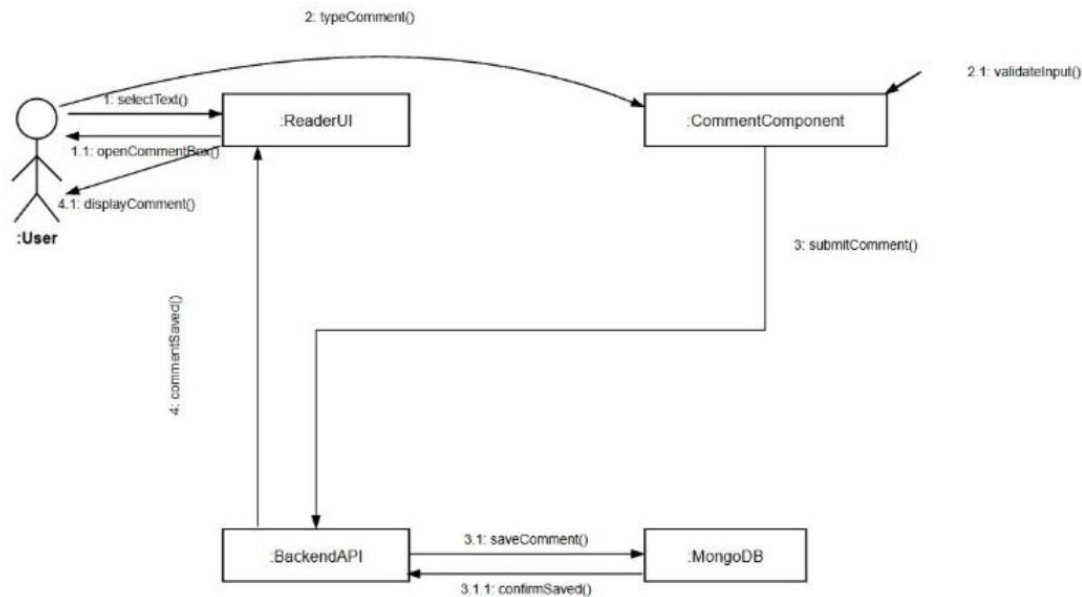


Figure 80 Comment

3.15.10 Bookmark

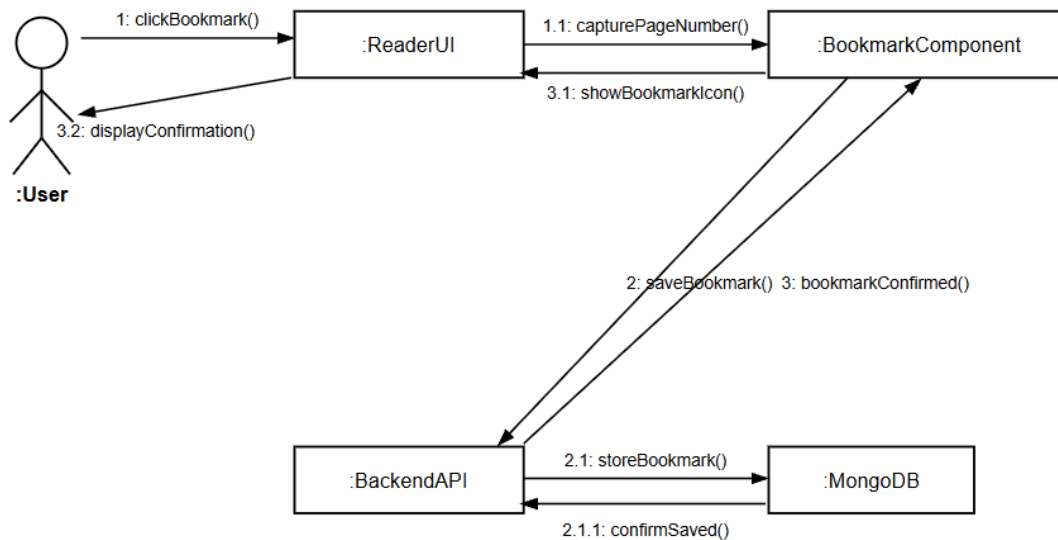


Figure 81 Bookmark

3.15.11 Payment

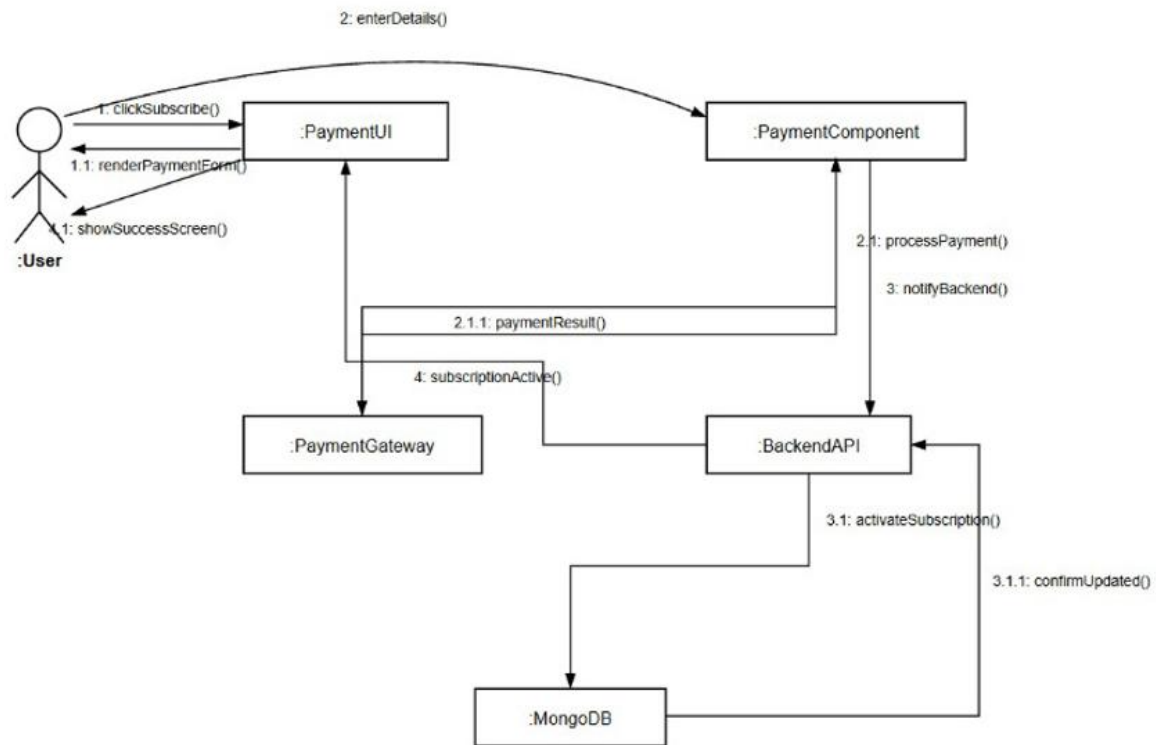


Figure 82 Payment

3.16 Activity Diagram

3.16.1 User Authentication

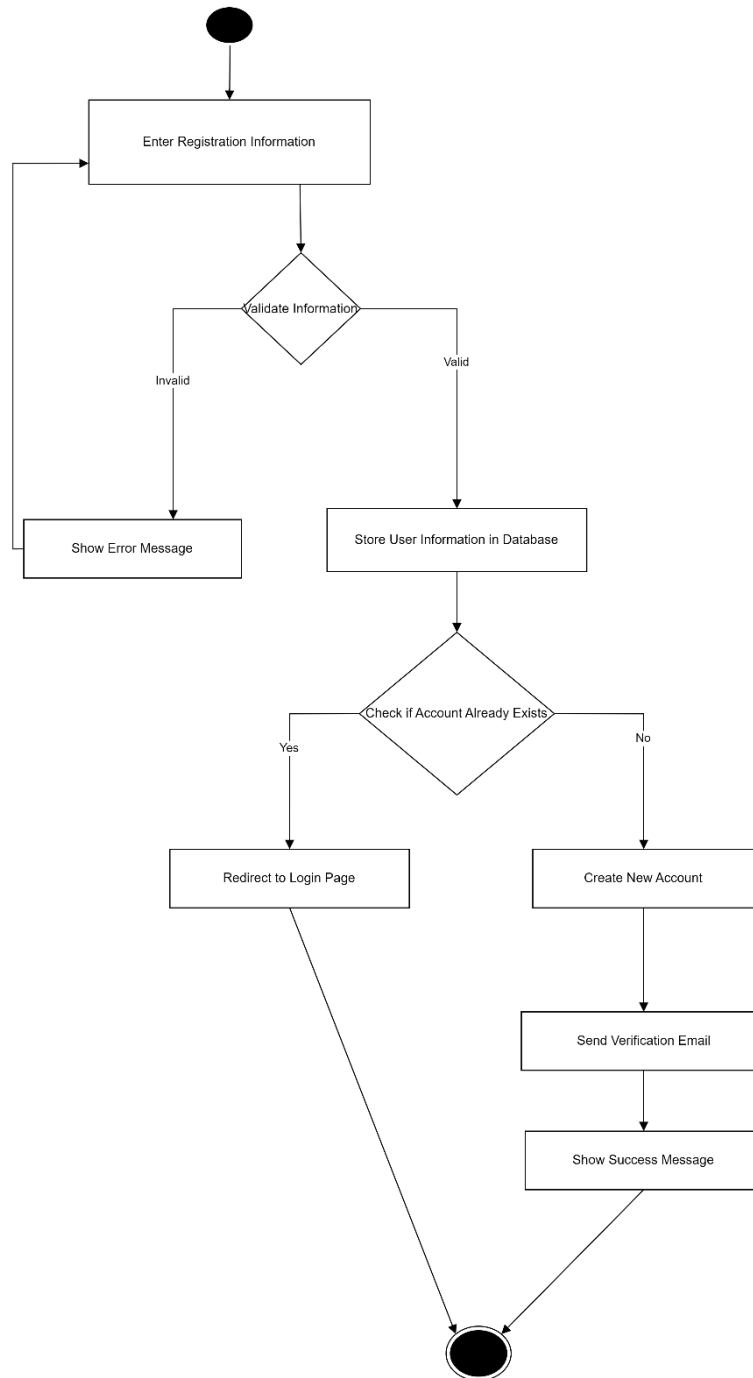


Figure 83 Activity Diagram of User Auth

3.16.2 Search

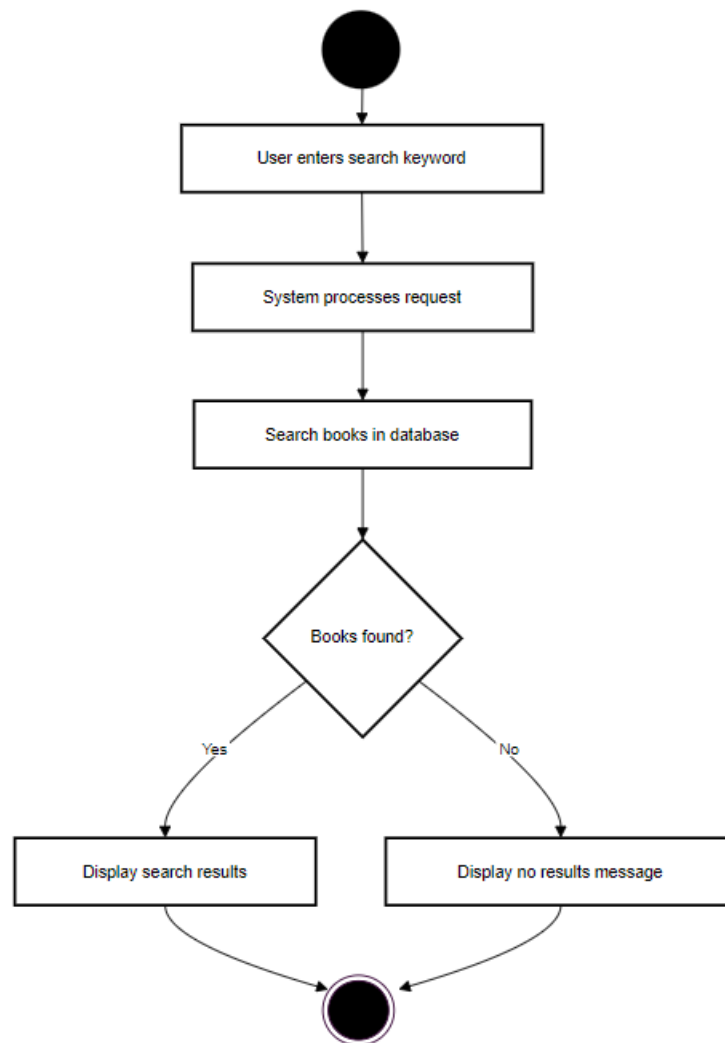


Figure 84 Activity Diagram of Search

3.16.3 In browser Read

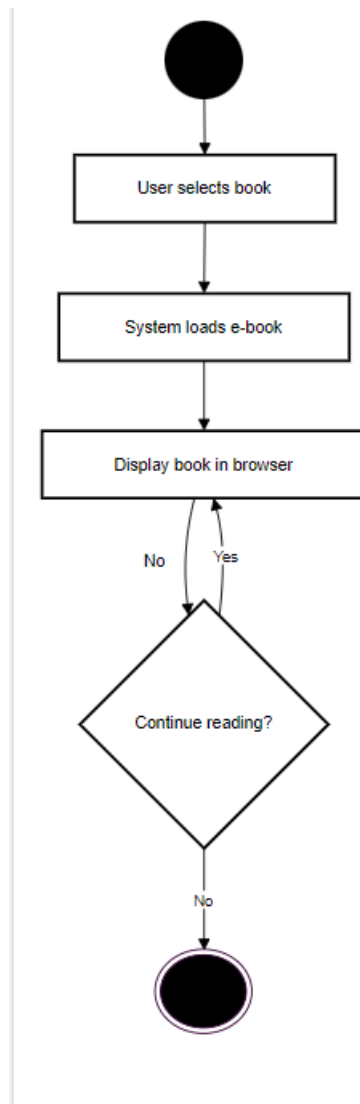


Figure 85 Activity Diagram of Inbrowser Read

3.16.4 Milestone

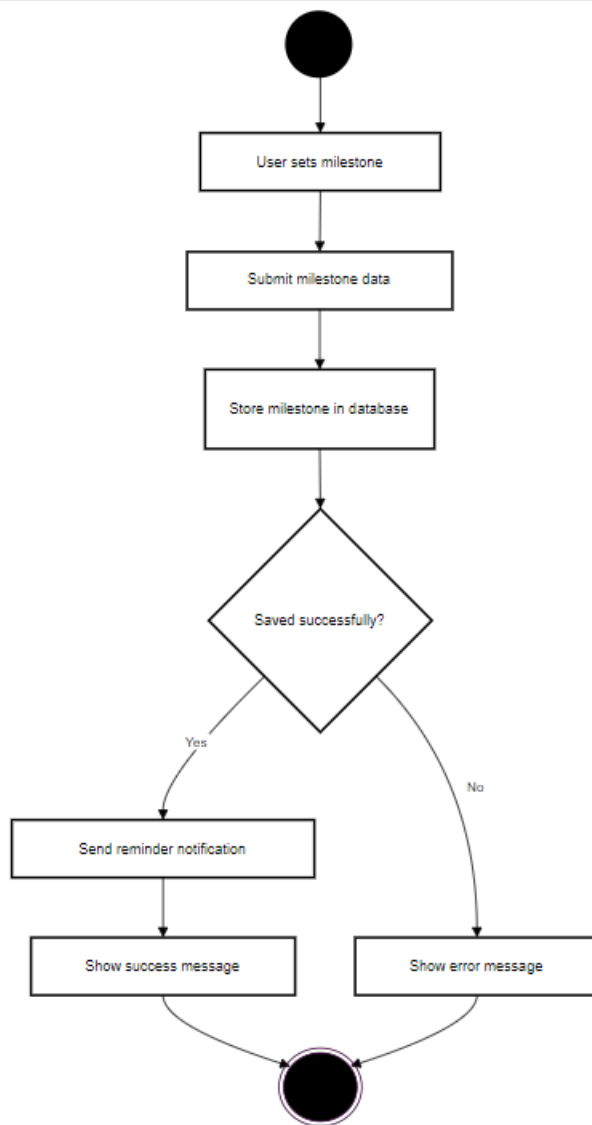


Figure 86 Activity Diagram of Milestone

3.16.5 Recommendation

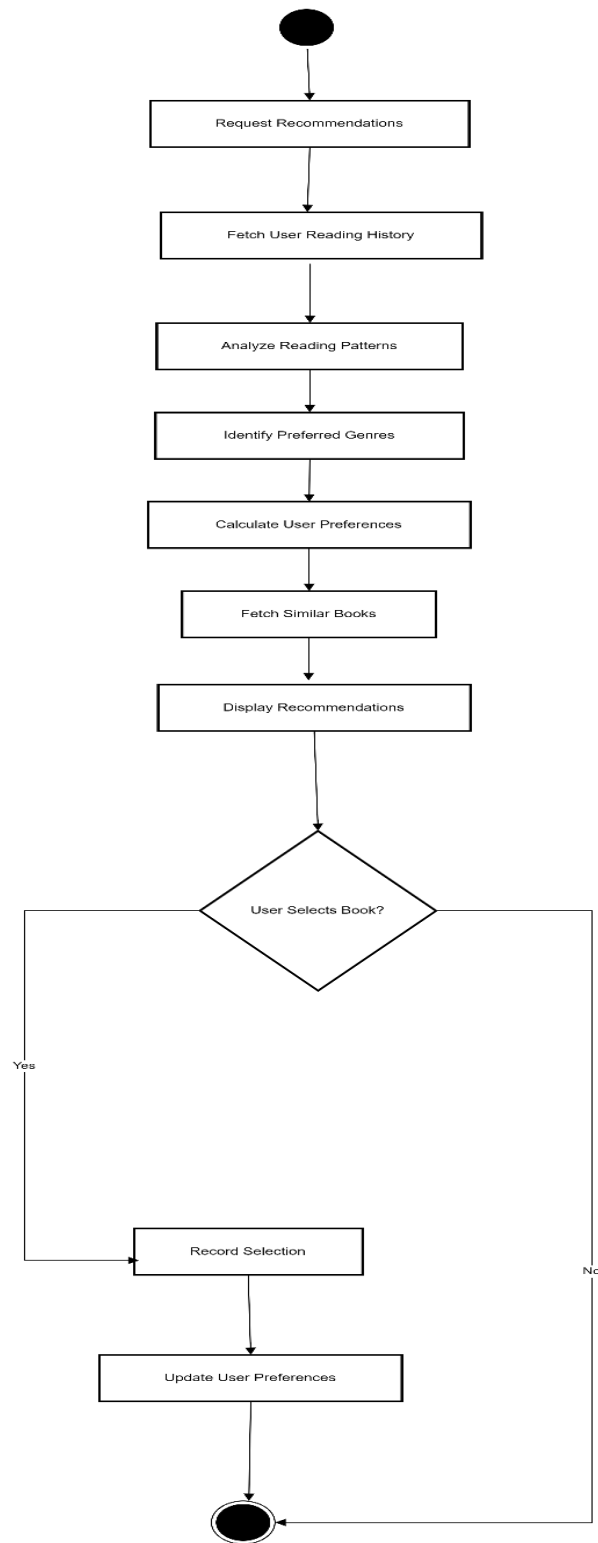


Figure 87 Activity Diagram of Recommendation

3.16.6 TTS

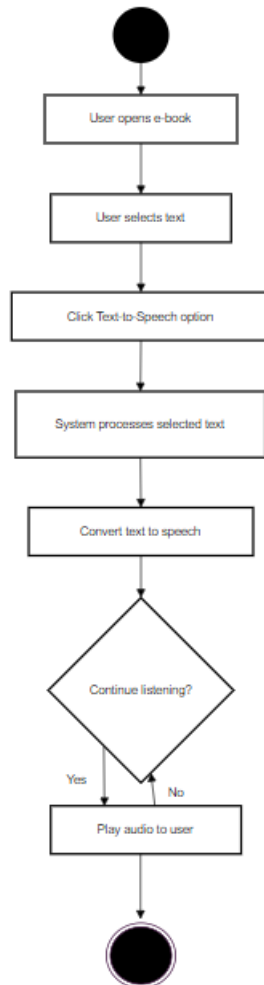


Figure 88 Activity Diagram of TTS

3.16.7 Admin CRUD

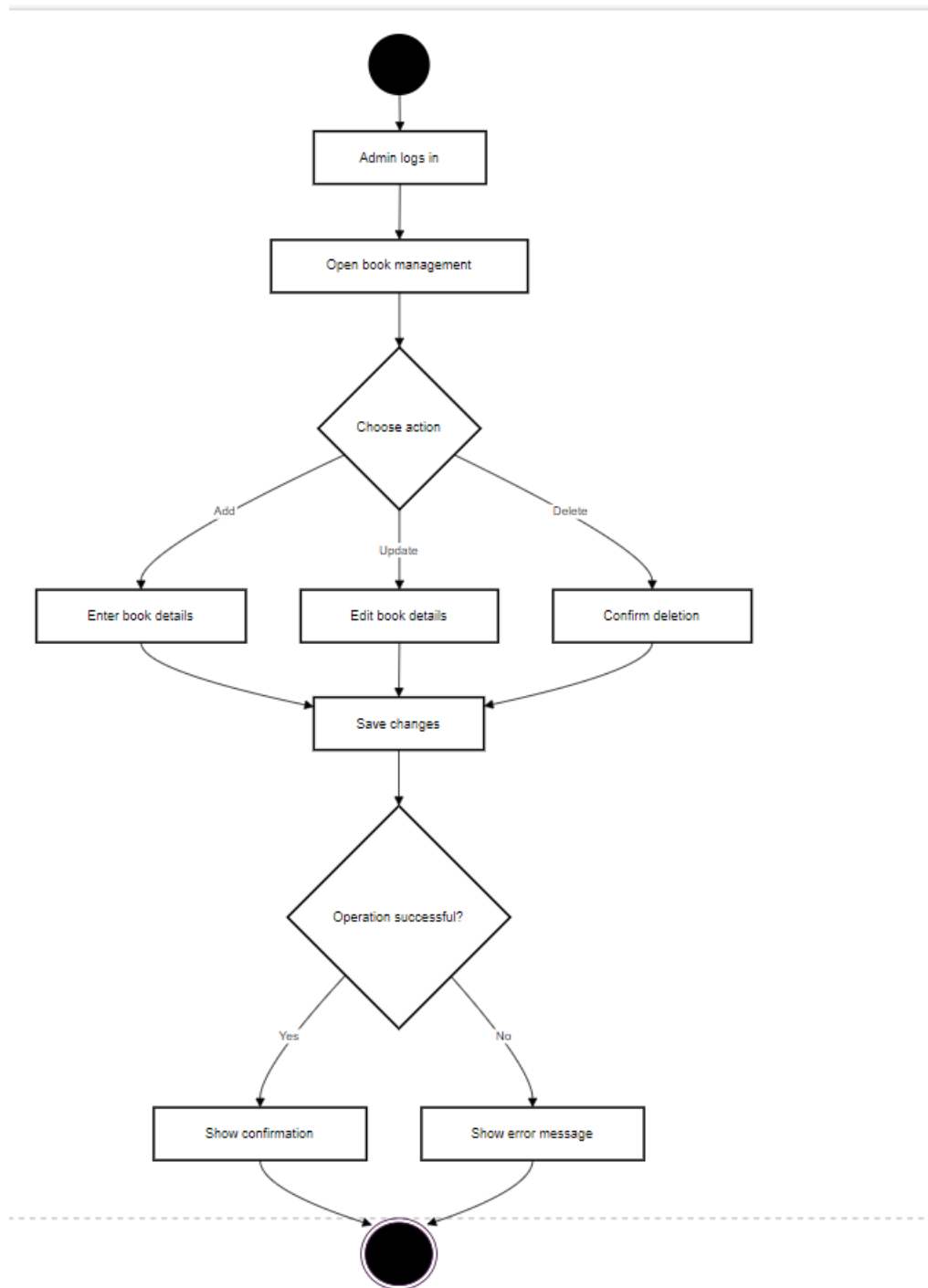


Figure 89 Activity Diagram of Admin CRUD

18

3.16.8 Admin User management

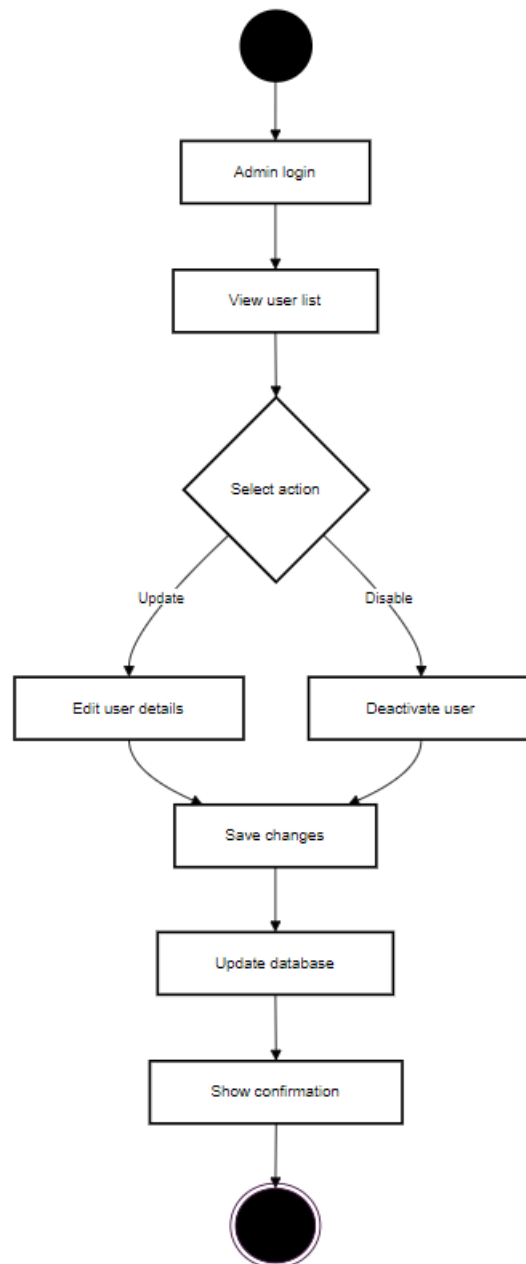


Figure 90 Activity Diagram of User Management

3.17 Class Diagram

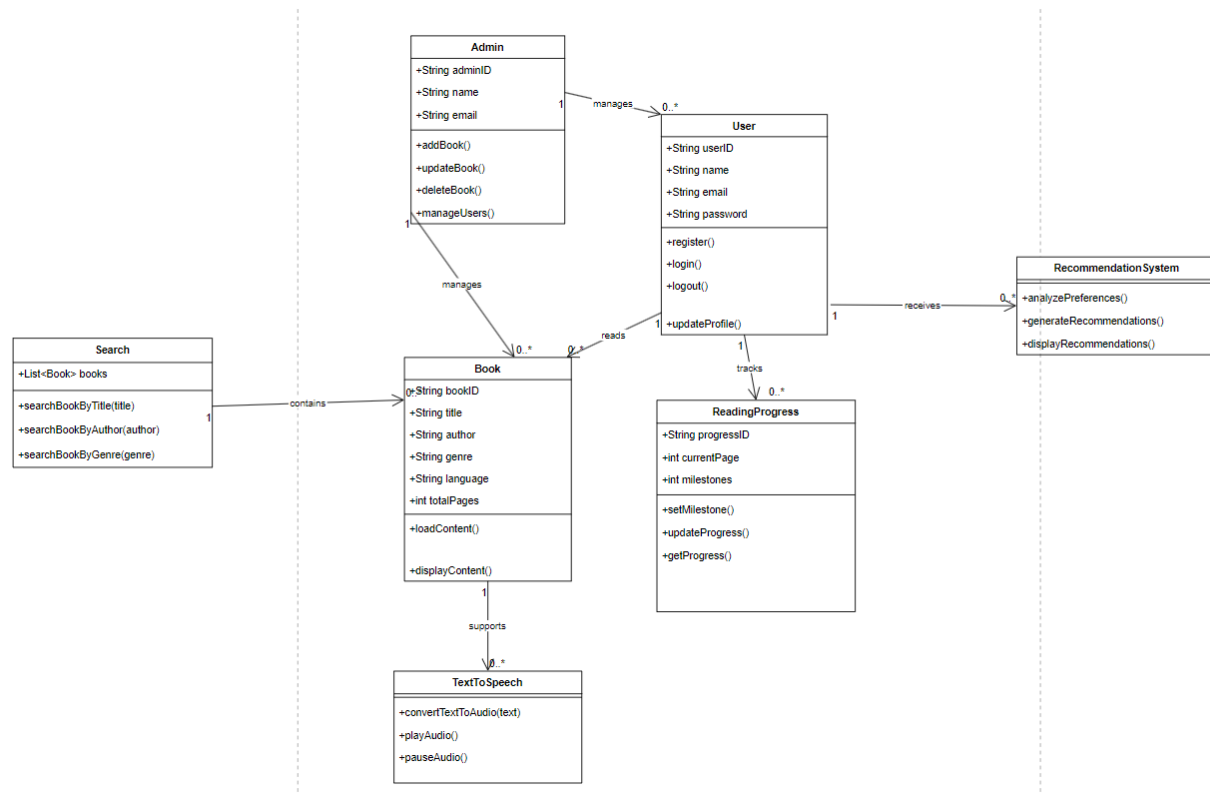


Figure 91 Class Diagram

3.18 Wireframe

3.18.1 Admin Web App Wireframes

SmartShelf Admin

Logout

Menu

- Dashboard
- Manage Books
- Manage Users

Dashboard Overview

Total Books
245

Total Users
1,523

Active Today
87

Recent Activity

Date	Activity	User
2025-12-20	New book added	admin@example.com
2025-12-20	User registered	user123@example.com

Figure 92 Wireframe Admin Dashboard

SmartShelf Admin

Logout

Menu

- Dashboard
- Manage Books
- Manage Users

Add New Book

< Back

Book Title *

Enter book title

Author *

Enter author name

Genre *

Select genre

Subject

Enter subject

ISBN

Enter ISBN number

Publication Date

YYYY-MM-DD

Description

Enter book description...

Upload Book File (PDF) *

Click to upload or drag and drop

* Required fields

Cancel Save Book

Figure 93 Wireframe of Admin CRUD

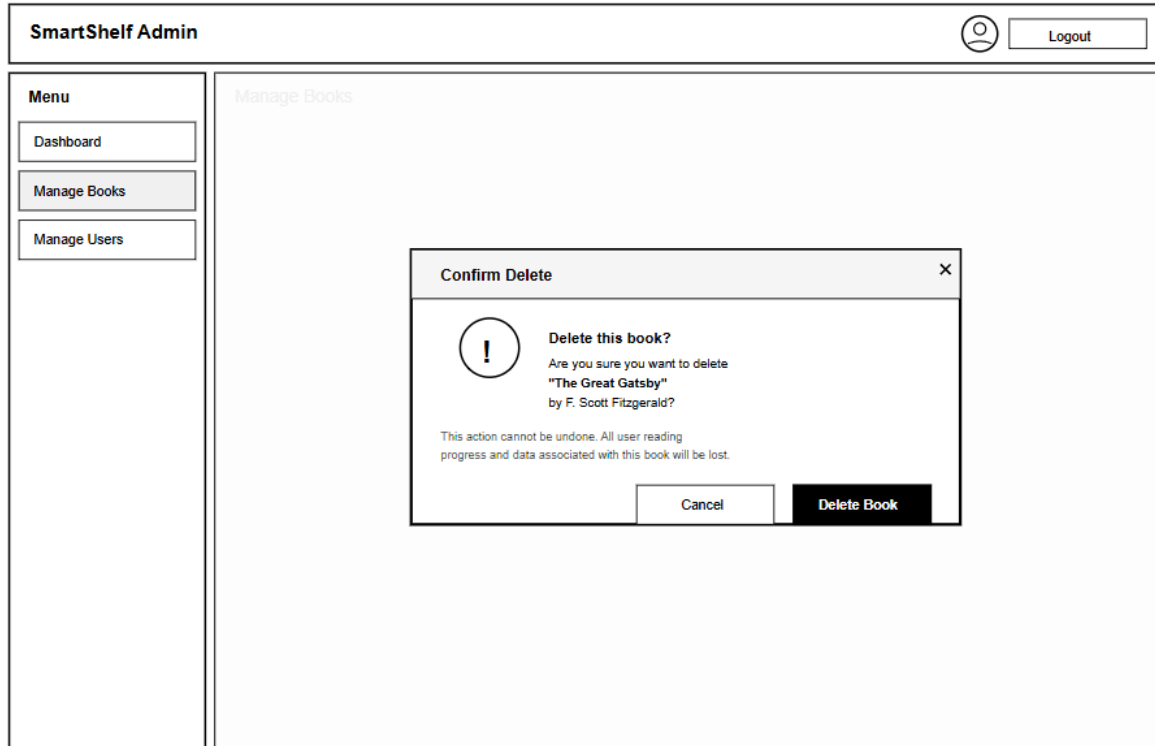


Figure 94 Wireframe of Admin Delete

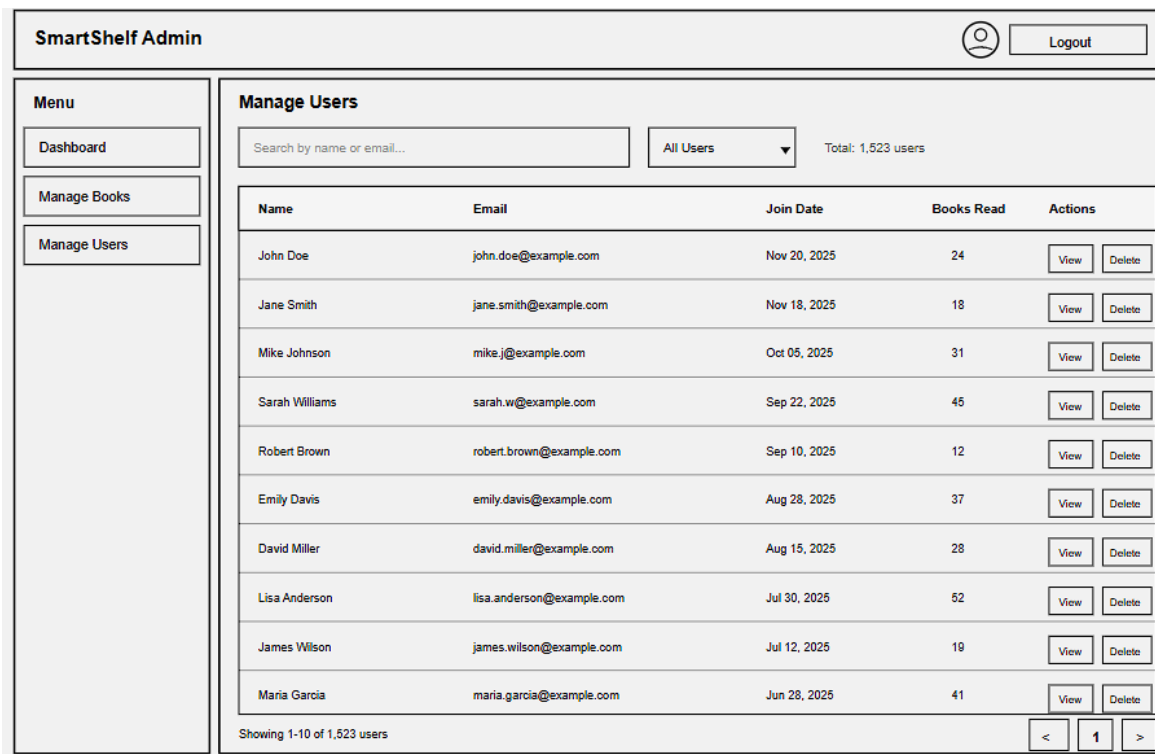


Figure 95 Wireframe of User Management

5

3.18.2 User Mobile App Wireframes

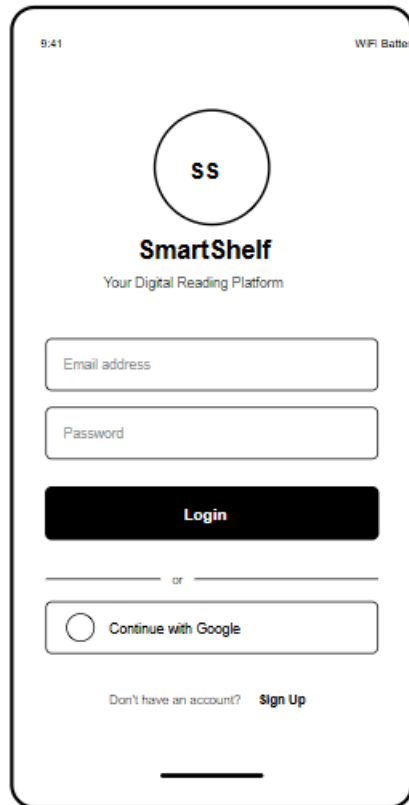


Figure 96 Wireframe of Login

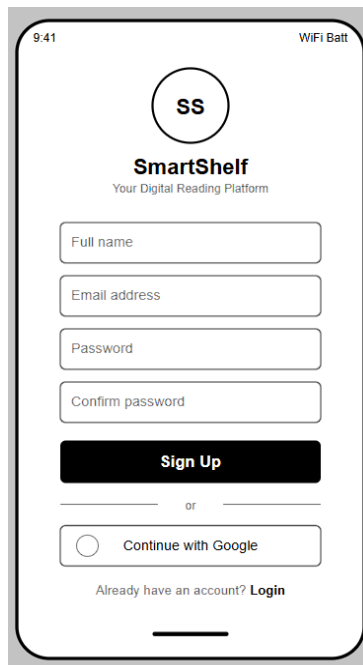


Figure 97 Wireframe of Signup

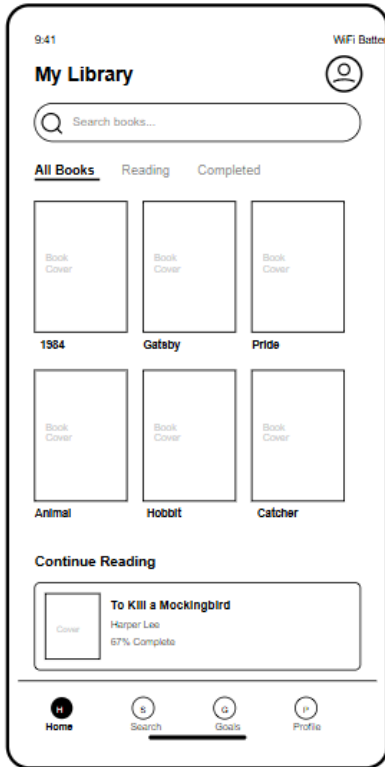


Figure 98 Wireframe of All Books

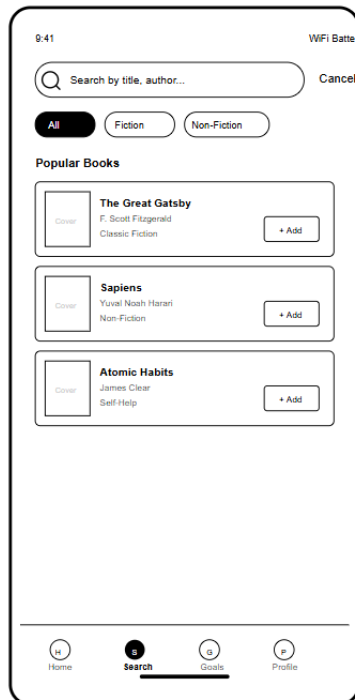


Figure 99 Wireframe of Search

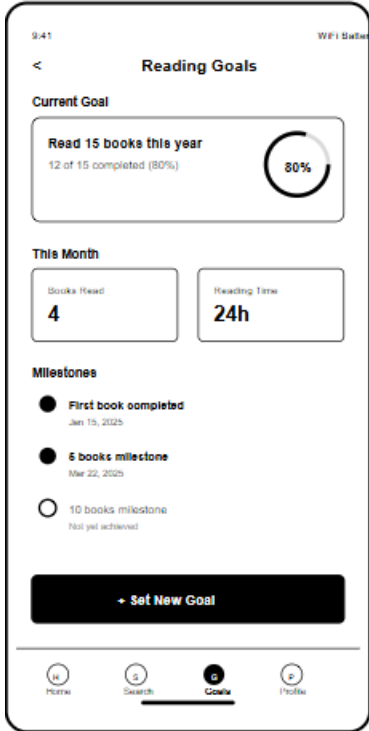


Figure 100 Wireframe of Milestone

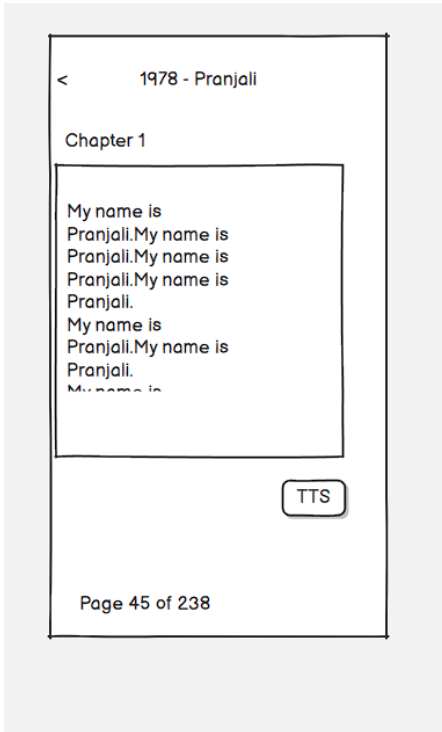


Figure 101 Wireframe of TTS

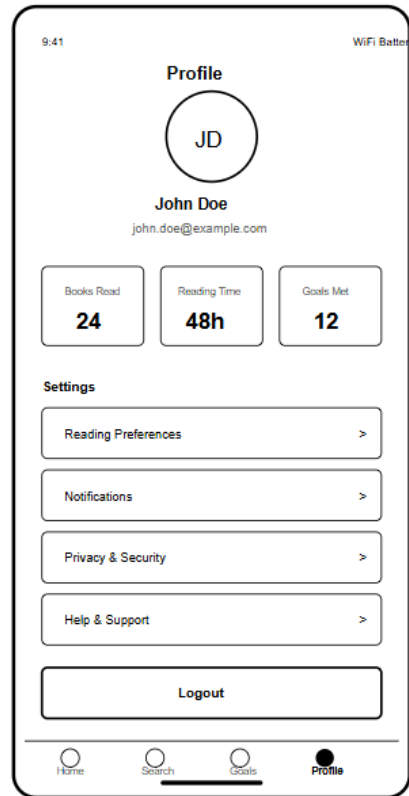


Figure 102 Wireframe of User Dashboard

3.19 UI

3.19.1 Admin Web App UI

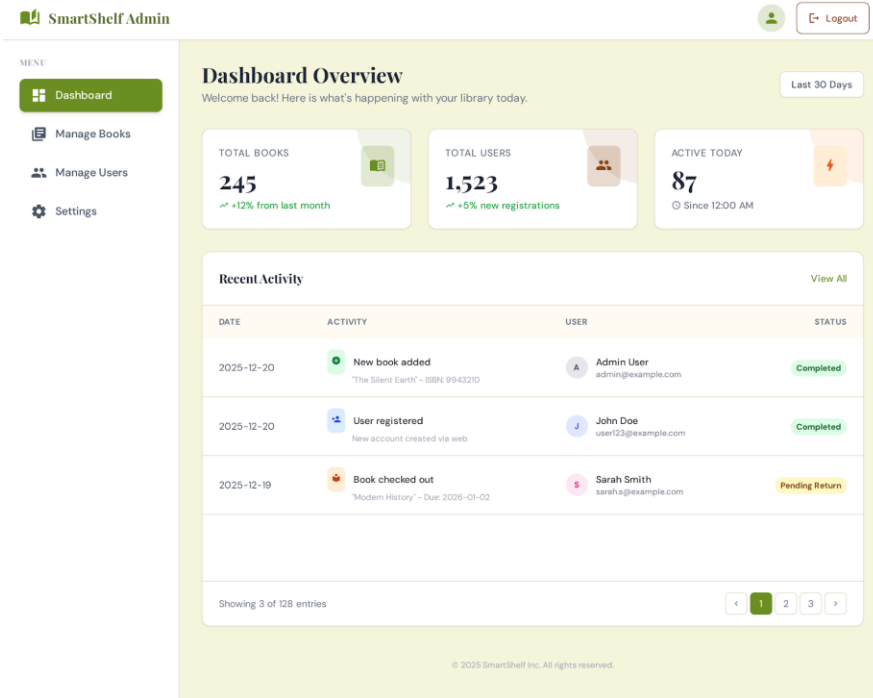


Figure 103 UI of Admin Dashboard

The screenshot shows the "Add New Book" form in the SmartShelf Admin application. The form is titled "Add New Book" and includes a "Back" link. It contains several input fields: "Book Title" (required), "Author" (required), "Genre" (a dropdown menu), "Subject" (required), "ISBN" (required), and "Publication Date" (required, with a date picker). There is also a "Description" field and an "Upload Book File (PDF)" section with a file upload icon and instructions. The form has "Cancel" and "Save Book" buttons at the bottom right. A footer note indicates "SmartShelf v2.4.0" and "Required fields".

Figure 104 UI of Admin CRUD

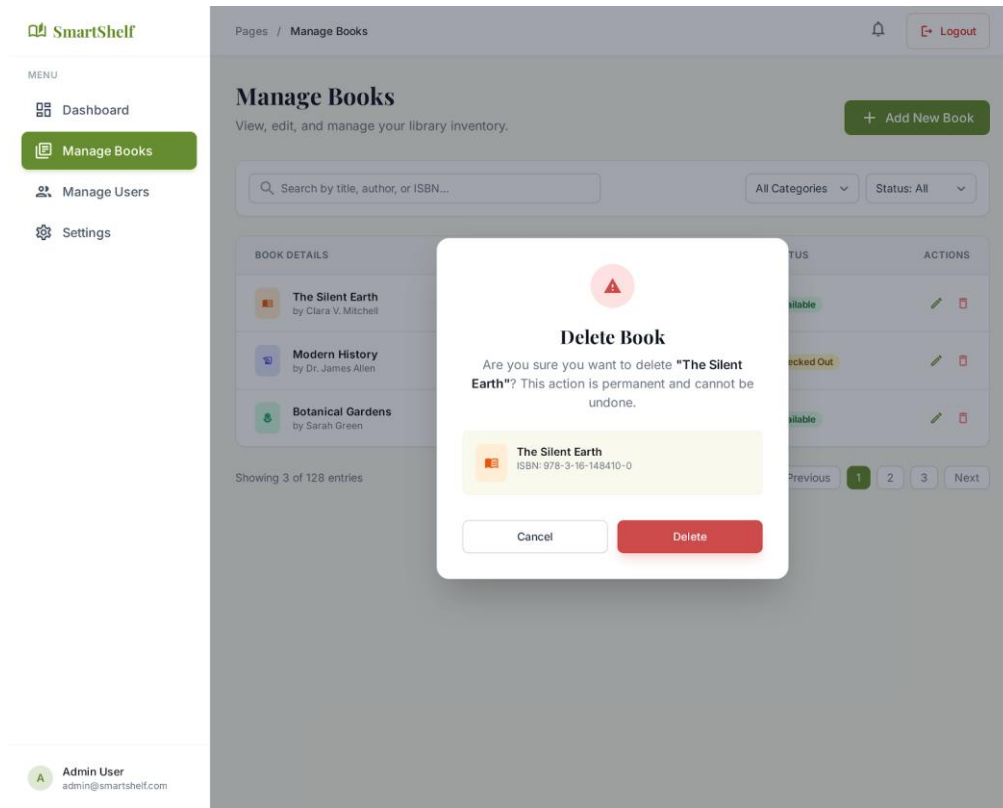


Figure 105 UI of Book Delete

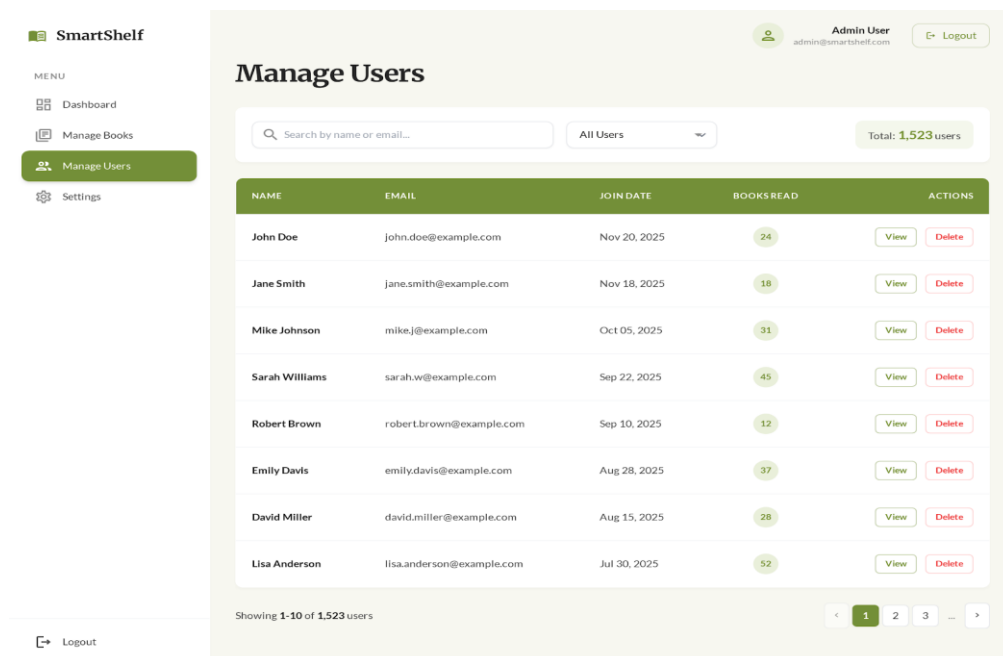


Figure 106 UI of User Management

19 3.19.2 Users Mobile App UI

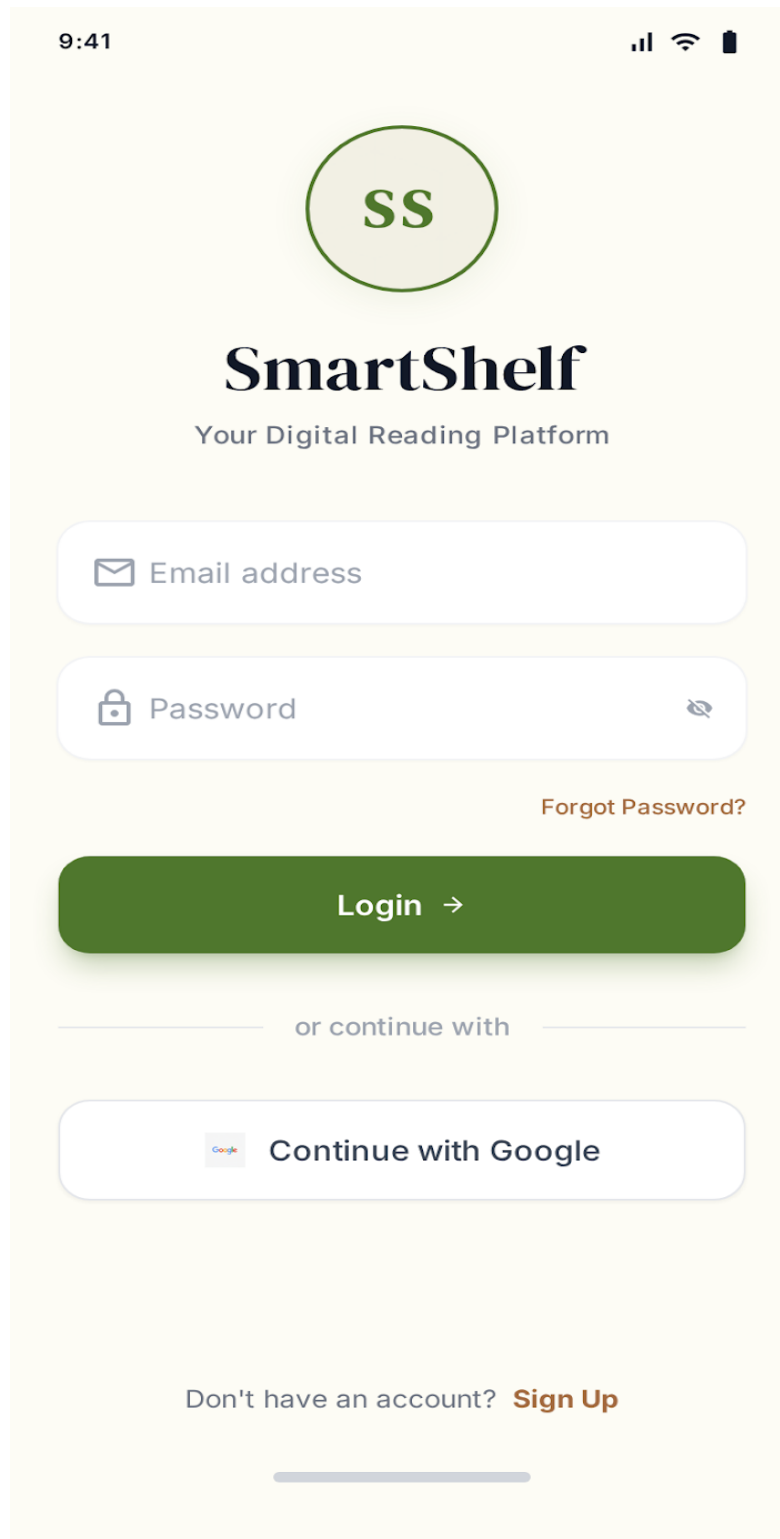
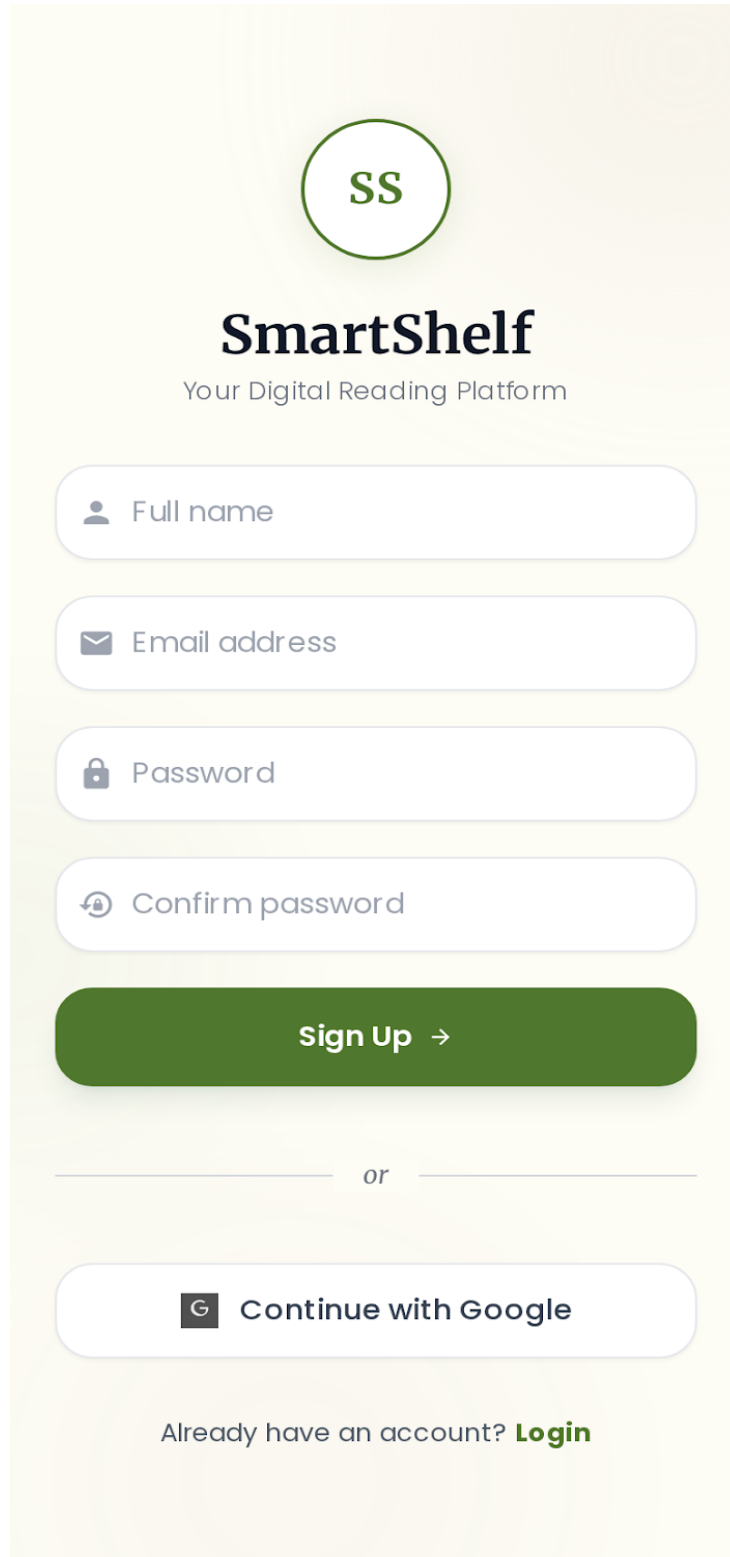


Figure 107 UI of Login



The image shows a mobile app interface for 'SmartShelf', a digital reading platform. At the top, there is a green circular logo with the letters 'SS'. Below the logo, the text 'SmartShelf' is displayed in a large, bold, black font, followed by the tagline 'Your Digital Reading Platform' in a smaller, grey font. The main section of the form consists of four rounded rectangular input fields stacked vertically. The first field is labeled 'Full name' with a person icon. The second field is labeled 'Email address' with an envelope icon. The third field is labeled 'Password' with a lock icon. The fourth field is labeled 'Confirm password' with a circular arrow icon. Below these fields is a large, dark green button with the text 'Sign Up →'. Underneath the button, there is a horizontal line with the word 'or' in the center. Below this line is a white button with a black 'G' icon and the text 'Continue with Google'. At the bottom of the form, there is a link that says 'Already have an account? Login', where 'Login' is in a bold green font.

Figure 108 UI of Signup

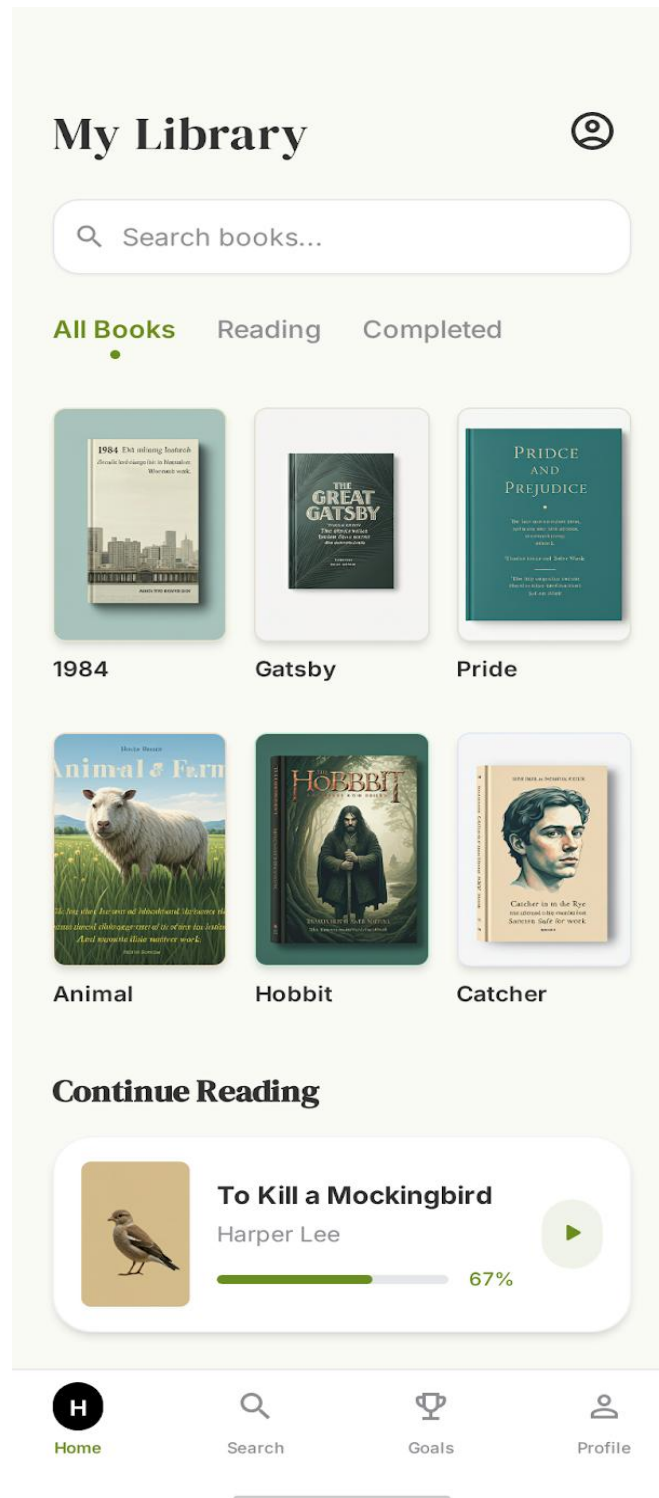


Figure 109 UI of Books

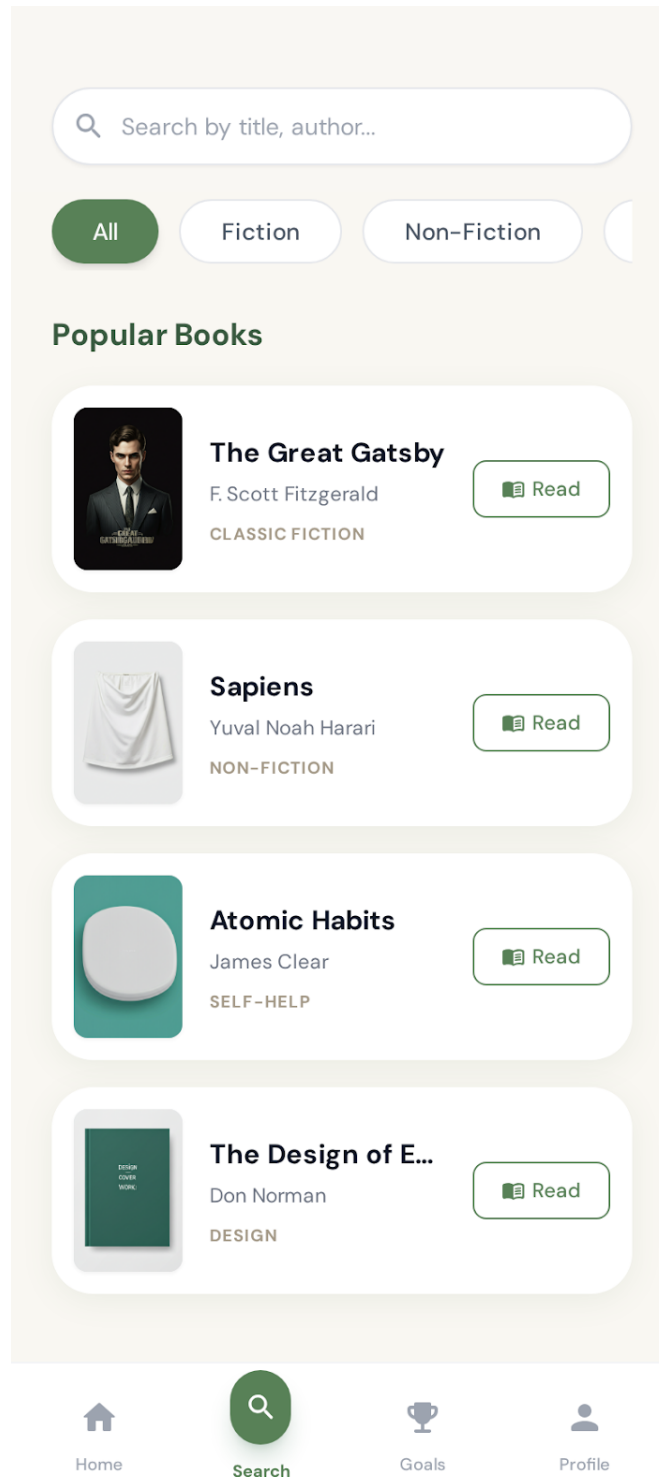


Figure 110 UI of Search

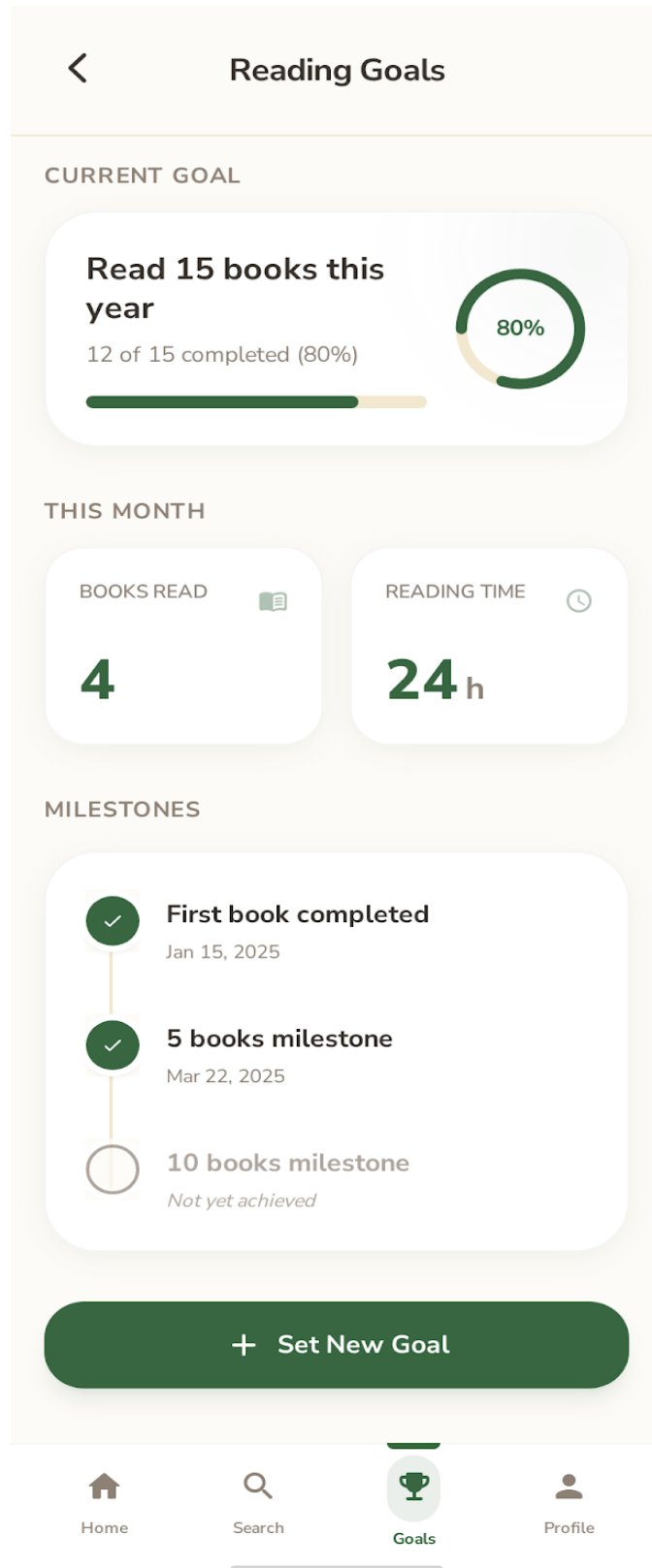


Figure 111 UI of Milestone

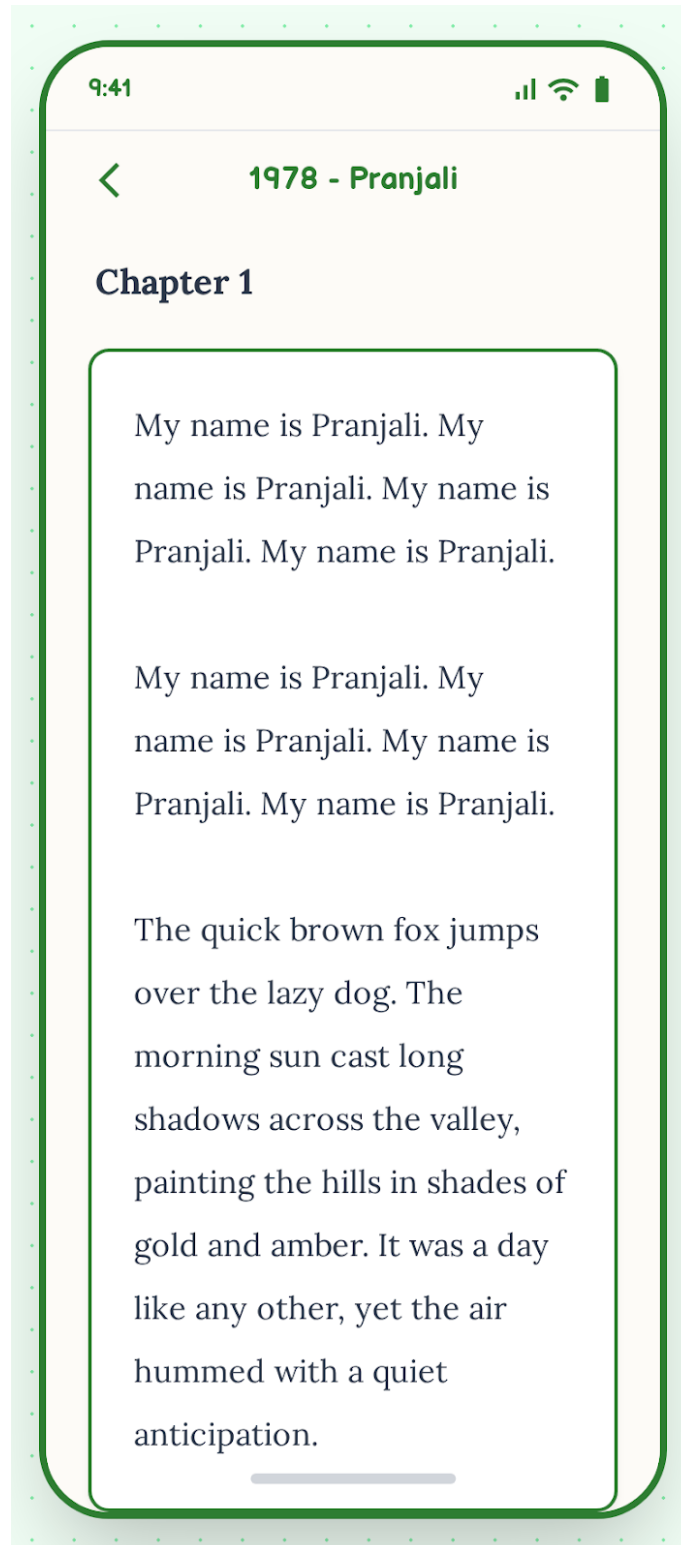


Figure 112 UI of TTS

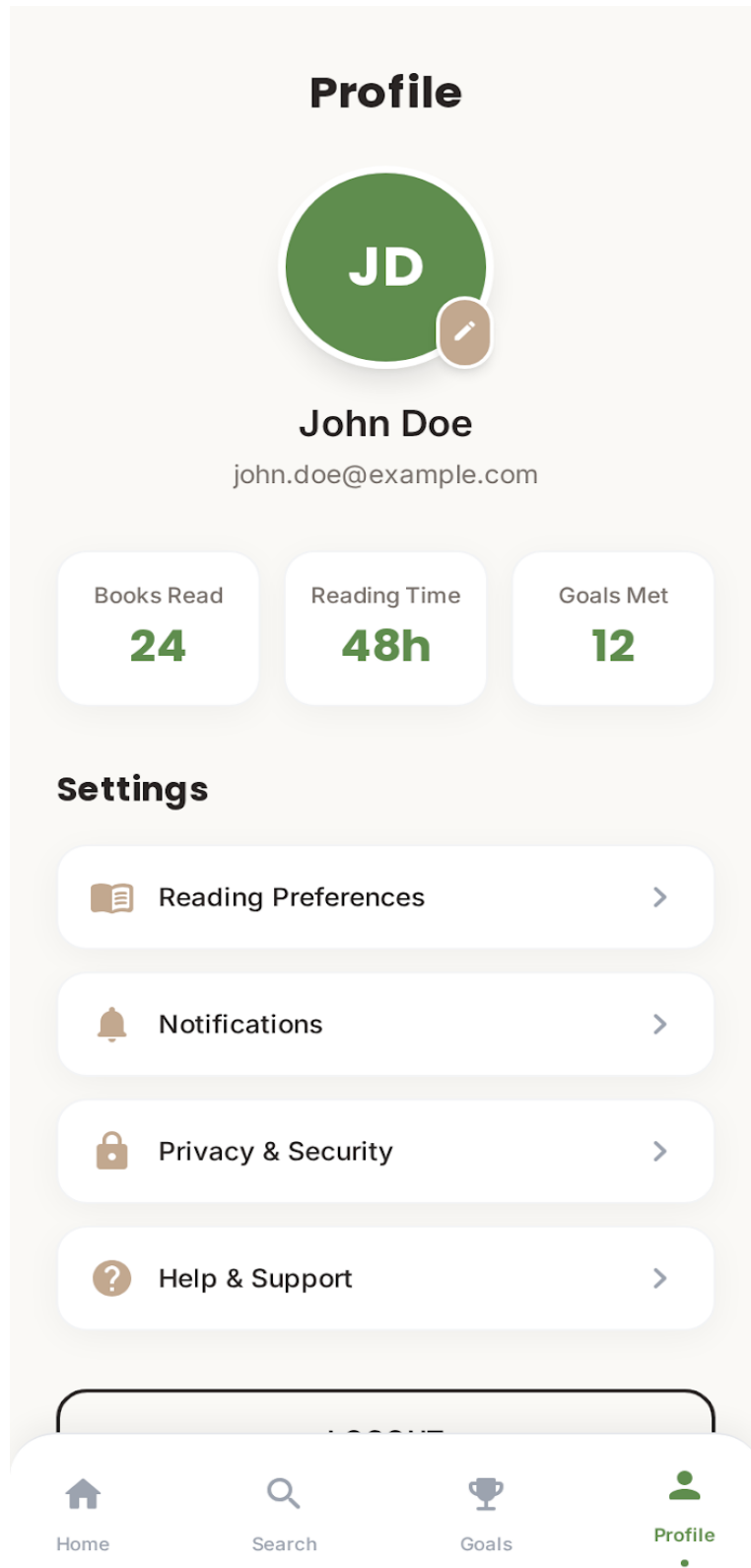


Figure 113 UI of User Dashboard

3.20 Implementation

Table 4 Implementation

Sprint No	Sprint Name	Duration	Key Activities	Deliverables
Sprint 1	User Authentication	Week 1–2	Design login & registration UI, implement user registration, login, Google authentication, password hashing	Working authentication system (login/register)
Sprint 2	Book Search	Week 3–5	Design search UI, integrate APIs (Google Books/OpenLibrary), implement search by title, author, genre	Functional book search feature
Sprint 3	E-Book Reader	Week 5–8	Design e-book reading interface, integrate EPUB reader, implement navigation and reading progress tracking	E-book reading system with progress tracking
Sprint 4	Text-to-Speech (TTS)	Week 8–10	Research TTS API, integrate TTS functionality, implement audio controls	TTS feature for reading books aloud
Sprint 5	Reading Goals	Week 10–13	Design goal-setting UI, implement reading goals and automatic progress updates	Personalized reading goals feature
Sprint 6	Book Recommendations	Week 13–14	Design recommendation logic, implement recommendation system based on genre and authors	Book recommendation feature
Sprint 7	Admin Module	Week 14–15	Design admin dashboard, implement book management	Admin panel with management features

			(CRUD), user management	
Sprint 8	Testing & Submit	Week 15–16	Perform unit, integration, and system testing, fix bugs.	Final tested and submit peoject

61 4. Testing and analysis

4.1 Test

Testing refers to the process of analysing the software does it meet specific requirements or not and worked as expected, it helps to identify the bug and error in the system which make developer easy to solve problem. It ensures quality and readability as well as help to detect buys and error, improve performance of whole system also at last provide better user experience (geeksforgeeks, 2025).

13 4.1.1 White box testing (Unit Test)

It is a software testing which include test of internal structure and working of software application, it makes sure the internal workings of an application, ensuring logic, structure, and flow operate as intended (geekforgeeks, 2025). Key area of focus in white box testing is:

- Code logic and flow
- Code coverage
- Data flow
- Internal function
- Error handling
- Boundary

1. User auth

User Authentication verifies users for providing access to the system, by allowing users to register, log in, and reset passwords securely using validation and OTP, helps protect user data and ensures only authorized users can access the system.

Table 5 User Auth Test Case

Test ID	Description
UT_AUTH_01	Login with valid credentials
UT_AUTH_02	Login with incorrect credentials
UT_AUTH_03	Login with unregistered email
UT_AUTH_04	Login with empty fields
UT_AUTH_05	Login with invalid email format
UT_AUTH_06	Forgot password and reset password using OTP
UT_AUTH_07	Register with non-Gmail email
UT_AUTH_08	Register with existing user email
UT_AUTH_09	Register with short password (less than 8 characters)
UT_AUTH_10	Register with empty fields
UT_AUTH_11	Register with valid details

Table 6 User Auth Case 1

Test	1
Description	Login with valid credentials
Action	Enter valid email Enter password click on login button
Expected Result	Users should log in and redirected to home page
Actual Result	User logged in and redirected to home page
Conclusion	Pass

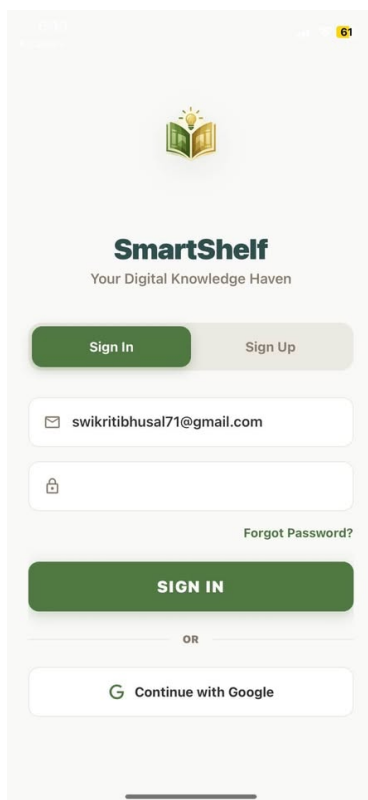


Figure 114 Valid Login

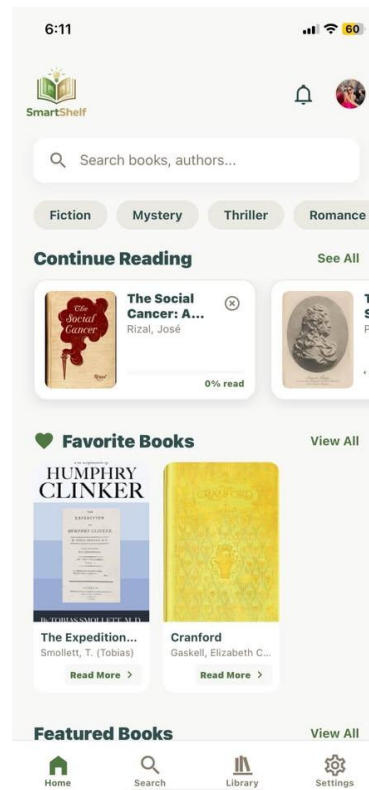


Figure 115 Valid Login 1

Table 7 User Auth Case 2

Test	2
Description	Login with incorrect credentials
Action	Enter invalid email Enter password click on login button
Expected Result	Error message "Invalid password" should be displayed
Actual Result	Error message displayed
Conclusion	Pass



Figure 116 Invalid login

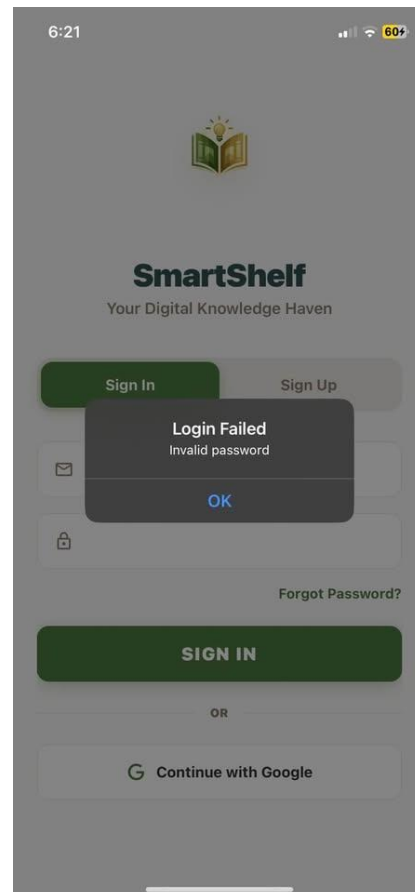


Figure 117 Invalid login 1

Table 8 User Auth Case 3

Test	3
Description	Login with unregistered email
Action	Enter unregistered email Enter password Click on login button
Expected Result	Error message "User not found " should be displayed
Actual Result	Error message displayed
Conclusion	Pass

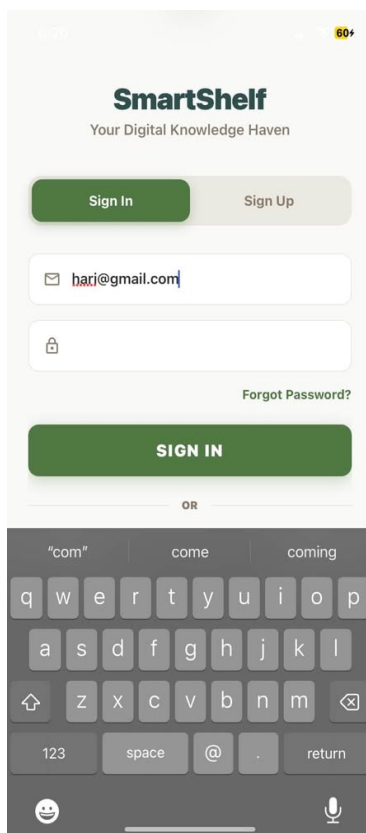


Figure 119 Unregister Login

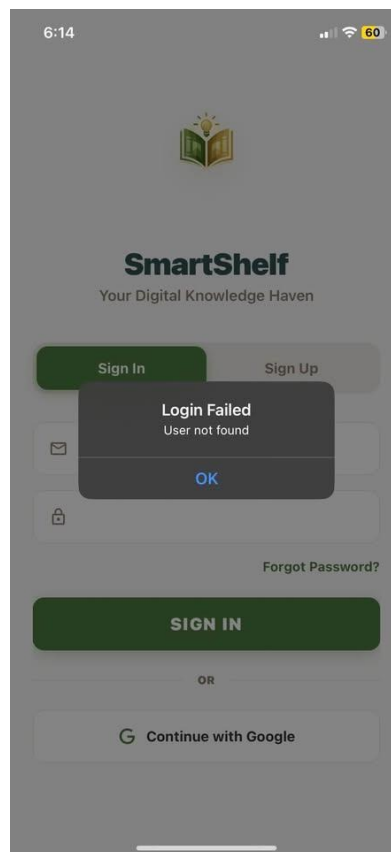


Figure 118Unregister Login 1

Table 9 User Auth Case 4

Test	4
Description	Login with empty field
Action	Leave email and password empty Click login
Expected Result	Error message " Please fill all fields" should be displayed
Actual Result	Error message displayed
Conclusion	Pass

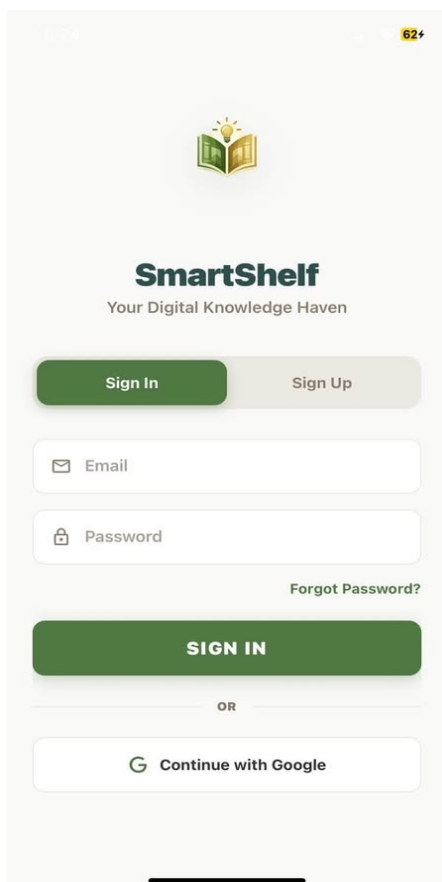


Figure 121 Empty Login

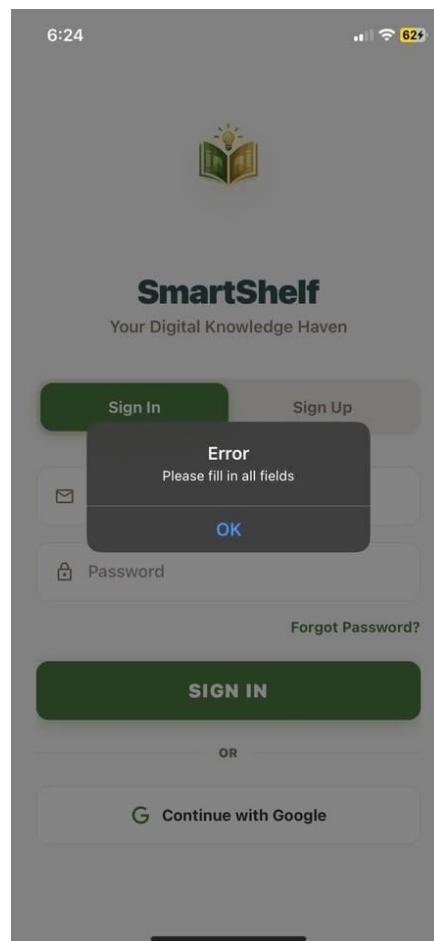


Figure 120 Empty login

Table 10 User Auth Case 5

Test	5
Description	Login with invalid email format
Action	Enter invalid email format Enter password Click login
Expected Result	Error message " User not found " should be displayed
Actual Result	Error message displayed
Conclusion	Pass

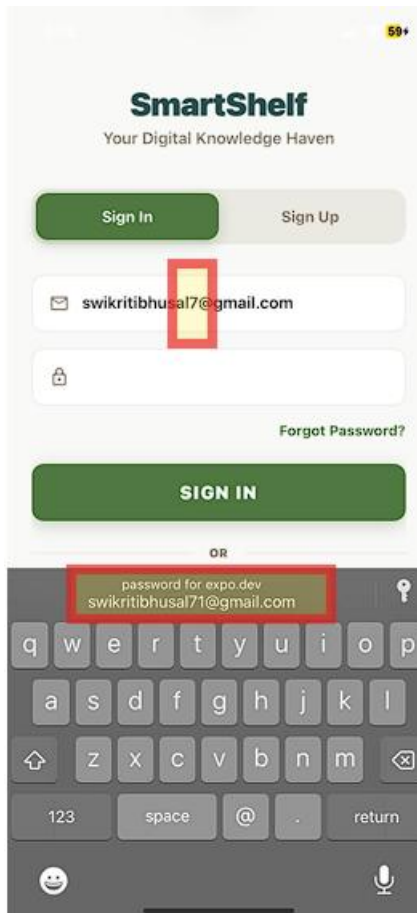


Figure 122 Login with Invalid Data

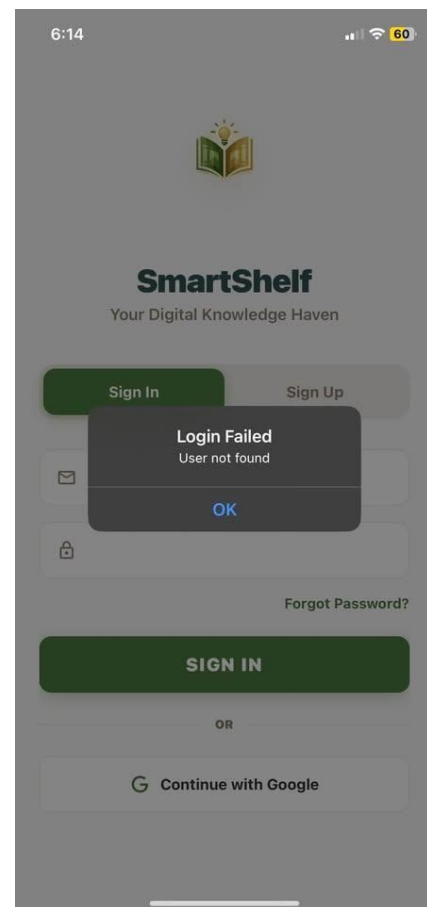


Figure 123 Login with Invalid Data 1

Table 11 User Auth Case 6

Test	6
Description	Forget Password and reset Password
Action	Enter email format Click on forget password Click on send otp Enter otp Enter new password Click on ok
Expected Result	Password should change successfully.
Actual Result	Password changed successfully.
Conclusion	Pass

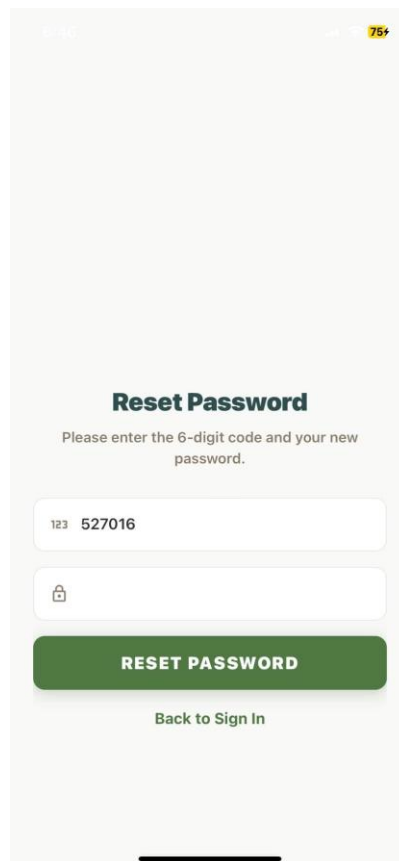


Figure 124 Forget Password

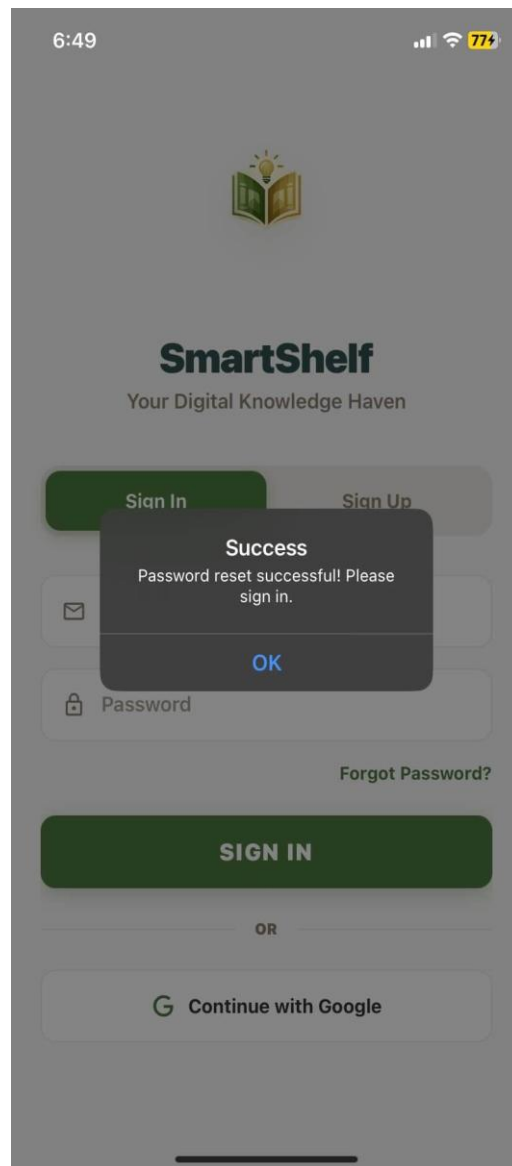


Figure 125 Reset Password

Table 12 User Auth Case 7

Test	7
Description	Register with non-Gmail
Action	Enter @hotmail.com email and valid password Click submit
Expected Result	Error "unregister email "
Actual Result	Error display
Conclusion	Pass

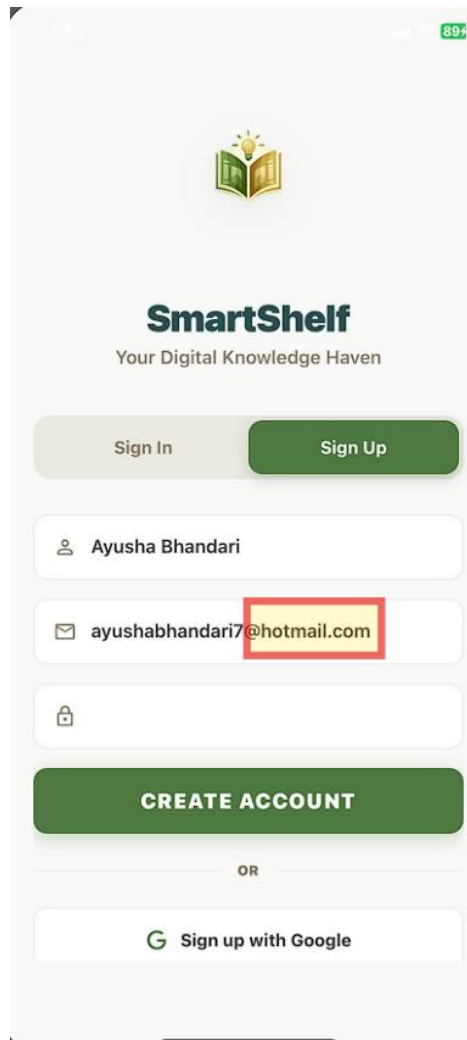


Figure 126 Register with non-Gmail

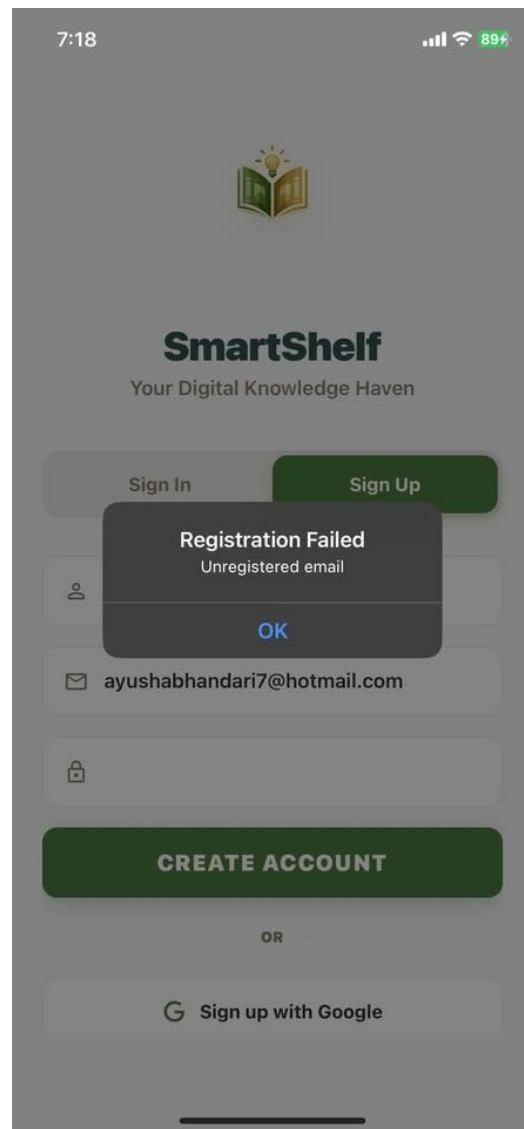


Figure 127 Register with non-Gmail 1

Table 13 User Auth Case 8

Test	8
Description	Enter email of exist user
Action	Enter email of existing user Click submit
Expected Result	Error "user already exist with this email "
Actual Result	Error display
Conclusion	Pass

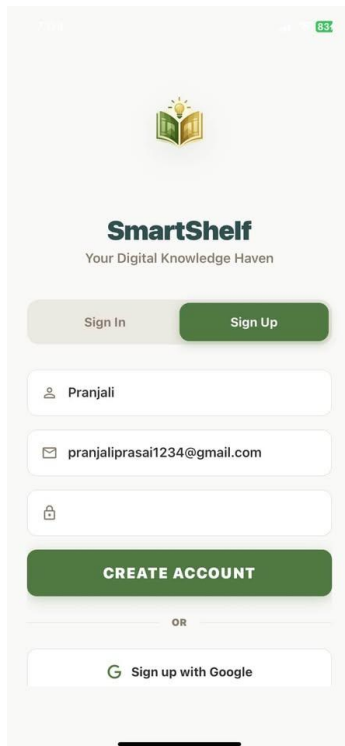


Figure 129 Exist User

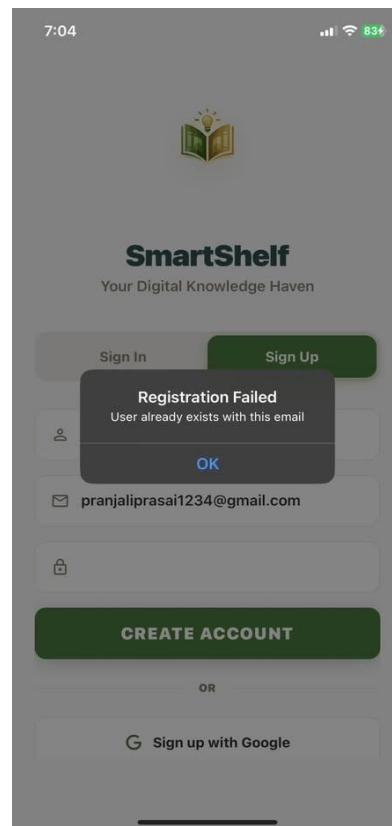


Figure 128 Exist User Register

Table 14 User Auth Case 9

Test	9
Description	Register with short password
Action	Enter valid Gmail but password less than 8 characters Click submit
Expected Result	Error "password must be at least 8 characters"
Actual Result	Error display
Conclusion	Pass

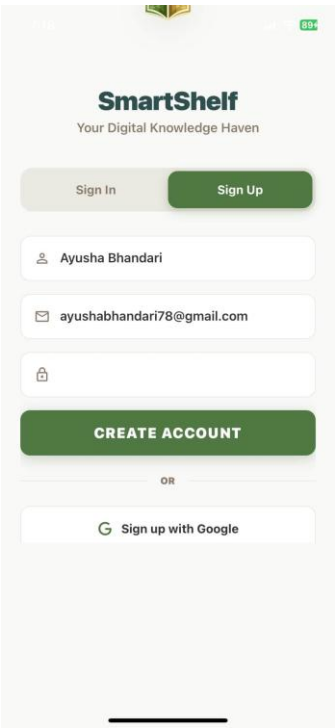


Figure 131 Short Password

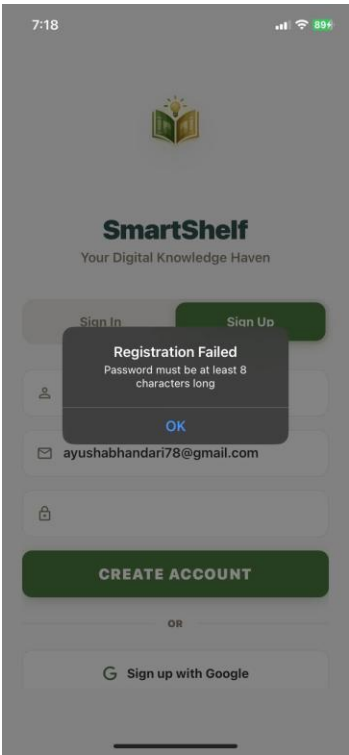


Figure 130 Short Password

Table 15 User Auth Case 10

Test	10
Description	Register with empty field
Action	Leave email or password empty Click submit
Expected Result	Error "validation error message"
Actual Result	Error display
Conclusion	Pass

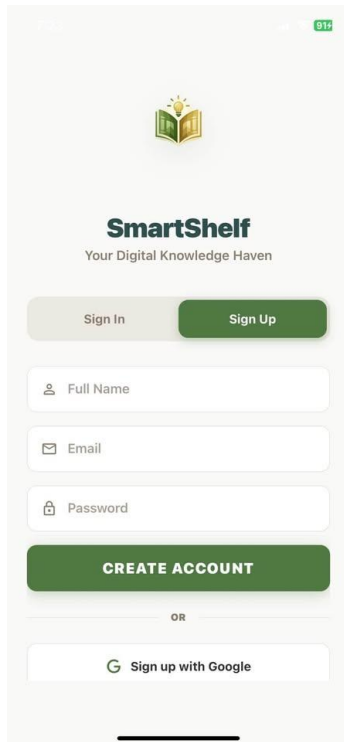


Figure 133 Empty Register

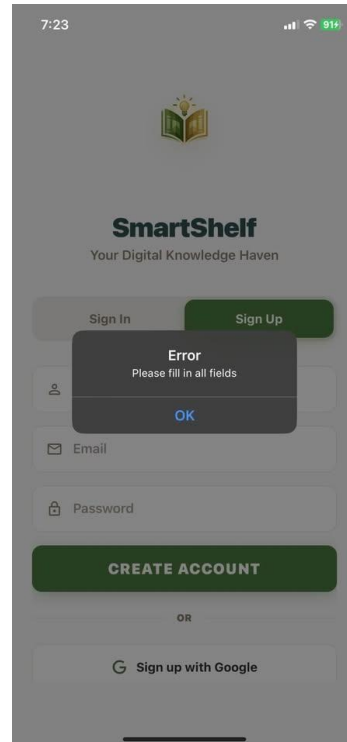


Figure 132 Empty Register

Table 16 User Auth Case 11

Test	11
Description	Register with valid details
Action	Enter valid gmail.com email + password (8+ chars) click on Register
Expected Result	Users should be registered successfully
Actual Result	User registered successfully
Conclusion	Pass

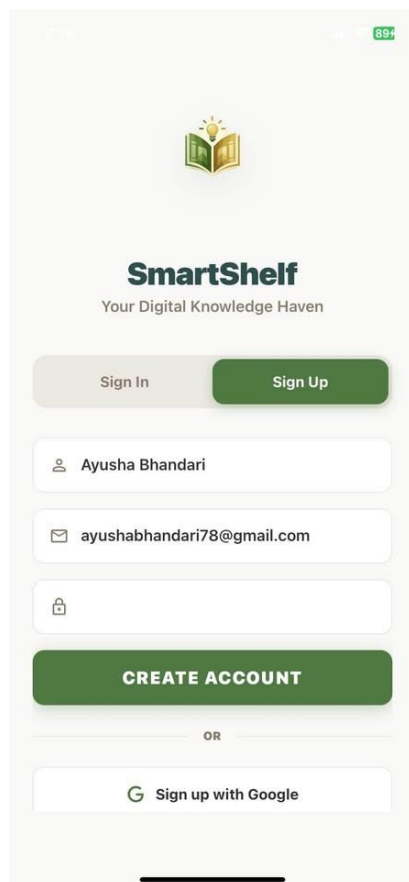


Figure 135 Register

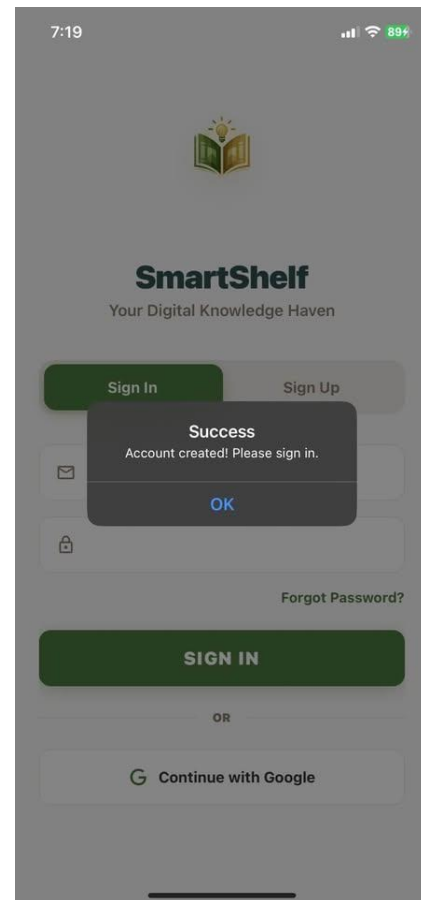


Figure 134 Register Success

2.Search Book

Table 17 Search Book Case

Test ID	Description
UT_SEARCH_12	Search by valid book title
UT_SEARCH_13	Search by author name
UT_SEARCH_14	Search with no matching results
UT_SEARCH_15	Search with empty input field

Table 18 Search Book Case 1

Test	12
Description	Search by valid book title
Action	Enter book title(pride and prejudice) Click search
Expected Result	Relevant books displayed
Actual Result	Books displayed correctly
Conclusion	Pass

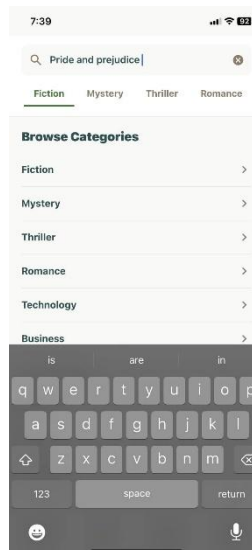


Figure 136 Search Book

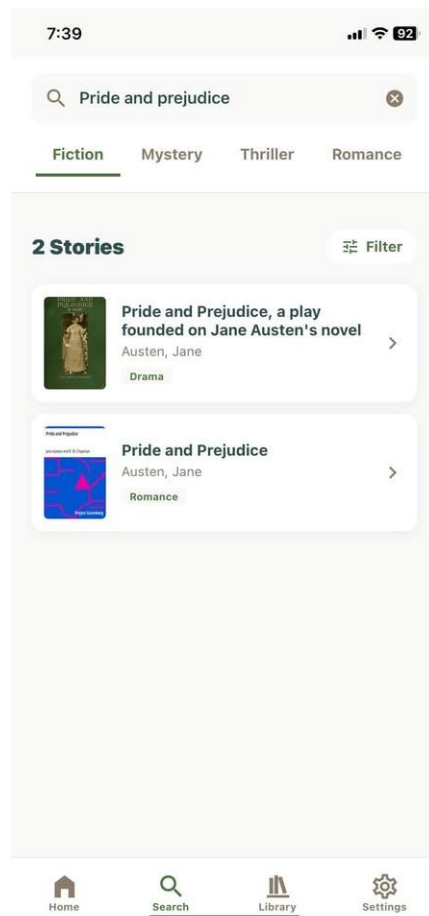


Figure 137 Search Book1

Table 19 Search Book Case 2

Test	13
Description	Search by author
Action	Enter author name (Jenny han) Click search
Expected Result	Books by author shown
Actual Result	Books displayed correctly
Conclusion	Pass

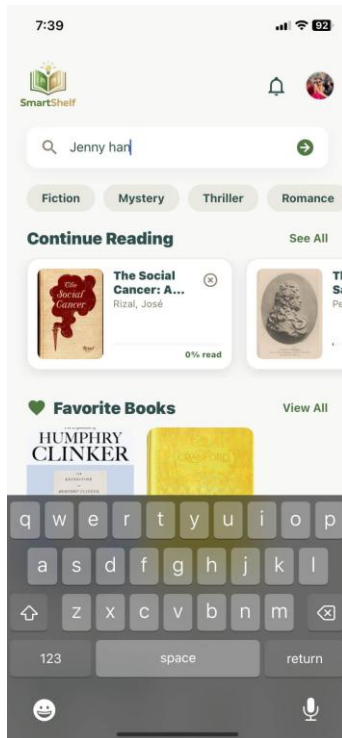


Figure 139 Search Author

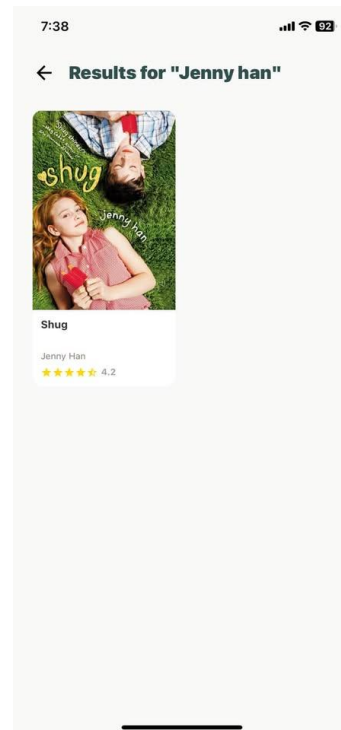


Figure 138 Search Author 1

Table 20 Search Book Case 4

Test	14
Description	Search with no results
Action	Enter random keyword(Apple) Click search
Expected Result	"No result found" message
Actual Result	Message displayed
Conclusion	Pass

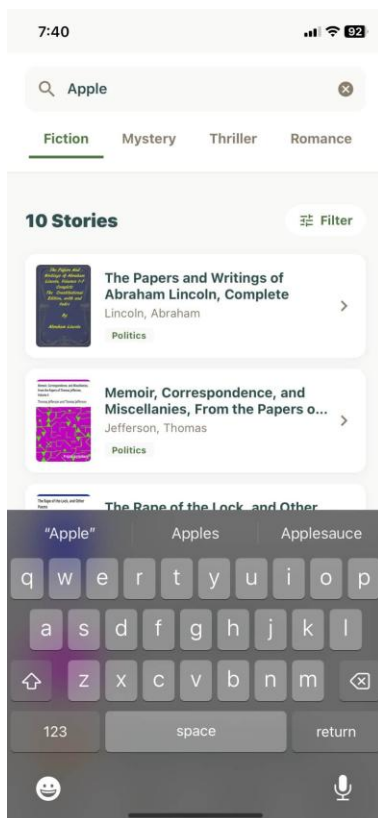


Figure 141 Invalid Search

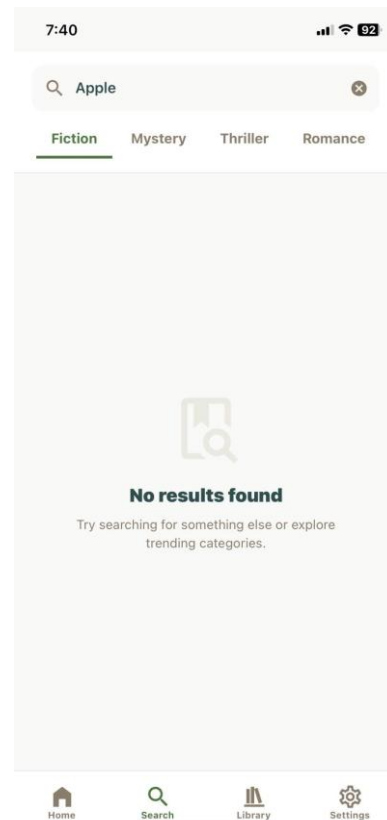


Figure 140 Invalid 1

Table 21 Search Book Case 5

Test	15
Description	Search with empty field
Action	Enter nothing Click search
Expected Result	Nothing happens, Show same page
Actual Result	Works correctly
Conclusion	Pass

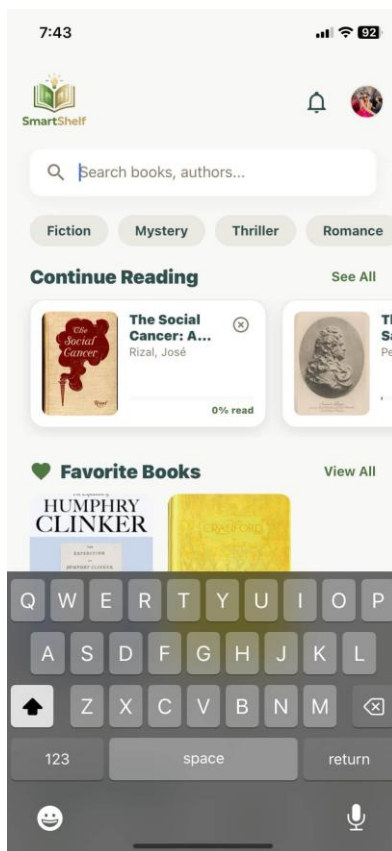


Figure 143 Empty Search

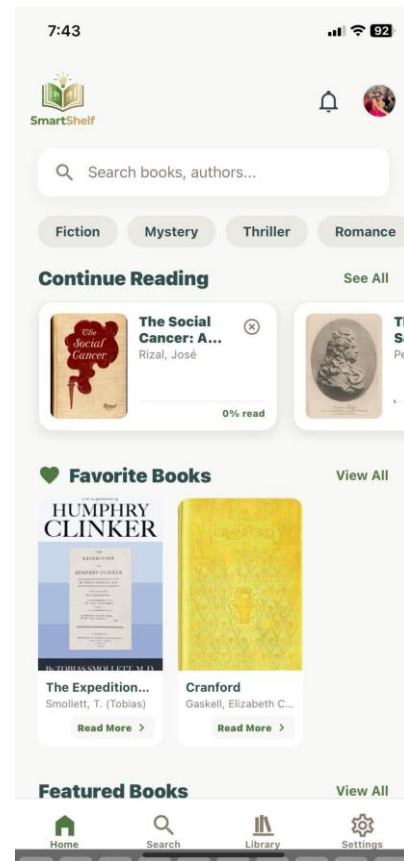


Figure 142 Empty Search 1

3.Comment

Table 22 Comment Test Case

Test ID	Description
UT_COMMENT_16	Add valid comment
UT_COMMENT_17	Add empty comment
UT_COMMENT_18	Delete own comment

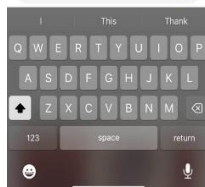
Table 23 Comment Test Case 1

Test	16
Description	Add valid comment
Action	Enter comment Click submit
Expected Result	Comment displayed
Actual Result	Comment saved and shown
Conclusion	Pass

No. 150 Me Tangere by José Rizal is a novel published in 1887 during the Spanish colonial period of the Philippines. Written by Filipino nationalist and polymath José Rizal, it exposes the injustices inflicted by the ruling government and Spanish Catholic friars on the Filipino people. The story follows Crisóstomo Ibarra, who returns home after studying in Europe to find his father dead under mysterious circumstances and his grave desecrated. As he attempts to build a school and improve his town, he faces hostility from powerful priests and deadly conspiracies that threaten his...

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No. 150 Me Tangere by José Rizal is a novel published in 1887 during the Spanish colonial period of the Philippines. Written by Filipino nationalist and polymath José Rizal, it exposes the injustices inflicted by the ruling government and Spanish Catholic friars on the Filipino people. The story follows Crisóstomo Ibarra, who returns home after studying in Europe to find his father dead under mysterious circumstances and his grave desecrated. As he attempts to build a school and improve his town, he faces hostility from powerful priests and deadly conspiracies that threaten his...

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Figure 144 Show Comment

Figure 145 Comment

Table 24 Comment Test Case 2

Test	17
Description	Add empty comment
Action	Click submit without text
Expected Result	Send button should unable
Actual Result	Send button is unable
Conclusion	Pass

Noli Me Tangere by José Rizal" is a novel published in 1887 during the Spanish colonial period of the Philippines. Written by Filipino nationalist and polymath José Rizal, it exposes the injustices inflicted by the ruling government and Spanish Catholic friars on the Filipino people. The story follows Crisóstomo Ibarra, who returns home after studying in Europe to find his father dead under mysterious circumstances and his grave desecrated. As he attempts to build a school and improve his town, he faces hostility from powerful priests and deadly conspiracies that threaten his...

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Figure 146 Empty Comment

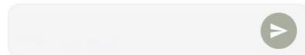
Table 25 Comment Test Case 3

Test	18
Description	Delete own comment
Action	Click on 3 dots and click on delete
Expected Result	Comment removed
Actual Result	Comment deleted
Conclusion	Pass

Noli Me Tangere by José Rizal" is a novel published in 1887 during the Spanish colonial period of the Philippines. Written by Filipino nationalist and polymath José Rizal, it exposes the injustices inflicted by the ruling government and Spanish Catholic friars on the Filipino people. The story follows Crisóstomo Ibarra, who returns home after studying in Europe to find his father dead under mysterious circumstances and his grave desecrated. As he attempts to build a school and improve his town, he faces hostility from powerful priests and deadly conspiracies that threaten his...

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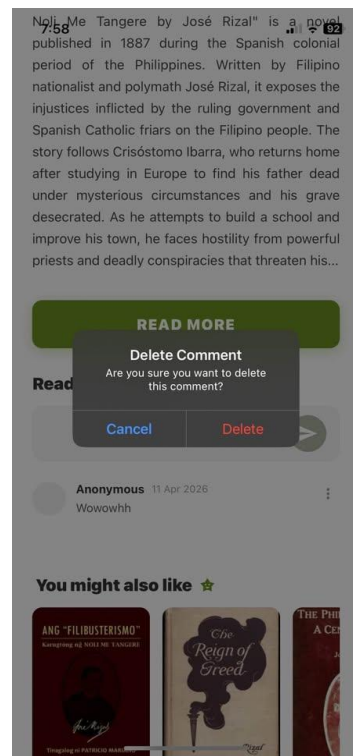


Figure 147 Delete Comment

Figure 148 Delete Comment

4.Favorite book

Table 26 Favourite Book Case

Test ID	Description
UT_FAV_19	Add book to favourites
UT_FAV_20	Remove book from favourites
UT_FAV_21	View favourite books list

Table 27 Favourite Book Case 1

Test	19
Description	Add to favorites
Action	Click on heart to add favorites
Expected Result	Book should added
Actual Result	Book added
Conclusion	Pass

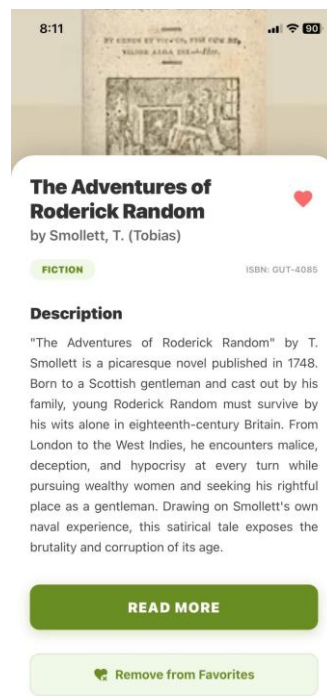


Figure 149 Favourite Book

Table 28 Favourite Book Case 2

Test	20
Description	Remove from favourites
Action	Click on remove
Expected Result	Book removed
Actual Result	Removed successfully
Conclusion	Pass



Figure 151 Remove Favourite

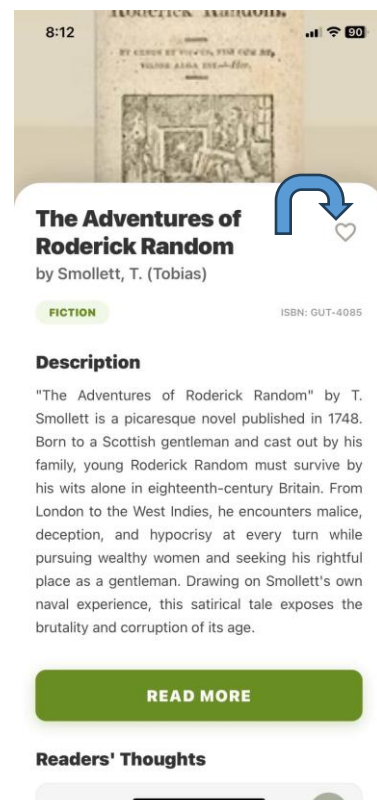


Figure 150 Remove Favourite 1

Table 29 Favourite Book Case 3

Test	21
Description	View favourites
Action	Open favourites list
Expected Result	Books displayed
Actual Result	List shown
Conclusion	Pass

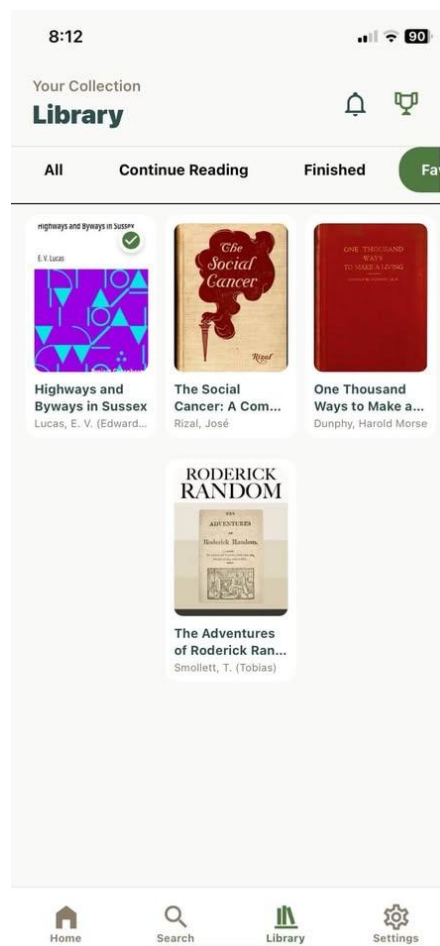


Figure 152 View Favourite

5.Highlight Tests

Table 30 Highlight Tests Case

Test ID	Description
UT_HIGHLIGHT_22	Add text highlight
UT_HIGHLIGHT_23	View highlighted text
UT_HIGHLIGHT_24	Delete highlighted text

Table 31 Highlight Tests Case 1

Test	22
Description	Add highlight
Action	Select text Click highlight
Expected Result	Highlight should show with yellow color
Actual Result	Highlight show successfully
Conclusion	Pass

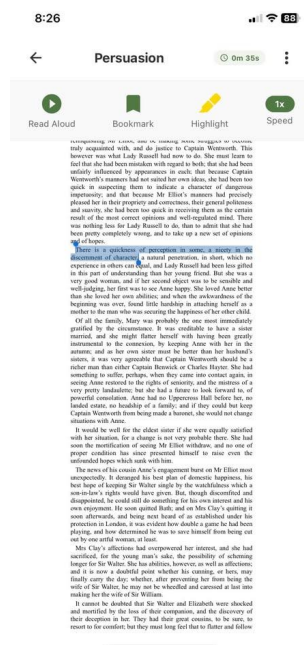


Figure 153 Select Text

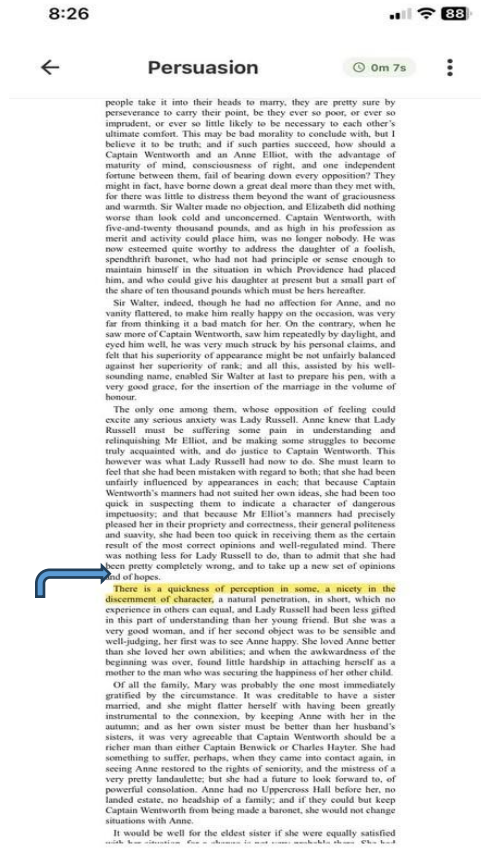


Figure 154 Highlight

Table 32 Highlight Tests Case 2

Test	23
Description	View highlight
Action	Reopen book ,show highlights section
Expected Result	Highlight should display
Actual Result	Highlight show successfully
Conclusion	Pass

8:26

Persuasion

people take it into their heads to marry, they are pretty sure by government to carry their point, so they rest as pass, or rest as impatient, or even as little likely to be necessary to each other's ultimate comfort. This may be had readily to conclude with, but I believe it to be truth, and if such parties succeed, how should a Captain Wentworth and an Anne Elliot, with the advantage of maturity of mind, consciousness of right, and one independent fortune between them, fall of being down every opposition? They might in fact, have borne down a great deal more than they met with, for Anne was little to distance them beyond the want of gracefulness and warmth. Sir Walter made no objection, and Elizabeth did nothing worse than look cold and unconcerned. Captain Wentworth, with five-and-twenty thousand pounds, and as high in his profession as merit and activity could place him, was no longer a nobody. He was now esteemed quite worthy to address the daughter of a foolish, spendthrift baronet, who had not had principle or sense enough to maintain himself in the situation in which Providence had placed him, and who could give his daughter at present but a small part of the share of ten thousand pounds which must be hers hereafter.

Sir Walter, indeed, though he had no objection for Anne, and no vanity flattered, to make him really happy on the occasion, was very far from thinking it a bad match for her. On the contrary, when he saw Anne of Captain Wentworth, saw him especially by daylight, and eyed him well, he was very much struck by his personal beauty, and felt that his superiority of appearance might be not unfairly balanced against her superiority of mind, and all this, assisted by his well-sounding name, enabled Sir Walter at last to prepare his pen, with a very good grace, for the insertion of the marriage in the volume of his love.

The only one among them, whose opposition of feeling could excite any serious anxiety was Lady Russell. Anne knew that Lady Russell must be suffering some pain in understanding and comprehending Mr Elliot, and he making some struggles to become truly acquainted with, and do justice to Captain Wentworth. This however was what Lady Russell had now to do. She must learn to feel that she had been mistaken with regard to both, that she had been unfairly influenced by appearance in such that became Captain Wentworth's manners had not raised for her own ideas, she had been too quick in supposing them to indicate a character of dangerous impetuosity; and that because Mr Elliot's manners had previously showed her in their propriety and correctness, their general politeness and civility, she had been too quick in receiving them as the certain result of the most correct opinions and well-regulated mind. There was nothing less for Lady Russell to do, than to admit that she had been pretty completely wrong, and to take up a new set of opinions and of hopes.

There is a quickness of perception in some, a nicety in the discernment of character, a natural penetration, in short, which no experience or observation can equal, and Lady Russell had been too often in this part of understanding than her young friend. But she was a very good woman, and if her second object was to be sensible and well-judging, her first was to see Anne happy. She loved Anne better than she loved her own abilities, and when the awkwardness of the beginning was over, found little hardship in attacking herself as a mother to the man who was securing the happiness of her other child.

Of all the family, Mary was probably the one most immediately gratified by the circumstance. It was allowable to have a sister married, and she might flatter herself with having been greatly instrumental to the connection. In keeping Anne with her in the autumn, and so her own sister must be better than her husband's sister. It was very agreeable that Captain Wentworth should be a richer man than either Captain Benwick or Charles Hayter. She had something to reflect, perhaps, when they came into contact again, in seeing Anne married to the rights of society, and the mistress of a very pretty residence; but she had a letter to Lady Russell as of powerful consolation. Anne had no Updown Hall before her, no limited estate, no husband of a family, and if they could but keep Captain Wentworth from being made a baronet, she would not change situations with him.

It would be well for the eldest sister if she were equally satisfied with her connection. But in absence to meet some amiable Mr. Elliot, who had

8:26 ON

100% 88

Description

"Persuasion" by Jane Austen is a novel published in 1817. Eight years after being persuaded to break off her engagement to naval officer Frederick Wentworth, twenty-seven-year-old Anne Elliot encounters him again when his sister rents her family's estate. Now wealthy and eligible, Wentworth mingles with Anne's social circle while she quietly observes, her feelings unchanged. As misunderstandings unfold and new attachments form, the story explores whether second chances at love are possible when pride and past regrets stand between two hearts.

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You might also like

Figure 155 Open Book

Figure 156 View Highlight

Table 33 Highlight Tests Case 3

Test	24
Description	Delete highlight
Action	Remove highlighted text
Expected Result	Remove highlighted
Actual Result	Highlight remove successfully
Conclusion	Pass

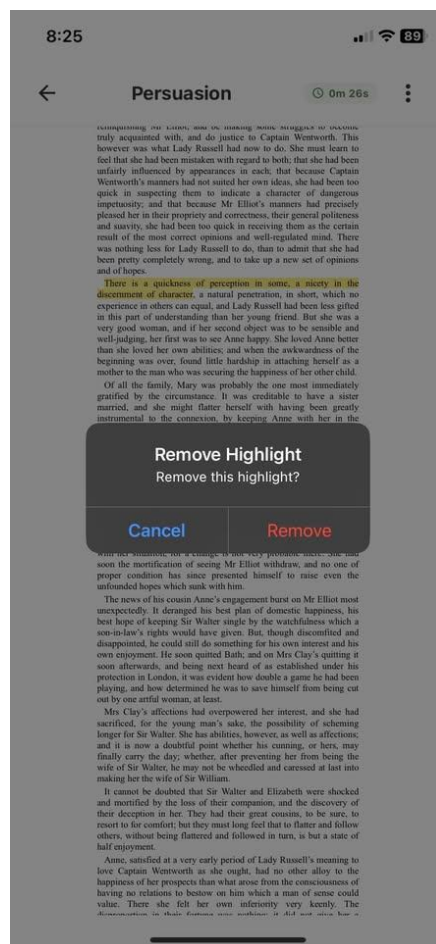


Figure 157 Remove Highlight

6.Milestone and Notification

Table 34 Milestone and Notification Test Case

Test ID	Description
UT_MILESTONE_26	Complete milestone successfully
UT_MILESTONE_27	Cancel milestone before completion
UT_MILESTONE_28	View completed milestones
UT_NOTIFICATION_29	Notification after milestone completion
UT_NOTIFICATION_30	Notification before milestone completion (1 minute prior)
UT_NOTIFICATION_31	Mark all notifications as read

Table 35 Milestone and Notification Test Case 1

Test	26
Description	Complete milestone
Action	Reach the defined milestone target during reading activity
Expected Result	System should mark milestone as completed and triggers notification
Actual Result	System marks milestone as completed and triggers notification
Conclusion	Pass

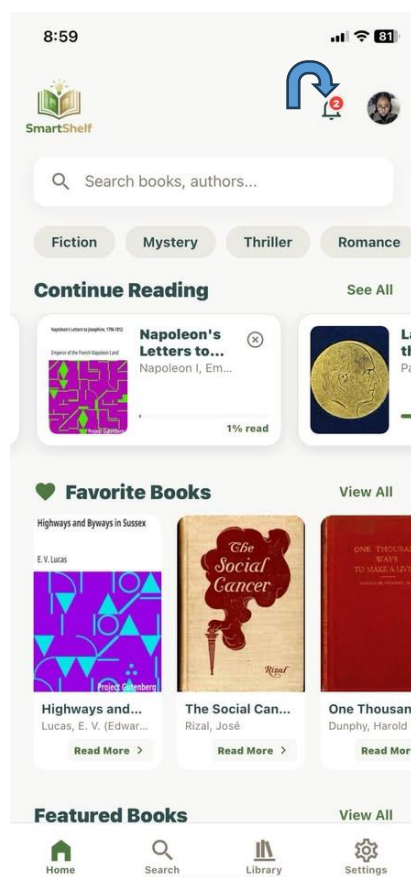


Figure 158 Complete Milestone

Table 36 Milestone and Notification Test Case 2

Test	27
Description	Cancel milestone
Action	Start milestones end it before completing
Expected Result	Milestone is not saved and cancel successfully
Actual Result	Successfully cancel and not save
Conclusion	Pass

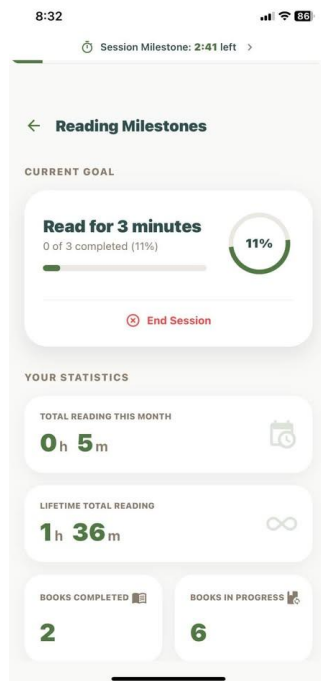


Figure 160 Milestone

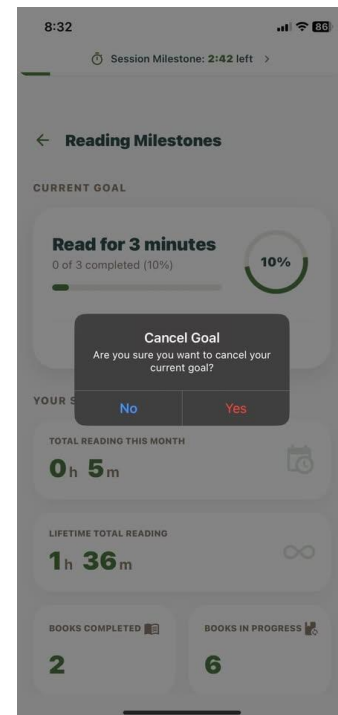


Figure 159 Cancel

1

Table 37 Milestone and Notification Test Case 3

Test	28
Description	View milestones
Action	Open milestone section from dashboard
Expected Result	All completed milestones are displayed in a number as total
Actual Result	Display successfully
Conclusion	Pass



Figure 161 View Milestone

Table 38 Milestone and Notification Test Case 4

Test	29
Description	Notification after completion
Action	Finish milestone successfully
Expected Result	User receives notification "Milestone completed achieved"
Actual Result	Notification receives successfully
Conclusion	Pass

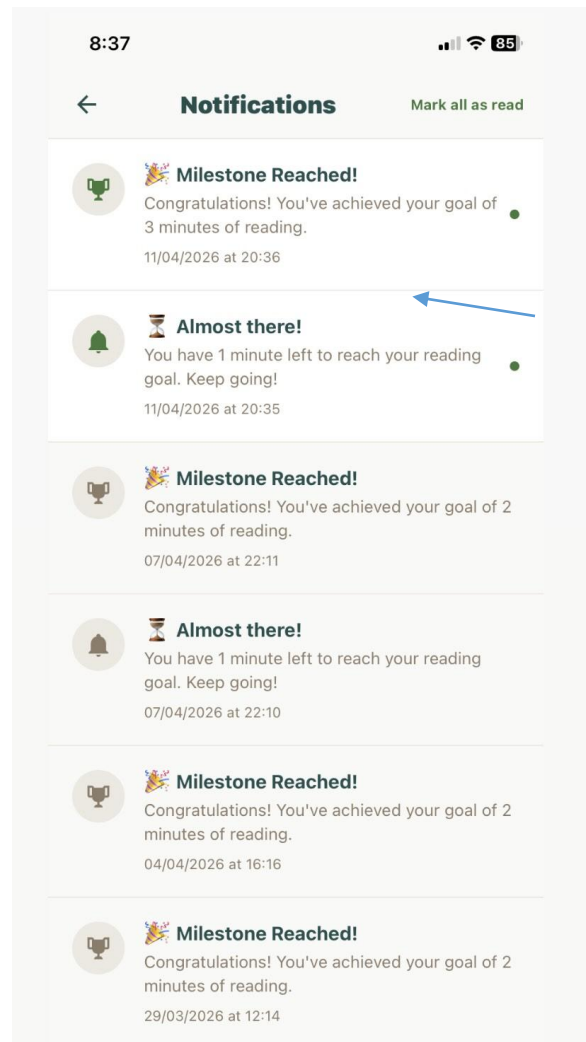


Figure 162 Notification Complete

Table 39 Milestone and Notification Test Case 5

Test	30
Description	Notification 1 min before Completing
Action	User is 1 minute away from completing milestone
Expected Result	System sends notification: "You have 1 minute left to reach your reading goal"
Actual Result	Notification received successfully
Conclusion	Pass

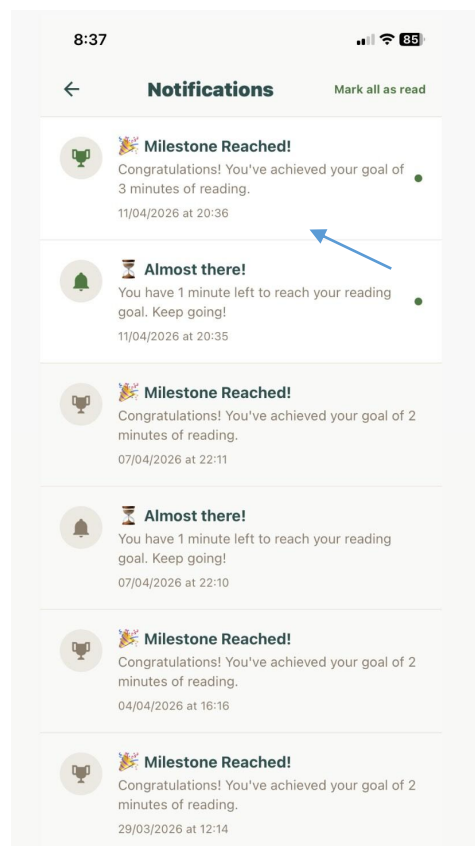


Figure 163 Milestone 1 before

Table 40 Milestone and Notification Test Case 6

Test	31
Description	Mark all notification as read
Action	Click "Mark all as read" button in notification panel
Expected Result	All unread notifications are marked as read and notification badge count becomes nothing
Actual Result	Notifications are marked as read and notification badge count becomes nothing successfully
Conclusion	Pass

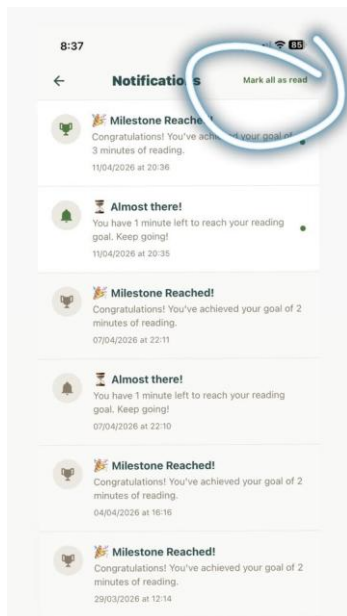


Figure 165 Markas read

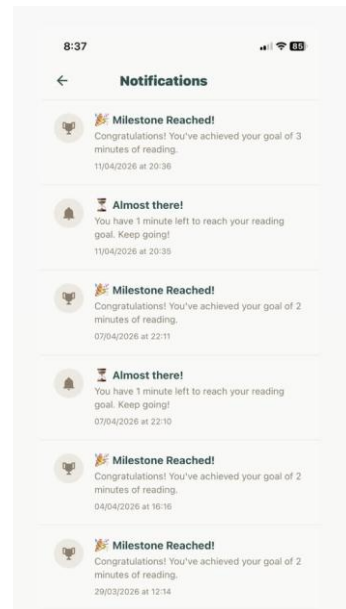


Figure 164 Worked

7.TTS (Text-to-Speech) Tests

Table 41 TTS (Text-to-Speech) Tests Case

Test ID	Description
UT_TTS_32	Play book content using TTS
UT_TTS_33	Pause TTS audio
UT_TTS_34	Resume TTS audio
UT_TTS_35	Stop TTS audio
UT_TTS_36	Change playback speed to 0.5x
UT_TTS_37	Change playback speed to 1x
UT_TTS_38	Change playback speed to 1.5x
UT_TTS_39	Change playback speed to 2x

Table 42 TTS (Text-to-Speech) Tests Case 1

Test	32
Description	Play book content using TTS
Action	Select paragraph and / TTS button on a book page
Expected Result	System converts text into audio and starts playing
Actual Result	Text convert in audio successfully
Conclusion	Pass

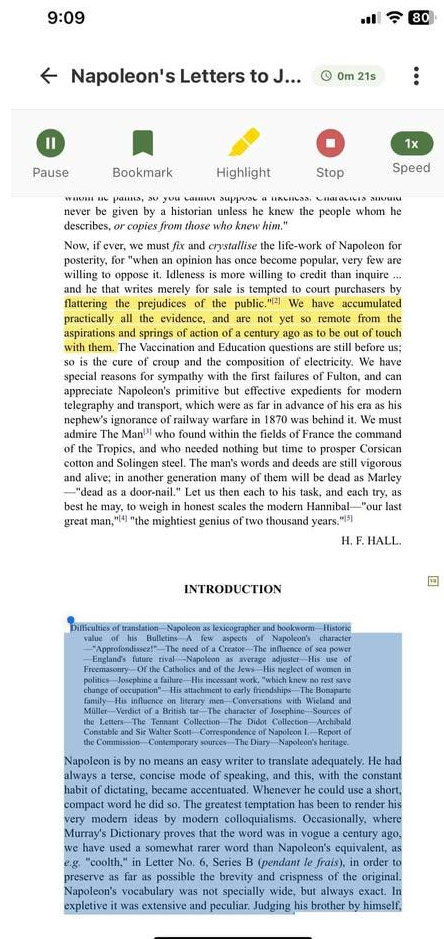


Figure 166 Play Content

Table 43 TTS (Text-to-Speech) Tests Case 2

Test	33
Description	Pause book content using TTS
Action	Click on 3 dot Click "Pause"
Expected Result	System should pause audio
Actual Result	System pause audio successfully
Conclusion	Pass

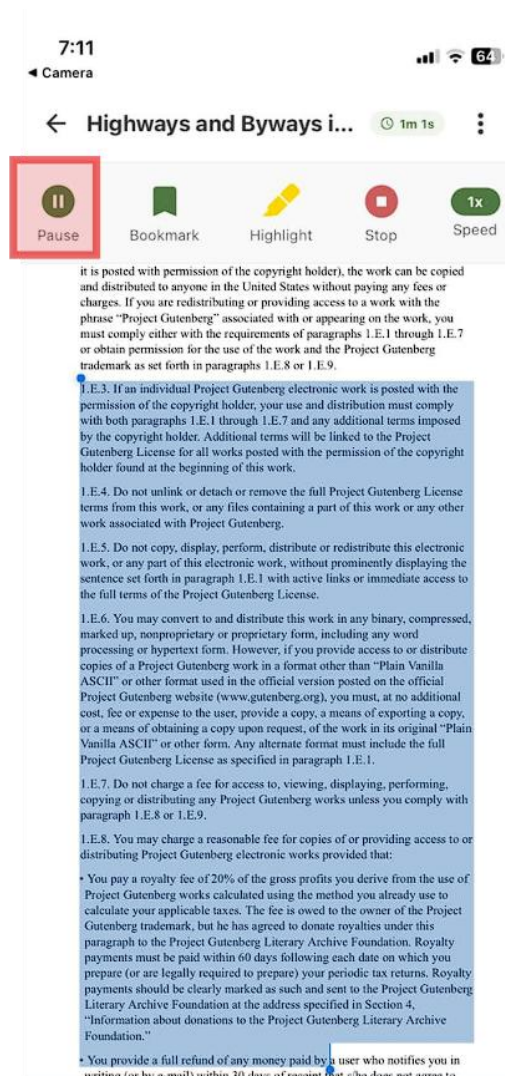


Figure 167 Pause

Table 44 TTS (Text-to-Speech) Tests Case 3

Test	34
Description	Pause book content using TTS
Action	Click on 3 dot Click on Resume
Expected Result	System should continue audio
Actual Result	System continue audio successfully
Conclusion	Pass

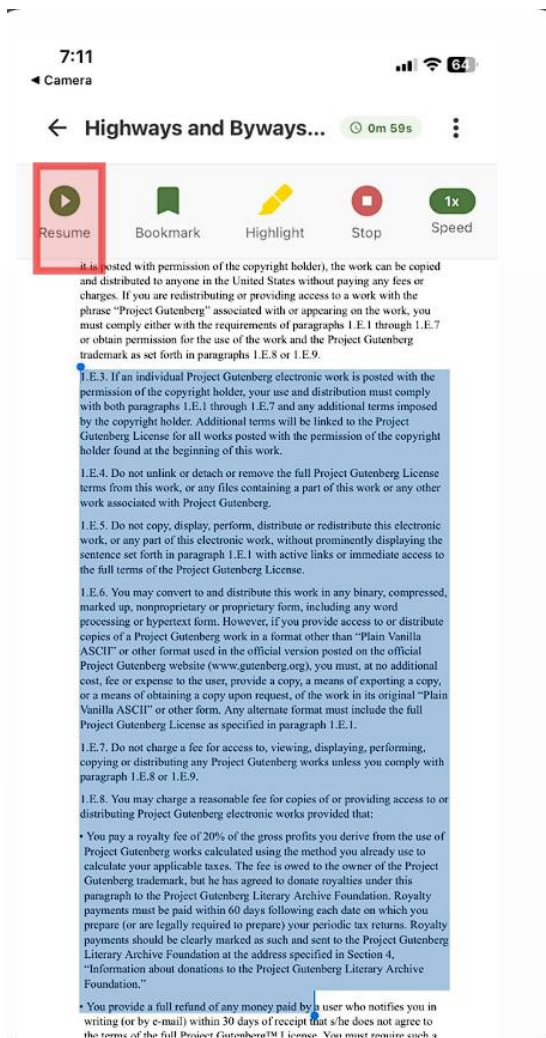


Table 45 TTS (Text-to-Speech) Tests Case 4

Test	35
Description	Stop TTS
Action	Click on 3 dot Click on Stop
Expected Result	System should Stop audio
Actual Result	System stop audio successfully
Conclusion	Pass

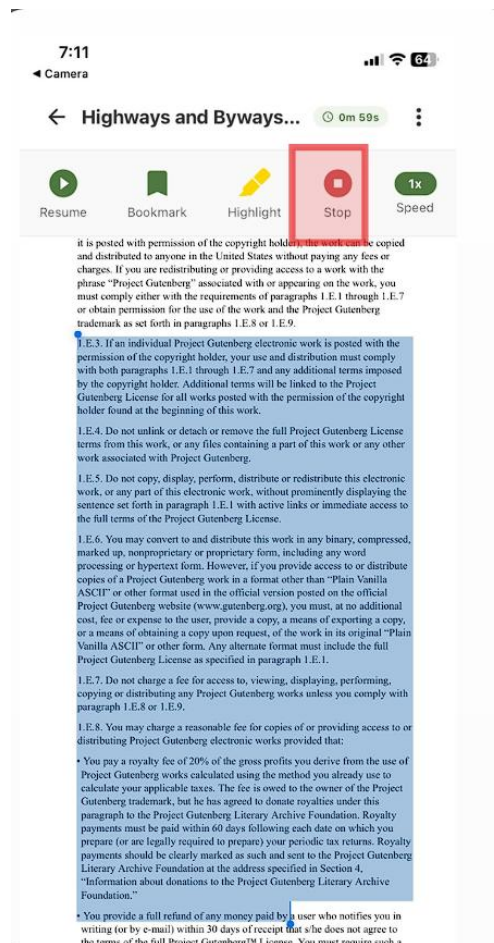


Figure 168 Stop

3

Table 46 TTS (Text-to-Speech) Tests Case 5

Test	36
Description	Change playback speed
Action	Click on 3 dot Click on 0.5x
Expected Result	System should changes audio speed accordingly
Actual Result	System changes audio speed successfully
Conclusion	Pass

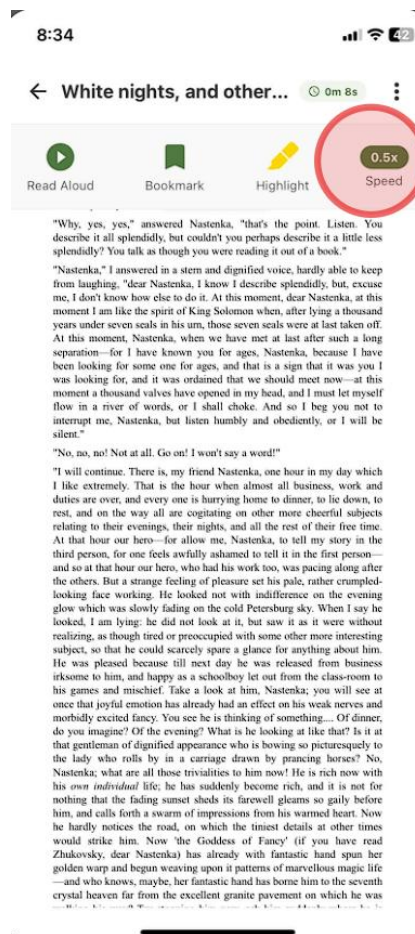


Figure 169 Speed 0.5

Table 47 TTS (Text-to-Speech) Tests Case 6

Test	37
Description	Change playback speed
Action	Click on 3 dot Click on 1x
Expected Result	System should changes audio speed accordingly
Actual Result	System changes audio speed successfully
Conclusion	Pass

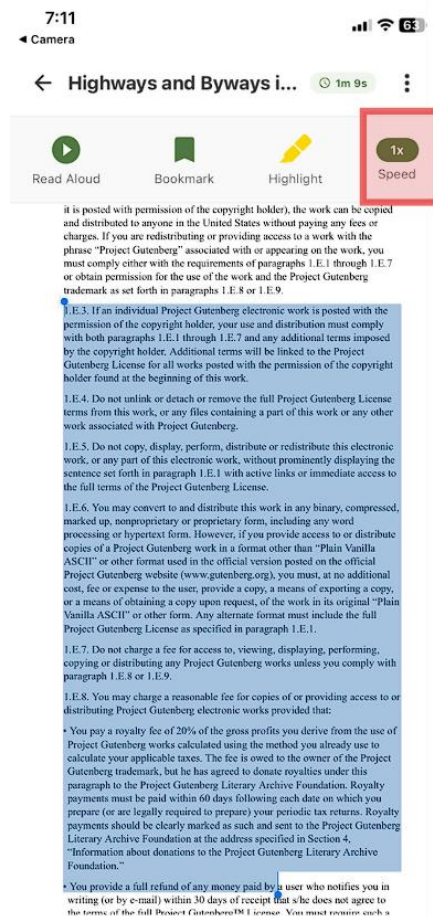


Figure 170 Speed 1

Table 48 TTS (Text-to-Speech) Tests Case 7

Test	38
Description	Change playback speed
Action	Click on 3 dot Click on 1.5x
Expected Result	System should changes audio speed accordingly
Actual Result	System changes audio speed successfully
Conclusion	Pass

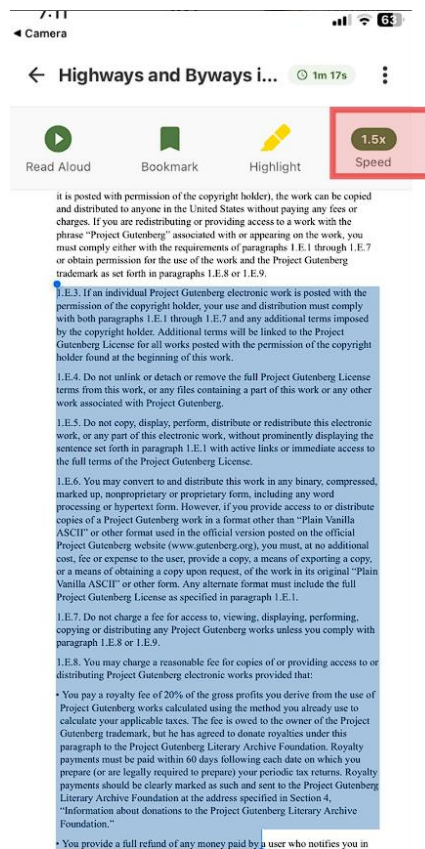


Figure 171 Speed 1.5

3

Table 49 TTS (Text-to-Speech) Tests Case 8

Test	39
Description	Change playback speed
Action	Click on 3 dot Click on 2x
Expected Result	System should changes audio speed accordingly
Actual Result	System changes audio speed successfully
Conclusion	Pass

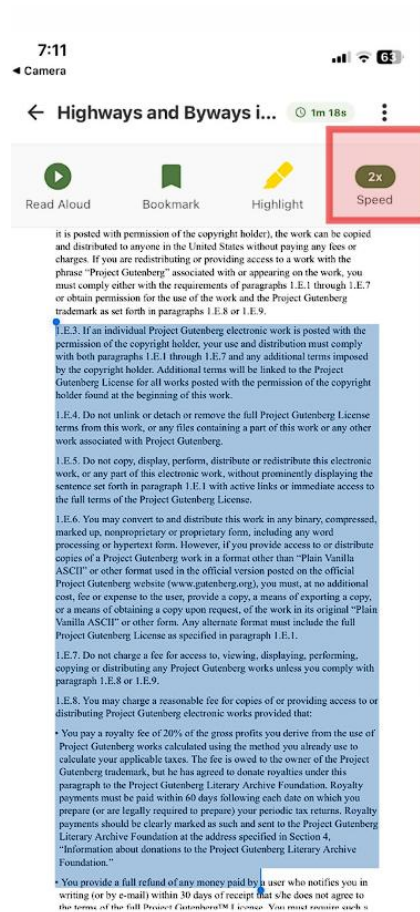


Figure 172 Speed 2x

8. Bookmark Tests

Table 50 Bookmark Tests Case

Test ID	Description
UT_BOOKMARK_40	Add bookmark to a book
UT_BOOKMARK_41	Navigate using bookmark

Table 51 Bookmark Tests Case 1

Test	40
Description	Add bookmark to a book
Action	Click "Bookmark" icon
Expected Result	Content should save to bookmarks and icon become active
Actual Result	Content saved to bookmarks and icon becomes active
Conclusion	Pass

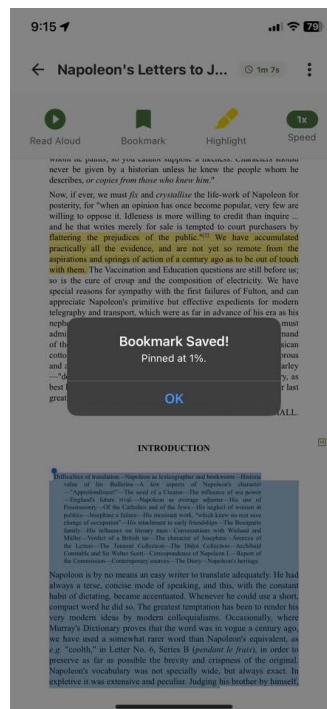


Figure 173 Bookmark

Table 52 Bookmark Tests Case 2

Test	41
Description	Bookmark Navigation
Action	Click a saved bookmark item
Expected Result	Users should be redirected to the exact saved page
Actual Result	User is redirected to the exact saved page
Conclusion	Pass

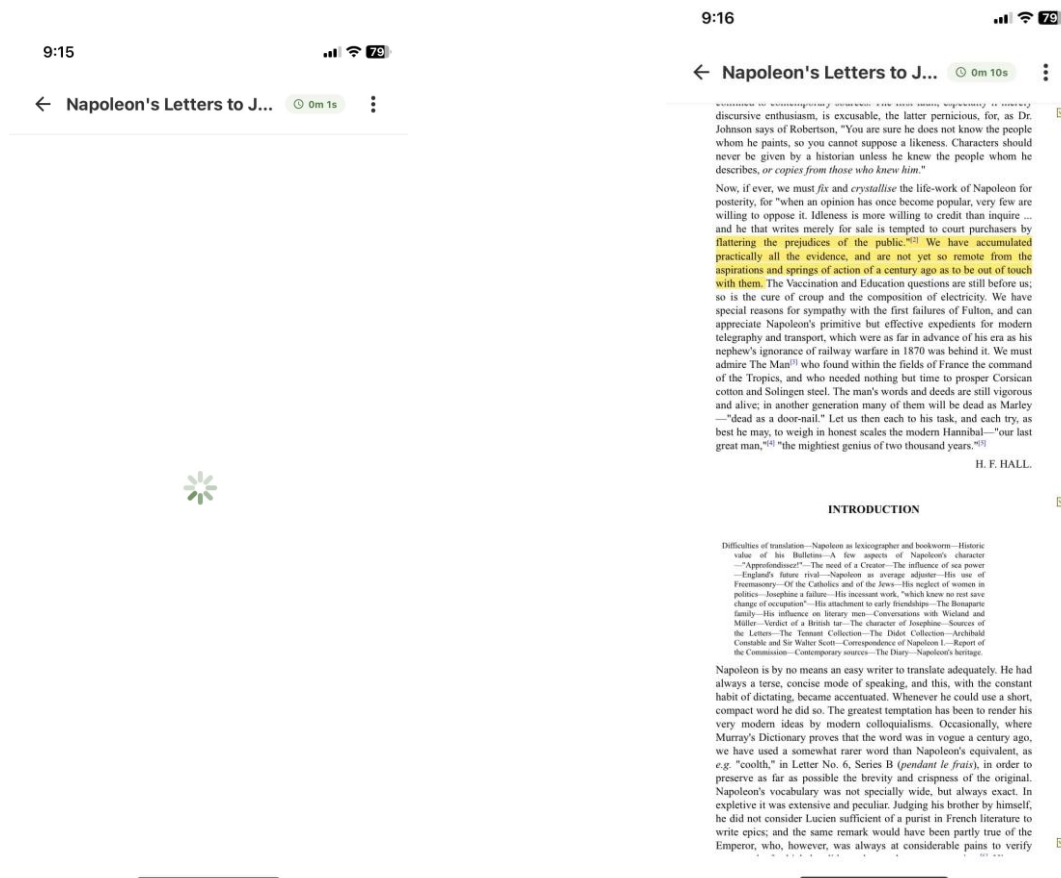


Figure 175 Load Book

Figure 174 Redirected Exact Page

9. Theme / Settings Tests

Table 53 Theme / Settings Tests Case

Test ID	Description
UT_THEME_42	Switch to pink mode
UT_THEME_43	Switch to brown mode
UT_THEME_44	Switch to green mode
UT_THEME_45	Switch to yellow mode

Table 54 Theme / Settings Tests Case 1

Test	42
Description	Switch to pink mode
Action	Go to Settings Select pink Mode
Expected Result	App theme should change to pink mode successfully
Actual Result	App theme changes to pink mode successfully
Conclusion	Pass

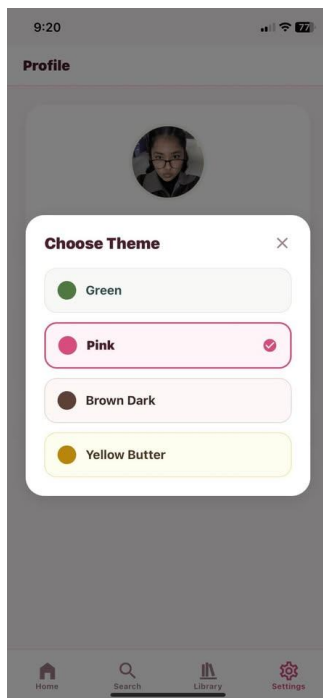


Figure 177 Pink Mode

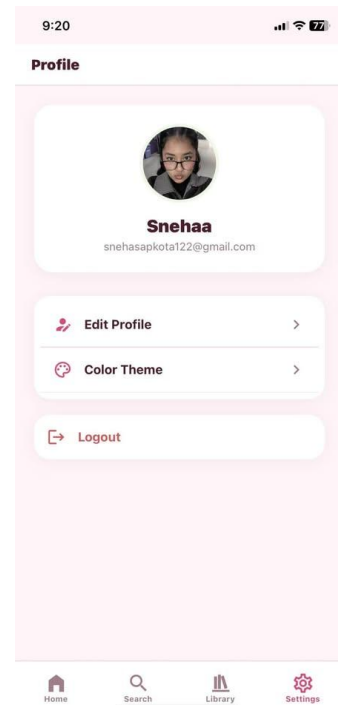


Figure 176 Change Pink

Table 55 Theme / Settings Tests Case 2

Test	43
Description	Switch to Brown mode
Action	Go to Settings Select brown Mode
Expected Result	App theme should change to brown mode successfully
Actual Result	App theme changes to brown mode successfully
Conclusion	Pass

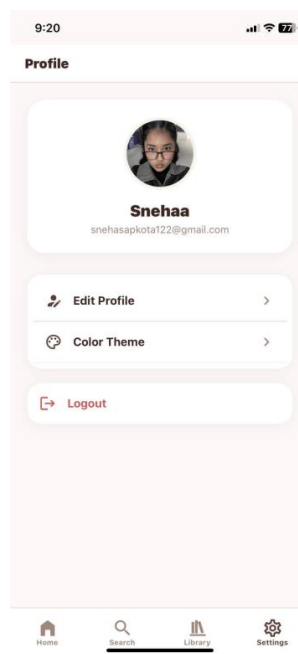
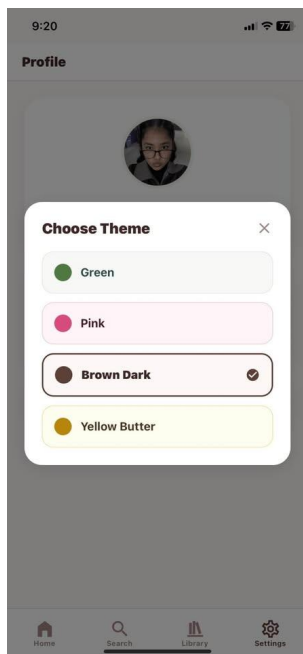


Figure 178 Change Brown

Figure 179 Brown Mode

Table 56 Theme / Settings Tests Case 3

Test	44
Description	Switch to Green mode
Action	Go to Settings Select green Mode
Expected Result	App theme should change to green mode successfully
Actual Result	App theme changes to green mode successfully
Conclusion	Pass

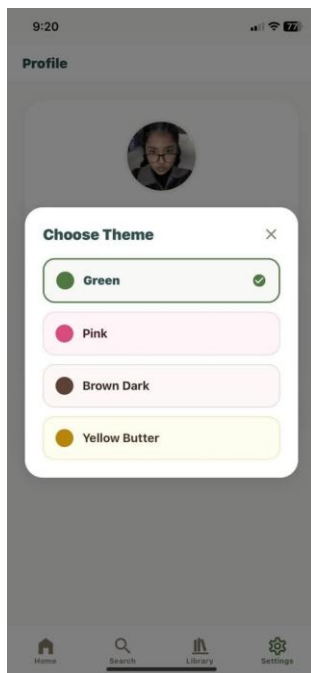


Figure 181 Green Mode

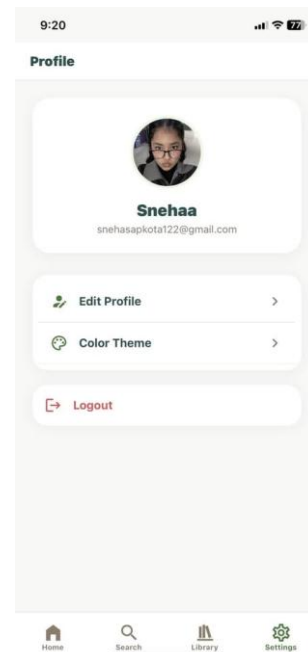


Figure 180 Change Green

Table 57 Theme / Settings Tests Case 4

Test	45
Description	Switch to yellow mode
Action	Go to Settings Select yellow Mode
Expected Result	App theme should change to yellow mode successfully
Actual Result	App theme changes to yellow mode successfully
Conclusion	Pass

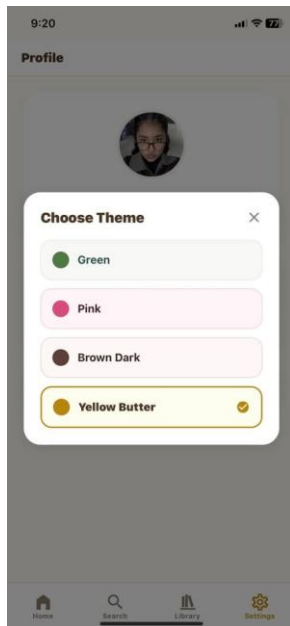


Figure 183 Yellow Mode

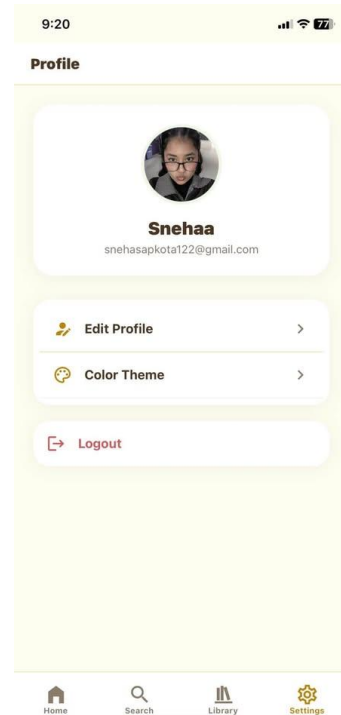


Figure 182 Change Yellow

Table 58 Theme / Settings Tests Case 5

Test	46
Description	Logout user
Action	Click on setting then "Logout" button
Expected Result	Users should log out and redirect to login page
Actual Result	User is logged out and redirected to login page
Conclusion	Pass

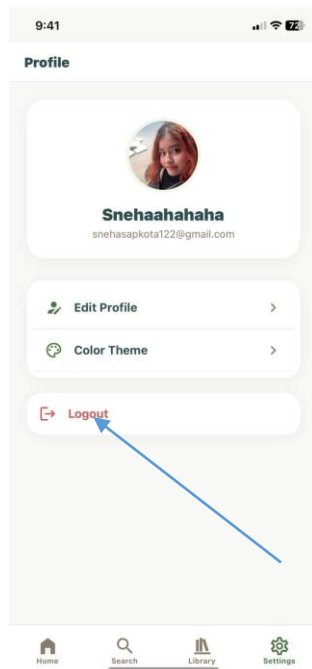


Figure 185 Logout

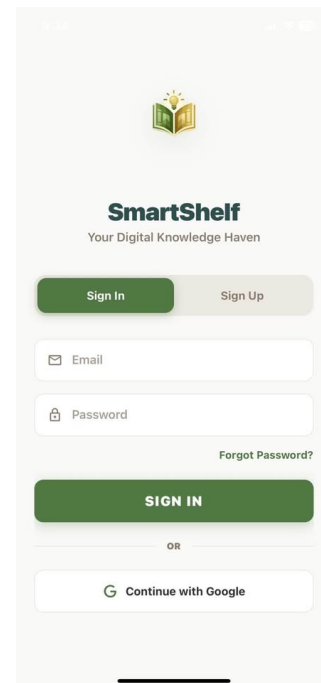


Figure 184 Redirect to Login

10. User Profile Tests

Table 59 User Profile Tests Case

Test ID	Description
UT_USER_46	Logout user
UT_USER_47	Change user password
UT_USER_48	Update profile information
UT_USER_49	Update profile picture

Table 60 User Profile Tests Case 1

Test	47
Description	User Change password
Action	Click on setting then, click on edit profile then new password
Expected Result	Password needs to update successfully
Actual Result	Password updated successful
Conclusion	Pass

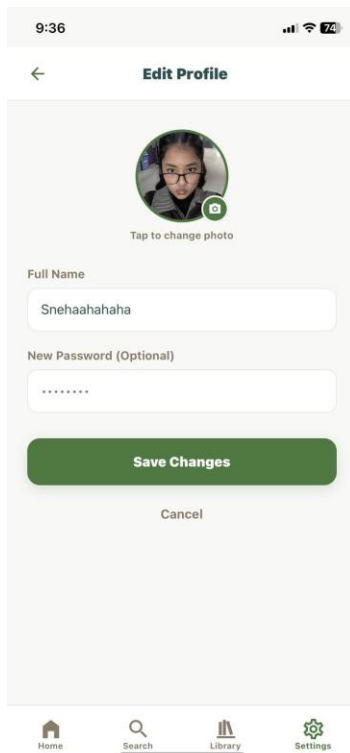


Figure 187 New Password

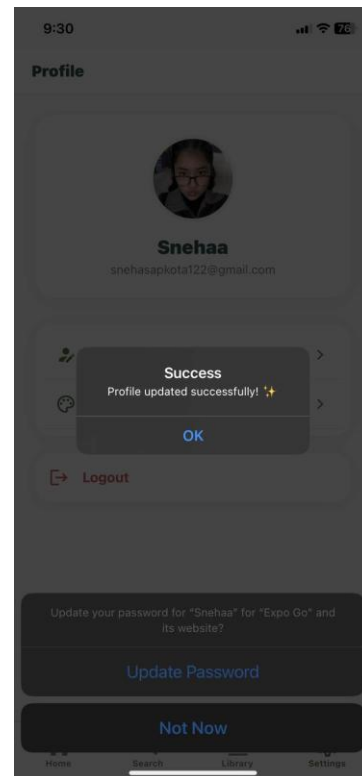


Figure 186 Password Set

Table 61 User Profile Tests Case 2

Test	48
Description	Update profile information
Action	Edit name and save changes
Expected Result	Profile data needs to be updated successfully and shown in UI
Actual Result	Profile data is updated successfully and shown in UI
Conclusion	Pass

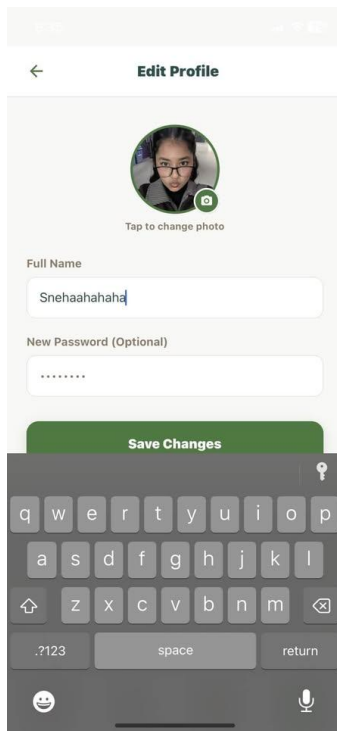


Figure 189 Name Set

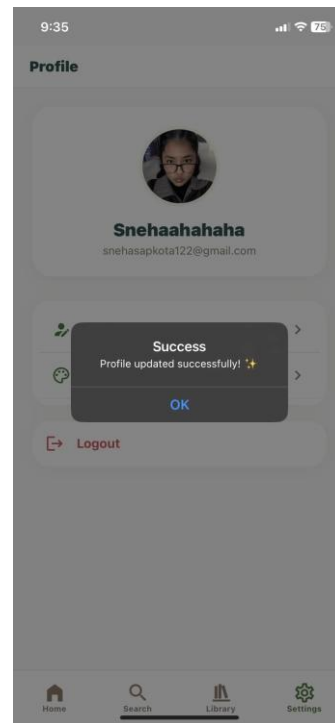


Figure 188 Name Update

Table 62 User Profile Tests Case 3

Test	49
Description	Update profile picture
Action	Go to setting, click profile picture then choose photo and save changes
Expected Result	Profile picture needs to be updated successfully and shown in UI
Actual Result	Profile picture update successfully
Conclusion	Pass

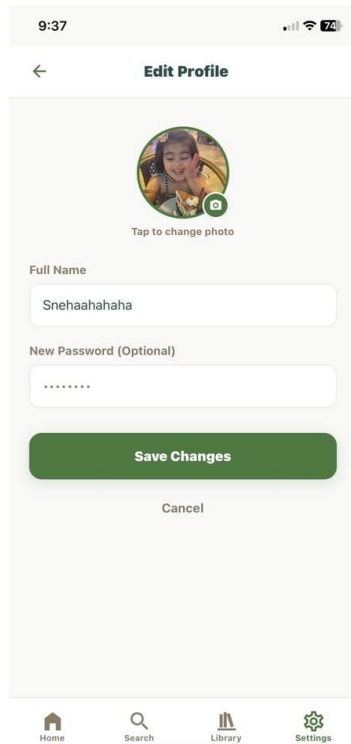


Figure 191 Profile Set

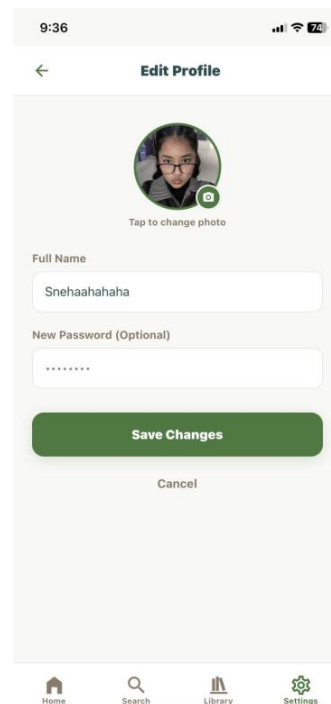


Figure 190 Success Profile Set

11. Content Based Recommendation

Table 63 Content Based Recommendation

Test ID	Description
UT_RECOMMEND_50	Content-based book recommendation

Table 64 Content Based Recommendation Case 1

Test	50
Description	Content Based Recommendation
Action	Search book Click "Book" Scroll down
Expected Result	System should suggest similar books based on genre
Actual Result	System suggests similar books based on genre successfully
Conclusion	Pass



Figure 192 Recommendation

12. Reading and Zoom Test

Table 65 Reading and Zoom Test Case

Test ID	Description
UT_READING_51	Zoom in book content
UT_READING_52	Zoom out book content
UT_READING_53	Save reading progress
UT_READING_54	Remove book from continue reading at 100%
UT_READING_55	Move completed book to finished section

Table 66 Reading and Zoom Test Case 1

Test	51
Description	Zoom in book content using pinch
Action	Open book Use two fingers and pinch-out (spread) on screen page
Expected Result	Text size should increase smoothly became more darker
Actual Result	Text size increases smoothly became more darker successfully
Conclusion	Pass

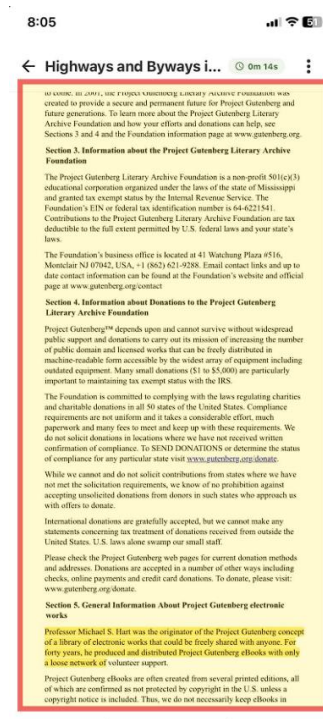


Figure 194 Before Zoom

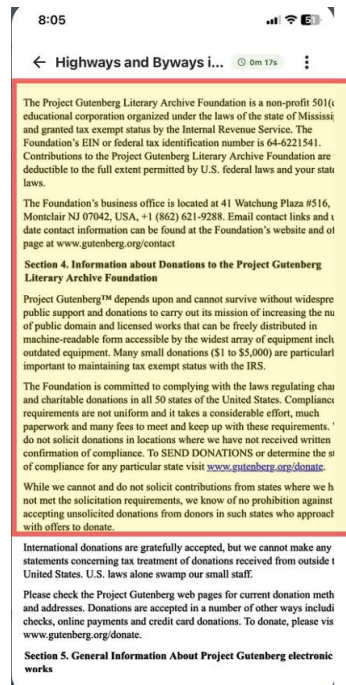


Figure 193 After Zoom

Table 67 Reading and Zoom Test Case 2

Test	52
Description	Zoom in book content using pinch
Action	Open book Use two fingers and pinch-in on screen page
Expected Result	Text size should decrease smoothly became more normal
Actual Result	Text size decrease smoothly became more normal successfully
Conclusion	Pass



Figure 195 Zoom out

Figure 196 Zoom in

Table 68 Reading and Zoom Test Case 3

Test	53
Description	Continue reading updates progress
Action	Read a book exit before finishing (e.g., 5%)
Expected Result	Progress needs to save and shown when user returns
Actual Result	Progress is saved and shown when user returns
Conclusion	Pass

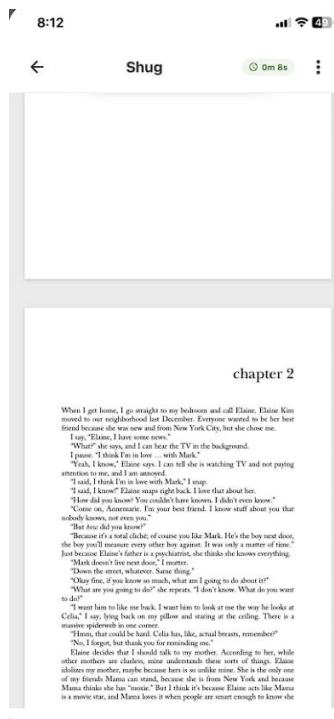


Figure 198 Read Book

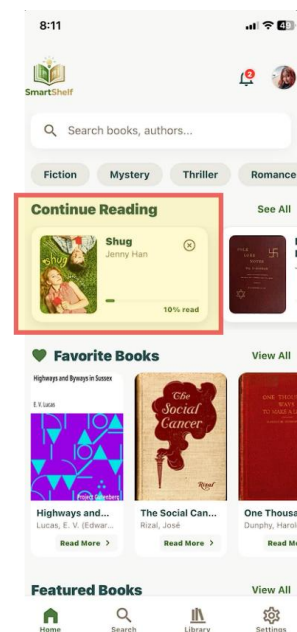


Figure 197 Update Progress

Table 69 Reading and Zoom Test Case 4

Test	54
Description	Remove from continued read as finished at 100%
Action	Read a book Until progress reaches 100%
Expected Result	Book should automatically remove from continue read
Actual Result	Book automatically remove from continue read
Conclusion	Pass

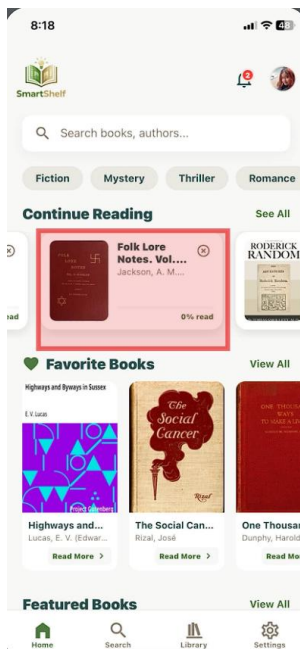


Figure 200 Complete Book

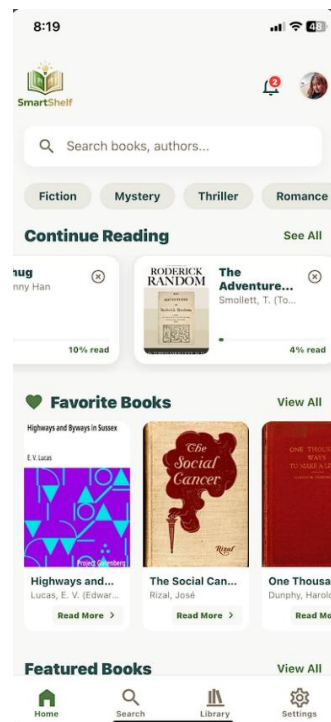


Figure 199 Removed Continue Read

Table 70 Reading and Zoom Test Case 5

Test	55
Description	100% progress book show in finished
Action	Read a book until progress reaches 100%
Expected Result	Book should automatically move to "Finished section"
Actual Result	Book automatically moved to "Finished section"
Conclusion	Pass

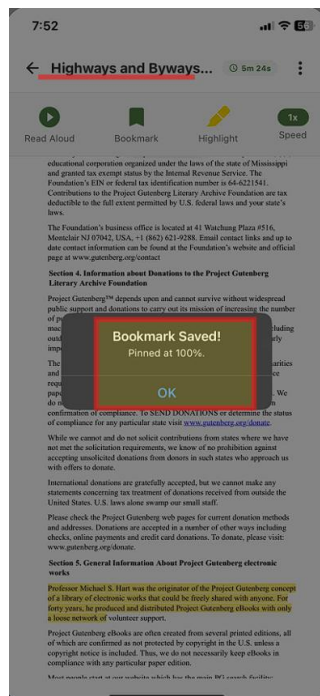


Figure 202 Bookmark 100

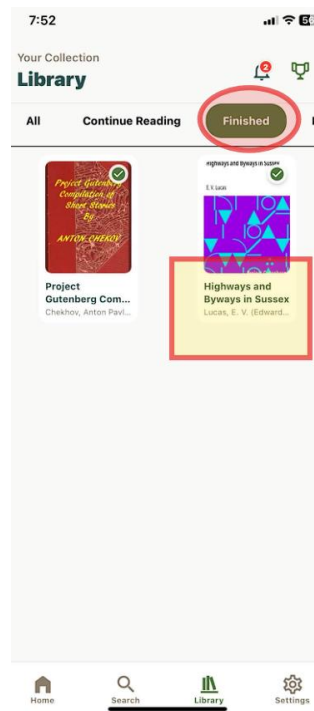


Figure 201 Finished

13. Subscription & Payment Tests

Table 71 Subscription & Payment Tests Case

Test ID	Description
UT_SUB_56	Restrict access without subscription
UT_SUB_57	Successful subscription via Stripe
UT_SUB_58	Cancel subscription/payment process
UT_SUB_59	Sync subscription status
UT_SUB_60	Handle failed payment

Table 72 Subscription & Payment Tests Case1

Test	56
Description	Access without subscription
Action	After reading 8 book, click on 9 books You see unlock the book
Expected Result	After 8 books, access show restricted and a subscription prompt is displayed
Actual Result	After 8 book access is restricted and a subscription prompt is displayed
Conclusion	Pass

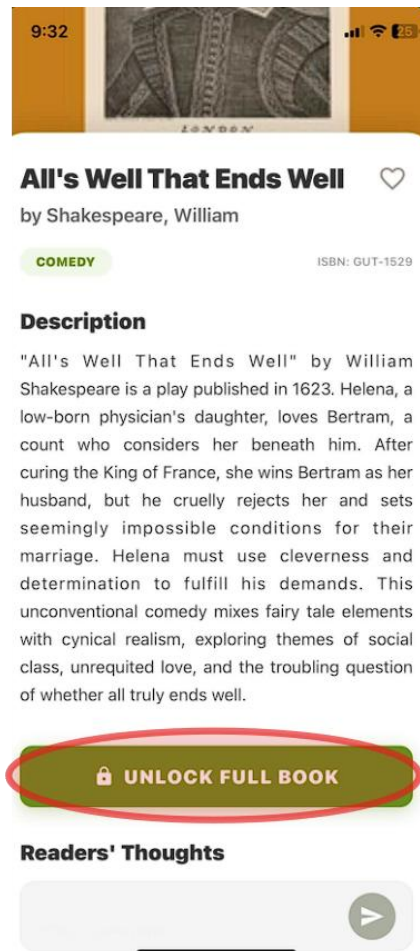


Figure 203 Lock

Table 73 Subscription & Payment Tests Case2

Test	57
Description	Successful subscription via stripe
Action	Open setting Click on upgrade to premium
Expected Result	Subscription should create and activated successfully
Actual Result	Subscription is created and activated successfully
Conclusion	Pass

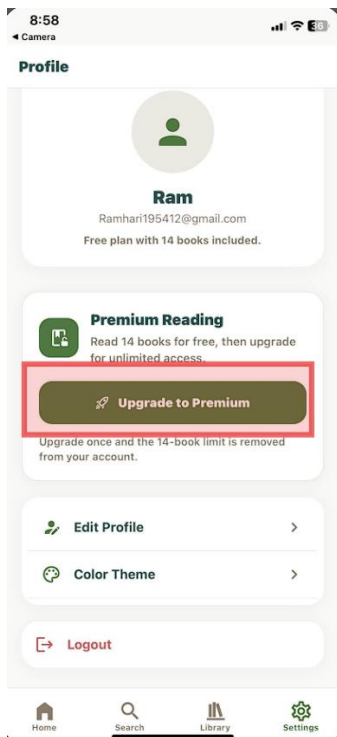


Figure 205 Subscription Upgrade

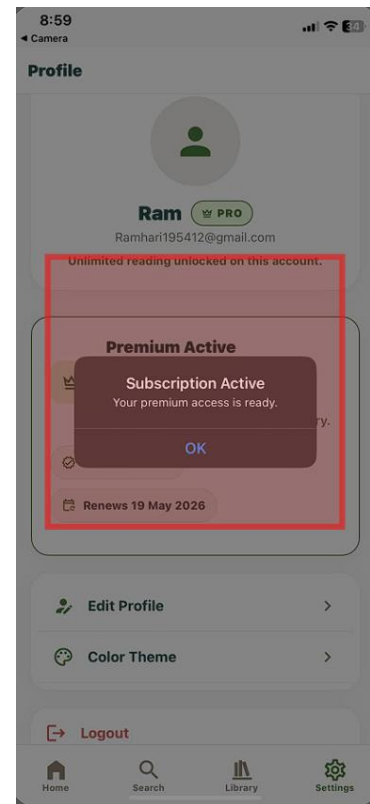


Figure 204 Active Subscription

Table 74 Subscription & Payment Tests Case3

Test	58
Description	Cancel payment
Action	Open unlock book Click on cancel
Expected Result	Should show dialog box saying cancel
Actual Result	Successfully show dialog box saying cancel
Conclusion	Pass

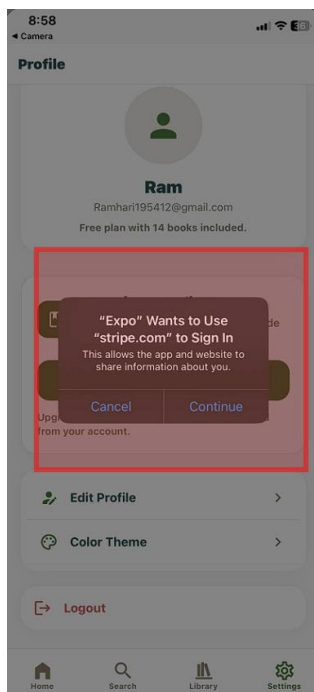


Figure 207 Payment

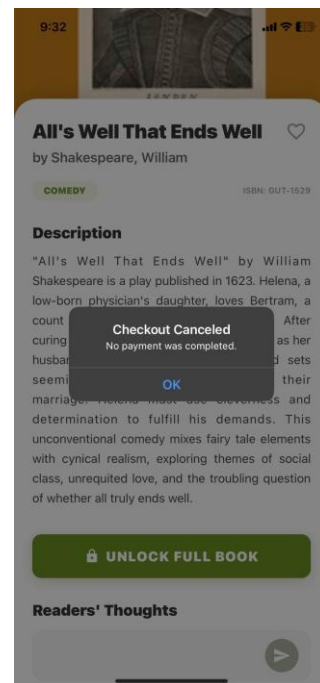


Figure 206 Check Out

Table 75 Subscription & Payment Tests Case4

Test	59
Description	Subscription status sync
Action	Check user profile after Stripe payment
Expected Result	Subscription status should show already updated
Actual Result	Subscription status should show already updated successfully
Conclusion	Pass

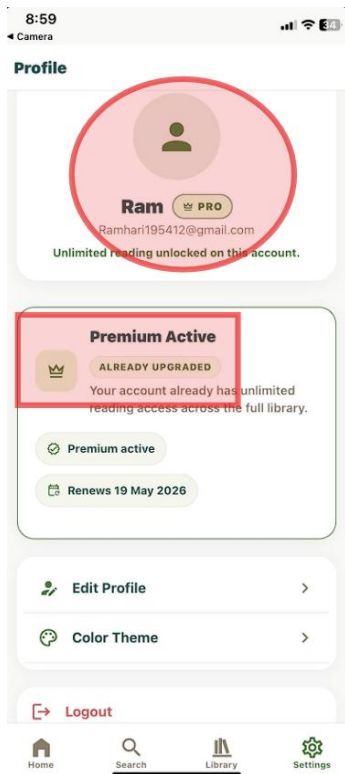


Figure 209 Subscription status sync

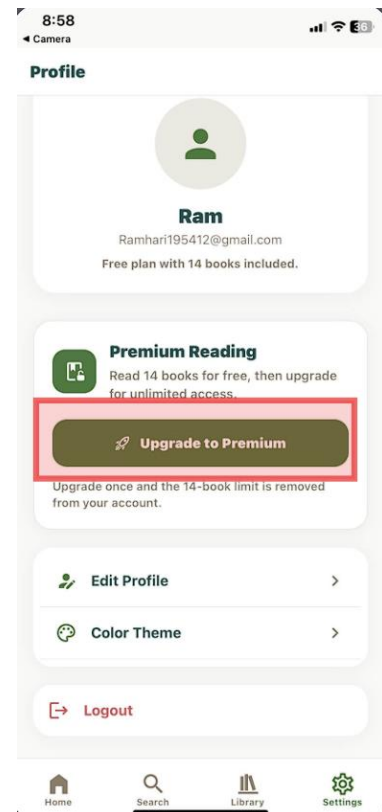


Figure 208 Before Subscription

Table 76 Subscription & Payment Tests Case5

Test	60
Description	Failed payment
Action	Put invalid card number
Expected Result	System should show invalid numbers, and subscribe is not working
Actual Result	System shows invalid number, and subscribe is not working
Conclusion	Pass

The screenshot shows a mobile browser view of a Stripe checkout page. The URL bar indicates 'checkout.stripe.com'. The page title is 'SmartShelf s...' and it includes a 'Sandbox' label. Under the 'Payment method' section, the 'Card' option is selected. The 'Card information' section is highlighted with a red box and contains the following details: Card number '3243 4343 4656 8868', Expiry date '02 / 96', and Cardholder name '5653'. Below this, a red error message states 'Your card number is invalid.' The 'Cardholder name' field contains 'Soskdidjdjx' and the 'Country or region' is set to 'Nepal'. The 'Apple Pay' option is also visible. At the bottom, there is a checkbox for 'Save my information for faster checkout' and a blue 'Subscribe' button, which is also highlighted with a red box.

Figure 210 Unable Payment

Admin Unit Test

1. Admin Login

Table 77 Admin Login Case

Test ID	Description
UT_LG_01	Valid login
UT_LG_02	Invalid password
UT_LG_03	Empty fields

Table 78 Admin Login Case 1

Test	61
Description	Valid login
Action	Enter valid Email and Password
Expected Result	Logged into system and should show admin dashboard
Actual Result	Logged into system and show admin dashboard successfully
Conclusion	Pass

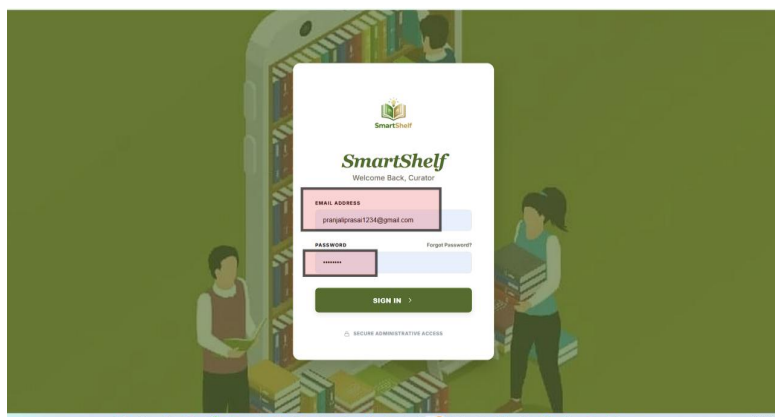


Figure 211 Admin Login

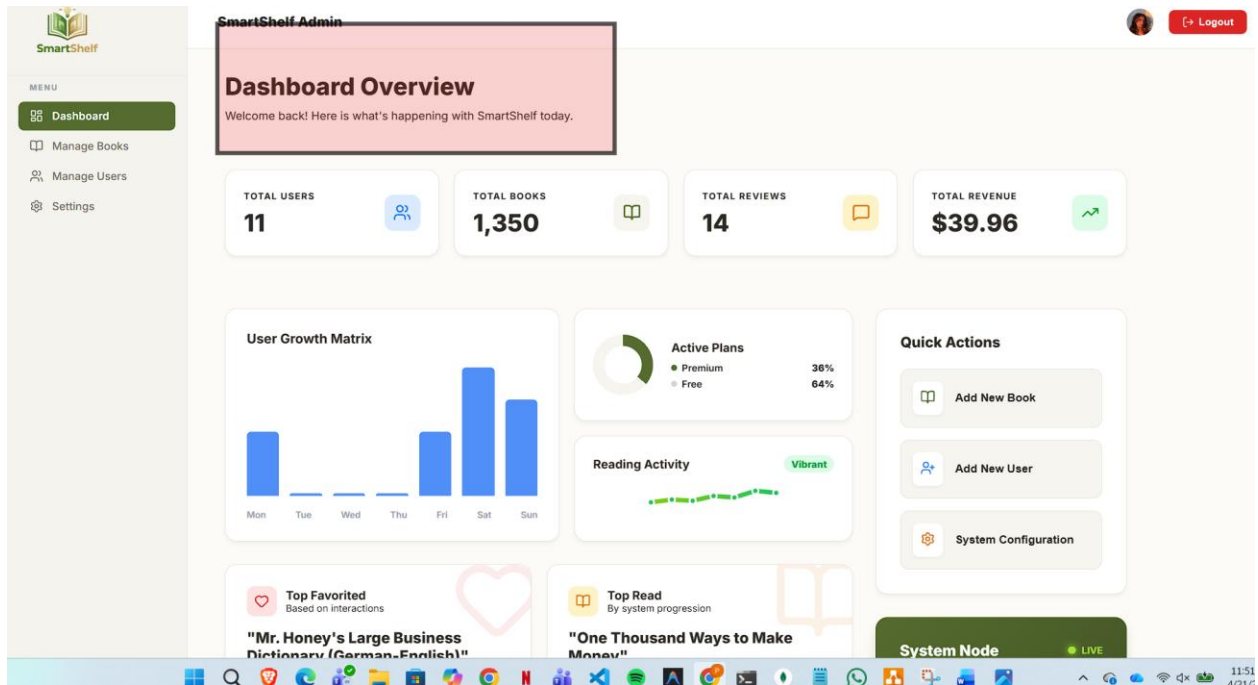


Figure 212 Dashboard

Table 79 Admin Login Case 2

Test	62
Description	Invalid login
Action	Enter invalid Email and Password
Expected Result	System should show invalid password
Actual Result	System show invalid password
Conclusion	Pass

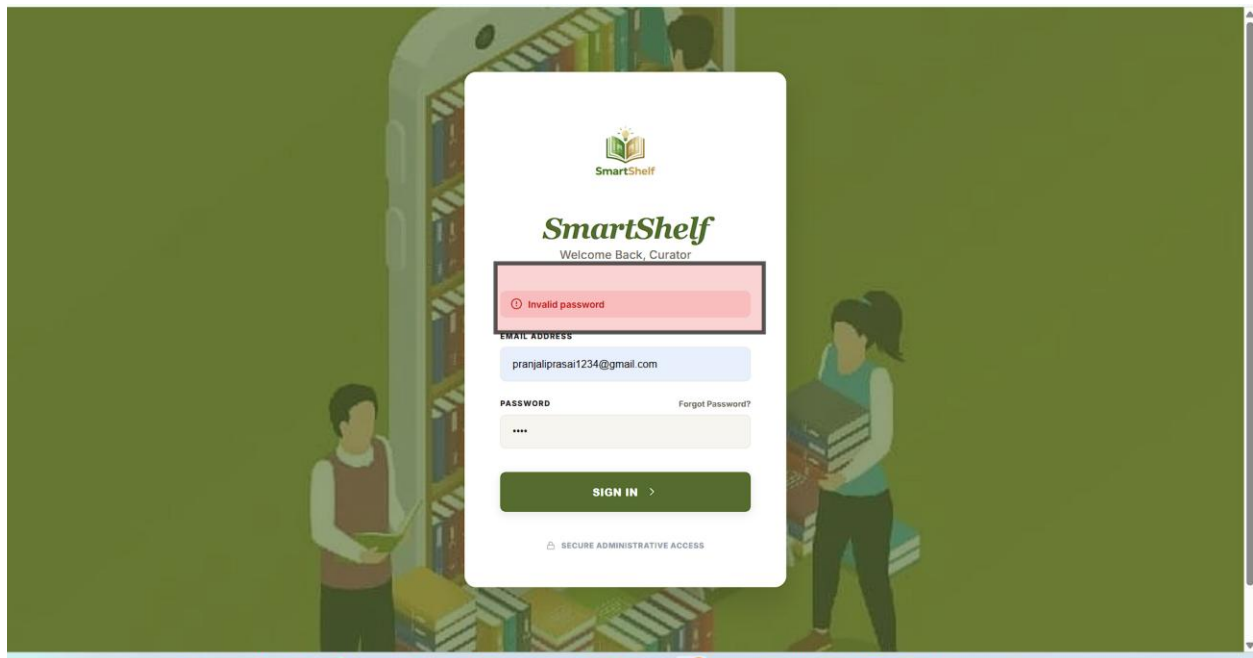


Figure 213 Invalid Login

Table 80 Admin Login Case 3

Test	63
Description	User login in admin
Action	Enter Email and Password
Expected Result	System should show access denied
Actual Result	System shows access denied
Conclusion	Pass

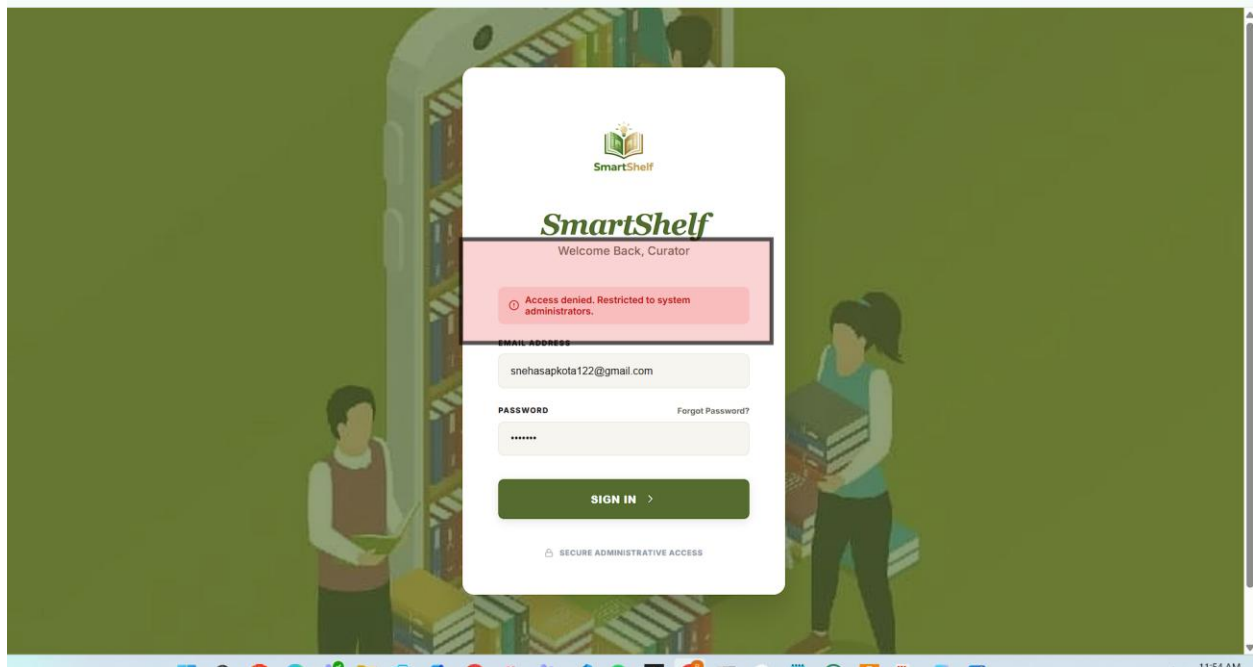


Figure 214 User login in admin

Table 81 Admin Login Case 4

Test	64
Description	Empty fields
Action	Enter nothing
Expected Result	System should show please fill field
Actual Result	System shows please fill field
Conclusion	Pass

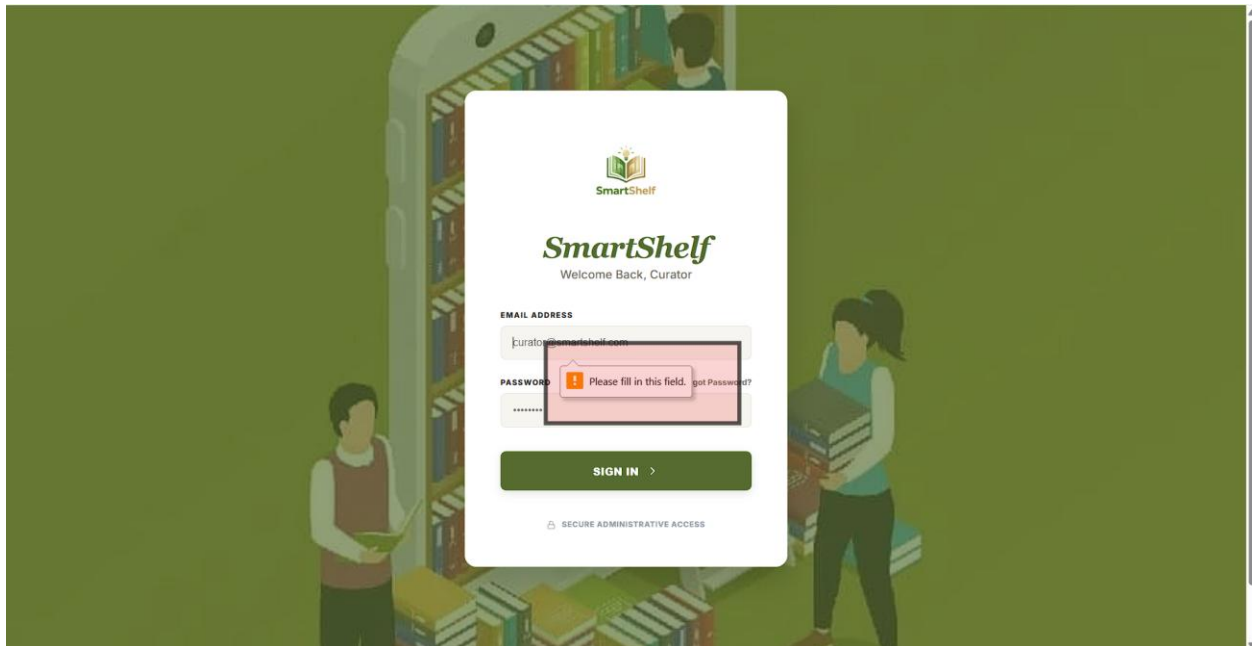


Figure 215 Empty Feilds Login

2. Admin Logout

Table 82 Admin Logout Case

Test ID	Description
UT_LO_01	Logout admin

Table 83 Admin Logout Case 1

Test	65
Description	Logout admin
Action	Click logout
Expected Result	System should show redirect to login page
Actual Result	System redirects to login page
Conclusion	Pass

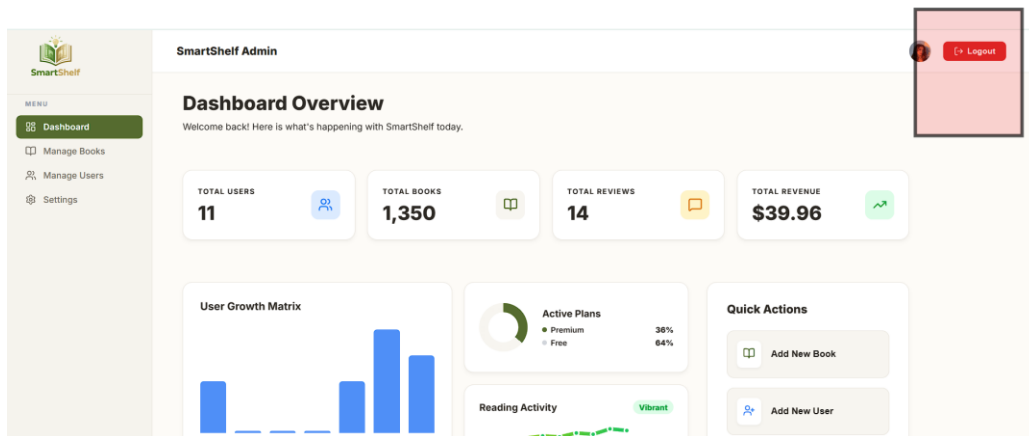


Figure 216 Admin Logout

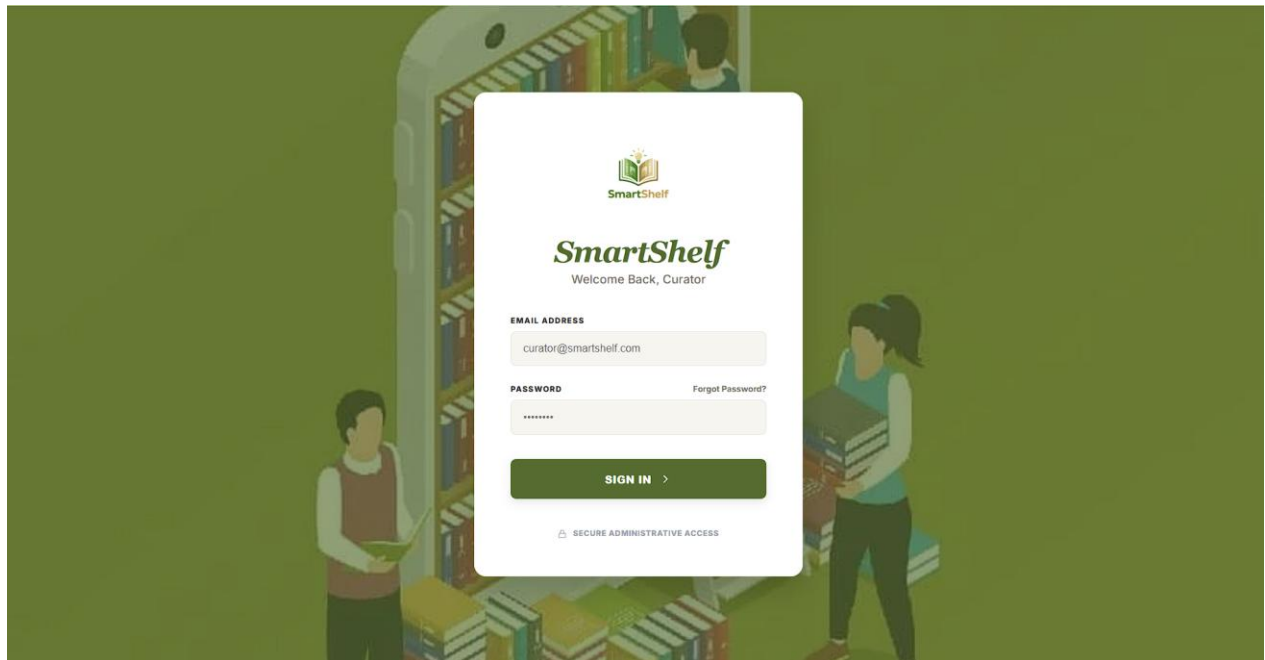


Figure 217 Redirect to Login

3. Admin Search Book

Table 84 Admin Search Book Case

Test ID	Description
UT_SB_01	Search existing book
UT_SB_02	Search non-existing book
UT_SB_03	Empty search

Table 85 Admin Search Book Case 1

Test	66
Description	Search Book
Action	Search Pride and prejudice
Expected Result	System should show book
Actual Result	System show book successfully
Conclusion	Pass

The screenshot shows the SmartShelf Admin dashboard. The 'The Collection' section is active, displaying search statistics and a search bar. The search bar contains the text 'Pride and prejudice'. Below the search bar, a table lists three entries for 'Pride and Prejudice' by Jane Austen, each with a cover image, title, meta, author, source (Gutenberg), ISBN-13, and actions.

Cover	Title & Meta	Author	Source	ISBN-13	Actions
	Pride and Prejudice, a play founded on Jane Austen's novel Published 1925 • Drama	Austen, Jane	Gutenberg	GUT-37431	> 🗑️
	Pride and Prejudice Published 1925 • Romance	Austen, Jane	Gutenberg	GUT-42671	> 🗑️
	Pride and Prejudice Published 1925 • Fiction	Austen, Jane	Gutenberg	GUT-1342	> 🗑️

Figure 218 Search Book

Table 86 Admin Search Book Case 2

Test	67
Description	Search non exist Book
Action	Search 12xyz
Expected Result	System should no book found
Actual Result	System show no book found successfully
Conclusion	Pass

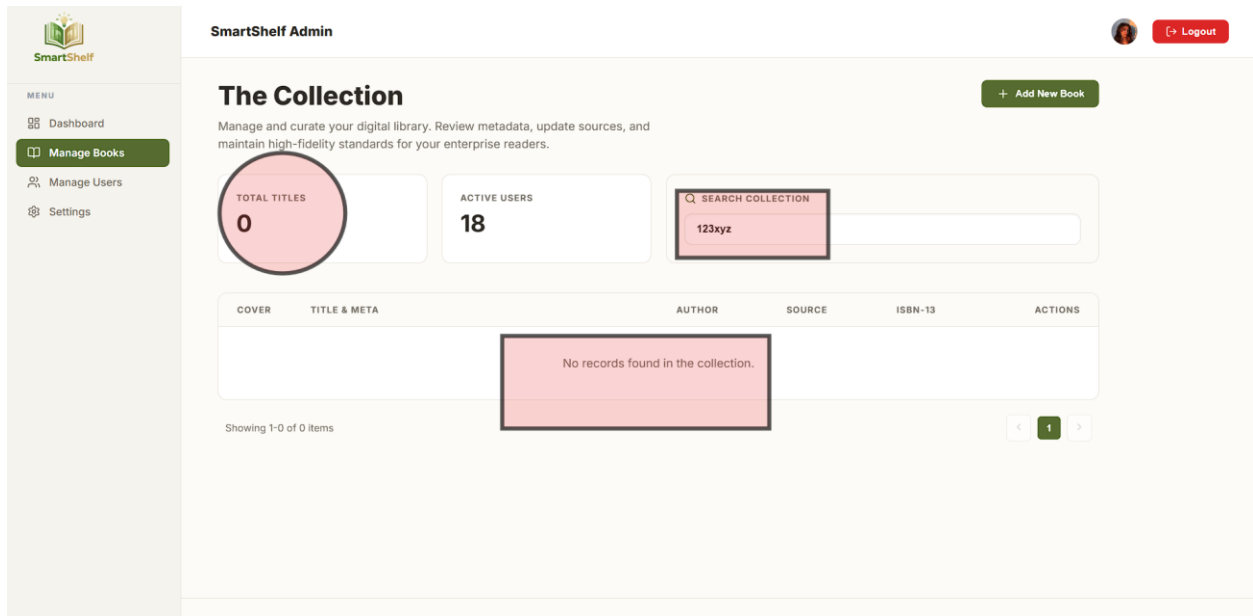


Figure 219 Non-Exist Search

28

Table 87 Admin Search Book Case 3

Test	68
Description	Empty search
Action	Search empty
Expected Result	System should all book
Actual Result	System show all book successfully
Conclusion	Pass

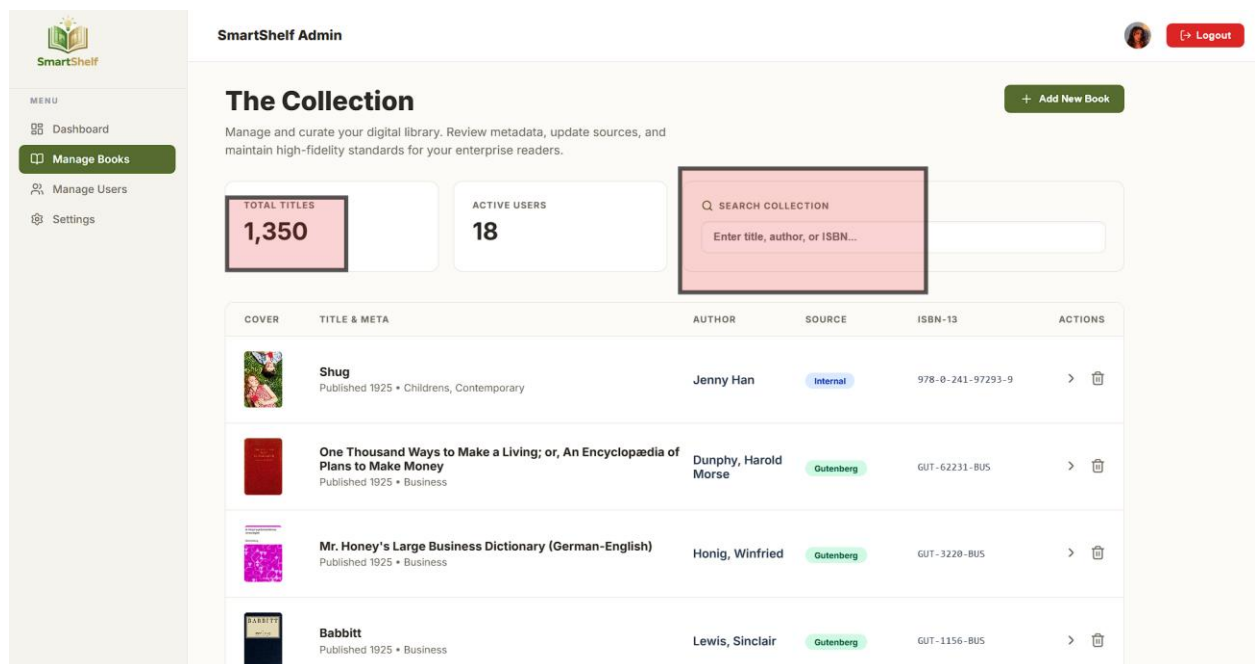


Figure 220 Empty Search

4. Admin Search User

Table 88 Admin Search User Case

Test ID	Description
UT_Su_01	Search user
UT_Su_02	Search for non-existing user
UT_Su_03	Empty search

Table 89 Admin Search User Case 1

Test	69
Description	Search user
Action	Search sneha
Expected Result	System should list of users name sneha
Actual Result	System show list of user successfully
Conclusion	Pass

The screenshot displays the SmartShelf Admin interface. On the left is a sidebar menu with options: Dashboard, Manage Books, Manage Users (highlighted), and Settings. The main content area is titled 'User Directory' with the subtitle 'Manage permissions and oversee curation team members.' and an 'Invite User' button. Below the title are two summary cards: 'TOTAL USERS' showing 2 with a +0% change, and 'ACTIVE NOW' showing 2. To the right is a card for 'TOTAL BOOKS' showing 1,350 Total Books. A search bar contains the text 'sneha', and a dropdown menu is set to 'All Users'. Below the search bar is a table with the following data:

USER PROFILE	ROLE	JOINED DATE	STATUS	ACTIONS
Sneha Sapkota snehasapkota2026@gmail.com	USER	Mar 02, 2026	Active	
Snehaahahaha snehasapkota122@gmail.com	USER	Dec 14, 2025	Active	

At the bottom of the table, it says 'Showing 1-2 of 2 users' and a pagination control shows '1'.

Figure 221 User Search

Table 90 Admin Search User Case 2

Test	70
Description	Search non exist user
Action	Search 12xyz
Expected Result	System should no user found
Actual Result	System show no user found successfully
Conclusion	Pass

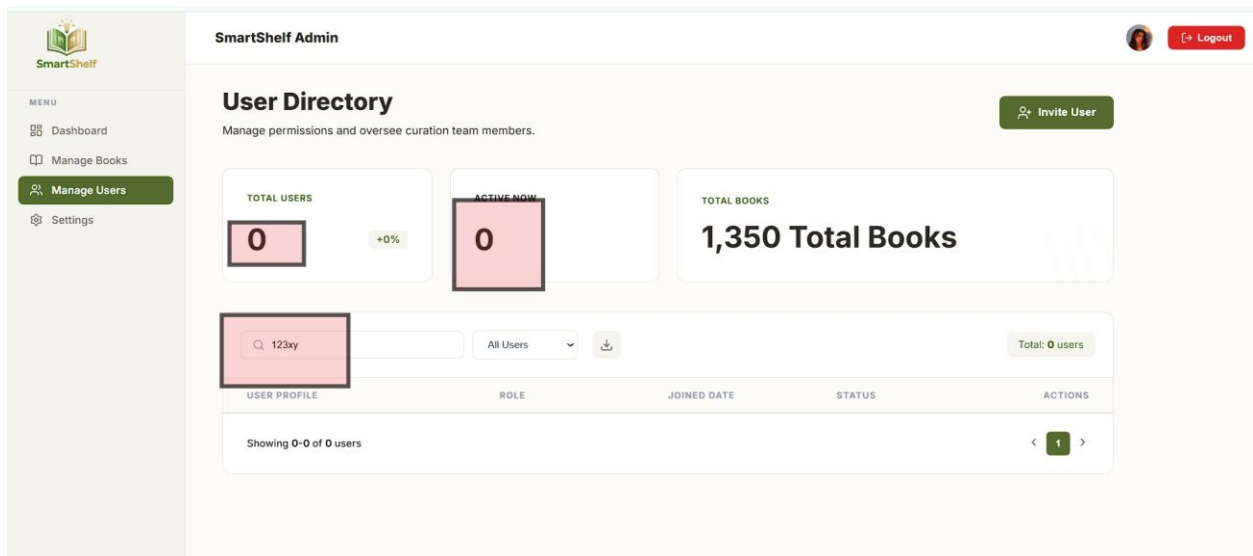


Figure 222 Search non exist user

Table 91 Admin Search User Case 3

Test	71
Description	Empty search in user
Action	Search empty
Expected Result	System should all users
Actual Result	System show all users successfully
Conclusion	Pass

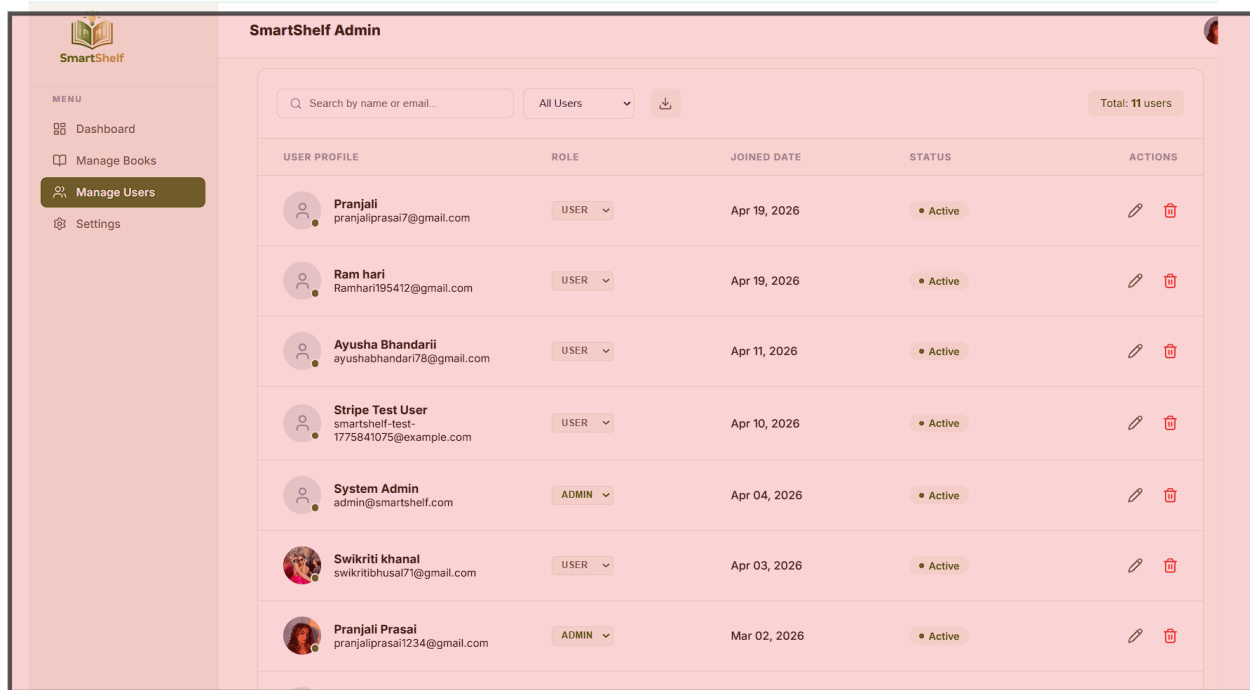


Figure 223 Empty search in user

5. Dashboard load

Table 92 Dashboard load Case

Test ID	Description
UT_DB_01	Load dashboard

Table 93 Dashboard load Case 1

Test	72
Description	Load dashboard
Action	Click on dashboard
Expected Result	System should load all data correctly
Actual Result	System show loads all correct data successfully
Conclusion	Pass

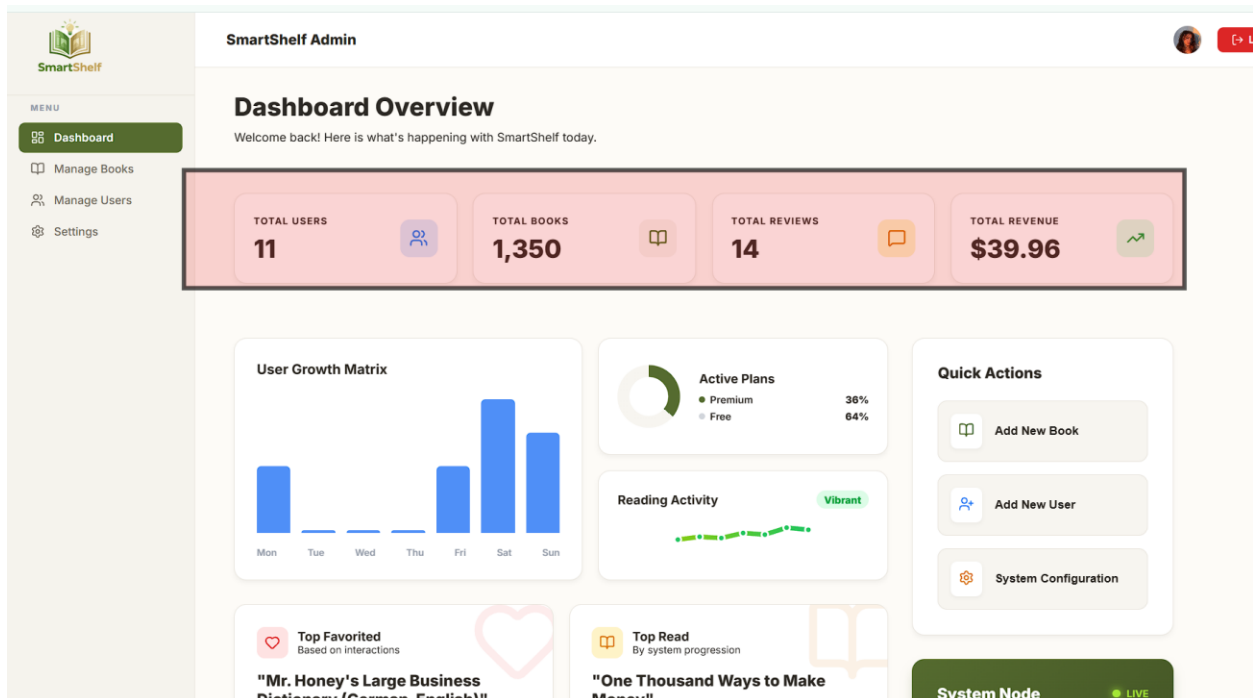


Figure 224 Load Dashboar

6. Pagination

Table 94 Pagination Case

Test ID	Description
UT_PG_01	Next page
UT_PG_02	Previous page
UT_PG_03	No more data

Table 95 Pagination Case 1

Test	73
Description	Next Page
Action	Click on next page
Expected Result	System should load next data
Actual Result	Systems show load next data successfully
Conclusion	Pass

Figure 225 Next Page

Figure 226 Work Pagination

Table 96 Pagination Case 2

Test	74
Description	Previous page
Action	Click on Previous page
Expected Result	System should load previous data
Actual Result	Systems show loads previous data successfully
Conclusion	Pass

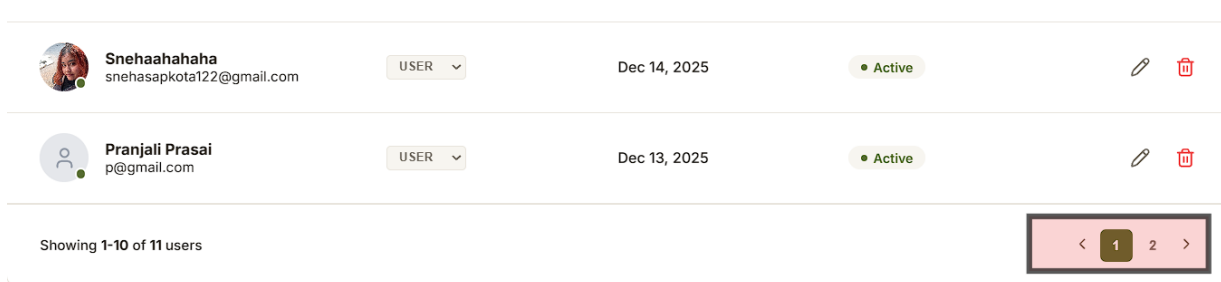


Figure 227 Back Pagination

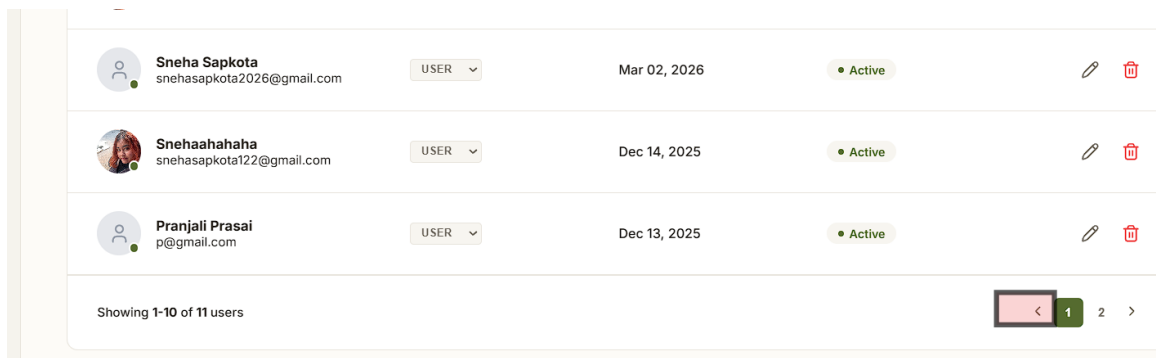


Figure 228 Back Pagination Work

Table 97 Pagination Case 3

Test	75
Description	No more data
Action	Click on next page
Expected Result	System should disable button
Actual Result	Systems disable button successfully
Conclusion	Pass

All Users

Total: 11 users

USER PROFILE	ROLE	JOINED DATE	STATUS	ACTIONS
pratiush prasain pratiushprasain@gmail.com	USER	Dec 13, 2025	Active	

Showing 11-11 of 11 users

< 1 2 >

Figure 229 Pagination Unable

7. Book CRUD

Table 98 Book CRUD Case

Test ID	Description
UT_BK_01	Add new book
UT_BK_02	Update book details
UT_BK_03	Delete book
UT_BK_04	View book list

Table 99 Book CRUD Case 1

Test	76
Description	Add new book
Action	Enter valid book details click add
Expected Result	Book should be added
Actual Result	book should be added successfully
Conclusion	Pass

Add New Book

TITLE: Shug

AUTHOR: Jenny Han

GENRE: love

ISBN: 090 123 566788 6777

DESCRIPTION: Lovely booknice book Lovely booknice book Lovely booknice book Lovely booknice book Lovely booknice book Lovely booknice book Lovely booknice book Lovely booknice book Lovely booknice book

BOOK FILE (PDF): Choose file Smartshelf...art (1).pdf

COVER IMAGE: Choose file Screensho...30326.png

Cancel Add Book

Figure 230 Add Book

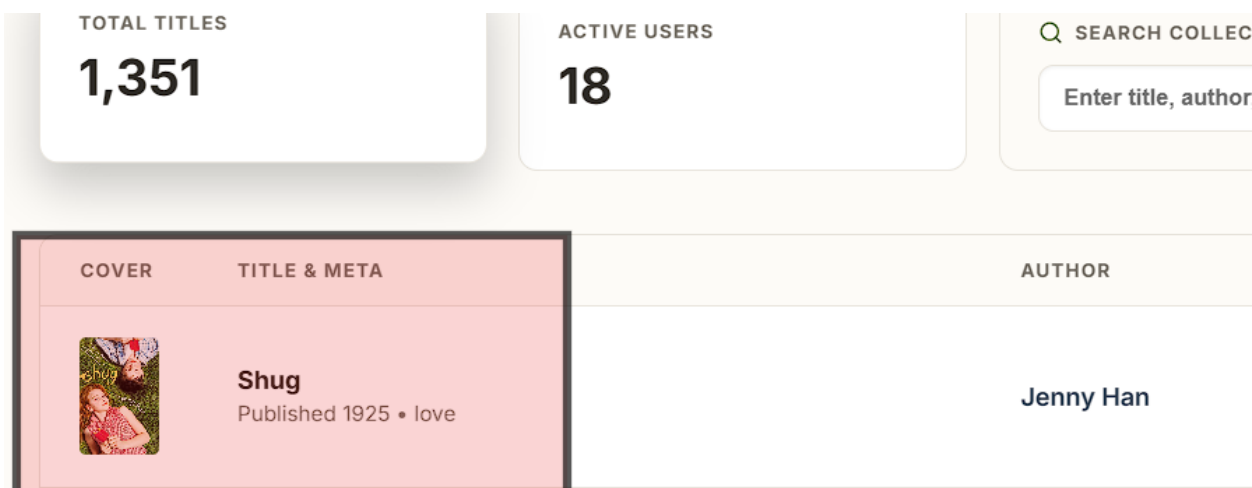


Figure 231 Successfully Added

Table 100 Book CRUD Case 2

Test	77
Description	Update book
Action	Edit book details click save
Expected Result	book should be update
Actual Result	book should be update successfully
Conclusion	Pass

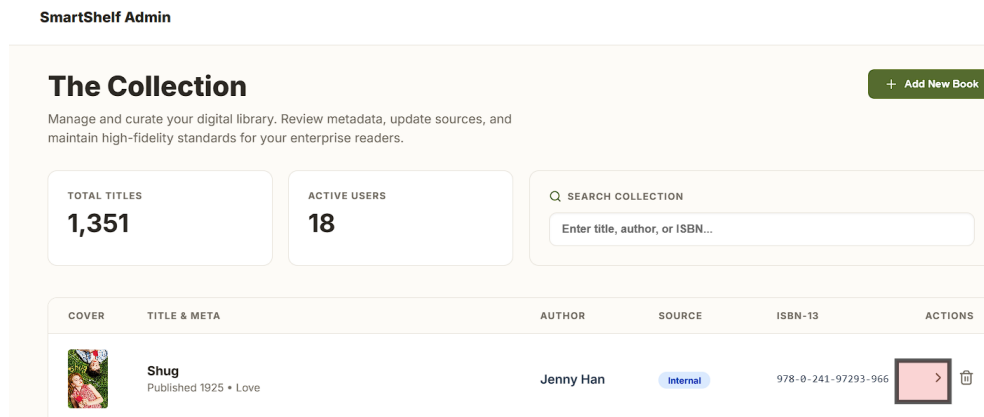


Figure 232 Update Book

×

TITLE

Shugggggg

AUTHOR

Jenny Han

GENRE

Love

ISBN

978-0-241-97293-966

DESCRIPTION

lovely lovely lovely lovely lovely lovely lovely lovely lovely lovely
lovely lovely lovely lovely lovely lovely lovely lovely lovely lovely
lovely lovely lovely lovely lovely lovely lovely lovely lovely lovely
lovely lovely lovely lovely lovely lovely lovely lovely lovely lovely

BOOK FILE (PDF)

Choose file No file chosen

COVER IMAGE

Choose file No file chosen

Cancel

Update Book

Figure 233 Edit Book

163

turnitin 23056303_Pranjali Prasai Page 232 of 311 - Integrity Submission

Submission ID trn:oid::3618:136149558

COVER	TITLE & META	AUTHOR	SOURCE	ISBN-13	ACTIONS
	Shuggggg Published 1925 • Love	Jenny Han	Internal	978-0-241-97293-966	> 

Figure 234 Successfully Added

Table 101 Book CRUD Case 3

Test	78
Description	Delete book
Action	Select book click delete
Expected Result	book should be deleted
Actual Result	book should be deleted successfully
Conclusion	Pass

COVER	TITLE & META	AUTHOR	SOURCE	ISBN-13	ACTIONS
	Shuggggy Published 1925 • Love	Jenny Han	Internal	978-0-241-97293-966	> 

Figure 235 Delete Book

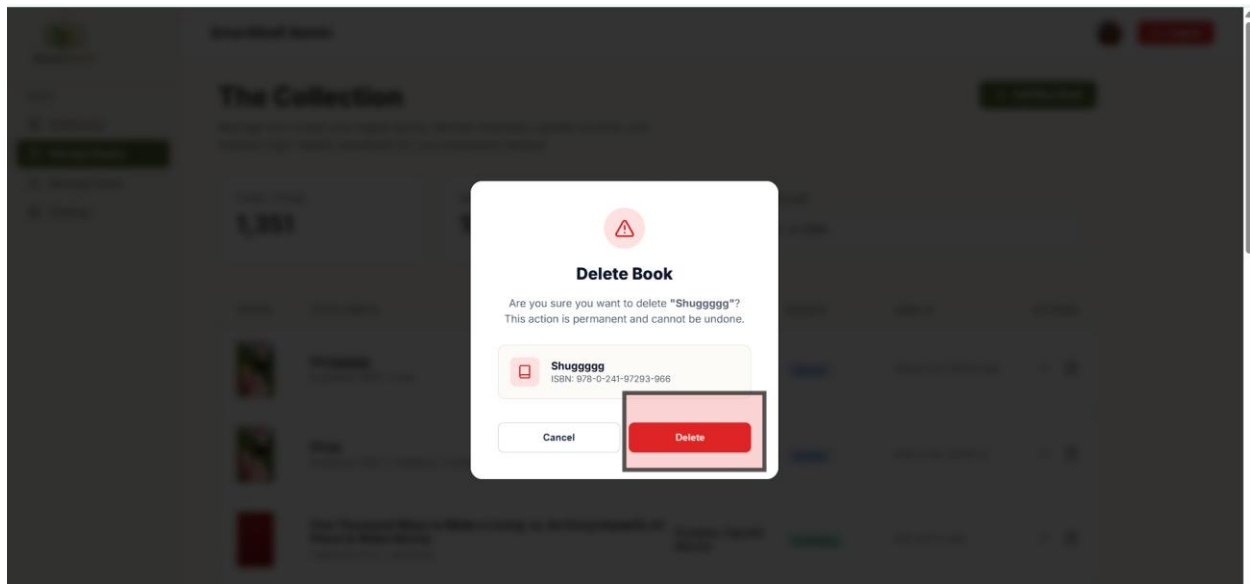


Figure 236 Book Delete by Admin

Table 102 Book CRUD Case 4

Test	79
Description	View book
Action	Open book management page
Expected Result	System should display all book
Actual Result	System displays all book
Conclusion	Pass

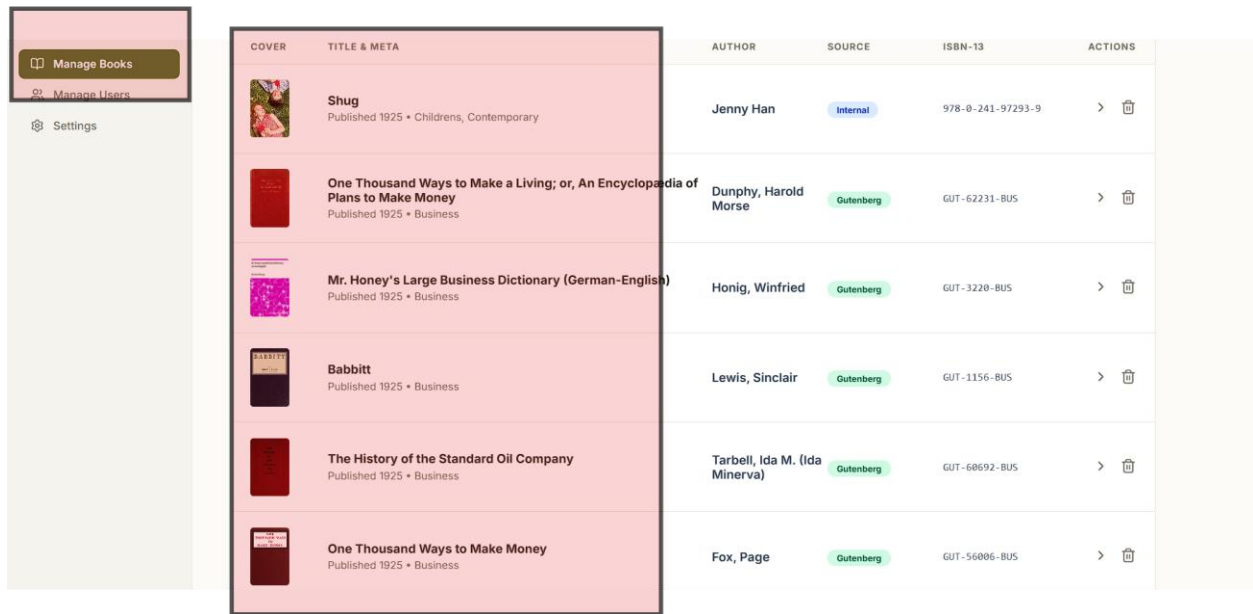


Figure 237 View Book

8. User CRUD

Table 103 User CRUD Case

Test ID	Description
UT_US_01	Add new user
UT_US_02	Update user
UT_US_03	Delete user
UT_US_04	View user list

Table 104 User CRUD Case 1

Test	80
Description	Add new user
Action	Enter valid user details click register
Expected Result	User should be created
Actual Result	User should be created successfully
Conclusion	Pass

Figure 238 Add New User

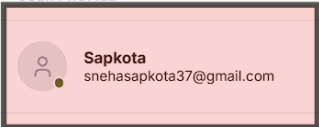
USER PROFILE	ROLE	JOINED DATE	STATUS	ACTIONS
  Sapkota snehasapkota37@gmail.com	USER ▾	Apr 21, 2026	● Active	 

Figure 239 Added User

Table 105 User CRUD Case 2

Test	81
Description	Update user
Action	Edit user details click save
Expected Result	User should be update
Actual Result	User should be update successfully
Conclusion	Pass

Edit User: Sapkota

FULL NAME
Sapkota Thapa

EMAIL ADDRESS
snehasapkota37@gmail.com

NEW PASSWORD (LEAVE BLANK TO KEEP CURRENT)

SYSTEM ROLE
Platform User (USER)

Cancel Save Changes

Figure 240 Edit User

USER PROFILE	ROLE	JOINED DATE	STATUS	ACTIONS
	USER	Apr 21, 2026	Active	

Figure 241 User Updated

1

Table 106 User CRUD Case 3

Test	82
Description	Delete user
Action	Select user click delete
Expected Result	User should be deleted
Actual Result	User should be deleted successfully
Conclusion	Pass

10

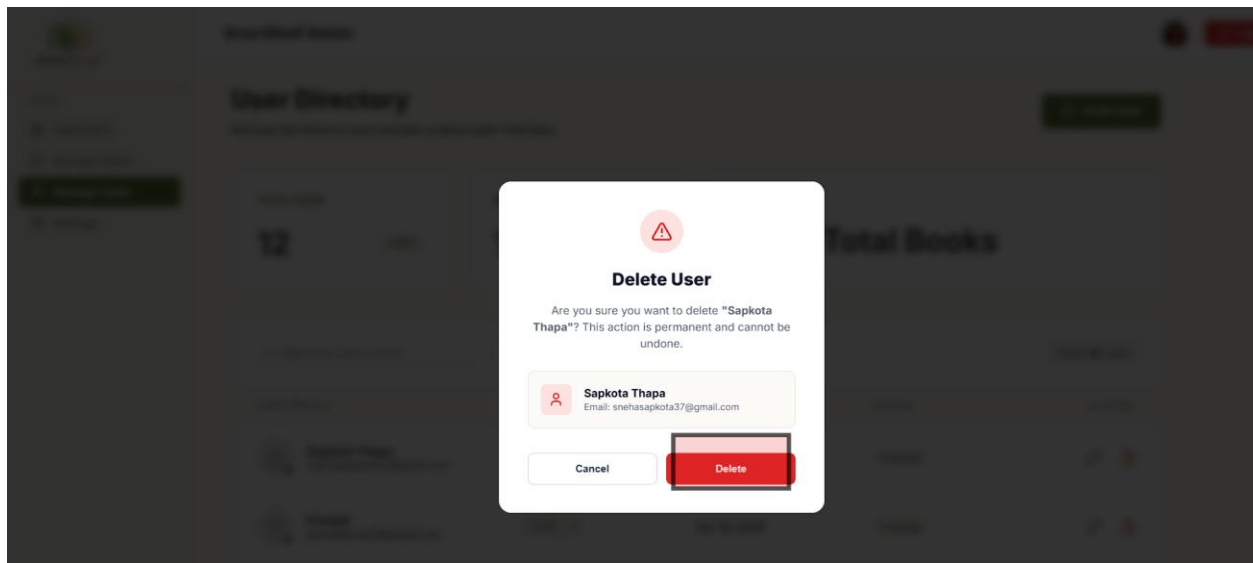


Figure 242 Delete User

Table 107 User CRUD Case 4

Test	83
Description	View user
Action	Open user management page
Expected Result	System should display all user
Actual Result	System display all user
Conclusion	Pass

USER PROFILE	ROLE	JOINED DATE	STATUS	ACTIONS
Pranjal pranjalprasa7@gmail.com	USER	Apr 19, 2026	Active	
Ram hari Ramhari195412@gmail.com	USER	Apr 19, 2026	Active	
Ayusha Bhandarii ayushabhandari78@gmail.com	USER	Apr 11, 2026	Active	
Stripe Test User smartshef-test-1775841075@example.com	USER	Apr 10, 2026	Active	
System Admin admin@smartshef.com	ADMIN	Apr 04, 2026	Active	
Swikriti khanal swikritibhusal71@gmail.com	USER	Apr 03, 2026	Active	
Pranjal Prasai pranjalprasa1234@gmail.com	ADMIN	Mar 02, 2026	Active	
Sneha Sapkota snehasapkota2026@gmail.com	USER	Mar 02, 2026	Active	

Figure 243 View User

9. Colour Mode

Table 108 Colour Mode Case

Test ID	Description
UT_CM_01	Switch to dark mode

Table 109 Colour Mode Case 1

Test	84
Description	Switch to sky mode
Action	Click on sky mode
Expected Result	System should change color mode
Actual Result	System change color mode sucessfully
Conclusion	Pass

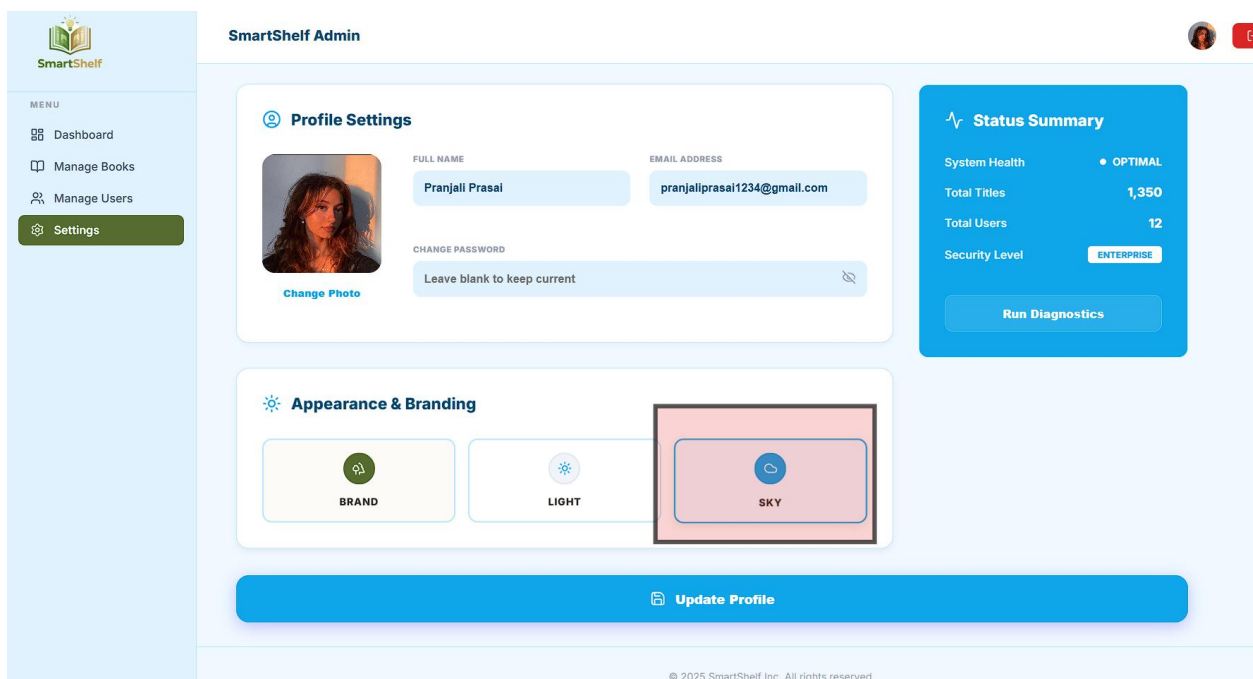


Figure 244 Color Mode Sky

38

4.1.2 Black Box Testing (System Test)

81

Black Box testing is a kind of software testing which the tester which is not interested in what the software knows internally or how it is implemented but instead what is the functionality of the software as per the specifications or requirements (geeksforgeeks, 2025).

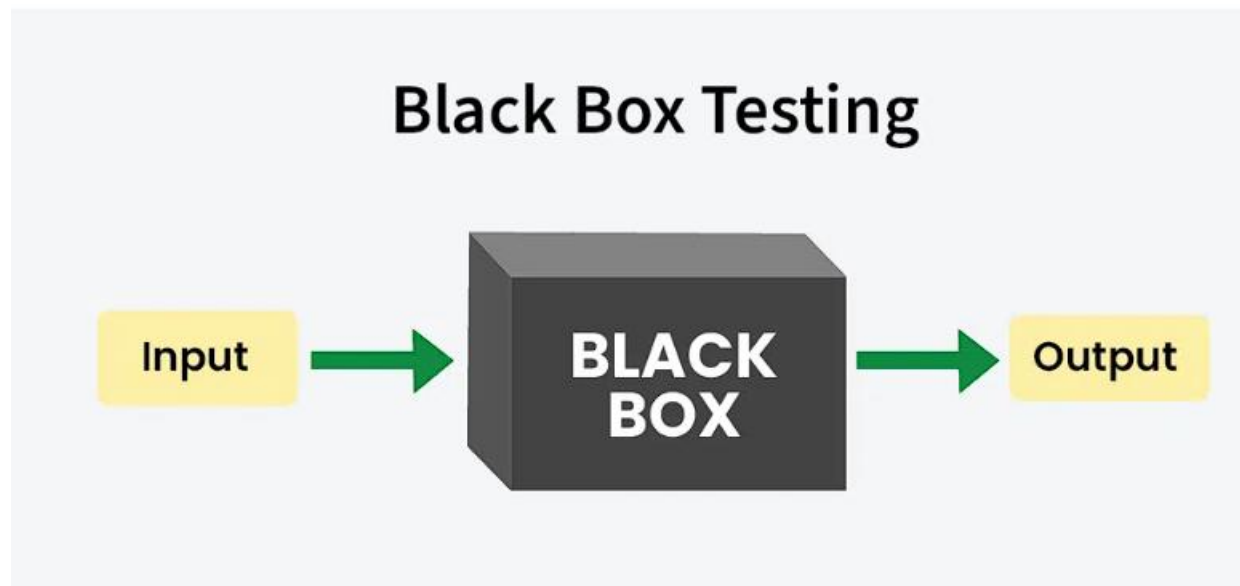


Figure 245 Black Box (geeksforgeeks, 2025)

Table: User Flow System Test Plan

Table 110 User Flow System Test Plan

Test Case ID	Test Scenario	Result
ST_USER_01	User registers and logs in successfully	Pass
ST_USER_02	User browses books and reads content	Pass
ST_USER_03	User highlights text and adds comment	Pass
ST_USER_04	User bookmarks a page while reading	Pass
ST_USER_05	User uses Text-to-Speech (TTS) feature	Pass
ST_USER_06	User tracks reading progress and milestones	Pass
ST_USER_07	User receives notification on milestone completion	Pass
ST_USER_08	User switches color mode (dark/light)	Pass
ST_USER_09	User gets content-based recommendations	Pass
ST_USER_10	User makes payment and unlocks premium book	Pass

1. ST_USER_01: User registers and logs in

Table 111 ST_USER_01: User registers and logs in

Field	Description
Objective	To verify that a new user can register and log in successfully
Test Steps	Navigate to register page → Enter user details → Submit registration → Login with credentials
Expected Result	User account created successfully and redirected to dashboard
Actual Result	User registered and logged in successfully
Conclusion	Pass

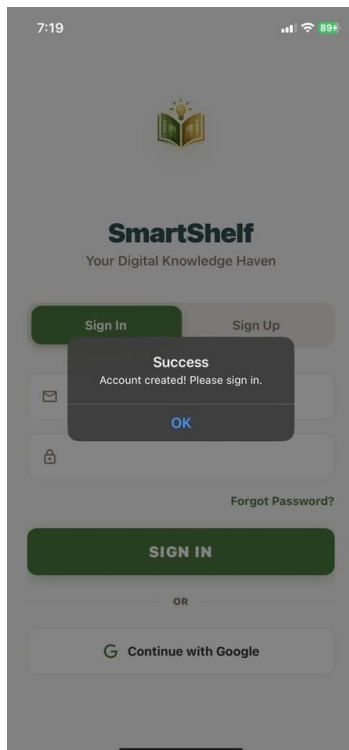


Figure 247 Register Account User

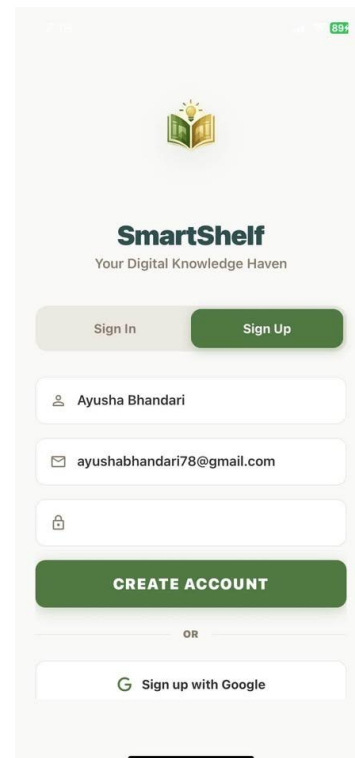


Figure 246 Register fill

2 ST_USER_02: User browses and reads book

Table 112 ST_USER_02: User browses and reads book

Field	Description
Objective	To verify that user can browse and read books
Test Steps	Login as user → Navigate to book list → Select a book → Click read
Expected Result	Book content displayed properly
Actual Result	Book opened and content displayed correctly
Conclusion	Pass

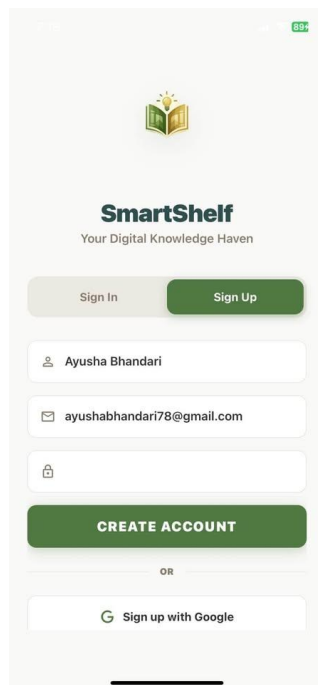


Figure 249 Login

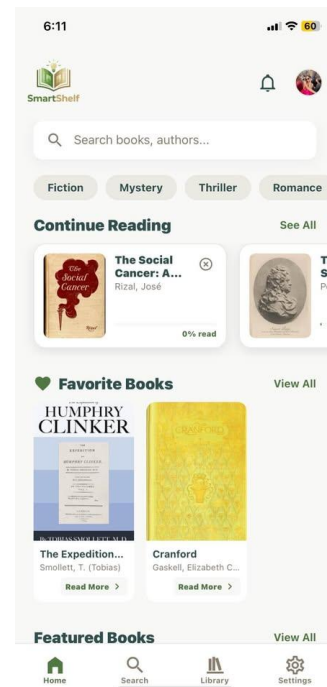


Figure 248 Open Home Page

3 ST_USER_03: Highlight and comment

Table 113 ST_USER_03: Highlight and comment

Field	Description
Objective	To verify highlight and comment functionality
Test Steps	Open a book → Select text → Click highlight → Add comment
Expected Result	Text highlighted and comment saved
Actual Result	Highlight and comment stored successfully
Conclusion	Pass

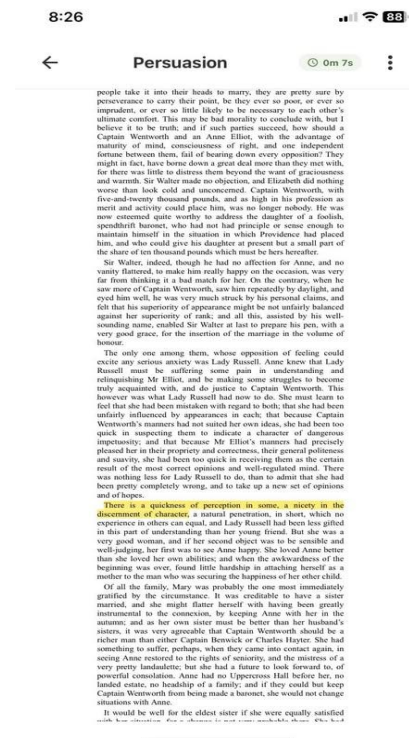
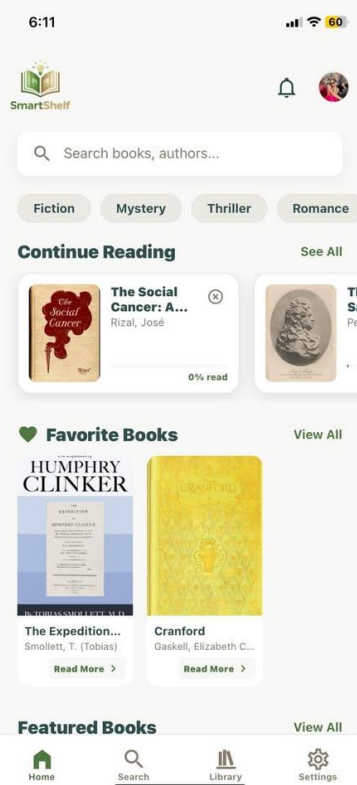


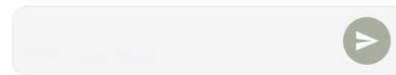
Figure 250 Set Highlight

Figure 251 Open Book

7:58 92
Noli Me Tangere by José Rizal" is a novel published in 1887 during the Spanish colonial period of the Philippines. Written by Filipino nationalist and polymath José Rizal, it exposes the injustices inflicted by the ruling government and Spanish Catholic friars on the Filipino people. The story follows Crisóstomo Ibarra, who returns home after studying in Europe to find his father dead under mysterious circumstances and his grave desecrated. As he attempts to build a school and improve his town, he faces hostility from powerful priests and deadly conspiracies that threaten his...

READ MORE

Readers' Thoughts



Anonymous 11 Apr 2026
Waaawww

You might also like ☆

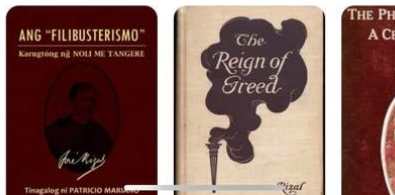


Figure 252 Add Comment

4. ST_USER_04: Bookmark page

Table 114 ST_USER_04: Bookmark page

Field	Description
Objective	To verify bookmarking feature
Test Steps	Open book → Navigate to a page → Click bookmark
Expected Result	Page saved in bookmarks
Actual Result	Bookmark added successfully
Conclusion	Pass

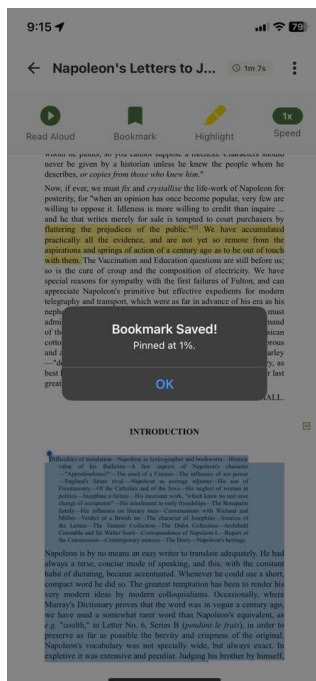


Figure 254 Bookmark

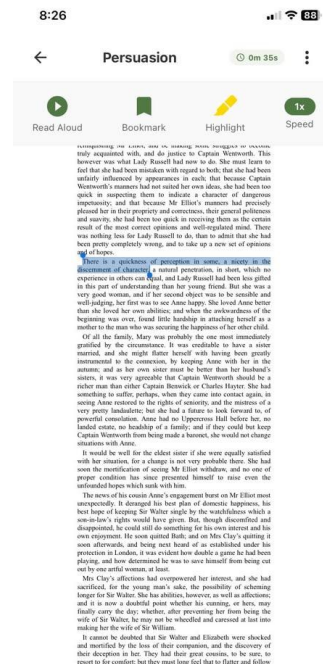


Figure 253 Set Mark

5. ST_USER_05: Text-to-Speech (TTS)

Table 115 ST_USER_05: Text-to-Speech (TTS)

Field	Description
Objective	To verify TTS functionality
Test Steps	Open book → Click TTS button → Play audio
Expected Result	System reads text aloud
Actual Result	Audio played successfully
Conclusion	Pass

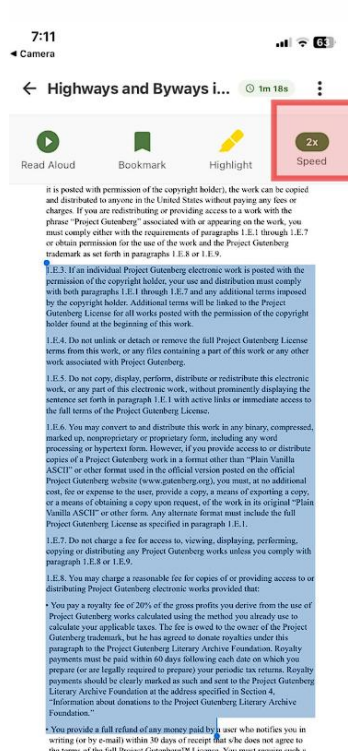


Figure 255 TTS Function

6. ST_USER_06: Reading progress & milestone

Table 116 ST_USER_06: Reading progress & milestone

Field	Description
Objective	To verify reading progress tracking
Test Steps	Read book up to 4% → Continue to 4%
Expected Result	Progress updated and milestones recorded
Actual Result	Progress tracked correctly
Conclusion	Pass

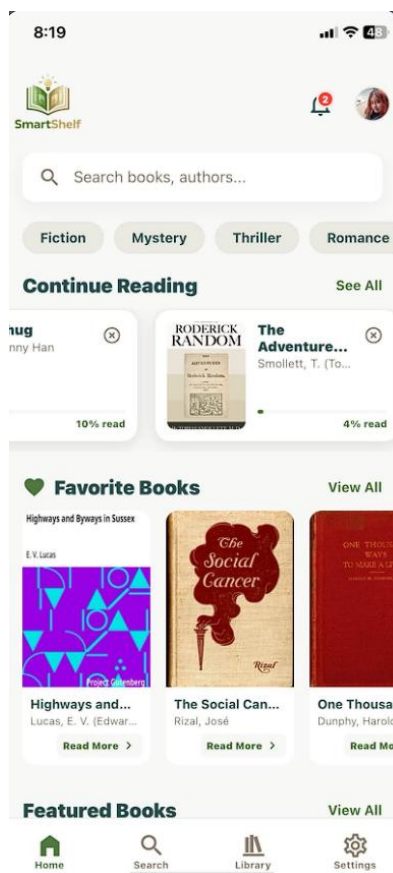


Figure 256 Reading Process

7. ST_USER_07: Notification system

Table 117 ST_USER_07: Notification system

Field	Description
Objective	To verify notifications on milestone
Test Steps	Complete milestone → Check notification
Expected Result	Notification displayed
Actual Result	Notification triggered successfully
Conclusion	Pass

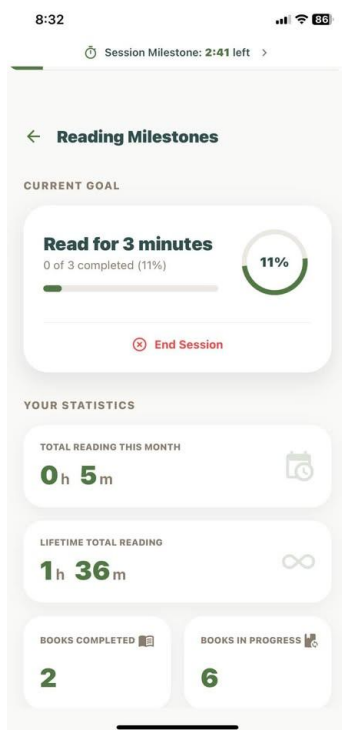


Figure 258 Milestone Set

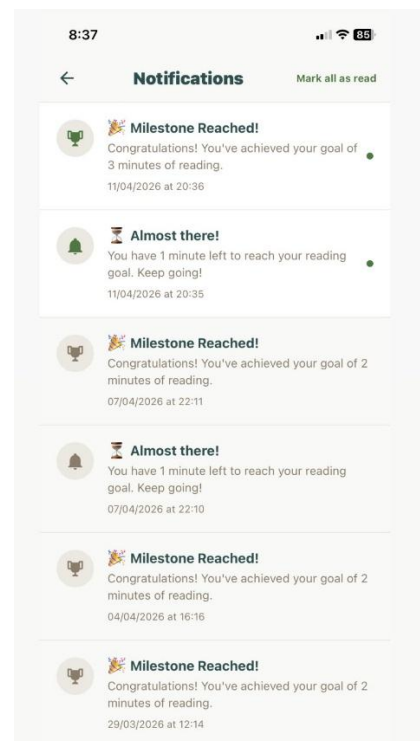


Figure 257 Notification Tigger

8. ST_USER_08: Color mode switch

Table 118 ST_USER_08: Color mode switch

Field	Description
Objective	To verify UI theme switching
Test Steps	Open reader → Switch to dark mode → Switch back to light mode
Expected Result	Theme changes correctly
Actual Result	Theme switched successfully
Conclusion	Pass

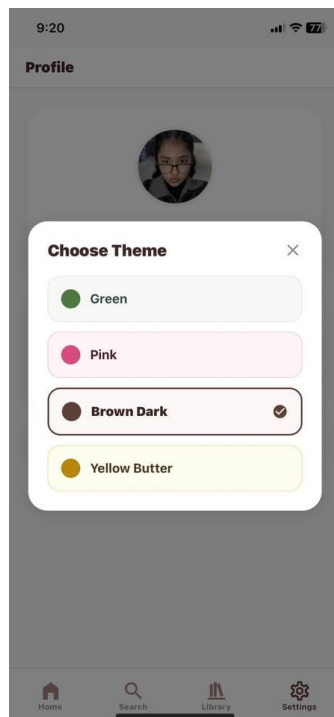


Figure 260 Color Mode

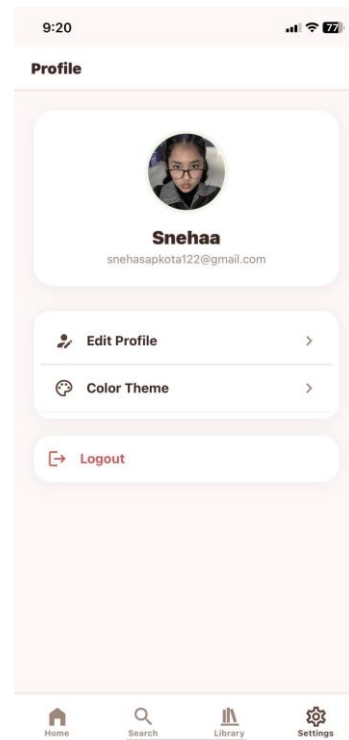


Figure 259 Color Mode Worked

9. ST_USER_09: Recommendation

Table 119 ST_USER_09: Recommendation

Field	Description
Objective	To verify recommendation system
Test Steps	Read a book → Scroll to recommendations
Expected Result	Relevant books displayed
Actual Result	Recommendations shown correctly
Conclusion	Pass

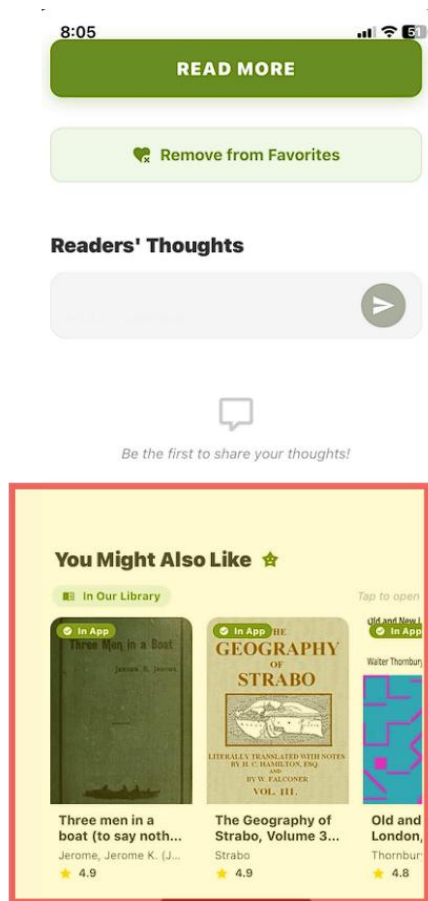


Figure 261 Recommendation

10. ST_USER_10: Payment system

Table 120 ST_USER_10: Payment system

Field	Description
Objective	To verify payment functionality
Test Steps	Select premium book → Proceed to payment → Complete payment
Expected Result	Book unlocked after payment
Actual Result	Payment successfully and book unlocked
Conclusion	Pass

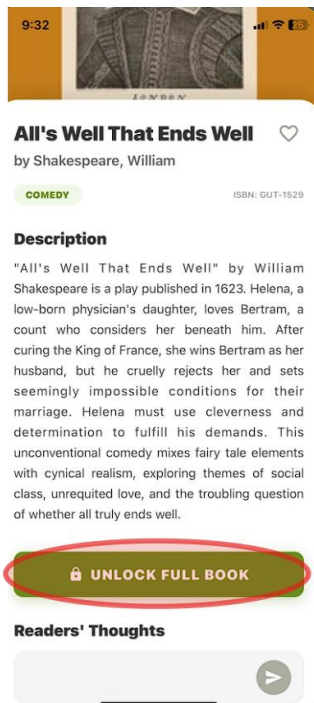


Figure 263 Unlock

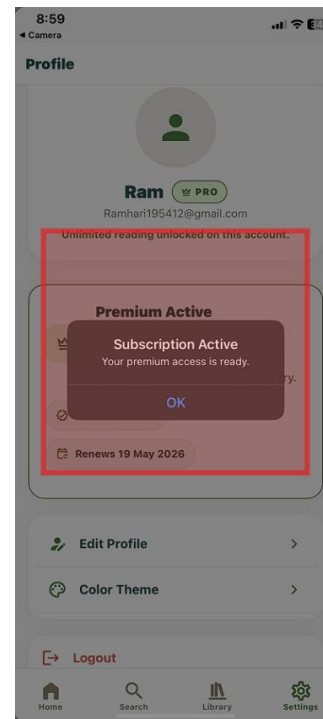


Figure 262 Subscription Active

Table: Admin Flow System Test Plan

Table 121 Table: Admin Flow System Test Plan

Test Case ID	Test Scenario	Result
ST_ADMIN_01	Admin logs in and accesses dashboard	Pass
ST_ADMIN_02	Admins manage user	Pass
ST_ADMIN_03	Admins manage book	Pass
ST_ADMIN_04	Admin logs out and session is cleared	Pass

1 ST_ADMIN_01: Admin logs in and accesses dashboard

Table 122 ST_ADMIN_01: Admin logs in and accesses dashboard

Field	Description
Test Case ID	ST_ADMIN_01
Objective	To verify that admin can successfully log in and access the dashboard
Test Steps	Navigate to admin login page → Enter valid credentials → Click login button → System redirects to dashboard
Expected Result	Admin should successfully log in and dashboard should be displayed
Actual Result	Admin logged in successfully and dashboard was displayed
Conclusion	Test was successful



Figure 264 Login Success

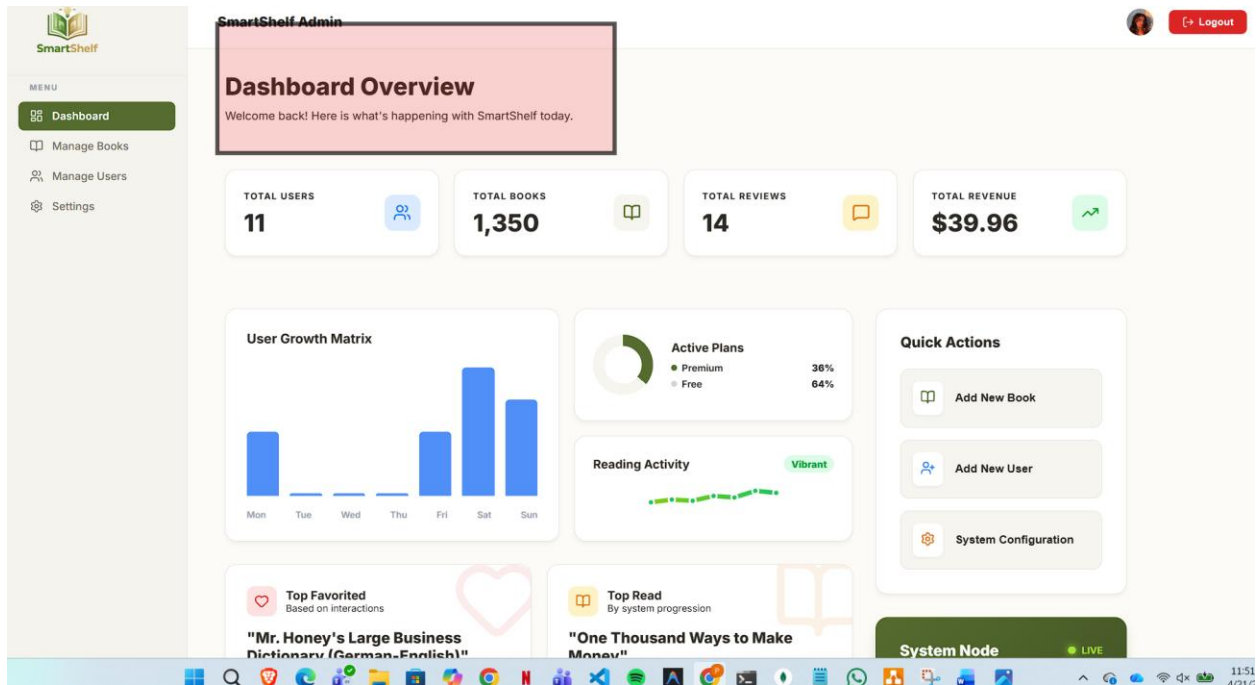


Figure 265 Open Dashboard

2 ST_ADMIN_02: Admin manages users

Table 123 ST_ADMIN_02: Admin manages users

Field	Description
Test Case ID	ST_ADMIN_02
Objective	To verify that admin can manage users (approve, view, delete)
Test Steps	Login as admin → Navigate to user management panel → View user list → Approve or block a user
Expected Result	Admin should be able to manage user accounts successfully
Actual Result	Admin successfully managed users (approve/block/view)
Conclusion	Test was successful

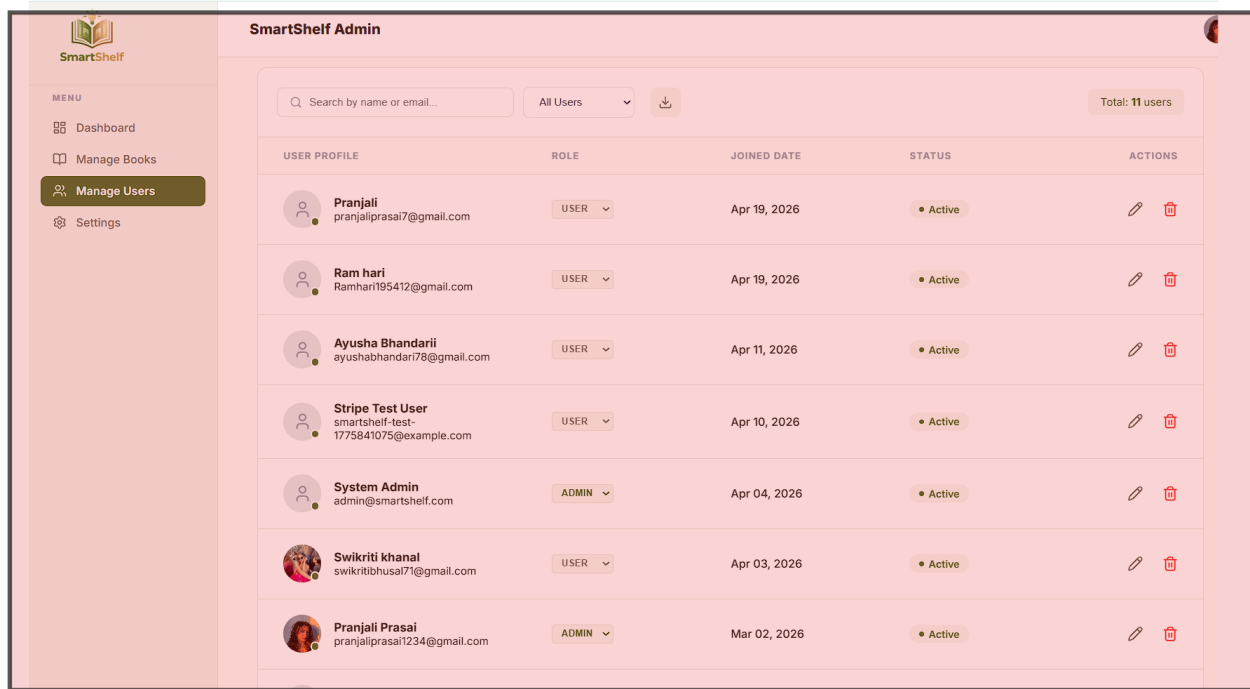


Figure 266 Manage User

The image shows a mobile application interface for editing a user profile. The form is titled "Edit User: Sapkota" and includes a close button (X) in the top right corner. The form contains the following fields:

- FULL NAME:** A text input field containing "Sapkota Thapa".
- EMAIL ADDRESS:** A text input field containing "snehasapkota37@gmail.com".
- NEW PASSWORD (LEAVE BLANK TO KEEP CURRENT):** A text input field with a masked password "*****".
- SYSTEM ROLE:** A dropdown menu currently showing "Platform User (USER)".

At the bottom of the form, there are two buttons: "Cancel" and "Save Changes". The "Save Changes" button is highlighted with a pink rectangular overlay.

Figure 267 Save Data

3 ST_ADMIN_03: Admin manages books

Table 124 ST_ADMIN_03: Admin manages books

Field	Description
Test Case ID	ST_ADMIN_03
Objective	To verify that admin can manage books (approve, reject, delete)
Test Steps	Login as admin → Navigate to book management panel → View submitted books → Approve, reject, or delete a book
Expected Result	Book status should be updated correctly
Actual Result	Books were successfully approved/rejected/deleted
Conclusion	Test was successful

The screenshot shows a web form titled "Edit Book" with a close button (X) in the top right corner. The form contains the following fields:

- TITLE:** A text input field containing "Shuggggg", which is highlighted with a red rectangular box.
- AUTHOR:** A text input field containing "Jenny Han".
- GENRE:** A text input field containing "Love".
- ISBN:** A text input field containing "978-0-241-97293-966".
- DESCRIPTION:** A text area containing multiple lines of the word "lovely".
- BOOK FILE (PDF):** A section with a "Choose file" button and the text "No file chosen".
- COVER IMAGE:** A section with a "Choose file" button and the text "No file chosen".
- Buttons:** At the bottom, there is a "Cancel" button and a green "Update Book" button.

Figure 268 Update book

COVER	TITLE & META	AUTHOR	SOURCE	ISBN-13	ACTIONS
	Shuggggg Published 1925 • Love	Jenny Han	Internal	978-0-241-97293-966	> 

Figure 269 Added Successfully

4 ST_ADMIN_04: Admin logs out and session is cleared

Table 125 ST_ADMIN_04: Admin logs out and session is cleared

Field	Description
Test Case ID	ST_ADMIN_04
Objective	To verify that admin can log out and session is cleared
Test Steps	Login as admin → Click logout button → Try accessing dashboard again
Expected Result	Admin should be logged out and session cleared
Actual Result	Admin was logged out successfully and session was cleared
Conclusion	Test was successful

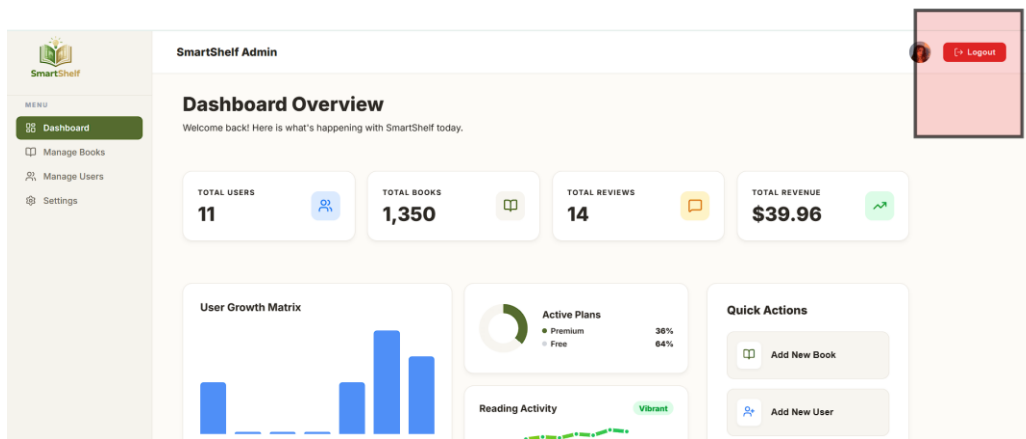


Figure 270 Logout

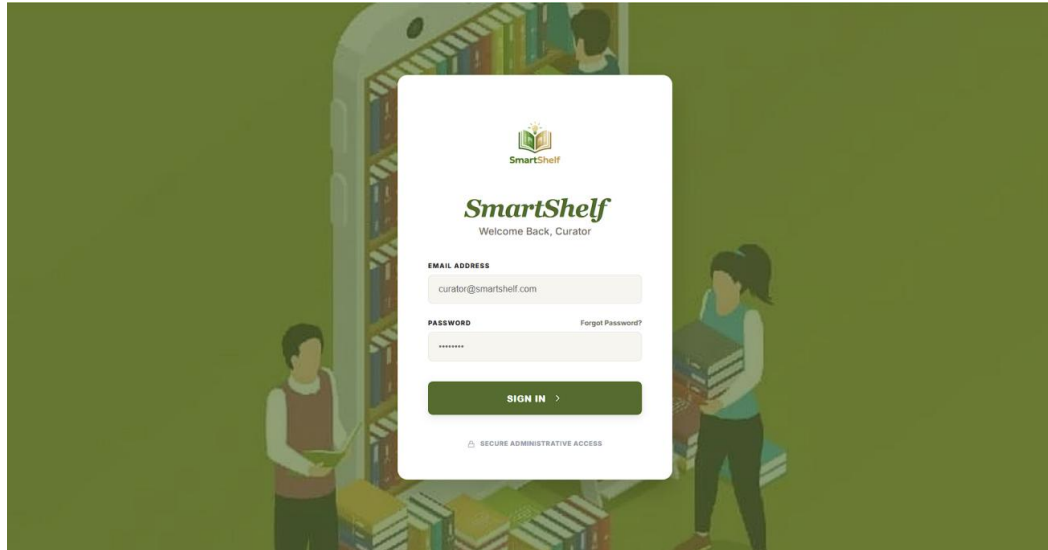


Figure 271 Redirect to Login

4.1.3 API Testing

An API (application programming interface) is “a set of routines and protocols that allow the two software systems to communicate with each other”. An API acts as a bridge that allows two software systems to communicate with each other (geeksforgeeks, 2026).

1 API_TEST_01: Register API

Table 126 API_TEST_01: Register API

Field	Description
Objective	To verify that user registration API creates a new user successfully
API Endpoint	/api/register
Method	POST
Test Data	name, email, password
Test Steps	Send POST request with valid user data
Expected Result	User is created and success response returned (status 200/201)
Actual Result	User registered successfully with response data
Conclusion	Pass

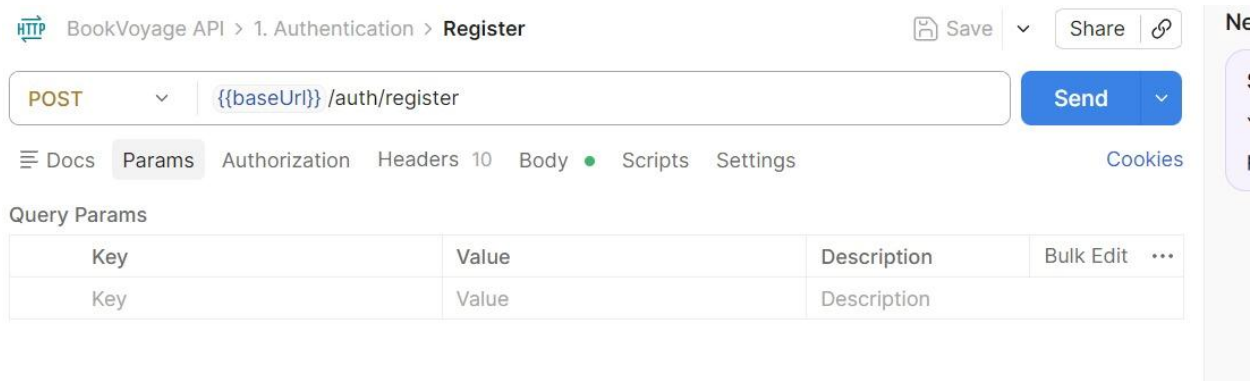


Figure 272 API Test 1

Body ▾ ↺

201 Created • 448 ms • 664 B • 🌐 | e.g. Save Response ...

{ } JSON ▾ ▶ Preview 🖼 Visualize ▾

```
1 {
2   "message": "User registered successfully",
3   "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY5ZTM4MmEwNjBmYjNkMDVlNzQzMDU3ZSIsIm1hdCI6MTc3NjUxUzNzc5MywiZmxwIjojbnZcMTIyNTkzfQ.tuzzQE1QcBks8-Cyech8KqAw4vZtBvT7lgy3izCFD04",
4   "user": {
5     "id": "69e382a060fb3d05e746057e",
6     "name": "Test User",
7     "email": "testuser@example.com",
8     "role": "user",
9     "profilePicture": null,
10    "createdAt": "2026-04-18T13:09:52.934Z"
11  }
12 }
```

Figure 273 API Test 2

2 API_TEST_02: Login API

Table 127 API_TEST_02: Login API

Field	Description
Objective	To verify user login API authentication
API Endpoint	/api/login
Method	POST
Test Data	email, password
Test Steps	Send valid login credentials
Expected Result	Token/session returned and login success message
Actual Result	User authenticated successfully and token generated
Conclusion	Pass

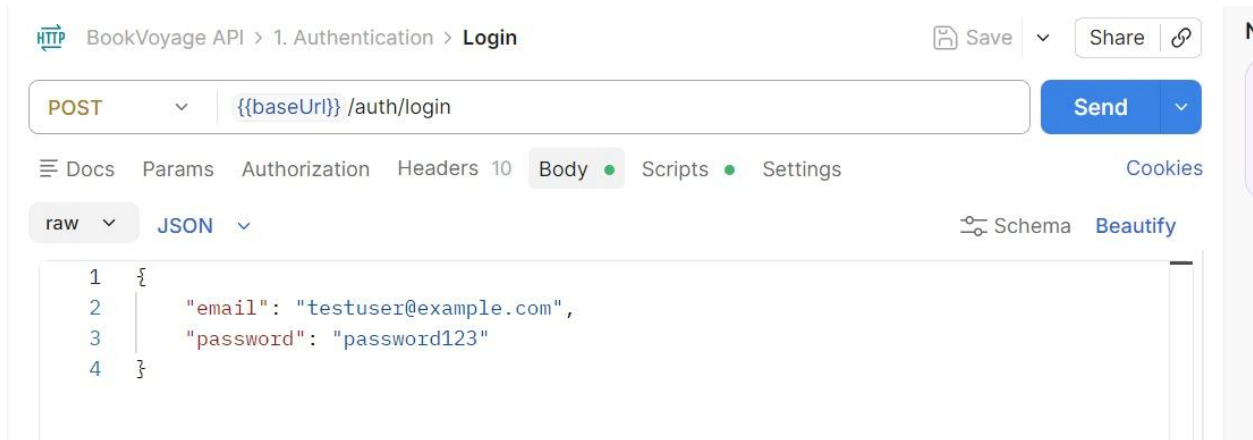


Figure 274 API Test 3

Body

200 OK • 206 ms • 647 B •

Save Response

JSON

Preview
Visualize

```

1  {
2    "message": "Login successful",
3    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY5ZTM4MmEwNjBmYjNkMDV1NzQ2MDU3ZSI6Im1hdCI6MTc3NjUxODAwMCwiZXhwIjoxNzc3MTIyODAwfQ.dxBWFGlpGZ6DEg-BKsc4_QgAbv_nZ2DPwCr7ZK63PWM",
4    "user": {
5      "id": "69e382a060fb3d05e746057e",
6      "name": "Test User",
7      "email": "testuser@example.com",
8      "role": "user",
9      "profilePicture": null,
10     "createdAt": "2026-04-18T13:09:52.934Z"
11   }
12 }

```

Figure 275 API Test 4

3 API_TEST_03: Search Book API

Table 128 API_TEST_03: Search Book API

Field	Description
Objective	To verify book search functionality via API
API Endpoint	/api/books/search
Method	GET
Test Data	query (book title/keyword)
Test Steps	Send search request with keyword
Expected Result	Matching books returned in response
Actual Result	Relevant books displayed successfully
Conclusion	Pass

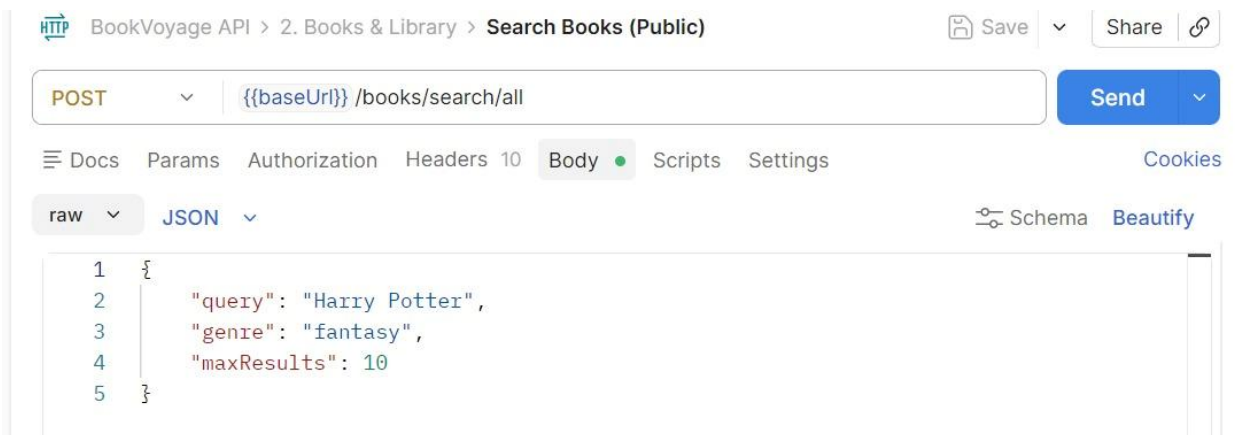


Figure 276 API Test 5

Body ▾ ↺

200 OK • 1.68 s • 4.43 KB • 🌐 | ⌨ Save Response ...

{ } JSON ▾ ▶ Preview 🖼 Visualize ▾

```

1  {
2    "message": "All sources search successful",
3    "books": {
4      "google": [],
5      "openlibrary": [
6        {
7          "title": "Harry Potter and the Philosopher's Stone",
8          "author": "J. K. Rowling",
9          "description": "No description available.",
10         "coverImage": "https://covers.openlibrary.org/b/id/15155833-M.jpg",
11         "publishedDate": "1997",
12         "publisher": null,
13         "isbn": null,
14         "pages": null,
15         "language": "lav",
16         "categories": [],
17         "rating": 0,
18         "source": "openlibrary",

```

iplete

Figure 277 API Test 6

4 API_TEST_04: Change Password API

Table 129 API_TEST_04: Change Password API

Field	Description
Objective	To verify password reset request API
API Endpoint	/api/change-password
Method	POST
Test Data	registered email
Test Steps	new password
Expected Result	Send POST request with valid old password and new password
Actual Result	Password updated successfully and confirmation response returned
Conclusion	Pass

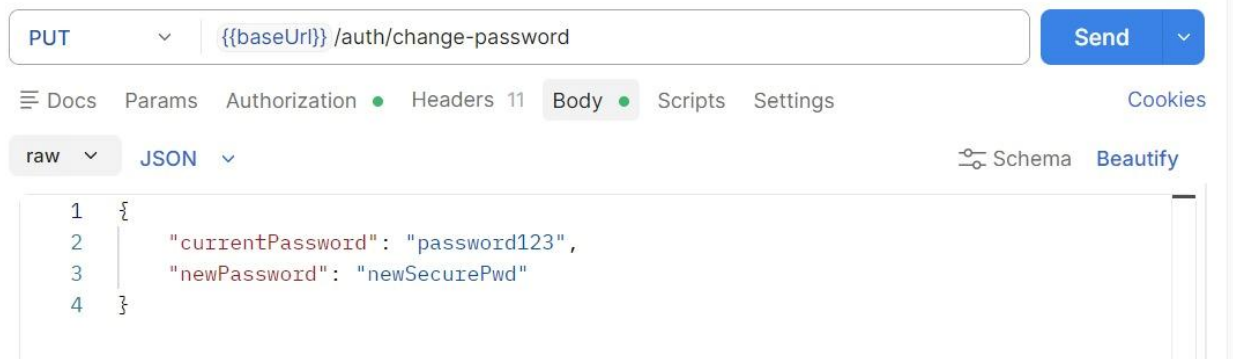


Figure 278 API Test 7

```
"message": "Password updated successfully"
```

Figure 279 API Test 8

4.2 Analysis

4.2.1 Test on selected methodology

Prior to the development of the SmartShelf system, three software development methodologies were assessed, including Waterfall, Prototype, and Agile Scrum, with Agile Scrum being chosen as the best-fit approach. SmartShelf was developed using Agile Scrum because it supports iterative development, continuous testing, and flexible adaptation to changing requirements, which were highly important for the project.

Waterfall follows a strict sequential process, and all phases like requirement analysis, designing, implementation, and testing should take place one after the other. This is because it is based on a sequential approach, making it impossible for developers to go back and change previous stages, making it ineffective for SmartShelf systems whose requirements change continuously during development. In waterfall, changes in clients' requirements following the previous phase are expensive to make and alterations difficult to implement, with testing taking place at the end of the development phase. As a result, there is a risk that major problems will be identified at a later stage of the project, which means that making the necessary corrections will be time-consuming.

The Prototype model was also looked at since it emphasizes developing the actual system to clarify the user's expectations. This helps to let the users know what to expect from the system. On the whole, adopting a prototype approach results in continuous changes and constant rebuilding of the system. Besides, the prototype is seldom to be fully structured for the system's long-term scalability and maintainability.

In contrast, Agile Scrum was the most appropriate methodology for SmartShelf given its iterative sprint-based structure. This approach enabled the development of a system in small increments with each sprint involving planning, development, testing, and review and also client feedback or requirement change could be incorporated without affecting other parts of the system. Bugs were also identified early and fixed immediately in the same development cycle, enhancing the reliability of the system.

Another major advantage of Agile Scrum in SmartShelf include the ability to always have a working version of the system. If any issue arise during development phase, the system

could be rolled back to the previous sprint, which allowed developers to detect bugs and correct them without jeopardizing and altering other completed modules.

Overall, Agile Scrum offered the best combinations of flexibility, control, and harnessing the benefits of continuous improvement for SmartShelf. It was the most effective methodology because it ensured that the system evolved according to the needs of the client, while guaranteeing stability during the development and the testing phases.

4.2.2 Failed test cases

Test Case ID	Description	Error	Fix Applied	Status
TC_ADM_01	Add book without PDF and cover image	Book was added without required files	Added validation to ensure PDF and cover image are required	Fixed
TC_ADM_02	User registration with invalid email	Invalid email format accepted	Added regex validation for email format in frontend and backend	Fix

1. Failed Test Case 4 Book Added Without File and Cover

Table 130 Failed Test Case 4 Book Added Without File and Cover

Field	Description
Objective	To identify and fix the issue where admin could add a book without uploading PDF and cover image
Error	Book was successfully added even when PDF file and cover page were not uploaded
Cause of Error	Missing validation for file inputs in backend and frontend form
Code Before Fix	Book saved without checking PDF and cover image

Fix Applied	Added validation to ensure both PDF file and cover image are required before saving
Code After Fix	Book does not save without checking PDF and cover image
Expected Result	System should prevent book creation and show error if PDF or cover is missing
Actual Result	Error message displayed and book not added without required files
Conclusion	Error fixed successfully

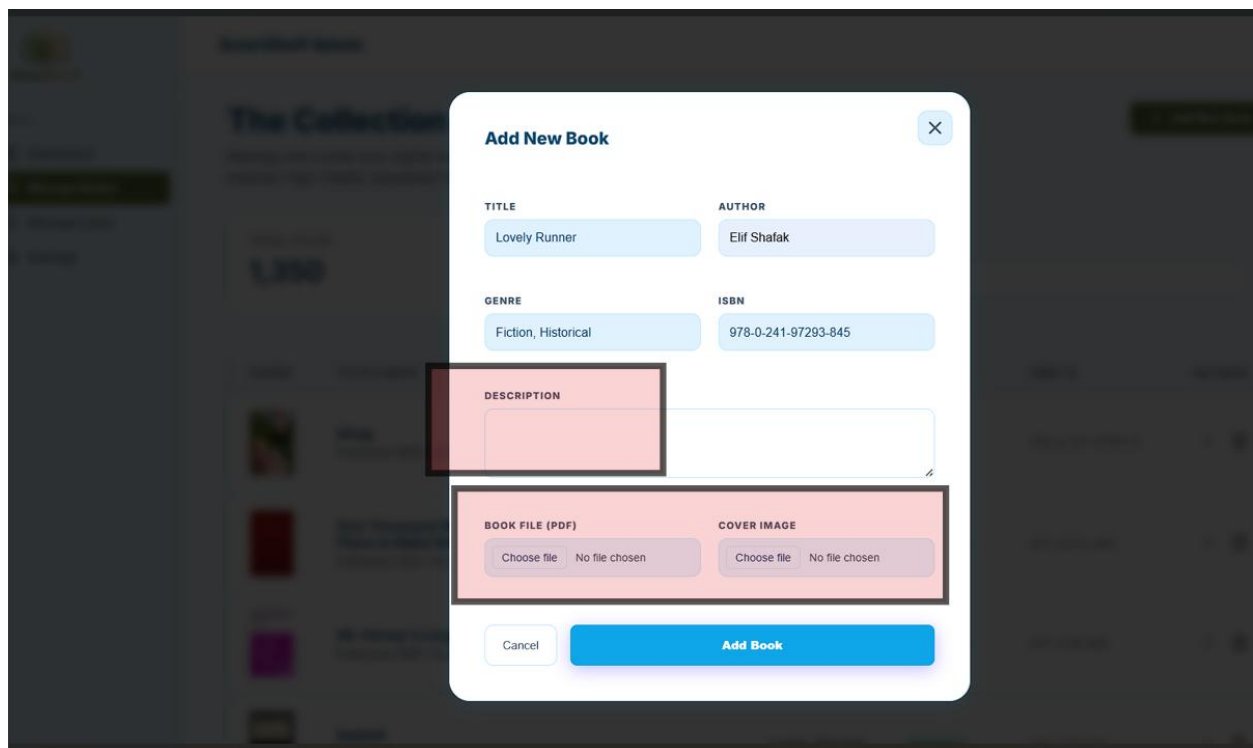


Figure 280 Book Added Without PDF

```
const handleBookFormChange = (e) => {
  setBookForm({ ...bookForm, [e.target.name]: e.target.value });
};

const resetBookForm = () => {
  setBookForm({ title: "", author: "", genre: "", description: "", isbn: "" });
  setEditingBook(null);
  setIsBookModalOpen(false);
};

const handleBookSubmit = async (data, files) => {
  if (!editingBook && !files.pdf) {
    triggerAlert("PDF file is required for new books.", "error");
    return;
  }

  const formData = new FormData();
  Object.keys(data).forEach((key) => formData.append(key, data[key]));
  if (files.pdf) formData.append("pdf", files.pdf);
  if (files.coverImage) formData.append("coverImage", files.coverImage);

  try {
    setLoading(true);
  }
};
```

Figure 281 Code Correction

Add New Book

TITLE: lovely

AUTHOR: Elif Shafak

GENRE: Childrens, Contemporary

ISBN: GUT-76878

DESCRIPTION:

BOOK FILE (PDF): Choose file No file chosen

COVER IMAGE: Choose file No file chosen

Please select a file.

Cancel Add Book

Figure 282 Correction

87

2. Failed Test Case 02 Invalid Email Accepted

Table 131 Failed Test Case 02 Invalid Email Accepted

Field	Description
Objective	To identify and fix the issue where system accepts invalid email format
Error	Email like archanasapkota@gmail.co was accepted and user was created successfully
Cause of Error	Missing proper email format validation in frontend and backend
Code Before Fix	// No email format validation applied
Fix Applied	Added regex validation to check correct email format before submission
Code After Fix	Email like archanasapkota@gmail.co not accepted and user cant be created
Expected Result	System should reject invalid email and show error message
Actual Result	Invalid email is rejected and proper message is displayed
Conclusion	Error fixed successfully

4

27

Search by name or email...		All Users		Total: 12 users
USER PROFILE	ROLE	JOINED DATE	STATUS	ACTIONS
Archana Sap archanasapkota09@gmail.co	USER	Apr 21, 2026	Active	

Figure 283 Email requires .com


```

</div>
<div className="field">
  <label style={{ fontSize: '0.7rem', fontWeight: 800, color: 'var(--muted)' }}>EMAIL ADDRESS</label>
  <input
    type="email"
    name="email"
    value={formData.email}
    onChange={onChange}
    required
    placeholder="elena.vance@smartshef.com"
    className="settings-input"
    pattern="[a-z0-9._%+-]+@[a-z0-9.-]+\.[a-z]{3,}$"
  />
  {formData.email && !/^[a-z0-9._%+-]+@[a-z0-9.-]+\.[a-z]{3,}$/i.test(formData.email) && (
    <span style={{ color: '#DC2626', fontSize: '0.65rem', fontWeight: 600, marginTop: '4px', display: 'block' }}>
      Invalid email format (must end in .com, .net, .org, etc.)
    </span>
  )}
</div>
<div className="field">
  <label style={{ fontSize: '0.7rem', fontWeight: 800, color: 'var(--muted)' }}>{editingUser ? "NEW PASSWORD (LEAVE BLANK TO KEEP CURRENT)" : "PASSWORD"}</label>
  <input type="password" name="password" value={formData.password} onChange={onChange} required={!!editingUser} placeholder="*****" />
</div>
<div className="field">
  <label style={{ fontSize: '0.7rem', fontWeight: 800, color: 'var(--muted)' }}>SYSTEM ROLE</label>
  <select name="role" value={formData.role} onChange={onChange} className="settings-input" style={{ background: 'var(--bg)', color: 'var(--text)' }}>
    <option value="user">Platform User (USER)</option>
    <option value="admin">System Curator (ADMIN)</option>
  </select>
</div>
<div className="modal-actions" style={{ marginTop: '20px' }}>
  <button type="button" className="btn-cancel" onClick={() => { onClose(); setEditingUser(&& setEditingUser(null)); }}>Cancel</button>
  <button type="submit" className="btn btn-primary" style={{ background: 'var(--primary)', color: 'var(--on-primary)', borderRadius: '8px' }}>Save</button>
</div>
</Form>

```

Figure 284 Correction Code 1

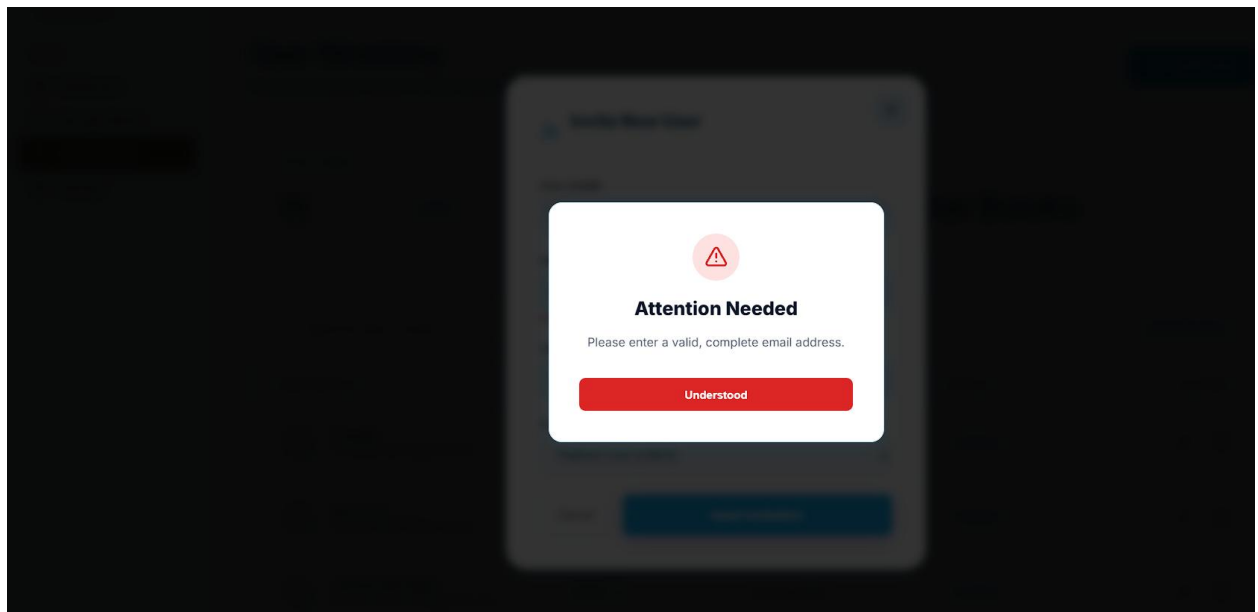


Figure 285 Correction of Error

4.2.3 UAT

[Click here for UAT](#)

4.2.4 Test summary

[Click Here for Test Summary](#)

5. Conclusion

[Click here for conclusion](#)

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6. Appendix

6.1 Overconsumption of resources

The environmental cost of producing books is huge, this production consumes large amounts of paper and energy, contributes to deforestation, and releases greenhouse gases. For example, in 2006, the US book industry used approximately 30 million trees and emitted 12.4 million metric tons of CO₂. A single 2.18 kg textbook generates 10.2 kg of CO₂, and the ink used in printing releases toxins such as toluene and xylene, which can pollute air, water and depletes the Earth's limited resources, threatens ecosystems and biodiversity, and accelerates climate change (Adhikari, 2020).

6.2 Digital Clutter and Storage Issues

Although e-books help in cutting down on physical wastage, downloading multiple book titles can overwhelm the device's memory space, thus creating digital clutter. This makes it inconvenient to manage collections, and it becomes less encouraging for the readers to navigate to more content.

6.3 Difficulty in Tracking Reading Progress

Readers often leads to the problem of tracking books, reducing motivation. Traditional books and most basic e-book platforms rarely provide effective progress-tracking tools. Research shows that e-readers with less experience may have difficulty with comprehension and engagement, which points to the need for supporting reading progress with digital tools for all users (Vuorinen, et al., 2024).

6.4 Lack of Engagement

Despite having digital access, numerous platforms do not provide interactive elements, reading objectives, or recommendations. This reduces a completely pleasurable reading experience by lowering user motivation and engagement.

6.5 Seamless digital reading

Readers using digital devices often face disruptions, slow loading times, and navigation challenges on e-book platforms which disrupts the flow of reading and is detrimental to maintaining concentration during reading. Regularity in reading habits is not easily

sustained by users. altogether reading engagement in addition enjoyment decline in the absence of a seamless, user-friendly experience.

[Click Here to go back Problem Statement](#)

6.6 Overconsumption of Resources

SmartShelf promotes e-books instead of printed books, which reduces the number of papers, inks, and electricity used, leading to a decrease in the carbon footprint and protection of trees and other natural habitat. 17trees, 4,000 kWh of energy, 7,000 gallons of water, and 60 pounds of air pollution can be saved for every ton of paper that gets recycled (Adhikari, 2020).The use of digital books means that SmartShelf eliminates printing, packaging, and shipping books, thus, reducing greenhouse gas emissions and environmental degradation. overall, e-books reduce toxins released from inks like toluene and xylene, hence by improving air and water quality and mitigating health risks linked to printed books. The survey supports this when the respondents were asked 'How often do you read e-books?' 62.4% confirmed that they already read e-books on a daily basis, thus a large majority of users have already made the shift towards digital reading as a regular practice, further complementing the aspect of a well-built digital platform that harnesses and nurtures such environmentally responsible behaviour.

2. How often do you read e-books?
173 responses

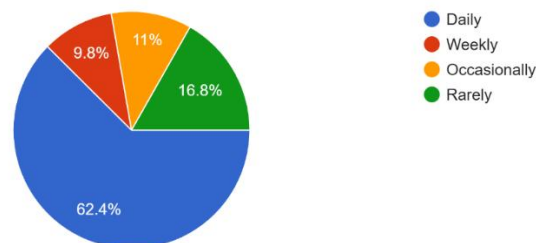


Figure 286 Question 2 Response

6.7 Digital Clutter and Storage Issues

Users can also directly read in the browser without downloading files, which is also called in browser read, helping to reduce storage problems and maintain book collections in order. SmartShelf also provides features like search which make it easy to navigate through large libraries. The developed platform makes sure smooth, efficient access to all e-books, encouraging readers to explore more content easily. The solution is directly related to a choice of devices, with 77.5% of respondents saying they prefer to read e-books on a mobile phone, where the internal memory is most limited compared to other devices. Tablets totalled 9.2%. Since the overwhelming majority of users read on mobile phones, the impact of downloading and storing multiple e-book files is keenly felt on this particular device, thus making the in-browser reading feature offered by SmartShelf a direct necessity and solution to everyday storage matters most familiar to the majority of its users.

4. Do you prefer reading e-books on?

173 responses

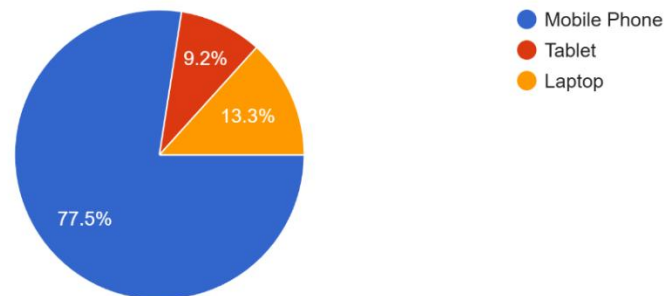


Figure 287 Question 4 Response

6.8 Difficulty in Tracking Reading Progress

SmartShelf allows users to create milestones, mark books as read and keep track of books they are currently reading. It also includes reminders, so that the reader can be more organized and complete more books, with proper motivation. The survey directly confirms the need for this solution when respondents were asked “Would additional features like Setting milestone enhance your reading experience?”, 83.2% responded yes and 13.9% responded maybe meaning that almost all users acknowledge the importance

of having structured progress-tracking and milestone features in an e-book platform. Because most of the user base reads on a daily basis ("How often do you read e-books?", 62.4% say daily), such a progress-tracking system becomes especially relevant.

7. Would additional features like Setting milestone enhance your reading experience?

173 responses

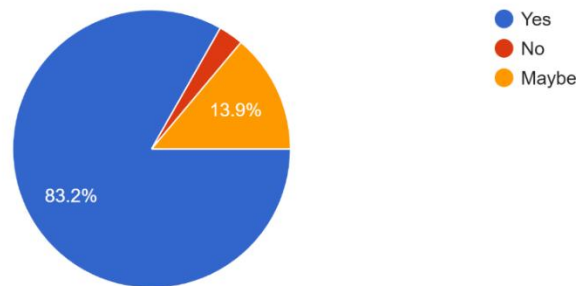


Figure 288 Question 7 Response

6.9 Lack of Engagement

SmartShelf has interactive features, goals, and recommendations for encouragement and engagement. The survey confirms that these features directly address what users need. When asked, "What feature do you find most important in an e-book system?" Text-to-Speech was identified as the single most important feature by 54.3%, and when asked "Are you aware of the Text-to-Speech (TTS) feature in e-book applications?", 82.8% said yes, showing an awareness and demand for TTS that the existing free platforms supply fail to meet. When asked "Would you like book recommendations in an e-book app?", 88.4% said yes, confirming that SmartShelf's recommendation feature meets a widely expected user need.

Additionally, in response to the question "How satisfied are you with current e-book platforms?", 79.2% reported only 'neutral' satisfaction. and when asked "would they recommend a well-designed e-book app to others?", 94.8% responded in affirmative, this statement helps to identify users' dissatisfaction with existing platforms and their readiness to embrace a more engaging solution like SmartShelf.

5. Are you aware of the Text-to-Speech (TTS) feature in e-book applications?

122 responses

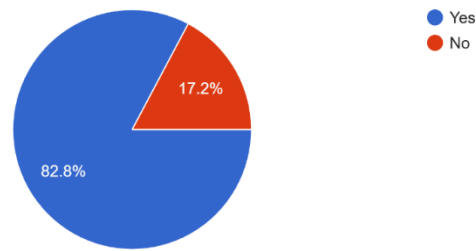


Figure 289 Question 5 Response

6. What feature do you find most important in an e-book system?

173 responses

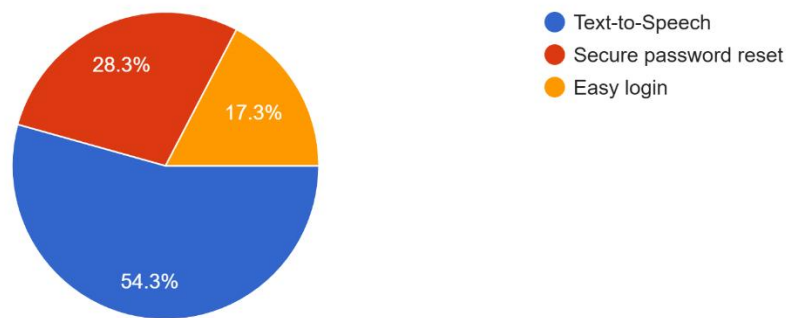


Figure 290 Question 6 Response

6.10 Seamless digital reading

SmartShelf provides the quick loading of pages, smooth navigation, and a user-friendly interface that maintains readers' attention and encourages regular reading. The survey questions most relevant to that solution prove its importance to the fullest. When asked, 'Have you ever stopped using an e-book app due to poor user experience?', 88.4% of respondents said yes, proving that poor usability directly causes users to abandon platforms entirely (ibid), a problem that the seamless interface of SmartShelf is specifically designed to solve. The most popular answer, Google Login, was chosen by 50.9% of users, showing that platform access needs to be quick and hassle-free from the first step. When respondents were asked if they would use an e-book app that offered secure logins and Google authentication, 41% said yes, and 56.6% said maybe.

[Click Here to go back Project as solution](#)

6.11 Waterfall

It is also known as Waterfall model which is sequential development process that flows like a waterfall through all phases of a project (for example, analysis, design, development, testing deployment and last maintenance) with each phase being entirely concluded before another phase starts. The main advantages of the Waterfall methodology include ensuring that requirements are well established and consistent at the beginning of each project and that it is easy to establish timeframes and measure progress (Adobe for Business, 2025).

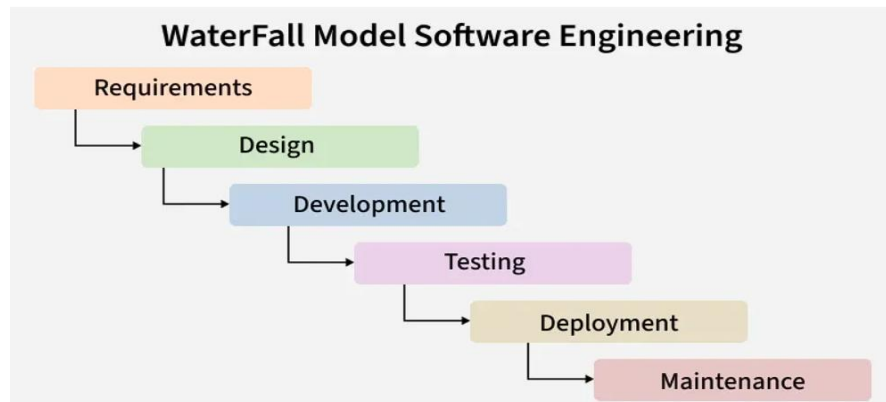


Figure 291 Waterfall Model (geeksforgeeks, 2025)

Waterfall is not ideal for SmartShelf because it is a rigid, sequential process. Once a stage is completed, we cannot go back to make changes, which makes complicated to fix mistakes and to accommodate new changes. Feedback from users also comes very late, hence the results can only be seen towards the end of the project, that way, there is a high chance that the final product will not satisfy the users' needs.

[To Go back click here Waterfall Methodology](#)

6.12 Prototype

Prototypes are meant to be used by the development team together with the customers and stakeholders to verify the product's design, architecture, and functionality (Yang, 2023).

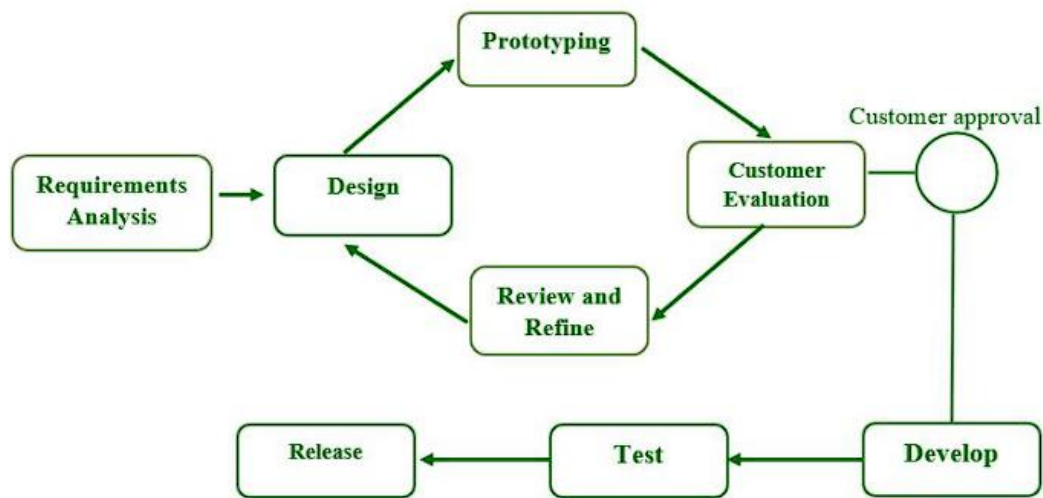


Figure 292 Prototyping Model (Geeksforgeeks, 2023)

Prototype methodology is not used for SmartShelf because this strategy involves building many representations of the system and wastes a lot of time. Although prototypes guarantee the understanding of the requirements, the continual construction and updating of prototypes may retard the development process as a whole. For a project such as SmartShelf, whose features can be developed incrementally and refined continuously through scrum, this approach is increasingly inefficient. Therefore, Prototype is not appropriate since it requires more time and workloads than scrum, which is flexible.

[To Go back click here Prototype Methodology](#)

6.13 Scrum Methodology

Scrum advocates empirical learning, self-organization while solving a problem and making new strategies concerning what went well and what did not regarding the last case. Scrum is a methodology commonly used by software developers and engineering teams to accommodate changing requirements and control project costs (Drumond, 2025).

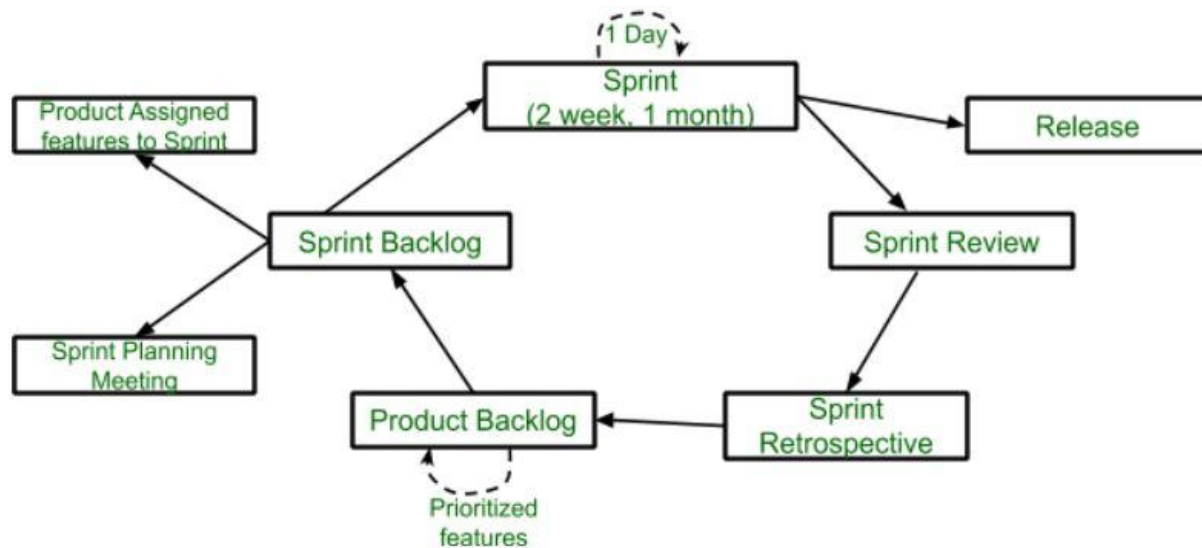


Figure 293 Scrum (GeeksforGeeks, 2025)

Advantages of Scrum

- They are more flexible and money efficient.
- Large projects are divided into easily manageable sprints.
- Helps to complete projects on time.
- For fast moving project it works smoothly (GeeksforGeeks, 2025).

Scrum was used because it works well on SmartShelf by allowing the project to be developed small in manageable chunks, which is known as sprints. Each sprint would focus on the completion of features such as login, book search, reading, text-to-speech, and recommendations. This makes tracking easier, frequent testing of the features, and quick changes in case something is not right. Scrum is the best suited because it allows for flexibility, continuous improvement, and timely delivery of quality features. Unlike others, which include Waterfall or Prototype, Scrum doesn't need to wait until the end to see results, thus keeping development efficient and responsive to user needs.

[To Go back click here Scrum Methodology](#)

6.14 High Level Usecase Description

The high-level use case is a brief depiction of the work in one paragraph of unstructured text. Its objective is to provide us with the necessary details to understand how difficult it is and to simplify the alloy of related use cases.

Register

- **Actor(s):** User
- **Description:** It helps users to make an account in.

Login

- **Actor(s):** User
- **Description:** It allows a registered user to access the SmatShelf.

Login Verification

- **Actor(s):** System
- **Description:** It verifies the login credentials which is provided by the user.

Login Failed

- **Actor(s):** User
- **Description:** It displays an error message when login is unsuccessful when user input incorrect credentials.

Search Books

- **Actor(s):** User
- **Description:** It helps users of smartshelf to search books which they wanted to read according to their input such as genre, author and title.

Read E-Book

- **Actor(s):** User
- **Description:** It allows users to read selected e-books in the browser without downloading.

Text-to-Speech (TTS)

- **Actor(s):** User
- **Description:** It converts e-book text into audio format for example user will select the text and they can use TTS in select sentence.

7

Track Reading Progress

- **Actor(s):** User
- **Description:** It tracks and saves the user's reading progress where user is read.

View Reading History

- **Actor(s):** User
- **Description:** It shows users' history, for example what user read.

Set Reading Goals

4

- **Actor(s):** User
- **Description:** It helps users to set their reading targets which help users to concentrate while reading book also set motivation.

Get Recommendations

- **Actor(s):** User
- **Description:** It helps to user by giving book recommendations to read book according to their history for example according to author and genre.

For Admin Use Cases

Add Books

- **Actor(s):** Admin
- **Description:** It helps to add new books.

Update Books

- **Actor(s):** Admin
- **Description:** It helps to update books.

Delete Books

- **Actor(s):** Admin
- **Description:** It helps to delete books.

Manage Users

- **Actor(s):** Admin
- **Description:** It helps to manage user accounts.

[For going back click here](#)

6.15 Expanded Level Usecase Description

Expanded level usecase description is a more detailed and structured than the high-level use case, which shows us the use case's start, the actors, the data input, and the case result.

1. Register

- **Actor(s):** User
- **Description:** It allows a new user to create an account in the SmartShelf.
- **Preconditions:** User is not already registered.
- **Postconditions:** A new user account has been successfully created.

Main Flow:

1. User selects the option of registering before logging in.
2. Users need to insert registration details.
3. System checks information provided by users.
4. System make a user account.

Alternative Flow:

- If invalid details are entered, the system displays an error message.

2. Login

- **Actor(s):** User
- **Description:** It allows a registered user to access the SmatShelf.
- **Preconditions:** User already registered.
- **Postconditions:** User logged into the system.

Main Flow:

1. Users enter login credentials.

2. System verifies the credentials.

3. User is get access.

Alternative Flow:

- If credentials are incorrect, the login fails.

3. Login Verification

- **Actor(s):** System
- **Description:** It verifies the login credentials which is provided by the user.
- **Preconditions:** User gives login details.
- **Postconditions:** Credentials validated successfully.

Main Flow:

- System checks the username and password.
- System confirms that given credentials are valid or not valid.

4. Login Failed

- **Actor(s):** User
- **Description:** It displays an error message when login is unsuccessful when user input incorrect credentials.
- **Preconditions:** User enters invalid login details.
- **Postconditions:** User remains logged out.

5. Search Books

- **Actor(s):** User
- **Description:** It allows users to search for books based on title, author, or genre.
- **Preconditions:** User is logged in.
- **Postconditions:** Match books display.

Main Flow:

- Users search the book based on author and genre.
- System looks the book database.
- Searches book for showing to the user.

6. Read E-Book

4

- **Actor(s):** User
- **Description:** It helps users to read selected e-books in the browser without downloading.
- **Preconditions:** User login and book exist.
- **Postconditions:** E-book is displayed.

7. Text-to-Speech (TTS)

- **Actor(s):** User
- **Description:** It converts e-book text into audio for example user will select the text and they can use TTS in select sentence.
- **Preconditions:** While user reading books and selects text.
- **Postconditions:** Selected text played as audio.

Main Flow:

- User selects text if he or she doesn't want to read.
- User use TTS.
- System converts selected sentences to tts.

8. Track Reading Progress

- **Actor(s):** User
- **Description:** It tracks as well as it will saves the user progress when user read.

- **Preconditions:** User is reading an e-book.
- **Postconditions:** Reading progress is saved.

9. View Reading History

- **Actor(s):** User
- **Description:** It displays past reading activities, for example what users read.
- **Preconditions:** Reading past data exists.
- **Postconditions:** Reading history will show.

10. Set Reading Goals

- **Actor(s):** User
- **Description:** It helps users to set their reading targets which help users to concentrate while reading book also set motivation.
- **Preconditions:** User is login.
- **Postconditions:** Reading milestones will save.

11. Get Recommendations

- **Actor(s):** User
- **Description:** It provides personalized book recommendations for user while help to read book according to their past read history for example according to author and genre.
- **Preconditions:** User is logged in.
- **Postconditions:** Recommended books are displayed.

Admin Expanded Use Cases

7

12. Add Books

- **Actor(s):** Admin
- **Description:** It adds new books.
- **Preconditions:** Admin is login.
- **Postconditions:** New book added.

13. Update Books

- **Actor(s):** Admin
- **Description:** It updates books.
- **Preconditions:** Book exists in the system.
- **Postconditions:** Book details updated.

14. Delete Books

- **Actor(s):** Admin
- **Description:** It deletes the books.
- **Preconditions:** Book exists in the system.
- **Postconditions:** Book is deleted.

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15. Manage Users

- **Actor(s):** Admin
- **Description:** It manages users.
- **Preconditions:** Admin is login.
- **Postconditions:** User account data is updated.

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6.16 Report Structure

Overall, this report is here to show us the SmartShelf project in a clear and logical way, by including the background of project, project development process, project progress analysis, and project future improvement.

6.16.1 Background

This part demonstrated an overall SmartShelf project by explaining its problem context and reason to develop SmartShelf. It has the client as an IT office environment and describes the intended end users. Additionally, this part reviews existing apps e-book reading to highlight their problems and justify why and what we need for the SmartShelf.

6.16.2 Development

This part demonstrated the development of the SmartShelf system. It talks about the software development methodology used, especially use of scrum methodology from agile, the part tells us how different system features were developed through iterative sprints.

6.16.3 Testing

This part analyses the overall progress of the SmartShelf project by comparing finished work with the planned milestones as well as SmartShelf Gantt chart. It points out finish task as well as ongoing tasks, then identifies any challenges that have been faced, and it describes how scrum were used to manage maintain as well as progress the SmartShelf project timeline also all the resting has benn completed.

6.16.4 Conclusion

SmartShelf project in a clear and logical way, by including the background of project, project development process, project progress analysis, and project future improvement. This part tells us that the incomplete tasks need to be fully complete on the SmartShelf and shows us potential future improvement. It mainly focuses on improving system performance, usability, security, and scalability beyond the current implementation.

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6.17 Test summary

49 Testing the SmartShelf system was done through three main testing levels, which included White Box Testing (unit testing), Black Box Testing (system testing), as well as API Testing, these testing methods were focused on the internal logic of the system and its external functionalities against the requirements. All core modules like user auth, search, comments, favourites, highlights, milestones, notifications, TTS, bookmarks, theme, user profile, recommendations, reading progress, subscription, admin operations was tested at a code-level during White Box Testing, More than 80 unit test cases have been run in total, all of which passed after the necessary minor issues were fixed, which makes sure the internal logic, flow of data, and error handling mechanisms were working as expected.

With Black Box Testing, entire workflows for users and administrators were tested without consideration of the internal codes. Key functionalities were also validated like registration of users, reading books, highlighting, bookmarking, payment, admin managing, and operation for the dashboard. All the system test cases were passed successfully, confirming the system according to users' requirements and expectations of the system's output.

Now, API Testing, important APIs such as register, login, search book, and change password with different inputs. These APIs returned correct responses with proper status codes and makes sure smooth communication between frontend and backend.

Failed test cases like add a book without a PDF or cover, incorrect email address being accepted (validation), and delete without refreshing the list (UI/behavioral). These issues solved by adding proper validation, proper responses from the API, and frontend logic changes.

After applying fixes, all failed test cases worked fine. As a whole, a success grade was given to the system after testing, with all system functionalities tested fulfilling the expected outcomes, post rectification of detected bugs. The testing process improved the

system's dependability, efficiency, and user-friendliness, by guaranteeing the end product is, safe, and ready.

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6.18 UAT

User Acceptance Testing (UAT) was held to verify that the developed system met all the functional and non-functional requirements defined by the client Grishal Dahal (Datum Pvt. The testing was done on a practical scenario to make sure the system is easy to use, dependable, and carried out all the required operations correctly.

During the UAT phase, all critical aspects of the system were tested, which include:

- Admin login and authentication.
- User login and registration
- Book management (add, update, delete, and view books).
- PDF uploading, cover image uploading validation
- Book listing and displaying.
- Search and view book details
- Comment and interactive features.
- Highlight and bookmark functionality.
- Reading progress tracking
- Notification.
- OTP Verification system.
- Payment integration system.
- User and content management by the admin.

All above features were tested to work as required without errors, validation checks, including required fields, email validity, and file upload restrictions, were also verified during testing. After the completing the testing process, the system was reviewed by the client. The client confirmed that all the required features were successfully implemented and that the system is functioning according to the expected requirements.

The system passed User Acceptance Testing (UAT). The client Grishal Dahal (DatumSystem Pvt. Ltd.) approved the system after confirming that all the features work correctly.

[Click here to go back to User Acceptance Testing](#)

6.18.1 Client Report

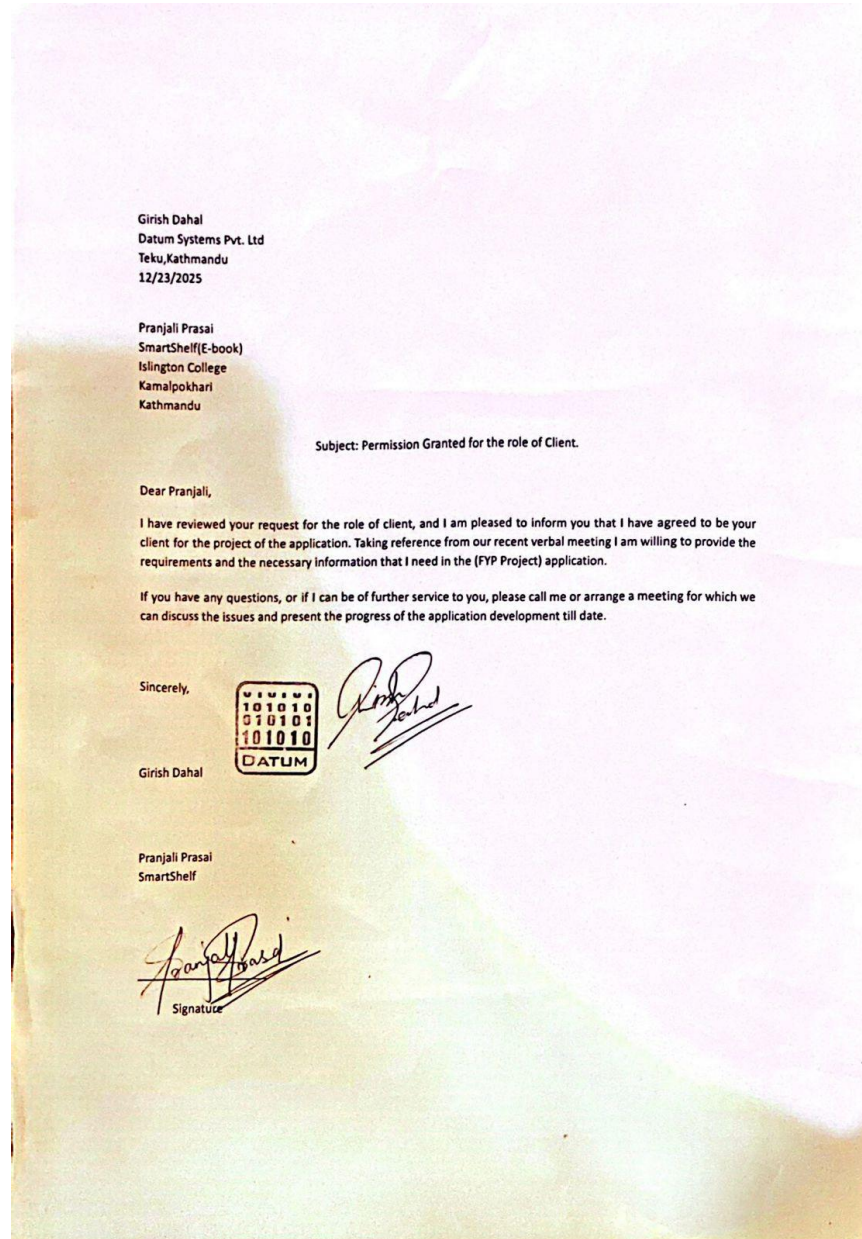


Figure 294 Client Report

6.19 Conclusion

6.19.1 Legal Issues

SmartShelf must follow legal requirements to operate safely and lawfully.

1. Copyright and Intellectual Property

The book which are available on SmartShelf are protected by copyright laws. System cannot upload or share books without proper permission from authors or publishers. If copyrighted content is used without authorization, it may violate laws such as the Copyright Act 1976 and international agreements like the Berne Convention, Copyright Act of 1976 provide legal rights to creators such as authors, artists, musicians, software developers, over their original works. It protects works like books, music as well as artwork from being copied or used without permission (copyright.gov, 2026). therefore, smart shelf used only licensed, or public-domain books only .

2. Data Protection and Privacy

SmartShelf gather user data which include user email, passwords, reading history, as well as payment details, therefore such data must be stored safe and securely and used responsibly. The system should follow regulations like the General Data Protection Regulation which is also known as GDPR. General Data Protection Regulation is a European Union law that governs how organizations and handle the personal data (IBM, 2026).

So Smartshelf always Encrypt user data and doesnot share do not share personal data without permission of users

3. Payment and Financial Security

In addition, because the platform has online payment functions Stripe, it is necessary to ensure the security of monetary transactions by relying on reliable payment gateways and meeting payment security standards.

- Use secure payment gateways
- Protect card details
- Ensure safe transactions

4. User Authentication and Security

SmartShelf must prevent unauthorized access to user accounts because weak security can lead to hacking or data breaches, we can see example of data breach of Yahoo which one of the worst and most infamous cases of a known cyberattack and currently holds the record for the most people affected (Chin, 2025). It affects more the 3 billion people. So SmartShlef use strong password encryption, implement secure login (JWT, OTP) and protect user accounts.

5. Terms and Conditions / User Agreement

The system should clearly define rules for users, including usage policies, subscriptions, and responsibilities by provide Terms & Condition and inform users about system usage and restrictions

6.19.2 Social Issue

SmartShelf has a lot of social implications.

On the positive side, it encourages digital literacy, and improves access to knowledge, especially for students and professionals. To enhance accessibility for users with impaired eyesight, assistive technologies like Text-to-Speech (TTS).

However, there are also challenges which is the digital divide, whereby people lacking access to smartphones or stable internet connections cannot benefit from the system.

This brings inequality in accessing educational materials, reduction in physical interaction is another concern since digital reading might substitute traditional library settings for learners, thus harming social learning.

6.19.3 Ethical Issue

Ethical considerations are paramount in the development and eventual use of the SmartShelf system. A major concern is the violation of the privacy rights, given that the system gathers sensitive user data, including personal details, reading patterns, as well as payment information. Such ethical issues can arise if the data is shared or misused without the users' consent, leading to loss of trust. Accordingly, there is a need to ensure data protection, confidentiality, and informed consent.

Another crucial issue is the violation of virtue theory based on moral principles such as honesty, fairness, and responsibility by Aristotle. If SmartShelf allows unauthorized access to copyrighted contents or manipulates recommendations for profits, it would violate these moral values (Christine Swanton, 2026).

In addition, there is bias creation, and the limitation of content diversity in recommendation systems that can affect fairness and user autonomy. Therefore, ethics in privacy protection, fairness, and responsible system design is crucial to developing a trustworthy system.

6.19.4 Limitations of the System

However, SmartShelf also has a few drawbacks here are some limitations:

- SmartShelf may cause digital fatigue due to prolonged screen time.
- SmartShelf needs internet connectivity, based on offline usability.
- SmartShelf has certain free access for user, example only 8 books before subscription are required.
- Performance depends on device capability and network speed.

These limitations highlight areas for improvement in future development.

6.19.5 Future Work

Future enhancements that can significantly improve the SmartShelf system are mentioned below:

- Offline reading mode
- Improve AI-based recommendation system for better personalization
- Multi-language support integration
- Improve security features such as biometric login and as well as advanced encryption.
- Integrate collaborative reading features such as shared notes, group discussions

These improvements will make the system more scalable, inclusive, and user-friendly.

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6.20 Trello

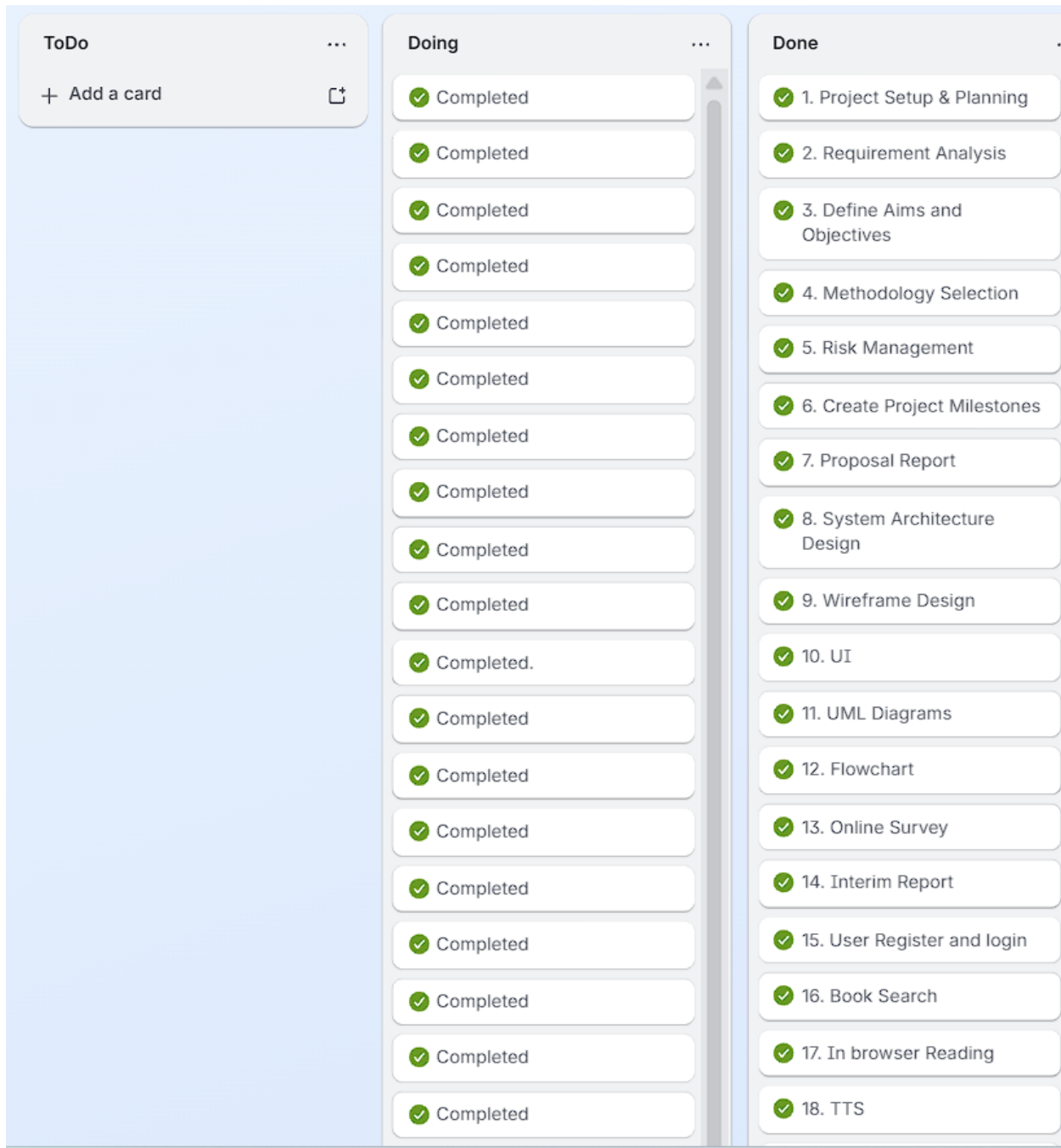


Figure 295 Trello

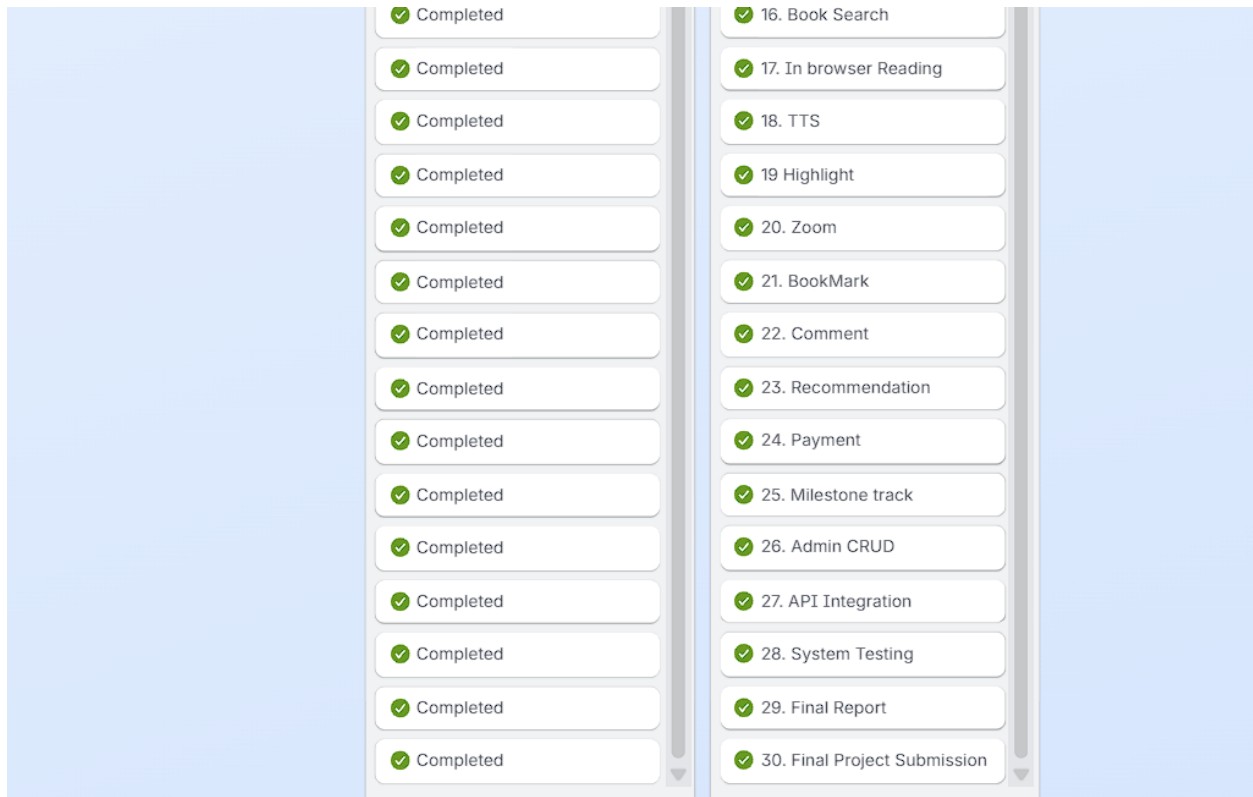


Figure 296 Trello 1

6.21 Code Logic

```

if (data.type === "READY") {
  if (webViewReadyRef.current) {
    setWebViewReady(true);
    setTimeout(() => restoreHighlights(highlightsRef.current), 500);
  }
} else if (data.type === "PDF_READY") {
  setPdfRendered(true);
} else if (data.type === "CONSOLE_LOG") {
  console.log("[WebView]:", data.data);
} else if (data.type === "TEXT_EXTRACTED" && data.text) {
  const clean = data.text.replace(/\\s+/g, " ").replace(/Loading PDF\\.\\.\\.\\/g, "").trim();
  if (clean) { setFullText(clean); setCurrentCharIndex(0); startSpeech(clean, 0, rate); }
} else if (data.type === "HIGHLIGHT_CREATED") {
  if (!bookId) return;
  const res = await saveHighlight({ bookId, text: data.text, rangeData: data.rangeData, color: "yellow" });
  if (res.success && res.data?.id) {
    setHighlights(prev => [...prev, res.data]);
    webViewRef.current?.injectJavaScript(`window._ssPatchId('${res.data.id}'); true;`);
  }
} else if (data.type === "DELETE_HIGHLIGHT_REQUEST") {
  if (!data.id || !highlightsRef.current.find(h => h.id === data.id)) return;
  Alert.alert("Remove Highlight", "Remove this highlight?", [
    { text: "Cancel", style: "cancel" },
    {
      text: "Remove", style: "destructive", onPress: async () => {
        setHighlights(prev => prev.filter(h => h.id !== data.id));
        const res = await deleteHighlight(data.id);
        if (res.success) {
          webViewRef.current?.injectJavaScript(`window._ssRemove('${data.id}'); true;`);
        } else { fetchHighlights(); }
      }
    }
  ], { cancelable: true });
},

```

Figure 297 Highlight

```

const maxPct = maxProgressRef.current;

// Accidental scroll: user scrolled back more than 2% from their furthest point
if (maxPct > 2 && pct < maxPct - 2 && !showingScrollDialogRef.current) {
  showingScrollDialogRef.current = true;
  Alert.alert(
    "SmartShelf: Accidental Scroll?",
    `You reached ${Math.round(maxPct)}% earlier, but are at ${Math.round(pct)}% now. Which should we save?`,
    [
      {
        text: `Save Current (${Math.round(pct)}%)`,
        onPress: () => {
          maxProgressRef.current = pct;
          saveCurrentProgress(pct, true);
          showingScrollDialogRef.current = false;
        }
      },
      {
        text: `Keep Max (${Math.round(maxPct)}%)`,
        onPress: () => {
          saveCurrentProgress(maxPct, true);
          showingScrollDialogRef.current = false;
        }
      },
      {
        text: "Cancel",
        style: "cancel",
        onPress: () => { showingScrollDialogRef.current = false; }
      }
    ],
    { cancelable: true }
  );
} else {
  maxProgressRef.current = Math.max(maxPct, pct);
  if (Math.abs(pct - lastSavedProgress.current) > 1) saveCurrentProgress(pct);
}

```

Figure 298 Bookmark

6.22 Git

<https://github.com/pranjaliprasai/e-book>